

Evidence Report:

Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes

Human Research Program

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V. PRD RISK TITLE: RISK OF CARDIOVASCULAR ADAPTATIONS CONTRIBUTING TO ADVERSE MISSION PERFORMANCE AND HEALTH OUTCOMES

Description: Given that exposure to the spaceflight environment can contribute to cardiovascular deconditioning, dysfunction, and remodeling, there is a possibility that crews will experience impaired performance and negative cardiovascular health outcomes during and after spaceflight and planetary operations.

VI. STATUS

- *Active:* Work/research is currently being done towards this risk.

VII. EXECUTIVE SUMMARY

The Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes is the integration of three previous risks defined by the National Aeronautics and Space Administration's (NASA) Human System Risk Board (HSRB): Risk of Orthostatic Intolerance During Re-Exposure to Gravity, Risk of Cardiac Rhythm Problems During Spaceflight, and Risk of Cardiovascular Disease and Other Degenerative Tissue Effects from Radiation Exposure. This integrated cardiovascular risk was officially accepted by the Human Systems Risk Board (HSRB) in 2021, and the Human Research Program (HRP) is aligning research priorities accordingly.

Manifestations of the cardiovascular risk vary depending upon the Design Reference Mission (DRM) category, specifically the length of mission and the distance from Earth. Development of dysrhythmias and venous thromboembolism (VTE), which has been reported recently, are the primary concerns during a spaceflight mission. However, while there have been anecdotal reports and some documented cases of cardiac dysrhythmias during and after spaceflight, it is inconclusive whether spaceflight exposure causes or increases the frequency of these. An incidental finding of VTE was reported, and NASA has initiated a targeted ultrasound surveillance protocol in International Space Station (ISS) crewmembers. Orthostatic intolerance after short- and long-duration spaceflight in low Earth orbit (LEO) is well-documented and is considered the primary risk during re-exposure to acceleration, including re-entry and landing on Earth. Orthostatic intolerance also may be a concern when astronauts are exposed to head-to-foot accelerations during descent to and ascent from the Moon and Mars. Cardiovascular disease (CVD) is considered the primary long-term health risk and concern for flight certification, especially for long-duration missions beyond LEO when exposure to space radiation will be greatest. A recent analysis of astronaut health records demonstrated that the risk of CVD death in astronauts is not greater than that of a well-matched cohort, but there is a statistically significant greater lifetime risk of a cardiovascular event. Current terrestrial data suggest that the lifetime risk of CVD is statistically greater than with normal aging for exposures above 0.5 Gy-Eq (e.g., Mars planetary DRM). However, risks to the cardiovascular system from space radiation exposure below 0.5 Gy-Eq are not well understood. The interaction between weightlessness or altered gravity and prolonged exposures to space radiation beyond LEO also are not yet known.

This evidence report will address the current state of knowledge in the arena of acute and chronic cardiovascular adaptations to spaceflight and in the context of mission phase (in- and post-mission), mission duration (weeks, months, and years) and mission characteristics (gravity transitions, altered gravity levels, and radiation exposure). This document is not intended to be inclusive of all cardiovascular findings in astronauts during and after spaceflight and in test subjects participating in spaceflight analogs but will address published scientific literature specific to the currently defined performance and health risks. Evidence reports evolve and expand based upon new findings as they become available.

VIII. INTRODUCTION

Among the concerns of early spaceflight mission planners and medical experts was how well the human body would tolerate launch and landing of the space vehicle and whether the cardiovascular system would effectively function during high accelerations and in weightlessness. Early spaceflight objectives were directed towards launching humans successfully into space and safely returning them home. Critical to these objectives were measurements of cardiovascular structure and function before, during, and soon after landing to characterize the effects of gravity transitions and adaptations to the spaceflight environment in preparation for increasingly longer missions in lower Earth orbit (LEO) and later to the Moon. Clinical outcomes and physiological assessments were the cornerstone of knowledge during the early spaceflight programs, which when combined with later targeted research in weightlessness and spaceflight analogs as well as terrestrial medicine yielded the foundation of knowledge in cardiovascular adaptation to spaceflight. Our knowledge base has expanded further as more sophisticated medical technology has become available, particularly ultrasound, and has been deployed in the spaceflight environment. More recently, medical specialists and research teams have turned their attention towards integrating this information with investigations regarding the radiation exposures in humans and animals to predict mission and health risks during extended duration missions beyond LEO and the long-term-health consequences of these spaceflight exposures. Cardiovascular function responds acutely to the spaceflight environment (e.g., cardiac chamber distension in response to the headward fluid shift) and adapts to the chronic exposure (e.g., cardiac atrophy in response to reduced plasma volume and cardiac work), contributing to functional outcomes (e.g., orthostatic intolerance and decreased exercise capacity) that affect mission critical tasks (e.g., egress), but less is known about the specific mechanisms of adaptation and the health effects across the lifetime of the astronauts.

The “Cardiovascular Risk” encompasses specific in-mission and long-term health risks, with the overall severity and consequences largely dependent upon the duration of the partial gravity or weightlessness exposure and the level of radiation exposure (**Figure 1**). Mitigating factors include screening of astronauts before selection to the mission (pre-mission); the availability of countermeasures, medical monitoring capabilities, and medical intervention readiness (in-mission); and the state of medical technologies and medical practices (in-mission and long-term health).

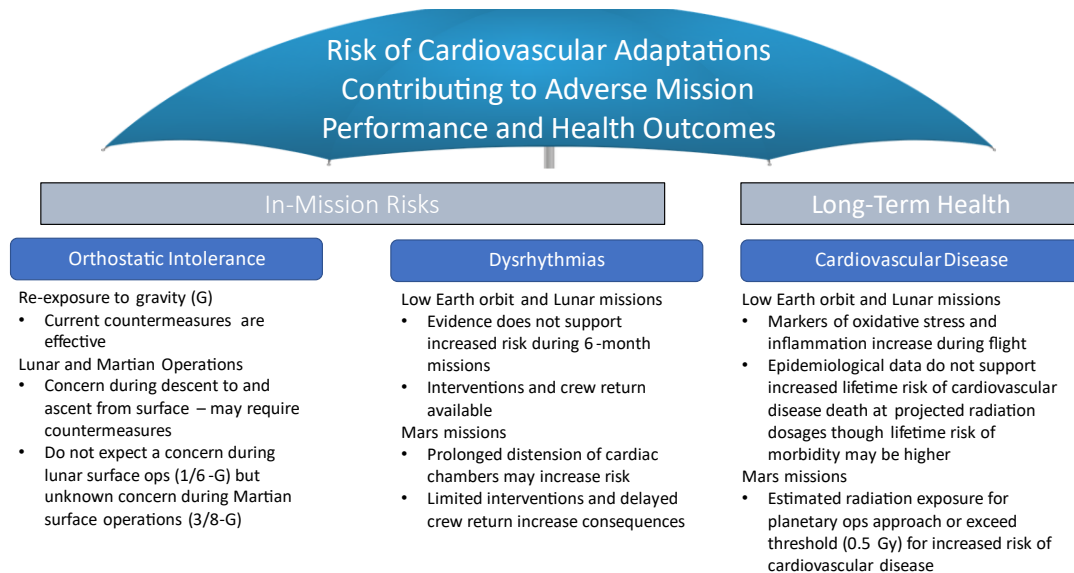


Figure 1. The newly-defined Cardiovascular Risk encompasses three previous HSRB-defined risks with the intent of developing a single integrated understanding. 1-G

The current spaceflight evidence of in-mission risks includes the potential for the development of dysrhythmias (**Section IX.A.1**) and venous thromboembolism (**Section IX.A.2**). Despite previous reports describing the occurrence of dysrhythmias during spaceflight, the notion that exposure to weightlessness is a specific causative factor has been challenged (Convertino and Cooke 2005; Convertino 2009). Since the last evidence report on cardiac rhythm problems (Lee et al. 2017), Khine et al. (Khine et al. 2018) reported that, although there may be some individuals who are more susceptible, there is insufficient evidence to conclude that 6-months in LEO causes the development of new, more frequent, or more complex dysrhythmias. These conclusions were based upon 48-hour Holter monitor recordings that encompassed the astronauts' normal activities but did not capture cardiac rhythm during extravehicular activities (EVA) or preflight EVA training. Also, there has been an incidental finding of a thromboembolism in the IJV in one astronaut (Marshall-Goebel et al. 2019; Auñón-Chancellor et al. 2020). Continued surveillance of the astronauts (Pavela et al. 2022) will aid in determining whether this thromboembolism was an isolated incident or a risk for the broader population of astronauts.

During re-entry and after landing on Earth after exposure to weightlessness, the primary cardiovascular risk is the development of hypotension leading to orthostatic intolerance (i.e., presyncope or syncope). This is particularly concerning if the astronauts are seated upright (e.g., during the Space Shuttle program), the astronauts are required to pilot the vehicle, or ground and medical support teams are not present at the landing site to help extract and assist the crew (e.g., off-nominal landing or landing on an extraterrestrial surface). The development of post-spaceflight orthostatic intolerance is well documented, although the mechanisms are not yet fully understood (**Section IX.A.3**). Since the release of the last evidence report on orthostatic intolerance (Stenger et al. 2015), two research teams have documented that wearing lower body compression garments is an effective countermeasure to orthostatic intolerance in the first 24 hours after landing from long-duration spaceflight (Fu et al. 2019; Lee et al. 2020c). However,

even with compression garment use some crewmembers were unable to complete postflight testing for a variety of other reasons.

Astronauts undergo rigorous selection testing so that those who are chosen to the crews, particularly long-duration exploration crews, have a lower overall risk of cardiovascular events during a mission. Given that astronauts represent an “above average” healthy population, have regular access to medical personnel and interventions, and that modifiable risk factors are closely monitored in the months leading up to launch, manifestation of CVD during a mission is unlikely. However, there is a concern that factors associated with spaceflight environment, such as exposure to psycho-social stress, impaired sleep, changes in musculoskeletal loading and in exercise routines, radiation exposure, and altered dietary practices, contribute to elevated levels of oxidative stress and inflammation (**Section IX.A.4**) and increase the lifetime risk of a CVD event or death. For example, although the prevalence of atrial flutter and fibrillation does not appear to be greater in astronauts than that seen in the general population, an expert panel convened by NASA (Barr 2010) suggested that astronauts present with these conditions at a younger age (**Section IX.A.5**). Atrial fibrillation is associated with an increased risk of all-cause mortality and stroke, as well as a range of other cardiovascular events (Odutayo et al. 2016). Further, while it is generally accepted that astronauts are not at an increased risk of death due to CVD (**Section IX.A.6**), a recent retrospective analysis comparing astronauts to a well-matched healthy population (Charvat et al. 2022) suggested that they have a higher risk of having a cardiovascular event. The size of the astronaut population is comparatively small and NASA’s experience with long duration missions is relatively recent, thus continued surveillance of long-term cardiovascular outcomes is required to fully characterize the risk of developing CVD following spaceflight.

Since the publication of the last evidence report describing the risk of CVD events (Lee et al. 2017), several studies have attempted to characterize the effects of spaceflight on the carotid artery as an index of atherosclerotic and stroke risk (Hughson et al. 2016; Arbeille et al. 2016, 2017; Lee et al. 2020d), with some conflicting results. Whether measures of the carotid artery are the appropriate biomarker to ascertain cardiovascular risk in an astronaut population is unclear. Determining the appropriate biomarkers to ascertain CVD risk in individual astronauts is particularly difficult given that most risk calculators are based on results from older and less fit populations. Two recent studies have demonstrated the utility of combining traditional biomarkers with other cardiovascular health-related measures, such as high sensitivity C-reactive protein (hs-CRP) and coronary artery calcium (CAC) scoring, in predicting the 3- (Gore et al. 2020) and 10-year (Khera et al. 2018) risk of a CVD event in an apparently healthy population not exposed to spaceflight.

The spaceflight environment is particularly challenging to conduct medical research. The operational demands of spaceflight missions limit the number of subjects and the crew time available to participate in studies and restrict the volume, mass, and power available to support science activities beyond that needed for vehicle functions and other mission-critical activities. Thus, evidence from ground-based investigations and spaceflight analogs (**Section IX.B**) contributes to the prediction of risks in each phase of the mission and of long-term health consequences (**Section IX.A.7**). Given that only the Apollo astronauts have ventured beyond LEO and that the durations of their exposures were short, most of the information about radiation

exposure and the long-term health consequences is the result of ground-based research (**Section IX.B.4**), including epidemiological studies of atomic bomb survivors, radiation professionals, and radiation-therapy patients as well as animal models. While several studies since the previous report have used wild-type animals to study structural and functional impacts of space radiation on the cardiovascular system (Ramadan et al. 2016; Seawright et al. 2019; Sridharan et al. 2020; Nemec-Bakk et al. 2022), the majority of observed effects were sub-clinical and not functional outcomes that are not linked causally to CVD, which limits the translatability of the results. The challenges of performing standard human cardiac measurements/techniques in small animal models cannot be discounted. Assays (e.g., transthoracic echocardiography) which may be gold standard for detection of cardiac effects in human or larger animals, such as mini pigs, may not be appropriate for small animal use. A recent study by Bishawi et al. demonstrated the benefit of performing multi-modal measurements longitudinally to detect meaningful and functional cardiac decrements (Bishawi et al. 2022). The results collected from this study represent an important milestone in demonstrating how newer and more sensitive modalities and multi-modal methodologies might improve translatability of animal research.

IX. EVIDENCE

A. Spaceflight Evidence

1. Inflight Risk: Dysrhythmias

The Mercury Program consisted of six single-person spaceflights, lasting as short as 15 minutes to as long as 34 hours, with the first mission occurring in May 1961 and the final mission completed in May 1963 (Berry 1968). The Gemini Program encompassed 10 spaceflights from March 1965 to November 1966 (Berry 1968). Two astronauts participated in each flight in which the shortest mission was just under 5 hours and the longest mission was almost 4 days. Six of the missions included extravehicular activity (EVA, “spacewalks”); the first EVA was performed on Gemini IV and lasted just 36 minutes. The longest EVA was performed on the final mission of the Gemini Program and was 5.5 hours long. There were no reports of significant dysrhythmias during these missions (Anzai et al. 2014).

The Apollo Program, NASA’s third human spaceflight program, consisted of 11 three-man missions between 1968 (Apollo 7) and 1972 (Apollo 17) with the specific goal of landing humans on the Moon. Nine of the missions carried humans beyond LEO, and of these, six missions landed on the Moon. Each lunar landing mission, beginning with Apollo 11, also included EVAs, with each successive mission incorporating longer and more complex mission objectives. It was in the course of the Apollo program that the first clinically significant dysrhythmia was reported (Anzai et al. 2014). One crewmember during Apollo 15 experienced a 22-beat nodal bigeminal rhythm, which was followed by atrial premature beats (Rossum et al. 1997). This crewmember reported extreme fatigue during the incident, but only reported the incident when questioned by crew surgeons. Twenty-one months later the crewmember suffered from coronary artery disease and a cardiac infarction (Leguay and Seigneure 1981).

Skylab was NASA's first space station; it was visited by three separate three-man crews between May 1973 and February 1974 with mission lengths of 28, 59, and 84 days. During the Skylab missions, several instances of premature ventricular complexes (PVCs), supraventricular PVCs, and nodal dysrhythmia were recorded. The dysrhythmias occurred during effort tests, EVAs, lower body negative pressure (LBNP) sessions, and were not specific to any particular phase of the mission or activity. These included two consecutive PVCs when an astronaut was exercising and an episode of atrioventricular dissociation preceded by sinus bradycardia in two astronauts (Leguay and Seigneuric 1981).

One hundred and thirty-five Space Shuttle missions occurred over the course of 30 years between April 12, 1981, and July 8, 2011, beginning with Space Transportation System (STS) 1 and ending with STS-135. During the first four flights (called test flights), ECG was monitored continuously, and frequent PVCs were observed in crewmembers during re-entry and landing (Charles et al. 1996). In subsequent missions, routine ECG monitoring was conducted only during EVA. According to Charles et al. (Charles et al. 1996), one third of the astronauts who flew before 1996 experienced either premature atrial complexes (PACs) or PVCs while participating in EVA. Notable ECG findings during EVA were reported in two crewmembers: one with 10 minutes of sustained ventricular bigeminy and the other with episodes of frequent PACs.

Two studies of cardiac rhythm disturbances were performed in response to medical reports of dysrhythmias occurring in 9 of 14 Shuttle EVA astronauts between 1983 and 1985. Rossum et al. (Rossum et al. 1997) reviewed 24-hour Holter recordings acquired during and after high altitude chamber training, underwater training for EVA (Weightless Environment Training Facility; WETF), 30 days before launch, during and after each EVA, and on return to Earth. The investigators observed no change in the number of PVCs or PACs per hour during flight compared to preflight or postflight. Fritsch-Yelle et al. (Fritsch-Yelle et al. 1996a) collected 24-hour Holter recordings from 12 astronauts before, during, and after six Space Shuttle missions lasting 5-10 days. The number of PVCs per hour decreased from pre- to in-flight, and there was a trend towards a decrease in the number of PACs per hour. Some astronauts experienced no dysrhythmias throughout the study.

In the NASA-Mir program, in which seven NASA astronauts participated in seven long-duration missions on the Mir Space Station with 12 Russian cosmonauts, ECG monitoring revealed episodic dysrhythmias in 10 astronauts and cosmonauts, with a wide range of incidence across crewmembers (Gontcharov et al. 2005). Atrial and ventricular dysrhythmias were observed when crewmembers were at rest, exercising, and performing EVA. All 10 astronauts and cosmonauts who experienced atrial or ventricular dysrhythmias during their mission had similar findings during their preflight training, although the episodes were perhaps less frequent or severe before launch. In-flight Holter monitoring recorded an isolated incident of a non-sustained 14-beat ventricular tachycardia in one of the crew member, with a maximum heart rate of 215 beats per minute (Fritsch-Yelle et al. 1998). This crewmember experienced a single occurrence of a similar event during a previous mission (Gontcharov et al. 2005). Although not part of a systematic scientific study, this case raised concerns of dysrhythmias during long-duration spaceflight (Baevsky et al. 2007b).

The ISS has been continuously inhabited since the arrival of Expedition 1 on November 2, 2000. In a prospective study Khine et al. collected 48-hour Holter recordings in 13 astronauts (9 men, 4 women) before, during, and after 6-month missions (Khine et al. 2018). There was no statistically significant increase in the number of premature atrial contractions (PACs) from preflight, although one astronaut had an increase in supraventricular ectopic beats, and another astronaut experienced an increase in ventricular dysrhythmias. Further, none of the astronauts developed atrial fibrillation, even though left atrial size increased, a risk factor for the development of atrial fibrillation, when measured with MRI within 5 days after landing. However, right atrial size was not different than preflight. The investigators have suggested that there could be a subset of individuals who are more susceptible to an increased incidence of dysrhythmias. A key element for future exploration missions may be to identify the phenotype and the stressors that could be responsible for the increases in dysrhythmias that occur in certain individuals (Romero et al. 2015). Further, there may latent, not yet understood, effects of spaceflight exposure that contribute to a higher risk of a cardiovascular event (Charvat et al. 2022).

Khine et al. (Khine et al. 2018) did not measure atrial dimensions during spaceflight but others have shown that the cephalad fluid shift upon entry into weightlessness (Thornton et al. 1977, 1987) initially results in distension of the cardiac chambers (Buckey et al. 1996a; Videbaek and Norsk 1997; Corsi et al. 2002; Caiani et al. 2006). This resolves somewhat (Arbeille et al. 2001) as the body fluids are redistributed and blood and plasma volume decrease (Leach et al. 1996; Meck et al. 2001). Reflecting this, stroke volume and cardiac output initially increase in weightlessness but then decrease over the first few days of flight (Prisk et al. 1993). Thus, stroke volume and cardiac output during spaceflight is similar to preflight resting supine levels (Herault et al. 2000; Hamilton et al. 2011) but greater than seated rest (Shykoff et al. 1996; Norsk et al. 2015). Further, although not a direct measure of atrial distension, IJV cross-sectional area also is increased during short- and long-duration spaceflight (Herault et al. 2000; Arbeille et al. 2001), and internal jugular venous pressure is increased during parabolic flight (Martin et al. 2016) and during long-duration spaceflight compared to the seated posture (Marshall-Goebel et al. 2019). Additionally, observations from parabolic flight and spaceflight (Summers et al. 2010; Khine et al. 2018) suggest that the cardiac chambers become more spherical, potentially contributing to cardiac remodeling and changes in cardiac electrophysiology.

A large amount of Soviet-era spaceflight results, including human research data, have not been published nor have the observations widely disseminated (Bogomolov et al. 2009). Until the Apollo-Soyuz Test Project, space medicine practices at NASA and in the Soviet (later Russian) space programs developed independently. Cooperation and exchange of information between the two space programs manifested as the US-USSR Joint Working Group on Biology and Medicine in the early 1970's, and the NASA and Russian space programs jointly prepared for the NASA-Mir Program in the early 1990's. A 1992 agreement between the United States (US) and the Russian Federation paved the way for a series of joint spaceflight missions from 1994 to 1998 (Bogomolov et al. 2009). Joint missions included seven Russian cosmonauts participating in Space Shuttle missions and seven NASA astronauts participating in long-duration missions on the Mir Space Station.

Romanov et al. (Romanov et al. 1987) reviewed electrophysiology results from Soviet and Russian spaceflight missions between 1964 and 1985 and reported observations of supraventricular and ventricular premature beats, with the highest reported frequency occurring during launch and the first hours of the mission. Whereas most in-flight dysrhythmias were not considered to be clinically significant, one cosmonaut was returned before the planned end of his mission due to persistent supraventricular tachycardia (Bogomolov et al. 2009). ECG changes that were observed after these missions generally resolved after flight, but three cosmonauts later demonstrated ECG changes consistent with ischemic heart disease (Romanov et al. 1987). In a separate study, two cosmonauts who completed a one-year mission on the Mir Space Station participated in a 24-hour Holter data collection in the first day after landing. Each cosmonaut had infrequent episodes of isolated PVCs (Grigoriev et al. 1991). A review of ECG records from 30 cosmonauts who participated in long-duration missions (73 to 197 days) on either Mir or ISS determined that the majority of the crewmembers (70%) did not exhibit dysrhythmias during preflight training, launch, or re-entry, and a few (10%) who had dysrhythmias preflight did not have dysrhythmias during launch or landing (Kotovskaia et al. 2008). However, a small number of individuals had increased frequencies of rhythm and conduction disturbances during re-entry that were not observed during preflight training.

Launching into and landing from space are significant physical and psychological stressors that historically were significant sources of concern. Kotovskaia et al. (Kotovskaia and Vil'-Villiams 2007) reported that individuals who exhibited dysrhythmias in the preflight period also had similar dysrhythmias during launch and entry into weightlessness. Comparing data from non-professional cosmonaut visitors (29 to 60 years old) and medically-qualified cosmonauts (31-49 years old) obtained during re-entry on the Soyuz after short stays on ISS (8 to 12 days), Kotovskaia et al. (Kotovskaia and Vil'-Villiams 2007) found the non-professional flyers, particularly the older ISS visitors, had higher heart rates and more frequent dysrhythmias than the cosmonauts. Less rigorous medical screening for non-professional space visitors may have contributed to this disparity.

There is not sufficient evidence to conclude that exposure to weightlessness alone precipitates cardiac dysrhythmias, although some factors that have been associated with dysrhythmias are modified by spaceflight. Some of the previously identified dysrhythmias resulted from pre-existing, undiagnosed disease. Additional pre-selection tests to screen astronauts and cosmonauts before selection and flight assignments, including coronary artery calcium scoring, have been added to reduce the likelihood such occurrences (Hamilton et al. 2005, 2006).

a. Q-T Intervals

The Q-T interval is a measure of the combined duration of ventricular depolarization (QRS) and repolarization (T-wave). The QRS complex is usually of fixed duration in healthy individuals. Thus, changes in QT duration represent alterations in ventricular repolarization. The Q-T interval of the surface ECG is a spatial and temporal summation of all cardiac cellular action potentials. Not all cells within the heart share identical action potentials; therefore, a certain

degree of variability, or inhomogeneity exists in their repolarization time. The degree of inhomogeneity during repolarization directly correlates with the overall morphology of the Q-T waveform (primarily the T-wave) and in most cases with the Q-T interval duration. A clear association between the magnitude of inhomogeneity of repolarization and the risk for the development of ventricular dysrhythmias has been established (Lux et al. 2001; El-Sherif 2001; Shusterman et al. 2006).

The Q-T interval is often corrected for heart rate and is shown as Q-Tc. Some conditions that can prolong the Q-Tc interval are ischemic heart disease, autonomic dysfunction, bradycardia, electrolyte abnormalities, cardiac remodeling, and medications that interfere with the cardiac potassium ion channels (Ishida et al. 1995; Lo et al. 1996; Savelieva et al. 1999; Haverkamp et al. 2000; Khan 2002). It is known that astronauts develop changes in the autonomic nervous system (Fritsch et al. 1992; Fritsch-Yelle et al. 1996b; Meck et al. 2001; Rossum et al. 2001), and that on long-duration flights astronauts have a relative bradycardia compared to astronauts on short-duration flights (Meck et al. 2001). Further, there is evidence of cardiac remodeling after long-duration flights (Perhonen et al. 2001a). Finally, there are medications available to astronauts aboard the ISS that prolong Q-Tc interval, including ciprofloxacin, haldol, inderal, verapamil, zithromax, Zoloft®, and nortriptyline. This combination of factors might cause or exacerbate the prolongation of the Q-T interval.

Prolongation of Q-Tc interval does not itself guarantee an increase in ventricular dysrhythmias. For example, sleep, hypothyroidism, and use of the anti-arrhythmic drug amiodarone all prolong Q-Tc without increasing the incidence of ventricular dysrhythmias. Prolongation of the Q-Tc interval in astronauts has been documented on several occasions but it is not clear if these findings have any clinical significance or portend risk (D'Aunno et al. 2003; Mitchell and Meck, 2004).

b. Electrolyte Disturbances

Cardiac rhythm abnormalities and exercise intolerance observed in the Apollo 15 Lunar Module Pilot were attributed to electrolyte disturbances, particularly hypokalemia (Leguay and Seigneure 1981). Inadequate potassium intake was suspected; thus, before Apollo 16 a 12-day bed rest study with low potassium intake was conducted with two subjects as a simulation of the Apollo 15 mission (Hyatt et al. 1975). Neither of the bed rest subjects developed symptomatic hypokalemia, only one demonstrated minor ECG abnormalities, and the decrease in exercise capacity and orthostatic tolerance was no more than expected for a bed rest of similar duration with normal dietary intakes. Serum and urinary potassium levels decreased by 7% and 47%, respectively, in both Apollo crewmembers (Leach et al. 1975)

A later review of the Apollo 15 mission suggested that magnesium deficiency contributed to the dysrhythmias and other symptoms (Rowe 1998). However, while not specific to Apollo 15, Smith and Zwart (Smith and Zwart, 2015) report that during long-duration ISS missions, urinary magnesium concentrations are elevated within the first 2 weeks of the mission and serum levels are elevated above preflight levels after 3-4 months of spaceflight. Both urinary and serum

magnesium decrease and are lower than preflight on landing day then return to preflight levels within 1 month after landing. Despite these dynamics, tissue levels of magnesium, measured in epithelial cells from sublingual scrapes, are not different from preflight on landing day or 1 month after landing. After short-duration spaceflight, neither urinary nor serum magnesium levels are different than preflight.

Prolonged Q-Tc intervals observed after long-duration spaceflight were not associated with clinically-relevant changes in electrolyte levels (D'Aunno et al. 2003). In five Shuttle and seven Mir astronauts, there was no pre- to postflight change in serum magnesium or calcium levels; although serum potassium measured on landing day was significantly less than preflight, the levels were still within clinical normative ranges.

c. Catecholamines

Elevated catecholamines and increased sympathetic activity have been associated with the development of dysrhythmias in terrestrial populations (Leguay and Seigneuric 1981). Leach et al. (Leach et al. 1996) reported no change in serum catecholamines from pre- to in-flight and that urinary norepinephrine levels were lower during Space Shuttle missions than before launch. Norsk et al. (Norsk et al. 2015) reported no change in norepinephrine and epinephrine levels from preflight (seated) to in-flight during 3-6 month ISS missions.

d. Cardiac Atrophy

Evidence suggests that apoptosis, or “programmed cell death” occurs in response to pathological, physiological, and/or genetic signals, and this may be a key mechanistic factor in the development of cardiac dysrhythmias. For example, apoptosis associated with atrophy and fibro-fatty replacement of right ventricular tissue has been identified as the likely mechanism for dysrhythmia development in arrhythmogenic right ventricular dysplasia, a condition which may lead to sudden death in otherwise healthy young individuals (Basso et al. 1996; Mallat et al. 1996). Therefore, a spaceflight-induced reduction in cardiac mass may increase the risk of a cardiovascular event.

Cardiac mass appears to be reduced even after short-duration spaceflight. Postflight cardiothoracic ratio, calculated from the posterior/anterior chest X-rays, decreased from preflight in 24 of 30 Apollo astronauts (Hoffler et al. 1974; Hoffler and Johnson 1975), representing a mean reduction of 5%. Left ventricular mass, measured using cardiac magnetic resonance imaging, decreased from pre- to postflight in three of four astronauts (**Figure 2**) after a 10-day Space Shuttle mission (Perhonen et al. 2001a). Summers et al. (Summers et al. 2007), using 2-dimensional echocardiography, reported decreased left ventricular mass on landing day after Space Shuttle missions, although there was a full recovery 3 days after landing (**Figure 3**). Tissue dehydration likely contributes to the reduction in mass after short-duration spaceflights (Perhonen et al. 2001a). Results from bed rest studies show the decreased ventricular mass can be prevented with exercise and/or nutritional countermeasures (Dorfman et al. 2007).

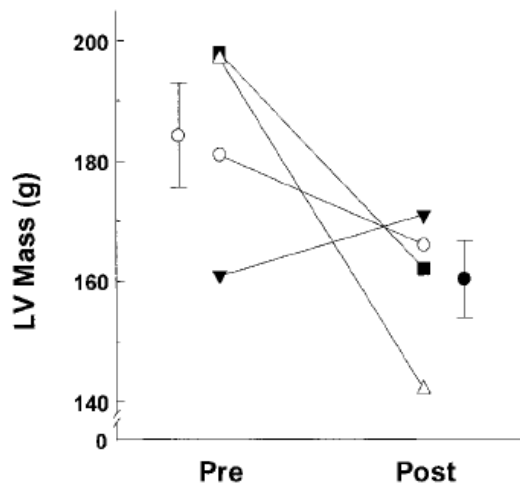


Figure 3. Left ventricular mass measured by magnetic resonance imaging before and after short-duration spaceflight. Lines represent results from individual astronauts (n=4). The average response (mean +/- SD) is represented by circles.

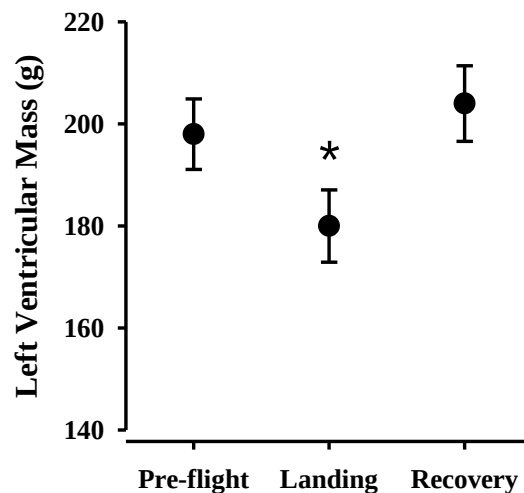


Figure 2. Left ventricular mass before and after short-duration spaceflight (n=38) adapted from Summers et al. 2005. * = $p \leq 0.05$.

One of the aims of the Integrated Cardiovascular Study was to investigate whether cardiovascular mass and volume in ISS astronauts are contributing factors for cardiac dysrhythmias (Levine and Bungo, 2015). Preliminary results presented in their final report to NASA indicate that, on average, ventricular mass was not different pre- to postflight in the 13 astronauts studied, but there was considerable variability. Additionally, there was no evidence of focal scarring or diffuse fibrosis on the MRI that might indicate cardiac injury. Unpublished

results from ultrasound (Platts et al., NASA Johnson Space Center, Cardiovascular and Vision Laboratory) suggest decreased left ventricular mass after 4-6 months of spaceflight during the early ISS missions. In these subjects, the reduction in left ventricular mass was double that observed after short flights, and ventricular mass did not fully recover by the third day after landing (**Figure 4**).

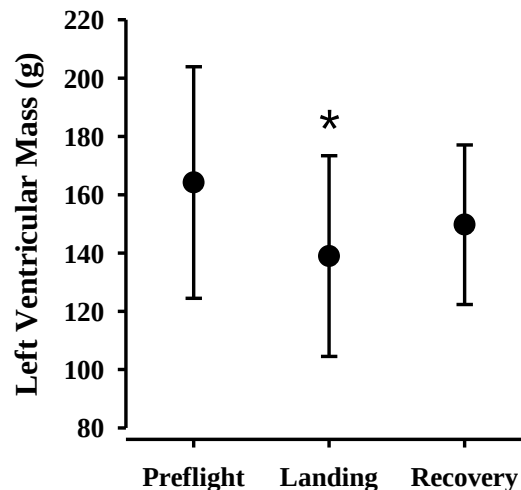


Figure 4. Left ventricular mass before and after long-duration spaceflight (n=6) drawn from unpublished data from Platts et al., NASA Johnson Space Center, Cardiovascular and Vision Laboratory. * = $p \leq 0.05$.

2. Inflight Risk: Venous Thromboembolism

Venous thromboembolism (VTE), represented by deep venous thrombosis (DVT) and pulmonary embolism (PE), is one of the most common terrestrial vascular diseases with approximately 100 cases per 100,000 people per year in the United States (White, 2003). Virchow's triad describes the three main categories of thrombosis risk factors, blood flow stasis, endothelial injury, and hypercoagulability, that might be present in astronauts during spaceflight (Kim et al. 2021; Harris et al. 2022).

The lack of gravity during spaceflight results in a relative headward fluid shift from the lower to the upper body with notable distension of the internal jugular veins (IJV) (Arbeille et al. 2015, 2021). During long duration spaceflight there have been reports of alterations in blood flow patterns in the left IJV after 50 and 150 days of spaceflight, including stagnant and/or retrograde IJV blood flow in six of eleven crewmembers (Marshall-Goebel et al. 2019). Stagnant blood flow is a known risk factor for the development of VTE, and indeed, two incidental cases of left IJV thrombosis were identified in the study cohort, including one occlusive thrombosis case that was

treated with anticoagulants while onboard the ISS (Marshall-Goebel et al. 2019; Auñón-Chancellor et al. 2020). Return of blood flow through the affected IJV was first noted 47 days after treatment with use of a Mueller maneuver (inspiration against a closed glottis) but spontaneous flow remained absent. Anticoagulants were stopped four days prior to return to Earth and ultrasound examination upon landing revealed spontaneous flow in the affected IJV with residual thrombus on the vessel wall. Retrospective review of IJV flow patterns led to identification of a heterogeneous partially occlusive thrombus in the left IJV in second crew member in the same study (Marshall-Goebel et al. 2019). However, it is unknown if other undiagnosed thrombosis occurred previously during human spaceflight.

NASA initiated an occupational surveillance protocol to determine if additional crewmembers aboard ISS have evidence of thrombus formation during spaceflight. Indeed, an additional six of 11 astronauts presented with decreased peak velocity and increased echogenicity in the left IJV (Pavela et al. 2022), but no thrombosis was detected. Together, these two reports suggest that while alterations in venous flow may contribute to the development of a DVT during spaceflight, it is not the only factor necessary to induce thrombosis formation. Stagnation of blood flow in some crewmembers during spaceflight also may contribute to changes in the local clotting environment. For example, these hemodynamic alterations can lead to elevations of hemostasis-activation factors, increase the blood cell-endothelium interaction, and result in local hypoxic endothelial activation. Additional common risk factors for the development of VTE on Earth include increasing age, active cancer, trauma, infection, thrombophilia, pregnancy or puerperium, and use of oral contraceptives (Previtali et al. 2011).

Recently, Zwart et al. demonstrated that female astronauts on combined estrogen-progestin oral contraceptives (cOC) had lower serum albumin levels than cOC non-users or male astronauts (Zwart et al. 2022). This link between cOC use and hypoalbuminemia is concerning due to the increased risk of developing venous thromboembolism. Patients on Earth with albumin levels <3.5 g/dL have a two-fold greater risk of developing venous thromboembolism compared to those with albumin concentrations >4.2 g/dL (Chi et al. 2019). When combined with venous flow alterations resulting from weightlessness during spaceflight, use of cOC and hypoalbuminemia may further increase the risk for thrombus formation. Of the six female crewmembers on cOC studied by Zwart et al., two had albumin levels <3.5 g/dL, including the crewmember who developed the asymptomatic thrombus (Zwart et al. 2022). Two additional cOC users had one or more serum albumin measures <3.8 g/dL, the low end of the normal clinical range. Women not on cOC (n=5) had normal serum albumin concentrations. Only 3 of the 41 men had clinically low serum albumin concentrations. In addition to a potential mechanism for hypoalbuminemia, cOC users demonstrated increased inflammation markers and elevated levels of transferrin, which terrestrially correlate to an increased risk of DVT or VTE. These data highlight the need to consider the type of cOC used during spaceflight since third-generation cOC impart a greater risk of developing VTE than second-generation cOC (Conard 1999; Kemmeren et al. 2001).

Further potential risks of VTE in the spaceflight environment include hematologic index alterations (reduced plasma volume and increased hematocrit and platelet concentration),

inflammation, thrombophilia, and changes in vascular structure/function (Kunz et al. 2017; Crucian et al. 2018). Moreover, abnormal flow characteristics in microgravity (reduced peak velocities, retrograde flow, and increased blood echogenicity), some of which may be due to compression of the jugular vein by the carotid artery in microgravity, may contribute to increased risk for thrombosis formation in some individuals, especially in the left IJV (Pavela et al. 2022). The limited data available at this point requires additional coagulation screening and monitoring biomarkers for risk of VTE, such as d-dimer, fragmin f1.2, coagulation cascade, V-CAM-1 and E-selectin, because some terrestrial predictors for VTE, including high body mass index and medical comorbidities, generally do not apply to astronauts due to fitness and healthy lifestyle (Jain et al. 2020).

Mechanical countermeasures to IJV distension and stagnant flow, such as venoconstrictive thigh cuffs, LBNP, centrifugation, and impedance threshold breathing, have not been implemented as a standard practice during spaceflight but research studies demonstrate that these modalities may mitigate the spaceflight-induced headward fluid shift (Marshall-Goebel et al. 2021). For example, LBNP reduces IJV area and volume and middle cerebral vein velocity (Marshall-Goebel et al. 2019; Arbeille et al. 2021). Venoconstrictive thigh cuffs used during Mir space station experiments found that “Braslets” reduced jugular vein cross-sectional area by 12-20% (Herault et al. 2000; Hamilton et al. 2012). However, these mechanical countermeasures address only one contributing component to the development of VTE. Successful countermeasures to VTE risk during spaceflight likely also will address an individual's thrombosis risk factors and hypercoagulability.

3. Post-flight Risk: Orthostatic Intolerance

Post-spaceflight orthostatic intolerance was evident, even after short stays in weightlessness, during the Mercury (1961 to 1963) and Gemini (1965 to 1966) missions. The Mercury-Atlas 9 pilot became hypotensive during a 70° head-up tilt test after only 34 hours of flight. Similarly, orthostatic intolerance has been observed after 24 hours of bed rest (Nixon et al. 1979; Blomqvist et al. 1980), a spaceflight analog (Fortney et al. 2010). Elevated heart rate, decreased pulse pressure, and increased fluid pooling in the lower extremities were consistently observed during tilt testing in Mercury and Gemini astronauts for up to 50 hours after splashdown following spaceflight missions of 3 to 14 days (1).

The larger spacecraft volume and increased mobility in the Apollo missions were predicted to reduce post-spaceflight orthostatic intolerance compared to the Mercury program but orthostatic intolerance still was prevalent (Hoffler et al. 1974; Hoffler and Johnson, 1975). Of the 21 Apollo astronauts that performed pre- and post-spaceflight orthostatic tolerance tests, 13 exhibited an increased heart rate when at rest postflight. Resting heart rate returned to preflight values by the third examination, two to three days after splashdown. During immediate postflight orthostatic evaluations, astronauts exhibited increased heart rate and decreased stroke volume, systolic and pulse pressures which were greater than those responses before flight. Additionally, body weight, calf circumference, and cardiothoracic ratio were all decreased immediately postflight. These measurements had not returned to their preflight values by the third postflight

examination, suggesting the changes were not entirely due to fluid loss (Hoffler and Johnson 1975). The findings of the Apollo Program aided the understanding of cardiovascular changes during spaceflight in preparation for longer duration spaceflight in the Skylab missions.

Each of the Skylab missions (1973 to 1974) set new records for amount of time spent in space (28, 59, and 84 days) by NASA astronauts. The larger spacecraft and longer duration of the missions allowed the Skylab program to assess the effects of spaceflight on more physiological parameters. During Skylab missions LBNP was used as an orthostatic tolerance test pre- and post-spaceflight, as well as during the mission. Inflight LBNP revealed the existence of orthostatic intolerance after 4 to 6 days of flight (Johnson et al. 1977b). Astronauts experienced a greater stress during inflight LBNP than preflight LBNP (Johnson and Bergman 1977), illustrated by their greater increases in heart rate and leg volume and greater decreases in systolic blood pressure (Johnson et al. 1977b). Inflight LBNP also served as a predictor of the degree of postflight orthostatic intolerance, which aided crew health care after long-duration missions.

Results from Gemini and Apollo had suggested a decrease in cardiac function due to spaceflight, raising concerns about potential detrimental effects of long duration spaceflight on the cardiovascular system. Postflight clinical data suggested there might be an impediment to venous return as well as a myocardial effect causing decreased cardiovascular function (Johnson and Bergman 1977). Two of the three astronauts in Skylab 4 had decreased stroke volume and cardiac output upon their return to Earth, yet the recovery of cardiac volume and mass in the first month after landing to preflight values led to the conclusion that 84 days in space is not a long-enough time to produce irreversible cardiac dysfunction (Henry et al. 1977).

The Extended Duration Orbiter Medical Project (EDOMP; 1989 to 1995) aimed to define the effects of short-duration spaceflight in a more controlled environment aboard the Space Shuttle Orbiter with a larger subject pool and develop appropriate countermeasures (Charles et al. 1999). Several studies included 24-hour Holter monitoring, blood pressure recordings, and two-dimensional echocardiography measured inflight, and heart rate and blood pressures were measured during launch and reentry. Inflight heart rate and systolic and diastolic blood pressure were decreased when compared to the preflight values (**Figure 5**). Upon re-entry, these values increased past their preflight baseline, reaching maximal values at peak gravity (Charles et al. 1999). During standing after touchdown, both systolic and diastolic pressures significantly decreased from the seated value, and the decrease in diastolic pressure was greater in the crewmembers who did not inflate their G-suits. Systolic pressure and heart rate returned to preflight values within an hour of landing, whereas all other spaceflight-induced cardiovascular changes were reversed within a week after landing.

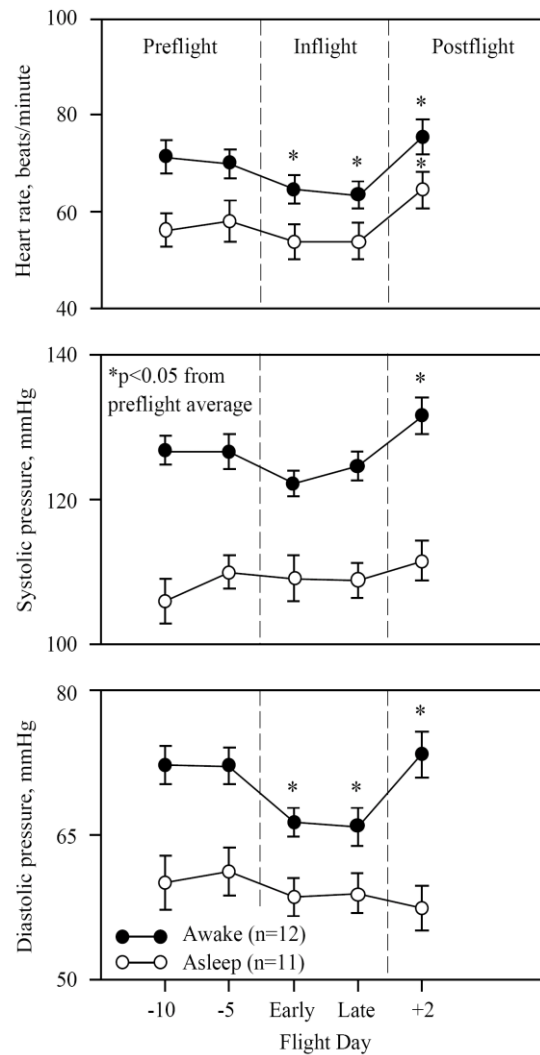


Figure 5. Heart rate and blood pressure during spaceflight. Reproduced from (Charles et al. 1999).

Four mechanistic studies were performed to explore the etiology of post-spaceflight orthostatic hypotension, concentrating on changes in autonomic control (Charles et al. 1999). The first three studies concluded that post-spaceflight cardiovascular responses were characterized by orthostatic intolerance, increased low-frequency R-R spectral power, decreased carotid baroreceptor response, and altered blood pressure and heart rate responses to Valsalva maneuvers. Catecholamine analyses revealed norepinephrine and epinephrine levels were increased when the astronauts were both resting and standing postflight. Three days after landing, the astronauts' norepinephrine levels when they were standing remained increased, while their epinephrine levels had returned to preflight values. The fourth mechanistic study reported that those who do not become presyncopal on landing day have a greater norepinephrine response upon standing, leading to greater peripheral vascular resistance. Analysis of preflight data yielded normal cardiovascular measures in both groups, yet the presyncopal group was characterized by lower diastolic and systolic blood pressure and peripheral vascular resistance when they were supine. However, plasma volume losses were not significantly different between the presyncopal and non-presyncopal groups (Charles et al. 1999). Additionally, increased heart rate, decreased systolic pressure, and reduced stroke volume were observed in 14 astronauts during a post-spaceflight 5-minute stand test (Buckey et al. 1996b) (Figure 6).

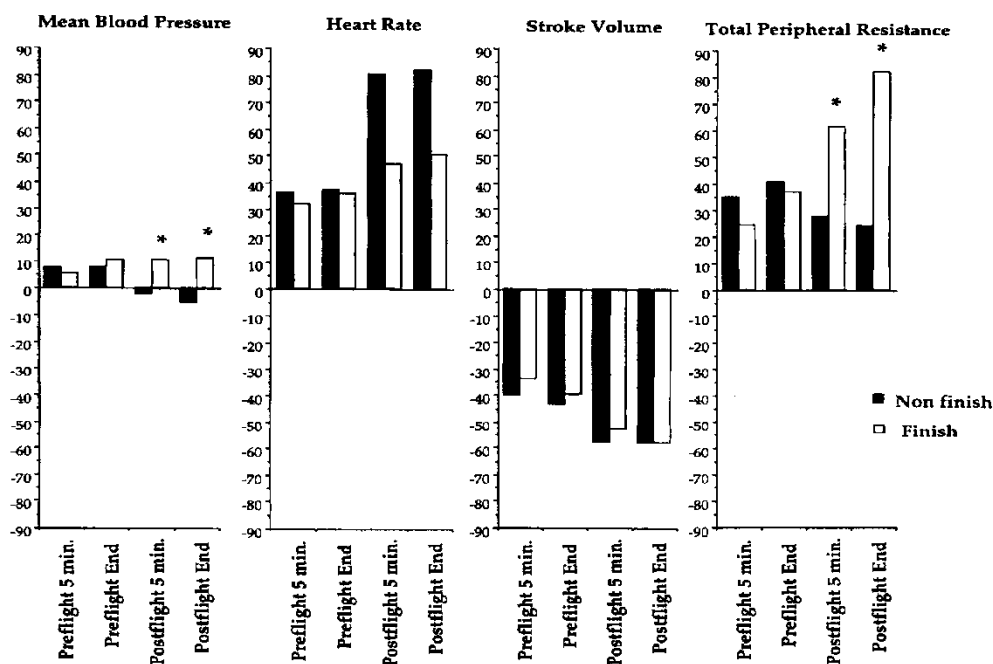


Figure 6. Hemodynamic responses to standing (5 finishers, 7-9 non finishers) before and after spaceflight (Buckey et al. 1996b).

The last goal of the EDOMP, evaluating countermeasures to increase postflight plasma volume, consisted of four studies implementing different LBPN protocols, salt and fluid loading, and fludrocortisone (Charles et al. 1999). The first protocol applied LBPN in a stepwise fashion ranging from 0 to -60 mmHg in 5-minute intervals (ramp). The treatment (soak) consisted of a

ramp to -50 mmHg, followed by a decompression at -30 mmHg for approximately 3.5 hours with a fluid and salt load during the first hour. The pre- and post-soak ramps were compared, and results showed the heart rate response post-soak was significantly less than that pre-soak, suggesting the soak treatment was effective for the first 24 hours. The second LBNP protocol required crewmembers to perform a soak within the 24 hours before landing. Upon landing, diastolic pressures and heart rate when the astronauts were seated and standing were lower in the LBNP group than in crewmembers who did not perform the soak. However, testing conducted one to three hours after landing showed no statistically significant differences in heart rate or blood pressures during a stand test and no significant differences in plasma volume losses. The third countermeasure study (which was instituted well before EDOMP), a mandatory fluid and salt load before reentry (6), did not allow for any conclusions due to a lack of control of fluid ingestion inflight and before postflight testing. The last potential countermeasure studied, fludrocortisone, proved unsuccessful since it was not well-tolerated by the crewmembers and did not result in any differences in plasma volume or orthostatic tolerance (Charles et al. 1999).

The data for long-duration crewmembers is more limited, but suggests a more severe spaceflight effect. The incidence of post-spaceflight orthostatic intolerance among US astronauts increased to greater than 80% on landing day following long duration (4 to 6 months) Mir spaceflights; five of six astronauts could not complete a 10-minute tilt test after Mir Space Station missions (Meck et al. 2001). **Figure 7** shows the hemodynamic responses of a single astronaut

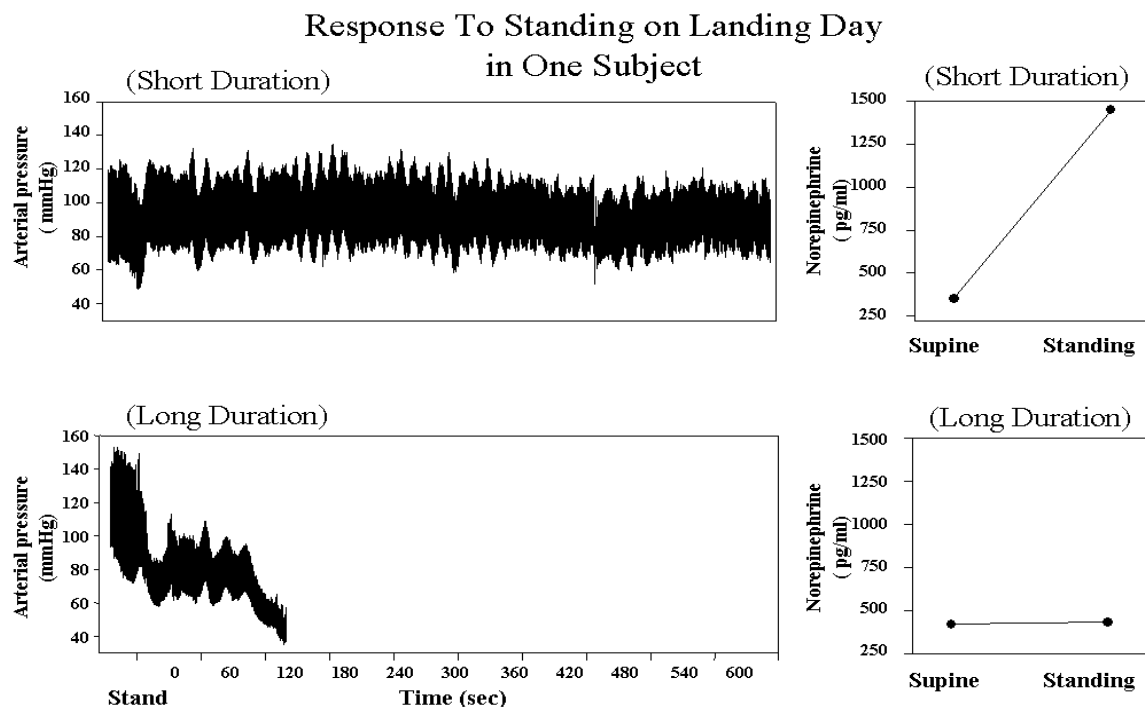


Figure 7. Effects of spaceflight on a single crewmember. Left panels show the blood pressure responses to an 80°head-up tilt. Right panels show the norepinephrine released during the tilt. This crewmember completed the tilt after short duration spaceflight with a normal norepinephrine response, while he failed the tilt test after ~2 minutes following long duration spaceflight and did not increase norepinephrine release upon exposure to tilt. Unpublished data from (Meck et al. 2001)

who completed both a short-duration flight (top) and a long duration flight (bottom). The astronaut did not become presyncopal on landing day after a Space Shuttle mission and demonstrated a normal norepinephrine increase to tilt. However, following the long-duration spaceflight the astronaut did not complete more than two minutes of tilt before hypotension ensued, and the test was terminated. This crewmember failed to mount any adrenergic response to tilt.

A review of tilt tests results from US astronauts after ISS and Space Shuttle missions also revealed a higher prevalence of presyncope after long-duration missions compared to short-duration flights (Lee et al. 2015). On landing day, 4 of 6 (66%) ISS astronauts were unable to complete a 10-min 80° head-up tilt test compared to 13 of 66 (20%) Shuttle astronauts. The rates of presyncope quickly decline after just one day after return from space; eight of nine ISS astronauts completed the tilt test on R+1. On R+3, 13 of 15 (87%) of the ISS and 19 of 19 (100%) of the Shuttle astronauts completed the 10-min test. Modeling of cardiovascular parameters associated with orthostatic hypotension, specifically stroke volume and diastolic blood pressure, suggest that recovery from long-duration space flight is prolonged compared to recovery following short-duration Space Shuttle missions. Astronauts are monitored closely during postflight activities, and flight surgeons guide astronauts to avoid activities that may provoke the onset of orthostatic hypotension. While orthostatic intolerance becomes less prevalent 24 hours after landing, partially through the restoration of plasma volume (Mulavara et al. 2018), some astronauts continue to experience symptoms for several days (Lee et al. 2015). It is not uncommon for some astronauts and cosmonauts to wear lower body compression garments in the first week after landing to manage symptoms, including lower leg swelling (Stenger et al. 2014).

The vast majority of astronauts have been male (Smith et al. 2020), and consequently any conclusions drawn regarding the physiological responses to spaceflight are male-biased. Some significant differences in how men and women respond to spaceflight exist (Harm et al. 2001; Platts et al. 2014), including sex differences in the effects of spaceflight on cardiovascular responses to orthostatic stress (Waters et al. 2002). Greater than 80% of female crewmembers became presyncopal during a postflight tilt test (Waters et al. 2002) compared to about 20% of men, even when pilots, whose occupational success includes the ability to tolerate high g-forces, were removed from the analyses.

In a recent study, Fu et al. (2019) observed no hypotensive events in 12 astronauts (four women) during the first 24 hours after landing following prolonged spaceflight with use of various countermeasures (suit cooling, fluid loading, compression garments, and inflight exercise) (Fu et al. 2019). However, interpretation of these results is limited because the analyses did not consider whether the astronaut was supine or upright or whether they were wearing compression garments. Further, most astronauts returning from the ISS receive intravenous fluids within hours of landing (Mulavara et al. 2018). A recent study of ISS astronauts showed that compression garments worn on landing day have a significant benefit to reducing the cardiovascular stress associated with the upright posture (Mulavara et al. 2018; Lee et al. 2020c), including the immediate post-landing period before intravenous fluids were administered.

a. Plasma and Blood Volume

The etiology of orthostatic intolerance is complicated and multifactorial. Decreased intravascular volume, secondary to the headward fluid shift that occurs in space (Thornton et al. 1977, 1987), is likely an initiating event. One of the first physiological changes noted during the Apollo program was the decrease in plasma volume, exhibited by the decrease in weight of the crewmen (Leach et al. 1975). One third of the average 5% weight loss was regained within 24 hours postflight, suggesting this fractional change was due to a loss of fluid. The remainder of the body weight loss was attributed to tissue loss, which is characterized by a longer recovery time (Leach et al. 1975). Apollo results suggested that the degree of plasma volume loss was independent of flight duration, later supported by a finding of no difference in plasma volume loss after two weeks and after six months of spaceflight. In contrast, average plasma volume losses were reported to be higher in the Skylab 2, 3, and 4 missions than in the Apollo program, with average losses of 8.4%, 13.1% and 15.9%, respectively (Johnson et al. 1977a). However, postflight orthostatic hypotension and presyncope are not dependent on the degree of postflight hypovolemia alone. For example, there was no difference in plasma volume loss between men who were presyncopal and non-presyncopal during tilt testing on landing day after short-duration missions (Waters et al. 2002). Further, despite no differences in plasma volume loss between short- and long-duration spaceflight (**Figure 8**), the incidence of orthostatic intolerance after long-duration spaceflight is much greater (Meck et al. 2001).

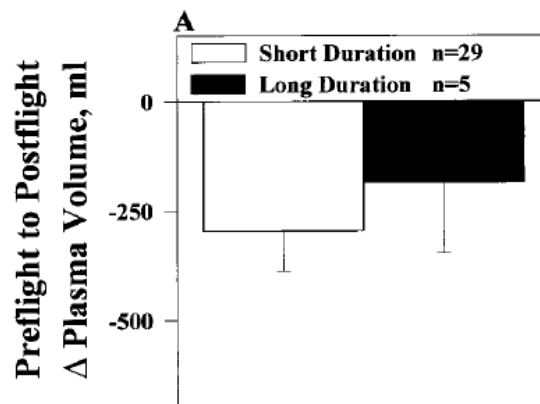


Figure 8. Plasma volume losses are not different between short- and long-duration spaceflight (Meck et al. 2001).

The mechanism of the plasma volume loss has been a matter of some debate (Norsk, 2005). There have been limited inflight studies of plasma volume. One study demonstrated a decrease in total body water during flight, suggesting but not proving a diuresis (Leach et al. 1991). A second study reported a decrease in plasma volume, but an increase in intracellular fluid, suggesting “3rd spacing” and not a diuresis (Leach et al. 1996). However, postflight studies from the Apollo (Leach et al. 1975) and Space Shuttle missions (Greenisen et al. 1999) programs do not show an increase in the intracellular fluid compartment. These disparate flight data reinforce the need for further flight studies.

Red blood cell count, hemoglobin concentration, and hematocrit are elevated during short- and long-duration spaceflight (Alfrey et al. 1996; Kunz et al. 2017), perhaps due to a disproportionate decrease in plasma volume (Leach et al. 1996). Hemolysis, particularly of young red blood cells (Alfrey et al. 1996), contributes to a decrease in red cell mass that persists throughout spaceflight such that astronauts are considered to be clinically anemic after landing (Trudel et al. 2022). However, these data must be considered within the dynamic of the restoration of plasma volume, which is more rapid (Leach et al. 1996) and actively manipulated by end-of-mission fluid loading and intravenous fluids after landing. Erythropoietin is decreased during short-duration spaceflight (Udden et al. 1995) though it appears to return to preflight levels in longer duration missions (Trudel et al. 2022). Within the first few days after landing, hemolysis is decreased (Trudel et al. 2022) and erythropoietin is increased (Alfrey et al. 1996) but restoration of red cell mass is not apparent until at least 30 days postflight (Johnson et al. 1977a). Red blood cell production and hemolysis abated but still were higher than preflight one year after landing (Trudel et al. 2022).

b. Other Factors

The downstream effects and the physiological responses (or lack thereof) are likely the primary drivers of orthostatic intolerance after weightlessness. While all crewmembers who have been tested are hypovolemic on landing day, only a fraction of them develop orthostatic intolerance during testing. In men who lost an equal amount of plasma volume, astronauts who had a hyper-adrenergic response to orthostasis did not become presyncopal (Waters et al. 2002). Dysfunction of the sympathetic nervous system (Goldstein et al. 2003), with or without failure of the renin-angiotensin-aldosterone system (Jacob et al. 1997) could contribute to post-spaceflight orthostatic intolerance.

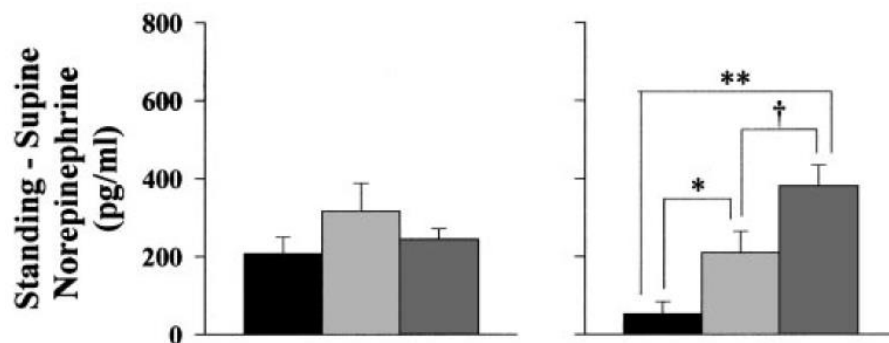


Figure 9. Plasma norepinephrine responses in women (n= 4; black bars), presyncopal men (n= 6; light gray bars), and nonpresyncopal men (n =22; dark gray bars) when tested preflight (left) and on landing day (right) (Waters et al. 2002).

It is widely accepted that the inability to elevate peripheral vascular resistance following spaceflight contributes to the orthostatic intolerance experienced by many astronauts. Lower body peripheral resistance vessels have a larger cross-sectional area (increased capacitance) (Tuday et al. 2007) and a diminished ability to constrict following exposure to simulated microgravity (Delp et al. 2000). Furthermore, in arteries from non-posture related muscles, the

lower constrictor function is related to a blunted response to vasoconstrictor stimuli (Delp et al. 1993; Behnke et al. 2008) and reduced myogenic tone (Purdy et al. 1998). Postflight data measuring muscle sympathetic nerve activity in six non-presyncopal male astronauts (Levine et al. 2002) shows that sympathetic responses in these crewmembers are appropriate. Unfortunately, there were no presyncopal subjects in this study and the postflight sympathetic dysfunction in that group of astronauts could not be duplicated.

Furthermore, astronauts who experienced both short- and then long-duration spaceflight were more likely to have a hypo-adrenergic response and become presyncopal during tilt testing after the long-duration flight despite similar plasma volume losses in both flights (Meck et al. 2001). Thus, it is not the plasma volume loss alone that causes presyncope, but the lack of an adequate physiological response to the hypovolemia. There also appears to be a correlation between the vestibular system, which is altered after long-duration spaceflight and the sympathetic reflex to orthostatic stress upon return from the ISS. This vestibulo-sympathetic response has been demonstrated to be damped after long-duration spaceflight (Deshpande et al. 2020).

Cardiac atrophy and the resulting decrease in stroke volume (Levine et al. 1997; Perhonen et al. 2001a) also may play a role in postflight orthostatic intolerance. Stroke volume is altered by mechanical and hydrostatic effects (Negishi et al. 2017) and serves as the primary stimulus to the baroreflex regulation of arterial pressure during an orthostatic stress (Levine et al. 1991). Orthostatic hypotension will ensue if the fall in stroke volume is of sufficient magnitude to overwhelm normal compensatory mechanisms or if the reflex increase in heart rate and total peripheral resistance is impaired by adaptation of the autonomic nervous system (Robertson et al. 1994). After adaptation to real or simulated microgravity, virtually all individuals studied have an excessive fall in stroke volume in the upright position (Buckey et al. 1996b; Fritsch-Yelle et al. 1996b), although there are conflicting data regarding changes in baroreflex regulation of heart rate and vascular resistance (Convertino et al. 1990; Eckberg and Fritsch 1992; Fritsch et al. 1992; Convertino et al. 1994; Hughson et al. 1994a; Fritsch-Yelle et al. 1994; Crandall et al. 1994; Iwasaki et al. 2000, 2004).

Heart rate variability (HRV) and blood pressure variability provide a means to noninvasively approximate autonomic activity through analysis of fluctuations in the R-R interval and systolic blood pressure (SBP) data. In particular, the high frequency component of the R-R interval series (HF-RR), in absolute or normalized units, is an index of parasympathetic modulation (Pomeranz et al. 1985; Pagani et al. 1986; Task Force of the European Society of Cardiology the North American Society of Pacing 1996), and the low frequency component of the SBP (LF-SBP) approximates sympathetic activity (Preiss and Polosa 1974; Pagani et al. 1986; Malliani et al. 1991; Tanaka et al. 2013). The interpretation of the low frequency to high frequency ratio of the R-R interval (LF/HF-RR) is more controversial. The LF/HF-RR is often used as a marker of sympathovagal balance (Pagani et al. 1986; Malliani et al. 1991; Montano et al. 1994; Task Force of the European Society of Cardiology the North American Society of Pacing 1996). However, this interpretation has been challenged by studies that question the concept and construction of the index (Eckberg 1997), suggest that variation in the LF/HF-RR is partially

due to changes in baroreflex function (Goldstein et al. 2011), or conclude that spectral HRV is mainly a measure of parasympathetic activity (Reyes Del Paso et al. 2013).

While a great deal of research has been focused on understanding and overcoming the cardiovascular deconditioning effects of the vasculature in the lower body, recent evidence suggests that changes in cerebral vascular conductance also may contribute to the incidence of orthostatic intolerance in astronauts. In general, cerebral blood flow is well protected against systemic changes in pressure and flow without compromising the high demands of brain tissue. Unfortunately, few spaceflight studies have been designed to examine the role of cerebral blood flow and cerebral autoregulation in humans. Much of the data which does exist argues against changes cerebral autoregulation and do not provide evidence for changes in vascular function. For example, Fritsch-Yelle et al. (Fritsch-Yelle et al. 1996b) measured blood flow velocity in the middle cerebral artery using ultrasound but did not find a difference in tilt response between presyncopal and nonpresyncopal astronauts. In a similar study, measures of cerebral blood flow, velocities, and beat-to-beat changes in arterial pressure were collected before and after 16 days in space. The authors of this study maintained that static autoregulation was not impaired and dynamic regulation was actually improved (Iwasaki et al. 2007). Greaves et al. reported similar findings in subjects following 60 days of head-down tilt bed rest (Greaves et al. 2007).

c. Current Countermeasures: Inflight Exercise

Inflight exercise was initiated by NASA as a test of exercise tolerance during the Mercury program and was recommended for future missions to maintain orthostatic responses and egress capability in the event of an emergency soon after landing (Moore et al. 2010). The exercise hardware that is available today began its development in the Space Shuttle program and more advanced systems have been employed on ISS (Hayes et al. 2013; Korth 2015). Inflight exercise time was allotted for the Space Shuttle crews but specific exercise was not prescribed by the NASA specialists until the NASA-Mir and ISS missions (Lee et al. 2019). Currently, ISS crews are scheduled 2.5 hours per day of time in which to prepare for exercise (don exercise clothes and set-up equipment), perform continuous or interval style “aerobic” exercise and resistive exercise, and clean-up (personal hygiene, stow exercise hardware). Exercise countermeasures are primarily intended to maintain aerobic and muscular fitness, musculoskeletal and cardiovascular health, and performance (Smith et al. 2012; Moore et al. 2014; Lee et al. 2019).

Few spaceflight studies have specifically addressed whether inflight exercise affects the post-spaceflight orthostatic responses. In 30 Space Shuttle astronauts who self-selected their inflight exercise habits, those who performed moderate to vigorous aerobic exercise regularly during their mission attenuated the postflight tachycardia normally seen with standing on landing day (Lee et al. 1999). However, increasing the volume or intensity of inflight exercise did not decrease the number of astronauts who became presyncopal while standing on landing day. In a separate investigation, cardiac output and stroke volume was better maintained during a 10-minute stand test in four Space Shuttle astronauts who performed a bout of maximal exercise 24 hours before landing than in four subjects who performed a minimal amount of exercise (Moore

et al. 2001). Maximal exercise was previously demonstrated to restore plasma volume (Convertino et al. 1996) and protect LBNP tolerance (Engelke et al. 1996) after 16 days of bed rest.

Exercise countermeasures, in general, are expected to protect cardiovascular function and postflight egress capability. In a recent study of ISS astronauts, the decrease in peak oxygen consumption was less (-6%) in astronauts who perform vigorous exercise (English et al. 2020) compared to that previously reported (-15%) (Moore et al. 2014). A previous long-duration bed rest study demonstrated that cardiac mass and function are protected by a vigorous exercise countermeasure protocol similar to that used by astronauts (Scott et al. 2018).

d. Current Countermeasures: Fluid Loading

All astronauts returning from space on a NASA vehicle are required to ingest a “fluid load” of an isotonic solution consisting of salt tablets and water or an electrolyte solution. The efficacy of this countermeasure following Shuttle flights was evaluated by Bungo et al. (Bungo et al. 1985). Heart rate and blood pressure were measured during a passive stand test before and after spaceflight (54 to 194 hours postflight). Both variables, as well as a cardiovascular index of deconditioning (CID), were significantly improved, but not fully protected to preflight levels, when astronauts used the fluid loading countermeasure compared to control subjects who did not use fluid loading (**Figure 10**). However, the efficacy of the fluid loading countermeasure to mitigate plasma and blood volume losses was not measured.

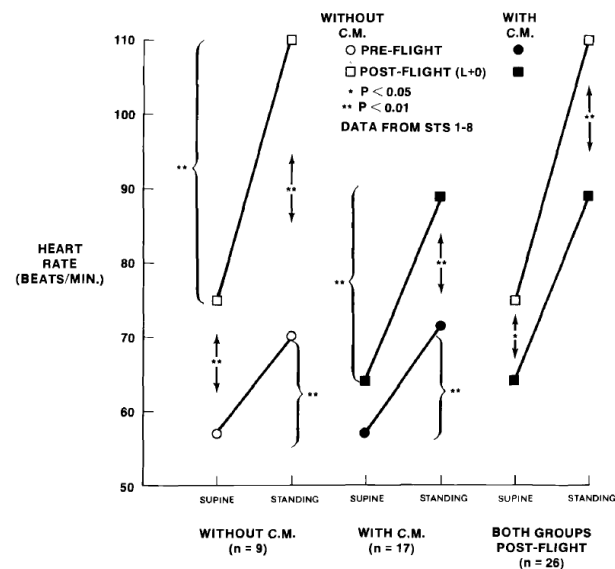


Figure 10. Heart rate response to standing with and without fluid loading countermeasure in Space Shuttle astronauts. Reproduced from (Bungo et al. 1985)

In 1985, consuming fluid and salt prior to landing (fluid loading) became a medical requirement, although the level of adherence to the countermeasure has not been clearly documented. Astronauts continued to return from space with plasma volume deficits ranging from 5 to 19% after Space Shuttle missions (Leach et al. 1996; Buckey et al. 1996b; Fritsch-Yelle

et al. 1996b; Waters et al. 2002; Meck et al. 2004). Additional confounding factors to accurate measurement of spaceflight-induced plasma volume loss include *ad lib* water ingestion following landing and IV fluid therapy given to the more severely affected crewmembers.

The fluid loading protocol currently used by cosmonauts and astronauts returning from ISS on the Soyuz differs from the protocol previously used by Shuttle astronauts. Both cosmonauts and astronauts increase salt intake with meals over the last several days of the mission. Between 18 and 20 ml/kg body weight of sodium chloride-water solution or dry salt (0.18-0.20 g/kg body weight) with water (18 -20 ml/kg body weight) are consumed three to four times in the last 12-20 hours before landing (Kozlovskaya et al. 1995). Russian cosmonauts also attempt to increase fluid levels by consuming fluids during Chibis sessions at the end of the mission. The water-salt additive normally taken with food before Chibis LBNP sessions consists of 0.9 g of sodium chloride and 300 ml of water (Kozlovskaya et al. 2010). Kozlovskaya et al. (Kozlovskaya et al. 2010) report that cosmonauts who participate in this form of end-of-mission fluid loading better tolerate the workloads associated with the final phase of the space flight mission and the post-flight reconditioning program.

We are unaware of any studies which specifically demonstrate the efficacy of the fluid loading countermeasure (i.e., measured plasma volume on landing day), particularly after long-duration missions. Meck et al. (Meck et al. 2001) reported that the change in plasma volume after long-duration spaceflight was not different than after short-duration spaceflight, but no details about the fluid loading protocols were provided and no control data were reported. Recent post-flight plasma volume data collected in ISS astronauts as part of NASA's Functional Task Test Study are confounded by the timing of the measurement (~1 day after landing) and by the common practice of intravenous administration of fluids soon after landing (Mulavara et al. 2018).

e. Current Countermeasures: Compression Garments

The Apollo 16 and 17 missions introduced countermeasures for orthostatic intolerance in the form of anti-hypotensive garments (Hoffler and Johnson 1975). Though the countermeasure appeared to provide moderate protection against orthostatic hypotension, testing in Apollo 16 was plagued with problems. The countermeasure for this flight was Jobst compression stockings with a pressure of 40-45 mmHg at the ankle and linearly decreasing pressure to 10 mmHg at the waist. The confined space inside the spacecraft prevented the crewmembers from donning the stockings inflight, and the stockings were only worn for a stand test after the postflight LBNP orthostatic tolerance test. Additionally, the stockings could not be fitted accurately for postflight testing due to the unquantifiable decrease in leg circumference. Finally, postflight testing was performed with ambient temperatures 10°C higher than preflight testing, augmenting the orthostatic stresses. Conversely, different countermeasure and testing conditions in Apollo 17 successfully prevented heart rate changes during LBNP. In this mission, the orthostatic test was performed in the air-conditioned Skylab Mobile Laboratory, and the anti-hypotension suit was an inflatable suit which applied lower body positive pressure from the ankles to the waist but was only used by a single crewmember. This crewmember donned the garment before reentry,

inflated it after splashdown while still in the spacecraft, and did not deflate the suit until ten minutes had elapsed in the passive stand test. Though the crewmember did not exhibit the typical change in heart rate during LBNP, it should be noted that he also did not follow the trend in cardiothoracic ratio change and postflight limb volume changes (Hoffler and Johnson 1975).

Astronauts and cosmonauts wear lower compression garments during re-entry to gravity (**Figure 11**). These garments compress the capacitance vessels, thereby promoting venous return. Lower body compression garments were not routinely worn by astronauts on NASA space vehicles until the Space Shuttle program. In astronauts (n=24) who inflated their anti-G suit (AGS) during re-entry landing, arterial pressure was less than those who did not inflate the AGS (n=8), although heart rate was not different (Perez et al. 2003) (**Figure 12**).



Figure 11. The NASA antigravity suit (left) contains bladders which are pressurized to produce compression. The Russian antigravity suit (Kentavr) utilizes a compressive material and lacings to produce compression.

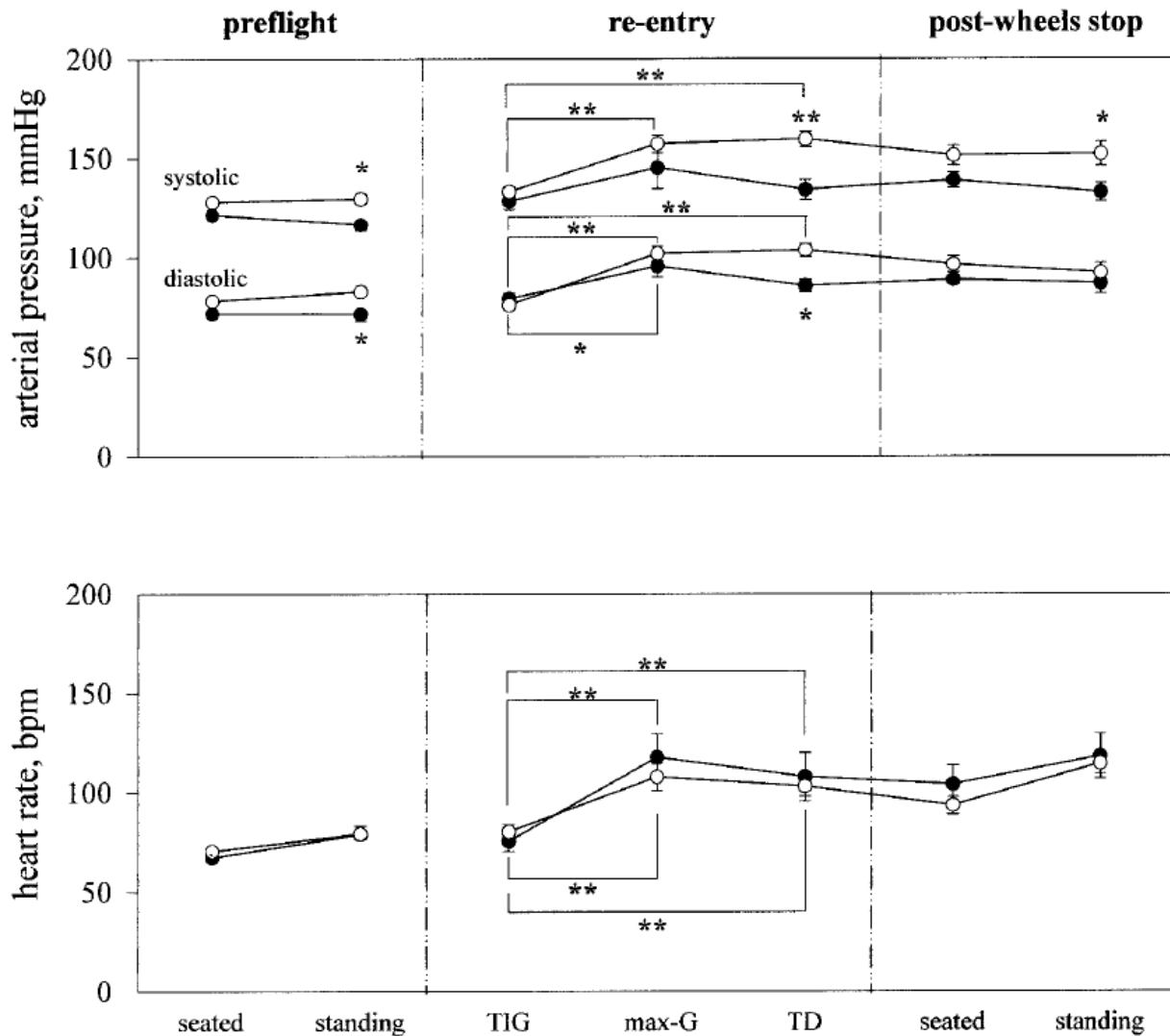


Figure 12. Heart rate and arterial pressure measured preflight, during re-entry, and after landing in astronauts who did (closed circles) and did not (open circles) inflate the anti-g suit (AGS). (Perez et al. 2003)

The Kentavr was worn originally by Soviet and Russian cosmonauts but also has been worn by NASA and other astronauts when returning from space on the Soyuz vehicle. In a review of data collected from 83 cosmonauts with and without compression garments during re-entry after spaceflight missions ranging from 8 to 438 days, (Vil-Viliams et al. 1998, p.) the cosmonauts who wore compression garments more accurately perceived the G-loads experienced during decent, while those without garments perceived the G-load to be up to 4 Gs higher than actual. The rise in heart rate during the high G portion of re-entry was not different during missions less than 1 month, but when missions exceed 1 month those without the Kentavr (or its predecessor the Karkas) had an exacerbated heart rate response.

There are some operational differences between the suits (Platts et al. 2009). One advantage of the AGS is that the pressure can easily be adjusted by the crewmember in increments of 0.5 psid (25.9 mmHg); the recommended pressure for re-entry is 1.5 psid (77.7

mmHg). A disadvantage of the AGS is that it must be connected to a pressure source to maintain compression. Once it is disconnected so that astronauts can egress the vehicle, the suit deflates as the subject moves. There is evidence that an inflated AGS more than doubles the metabolic cost of ambulation and may impede an astronaut's ability to safely egress a vehicle in the event of an emergency (Bishop et al. 1999). In addition, some crewmembers find the high pressure over the lower abdomen uncomfortable, particularly after fluid loading. The Kentavr is a non-inflatable elastic garment (nominal compression ~30 mmHg) that maintains protection after egress. In fact, cosmonauts often continue to wear the Kentavr for several days after landing (Lee et al. 2015). Crewmembers can adjust the level of compression by loosening or tightening the laces on the lateral portions of the Kentavr; the crewmembers reduce the compression as they readapt to their preflight condition. However, one disadvantage of the Kentavr is that uncovered areas of the body (for example, knees, feet and groin) may swell uncomfortably if the garment is worn for an extended period of time.

The Cardiovascular and Vision Laboratory at NASA Johnson Space Center has been testing different configurations of lower body compression garments (e.g., area of coverage, amount of compression) for use during exploration missions. Thigh-high garments (55 mmHg of compression at the ankle, decreasing linearly to 18 mmHg at the knee and to 6 mmHg at the top of the leg) were custom fitted to five Space Shuttle crew members (Stenger et al. 2010). Thigh-high garments provided some protection to orthostatic intolerance after a short duration Space Shuttle mission, but the lack of abdominal compression may have limited their efficacy (Denq et al. 1997). Despite evidence that leg-only compression can ameliorate vasovagal syncope (Dos Santos et al. 2013), it is generally accepted that leg-only compression is insufficient (Protheroe et al. 2011) and that abdominal compression alone is superior to leg only compression. As such, compression of both the legs and abdomen may be the ideal configuration (Denq et al. 1997; Smit et al. 2004). The JSC Cardiovascular and Vision Laboratory developed a 3-piece gradient compression garment (GCG) consisting of thigh-high stockings with overlapping biker-style shorts that provide 55 mmHg at the ankle decreasing to 35 mmHg at the knee, 18 mmHg at the top of the leg, and 16 mmHg over the abdomen. The GCG has zippers over the calf (area of highest compression) and along the sides of the biker shorts to aid in donning. The GCG prevented the tachycardia and drop in stroke volume during orthostasis normally seen after space flight and were comfortable to wear in a group of seven astronauts tested after four different Space Shuttle missions (Stenger et al. 2013).

The GCG also has been evaluated after long duration (~6 months) ISS missions (Lee et al. 2020c). After 2 minutes of prone rest, crew members were asked to stand as quickly as possible and maintain a quiet stand for 3.5 minutes. During this test, the GCG effectively mitigated increases in heart rate and decreases in mean arterial pressure when transitioning from the prone to upright posture. Overall, the GCG attenuated the postural change in heart rate and mean arterial pressure from pre- to post-spaceflight. However, three crew members were unable to start the stand test immediately following landing and one other subject terminated to test early, likely due to motion sickness.

f. Current Countermeasures: Cooling

The Launch and Entry Suit (LES) and Advanced Crew Escape Suit (ACES) that were worn by the Space Shuttle astronauts were designed to protect crewmembers against a loss in cabin pressure, contaminated atmosphere, and temperature extremes in case of bail out during re-entry. Given that the garments were intended to insulate the astronaut from the cold, the LES and ACES increased the heat stress to the astronauts during nominal operations. Elevated skin and core temperatures cause a shunting of blood to cutaneous vascular beds to dissipate heat, a decrease in blood volume due to sweating, and an increase in heart rate (Abraham et al. 1994). It was postulated that the increased heat load from the suits contributed to the increased incidence of orthostatic intolerance following resumption of Shuttle flights after the Challenger accident (Nicogossian et al. 1995). The addition of an in-suit fan that circulated cabin air across the upper torso of the astronauts was ineffective (Sawin et al. 1998), and crewmembers continued to report heat discomfort. A liquid cooling garment (LCG), a two-piece garment worn under the LES and ACES, covers the arms, torso, and legs, and circulates chilled water (thermoelectrically cooled using Shuttle power) through a network of small vinyl tubing sewn into the fabric, was added to the suit ensemble. In a group of Shuttle astronauts (n=3) who inflated anti-G suits and also wore the LCG, heart rate was lower than those who did not wear the LCG during re-entry and landing (Perez et al. 2003). Even with liquid cooling, core temperature is as much as 0.5 °C higher during re-entry while wearing the re-entry suit than the same time of day inflight without the garment (Rimmer et al. 1999).

g. Current Countermeasures: Pharmacotherapy

Several different pharmacotherapeutic including fludrocortisone, midodrine, and octreotide have been considered for use as countermeasures to post-flight orthostatic intolerance. However, none have been found to be suitable due to ineffectiveness in protecting against orthostatic challenge, the risk for adverse effects, or the difficulty in administering the drug in operational environments.

Fludrocortisone is a synthetic mineralocorticoid commonly used in Earth-bound patients to treat orthostatic intolerance. While administration of a single dose of fludrocortisone helped preserve plasma volume in crew returning from short-duration shuttle flights, it failed to protect against orthostatic intolerance (Shi et al. 2004).

Midodrine, an alpha-1 receptor agonist, also is used to treat orthostatic intolerance terrestrially and has been shown to effectively protect subjects from presyncope after two-week head-down tilt bed rest (Ramsdell et al. 2001). In a study of a female astronaut with presyncopal symptoms after her first spaceflight mission, midodrine administered 1 hour prior to landing on her second mission enabled her to complete a post-flight tilt test for the full duration while maintaining systolic blood pressure and cardiac output in contrast to her first flight (Platts et al. 2004). Moreover, midodrine has excellent oral bioavailability and its peak therapeutic effect occurs 1 to 2 hours after administration, offering the operational advantage of being able to initiate therapy once a decision to land has been made. However, midodrine may cause an increased akathisia response when co-administered with promethazine, a common drug used for treatment of space motion sickness, due to shared metabolism by cytochrome p450 isozyme 2D6

(Platts et al. 2006). It also may contribute to QT prolongation, potentially exacerbating known effects of spaceflight and drugs such as promethazine. Due to these concerns, midodrine is not used as a countermeasure for post-flight orthostatic intolerance.

Octreotide is a synthetic peptide (octomer) that is an analog of the naturally occurring hormone, somatostatin. It is a FDA-approved drug and is used for, among other indications, the treatment of hypotension in patients with autonomic dysfunction (Hoeldtke et al. 1991, 1998). Octreotide was shown to be superior to midodrine in preventing both post-prandial and orthostatic hypotension in 16 patients with autonomic neuropathy (Hoeldtke et al. 1998). A double-blind, placebo-controlled crossover study showed that octreotide improved orthostatic tolerance to a 45-minute tilt test (Jarvis et al. 2012). However, octreotide must be administered as a slow infusion or given subcutaneously to prevent reflex bradycardia and syncope, limiting its utility as pre-landing prophylactic countermeasure.

h. Current Countermeasures: Thigh Cuffs

Cosmonauts often wear cuffs around the upper thigh, particularly during the early adaptation to spaceflight, to reduce symptoms of head congestion, facial puffiness, and vestibular disorders (Kozlovskaya et al. 2010). The thigh cuffs, called “Braslets”, are composed of non-elastic and elastic sections to conform to the upper thigh and are tightened to a level that manages symptoms. The cosmonauts learn to adjust the compression of the Braslets in preflight training when placed 30°-head-down tilt (Hamilton et al. 2012). Cosmonauts target the compression that relieves or reduces symptoms of head congestion and achieves cranial impedance rheography waveforms similar to those observed when supine. The Braslets nominally apply 30-40 mmHg of compression and are prescribed to be worn for up to 10 hours/day continuously (Fomina et al. 2004, 2007) or in 2-3 hour segments with 30 minutes of rest (Robin et al. 2020). They typically are not worn during exercise or sleep.

The Braslets were first used in 1984 during a long-duration Mir mission (232 days) and were adopted as part of the cosmonaut countermeasure system in 1990. While the application of the Braslets decreases central blood volume (lower internal jugular vein diameter, stroke volume and cardiac output) and increases fluid distribution to the legs (increased femoral vein cross-sectional area) (Herault et al. 2000; Hamilton et al. 2012), their use does not appear to prevent signs of orthostatic intolerance during post-flight LBNP tests (Fomina et al. 2004, 2007). Similarly, the pilot on the 14-day Gemini 7 mission became presynopal during post-flight orthostatic testing despite intermittent use of thigh cuffs during his mission (Stevens et al. 1966). Interestingly, Herault et al. reported that the acute effects of Braslet application diminished later in the mission (Herault et al. 2000), perhaps due to ill-fitting thigh cuffs due to muscle atrophy (Hamilton et al. 2012).

4. Long-term Health: Cardiovascular Risk Factors

a. Age

As of March 2020, 562 astronauts and cosmonauts had flown in space (Smith et al. 2020). As of 2021, 350 individuals have been selected to be NASA astronauts, with an additional 10

selected at the beginning of 2022 (Lifetime Surveillance of Astronaut Health [LSAH], Unpublished data). The average age of an individual selected for the astronaut corps is 35 and 33 years for men and women, respectively (Hamilton et al. 2005). Men who have flown in space typically participated in their first mission in their early 40s and women typically had their first mission in their late 30s (Smith et al. 2020). The average age of the active astronauts is currently 52 and 45 years for men and women, respectively. Thus, most astronauts are “middle-aged” by the time they fly in space, and likely to begin to experience age-related health effects during their active careers. Whereas spaceflight crews now include more women than ever (women now constitute nearly 36% of the current active NASA astronaut corps), the majority of the astronauts and cosmonauts who have flown in space are middle-aged men (Harm et al. 2001; Platts et al. 2014; Smith et al. 2020), who, because of their age and sex, are more likely to experience dysrhythmias and have subclinical manifestations of CVD (Tsao et al. 2022).

b. Circadian Desynchrony

Crewmembers sleep significantly less before and during Space Shuttle missions relative to their return to Earth (Barger et al. 2014). Adequate sleep is not only essential for cognitive and operational performance during EVAs and tasks in spaceflight, but it also is necessary for the maintenance of cardiovascular health (Mullington et al. 2009). Variable light-dark cycles, Space Motion Sickness (SMS), adaptation to weightlessness and several other factors contribute to sleep reduction during space missions. Moreover, adaptations to local time zones immediately before and after missions may alter sleep structure and result in changes in circadian rhythm. Previous studies have analyzed sleep, performance, circadian rhythms, and light-dark cycles as well as pharmaceutical countermeasures during Space Shuttle flights to assess changes in sleep quality before, during, and after spaceflight (Putcha et al. 1999; Dijk et al. 2001). It is important to evaluate the effects and countermeasures of sleep deprivation during spaceflight not only to mitigate errors and accidents but also to consider the implications of CVD, especially for longer duration missions to the Moon and Mars.

Circadian rhythms are physical, mental, and behavioral cyclical changes of natural processes that respond to 24-hour light-dark sequences. Environmental disruptions in one's circadian rhythm, such as artificial lighting, jet lag, or even weightlessness during spaceflight may contribute to adverse health implications (Brainard et al. 2015). Myocardial infarction, sudden cardiac death, and transient ischemia are all strongly associated with circadian patterns, so there is reason to believe that circadian desynchronization may promote risk for cardiovascular events (Shaw and Tofler 2009; Thosar et al. 2018). Moreover, shift workers who experience circadian misalignment have increased 24-hour blood pressure, inflammatory markers (i.e. IL-6, C-reactive protein, and TNF- α), and cardiovascular risk (Mullington et al. 2009; Morris et al. 2016). Animal studies with Syrian (*Mesocricetus auratus*) hamsters have also demonstrated that disruptions in normal circadian rhythmicity significantly reduced longevity and increased susceptibility to early mortality due to cardiomyopathy (Penev et al. 1998).

Sleep reduction is associated with spaceflight, and astronauts have reported decreased perceived quality of sleep, although objective parameters do not indicate major abnormalities in

sleep efficiency or structure (Dijk et al. 2001). The average duration of sleep for crewmembers during flight was 6-6.5 hours, as compared to the first week post-mission was 7-8 hours (Barger et al. 2014). It is important to note, however, some crewmembers may have delayed onset of sleep episodes due to increased operational demands or voluntary leisure activities. Crewmembers also experienced shorter REM and sleep latencies and higher percentages of REM sleep upon return. Case studies on Earth have reported that sleeping less than six hours per night significantly increases chances of developing hypertension along with future risks for metabolic syndrome (Mullington et al. 2009). Since crewmembers on average receive more than six hours of sleep per night and are in peak physical condition, they likely are not as high of risk as short sleep duration individuals.

Ninety-four percent of astronauts have recorded taking some medication during their missions, mostly including anti-SMS formulations and sleep aids (Putcha et al. 1999; Barger et al. 2014). Promethazine was the predominant choice for SMS during the first mission day, but sleep medications, frequently zolpidem and flurazepam, were taken throughout the entire mission duration (Wu et al. 2018). Zolpidem users have a tendency for decreased risk of heart disease; however, flurazepam has been observed to significantly elevate heart rate and inhibit contraction of smooth muscle, which may increase risk of heart failure with regular use (Ishii et al. 1983; Kim et al. 2018). Both are highly addictive and have a risk for life-threatening sleep behaviors, so supervision and trial use before missions is encouraged.

Melatonin also is used occasionally as a sleep-aid. Dijk et al. (Dijk et al. 2001) observed that melatonin (0.3 mg) on alternate nights did not significantly improve sleep on actigraphic measurements; however, polysomnography detected that melatonin significantly reduced sleep latency. Melatonin has also been found to provide protective effects for some CVDs (Sun et al. 2016), and adverse arrhythmia side effects are rare for promethazine users, especially for infrequent doses. Overall, use of promethazine and sleeping-aid medications under supervision by medical professionals for treating SMS and sleep deprivation likely do not pose substantial cardiovascular health threats.

As more crewed missions launch and LEO commercialization gains popularity, an increasing number of voyagers will experience desynchronization and subsequently cardiovascular stressors. Humans have evolved to respond to 24-hour light-dark sequences, however, the Moon's light-dark cycle is 29.5 hours. With plans to inhabit the Moon through NASA's Artemis Program, it is important to consider that differences in light-dark cycles will disrupt astronauts' circadian rhythms and may potentiate cardiovascular events where the environment is much less forgiving.

c. Traditional Biomarkers: In-Mission

Whether spaceflight is associated with changes in circulating cholesterol levels and proportions is unclear. Lee et al. observed that total cholesterol and LDL increased during spaceflight, whereas HDL was unchanged; thus, total cholesterol-to-HDL ratio and LDL-to-HDL ratio were elevated (Lee et al. 2020d). Similarly, in the Garrett-Bakelman (2019) study of the

astronaut twin who flew for 340 days on ISS, the ratio of apolipoprotein (APO)-B to APO-A was elevated during spaceflight (Garrett-Bakelman et al. 2019). In contrast, Smith et al. observed no change in total cholesterol or triglycerides from preflight to postflight (Smith et al. 2005), and Hughson et al. observed no change in total, HDL, or LDL cholesterol from preflight to inflight, although plasma triglycerides were lower (Hughson et al. 2016).

Some evidence exists that insulin resistance is increased by spaceflight exposure. Stein et al. reported that C-peptide, a metabolite of insulin, was significantly greater than preflight in four astronauts after a 9.5-day mission, and they observed similar findings during 12 days of bed rest (Stein et al. 1994). More recently, Hughson et al. observed that blood glucose levels were not different after long-duration spaceflight but circulating insulin was increased by ~50% and HOMA-IR was elevated by almost 60% (Hughson et al. 2016).

Studies of the carotid artery using ultrasound have yielded inconsistent results across three different groups of astronauts. Common carotid artery intima-media thickened in 10 astronauts within two weeks after launch to the ISS and remained thickened when measured late in their 6-month mission and 4 days after landing (Arbeille et al. 2016). In those same astronauts, carotid artery stiffness increased and distensibility decreased from pre- to inflight but was not different than preflight after landing (Arbeille et al. 2017). In a separate group of eight ISS astronauts, carotid artery stiffness increased and distensibility decreased when measurements were obtained within 38 hours after landing after a 6-month mission (Hughson et al. 2016). More recently, in- and postflight measures of carotid artery stiffness, distensibility, and intima-media thickness (cIMT) were not different than preflight in 13 astronauts who participated in ISS missions up to 11 months (Lee et al. 2020d). In these studies, there were no reports of incidental findings associated with plaque and no cases where cIMT was considered abnormal (>0.9 mm) (Mancia et al. 2013) or suggestive of diffuse plaque (>1.5 mm) (Johri et al. 2020).

Baevsky et al. reported that pulse transit time, another measure of vascular stiffness, decreased in cosmonauts after 6-month spaceflights when measured at both three and six days after landing (Baevsky et al. 2007a), in agreement with the findings from previous from ultrasound measures (Hughson et al. 2016). A more recent report in a separate group of cosmonauts did not replicate these findings (Hoffmann et al. 2019), though a wide range of postflight responses were reported.

d. Oxidative Stress and Inflammation: In-Mission

Several studies have suggested that biomarkers of oxidative stress are elevated during spaceflight (Buchheim et al. 2019; Garrett-Bakelman et al. 2019; Lee et al. 2020d), although the specific source of the oxidative stress is yet to be determined (Smith et al. 2009). In the course of a spaceflight mission, astronauts can experience oxidative stress induced from a variety of conditions, including changes in exercise (Moore et al. 2010) and dietary habits (Smith et al. 2005), exposure to ionizing radiation, psychological stress, and hyperoxic conditions associated with extravehicular activity (EVA) and EVA pre-breathe protocols.

Increased levels of inflammation biomarkers during spaceflight have been observed (Crucian et al. 2014; Lee et al. 2020d). Lee et al. (2020d) reported that biomarkers of inflammation that were increased from pre- to inflight were elevated by flight day 15 and were consistently high during the mission. Interestingly, hs-CRP, a biomarker that has predictive value in CVD risk (Khera et al. 2018; Gore et al. 2020), decreased for pre- to inflight. Crucian et al. (2014) reported that fewer biomarkers increased from pre- to inflight, although their ability to detect changes may have been hampered by the assays' minimum threshold for detection and the statistical treatment.

Biomarkers of oxidative stress and inflammation that are elevated during spaceflight generally return to preflight levels in the days and weeks after landing (Crucian et al. 2014; Garrett-Bakelman et al. 2019; Lee et al. 2020d). Notably, hs-CRP and IL-1ra are increased in the week after landing, perhaps related to muscle damage (Isaacs et al. 2019) secondary to reloading after re-exposure to gravity. The dynamic nature of the readaptation process makes interpretation of biomarker data difficult (Lee et al. 2020a); therefore longitudinal surveillance of these biomarkers may be more informative.

e. Traditional Biomarkers: Longitudinal Changes

Minimal data on CVD biomarkers across the astronaut lifespan has been published. Most reports of traditional CVD biomarkers such as cholesterol levels and glucose metabolism variables, are only reported inflight and near flight. Recent CVD analyses (Ade et al. 2017; Charvat et al. 2022) only include baseline levels of these biomarkers and do not report any longitudinal results. Across an individual's lifespan in the general US population, increasing blood pressure, cholesterol levels, and impaired glucose metabolism are noted (Tsao et al. 2022).

f. Coronary Artery Calcium: Longitudinal Changes

Coronary Artery Calcium (CAC) scoring measures atherosclerotic burden in the coronary arteries using CT scan, with scores computed using the Agatston algorithm ranging from 0 to over 1000. Higher scores are indicative of a markedly increased risk of coronary heart disease (CHD) (4- to 10-fold higher) independent of other risk factors (Radford et al. 2018). Increased CAC scores also have been associated with atrial fibrillation and congestive heart failure (O'Neal et al. 2015).

As astronauts tend to be middle-aged, they already have been exposed to risk factors and lifestyle behaviors which along with predispositions due to genetics contribute to accumulation of coronary artery calcium. Although CAC scores are considered during astronaut selection and selection to spaceflight missions (Hamilton et al. 2006) longitudinal analyses of CAC scores have not been published. Inclusion of CAC scores in the Astronaut Cardiovascular Health and Risk Modification (AstroCHARM) risk calculator improves the prediction of future cardiovascular events in apparently healthy individuals (Khera et al. 2018; Gore et al. 2020).

While adults who are free of CAC at an initial exam many develop subsequent CAC, the utility of serial measurement of CAC in clinical practice is currently being debated in the medical

literature. Initially it was hypothesized that evaluation of the change in CAC score could enhance cardiovascular risk prediction, with an early study from 2004 suggesting that change in CAC may be a meaningful and additive contribution to predicting an individual's risk (Raggi et al. 2004). However, in the last several years population-based studies that have paired scan data and adequate numbers of atherosclerotic cardiovascular disease (ASCVD) events have been reported. A large change in CAC score (> 100 U/y) was associated with hard CVD events independent of CHD risk factors and baseline CAC score. However, the change in CAC score was only the 5th strongest risk factor for CHD, following CAC baseline score, being male, SBP, and total cholesterol (Budoff et al. 2013). A report using data from the Cooper Center Longitudinal Study (CCLS) found similar results, that while CAC score is associated with asymptomatic CVD, the addition of change in CAC scores only modestly improved the prediction of CVD events when compared to a model using baseline CAC score (Radford et al. 2018). Most recently, a report from the Heinz Nixdorf Recall Study with over 3000 subjects with paired CAC scans over five years and eight years of follow-up after both scans found that although various measures of CAC progression were independently associated with outcomes, none of the CAC progression algorithms they tested demonstrated an improvement in determining risk for hard CHD or hard asymptomatic CVD compared with models including risk factors and baseline CAC score (Lehmann et al. 2018).

5. Long-term Health Consequence: Atrial Dysrhythmias

In 2010, NASA convened the “Atrial Arrhythmia Summit” and solicited expert opinion in the fields of electrophysiology, cardiology, aerospace medicine, and epidemiology. Panel members were given the charge of reviewing cases of atrial dysrhythmias in astronauts with the intent of providing recommendations regarding screening, diagnosis, and treatment of the astronaut corps, determining lifestyle and space-flight related factors that might predispose astronauts to atrial dysrhythmias, and assessing risks to astronauts who participate in long-duration spaceflight or extravehicular activities after treatment. At that time, 17 astronauts had been diagnosed with an atrial dysrhythmia out of a total cohort of 317 active and retired astronauts, representing a prevalence of approximately 5% (Barr 2010; Khine et al. 2018). Most cases were asymptomatic, and no occurrences of atrial fibrillation or flutter were recorded during flight. Of the cases, 5 individuals presented with atrial dysrhythmias between 2001 and 2010 and received radiofrequency ablation treatment.

While the prevalence of atrial dysrhythmias was not stated as a significant concern, the age of presentation was lower in astronauts than in the general population (~40 vs. ~60 years old (Abdulla and Nielsen 2009; Barr 2010). The panel suggested two factors that could account for earlier than expected presentation of atrial dysrhythmias: more frequent and comprehensive screenings in the astronaut corps in the form of annual physicals leading to a higher discovery rate and greater levels of vagal tone associated with a high frequency and duration of aerobic exercise performed by astronauts (Wilhelm 2014). In middle-aged non-elite distance runners of similar age range (40 to 50 years old) lifetime training volume is associated with increased vagal tone, prolongation of signal-averaged P-wave duration, and an increase in left atrial volume (Wilhelm et al. 2011a). Adaptations such as these, coupled with a higher systolic blood pressure,

are more pronounced in men than women and may explain the different rates of atrial fibrillation associated with sex (Wilhelm et al. 2011b). Other investigators have noted an association between exercise volume and risk of atrial fibrillation in men but not in women (Zhu et al. 2016).

While the Atrial Arrhythmia Summit panel concluded that there was no evidence that spaceflight-related factors contribute to the development of atrial dysrhythmias during spaceflight, they speculated as to which specific factors might be of concern, particularly for future missions: weightlessness-induced cephalad fluid shifts resulting in prolonged atrial distension, spaceflight-induced increase in sympathetic activation, plasma volume expansion secondary to elevated dietary sodium intake in salt-sensitive individuals due to the use of sodium as a food preservative for spaceflight missions, and chronic radiation exposure.

Continuing to monitor the cardiovascular diagnoses among astronauts, Charvat et al. (2021) reported that through 2019, 36 NASA astronauts had been diagnosed with atrial fibrillation or flutter for a prevalence of 10%. However, only nine crewmembers were diagnosed during their active career. Nine of 36 crewmembers underwent radiofrequency ablation for treatment. Compared to a primary care population (Barrios et al. 2012), astronauts had a lower prevalence in all age categories except from the ages of 43-49, supporting the previous suggestion that atrial arrhythmias may be diagnosed at a younger age among astronauts.

6. Long-term Health Consequences: Cardiovascular Disease

As of the end of 2021, the average age of living retired astronauts is 69 years (range: 39 to 94 years), with nearly 30 astronauts over the age of 80. Prior to now, few astronauts presented with chronic diseases, and analyses for these chronic diseases have been limited to mortality. Only recently have enough cardiovascular diagnoses accumulated to begin studying morbidity. Interpretations of the data should be viewed with caution considering the low numbers of astronauts and the small number of astronaut deaths attributable to CVD. Most analyses have been performed to determine the effects of spaceflight on long-term cardiovascular health (Appendix A). The majority of these have compared astronaut mortality to other cohorts, including the general population, NASA civil servants, and professional athletes, though two studies have attempted to assess morbidity.

The first studies on astronaut mortality were conducted within the Longitudinal Study of Astronaut Health (Peterson et al. 1993; Hamm et al. 2000). Peterson et al. examined records from 195 astronauts selected between 1959 and 1991 compared to the US General population (Peterson et al. 1993). Hamm et al. examined the same astronauts compared to a JSC Civil Service sample, for a longer follow-up period (Hamm et al. 2000). Cause of death in the astronauts was categorized based upon death certificate and autopsy reporting. Peterson et al. found that circulatory disease accounted for only two of 20 (10%) reported deaths, and the standardized mortality ratios (SMR) in astronauts were 47 (95% CI: 5-168) and 64 (95% CI: 7-230) for circulatory disease and ischemic heart disease, respectively (Peterson et al. 1993). The small number of CVD deaths in this study yielded large confidence intervals that were not statistically significant. Hamm et al. found three of the 195 astronauts and seven of the 575 comparison subjects had

died from CVD, which resulted in a relative risk of 1.20 (95% CI: 0.27-5.28) (Hamm et al. 2000). Accidents accounted for the largest proportion of the deaths in both analyses. Because these data were from early in NASA's space program, most astronauts were not at an age at which CVD would be expected to have developed.

More recently, astronaut mortality data has been curated from publicly available records by Reynolds and Day to compare the astronauts to other groups of individuals. When astronaut mortality was compared to the general population of Harris County, Texas (the county in which Johnson Space Center is located), astronauts had lower CVD mortality (SMR = 27, 95% CI 9-63) (Reynolds and Day 2010). This analysis did not look at the difference in astronauts to the US general population for CVD mortality, only all-cause mortality. Another study by Reynolds and Day (Reynolds et al. 2014) compared mortality between NASA astronauts and Russian cosmonauts. This report found that cosmonauts have a lower risk of death due to any cause compared to US astronauts; however, cosmonauts have a higher risk of circulatory disease deaths. It is possible, however, that the comparisons between astronauts and cosmonauts are not appropriate given that historically cosmonauts have engaged in longer duration missions, through the Salyut and Mir Space Station programs, than NASA astronauts and other culture-related health behavior differences exist (Shkolnikov et al. 1998). This highlights again the potential heterogeneity found within different populations that can contribute to confounding factors in epidemiological analyses.

Reynolds and Day also compared CVD among the Apollo astronauts to address if those individuals who flew a mission to the Moon had higher rates of CVD mortality (Reynolds and Day 2017). Results showed that there was no evidence of increased risk of CVD death in astronauts who traveled to the Moon compared to those who did not (SMR = 72, 95% CI: 15-210). There continued to be significantly lower CVD death among astronauts compared to the general US population (15, 95% CI: 3-43), similar to findings from their other studies.

A final analysis by Reynolds and Day compared NASA astronauts to professional baseball (Major League Baseball) and basketball (National Basketball Association) players (Reynolds and Day 2019). Astronauts had lower rates of CVD death than NBA players but could not be compared to MLB players on cause specific deaths due to incomplete data. Astronauts also had lower rates of death due to any cause than the professional sportsmen. However, similar to other analyses, astronauts had higher cause-specific SMR for accidental deaths.

An analysis performed by Elgart et al. found that among astronauts selected before 1970, the SMR for CVD mortality was 33 (95% CI: 14-65) compared to the US white male population age and calendar year matched (Elgart et al. 2018). This result was consistent with the findings of Reynolds et al. showing astronauts having lower risk of death due to CVD than the general population (Reynolds and Day 2010). Additionally, in an internal logistic regression analysis comparing astronauts with higher radiation exposure to those with lower radiation exposure, there was no evidence that radiation exposure was associated with CVD. Further Elgart et al. presented a power analysis detailing the number of astronauts needed per level of radiation effectiveness to achieve a power of 80% (Elgart et al. 2018). Sample sizes of over 400,000 deaths

due to CVD would be necessary to achieve a power of 80% for radiation exposures consistent with that of early astronaut cohort. As this number of astronauts will not be likely in the near future, this study underscores the need to temper any significant results with an understanding of statistical and cohort limitations.

As stated by Elgart et al., “the most promising approach for future radiation research in astronaut populations would be a longitudinal study with a suitable control population” (Elgart et al. 2018). The Lifetime Surveillance of Astronaut Health occupational surveillance program has been working to find appropriate cohorts for comparison to NASA astronauts. One promising comparison is to the Cooper Center Longitudinal Study (CCLS), a study that began in 1971 with nearly 400,000 records from over 100,000 participants. The CCLS is a repository for one of the largest and most comprehensive datasets of health-related information in the world collected as de-identified clinical data from the Cooper Clinic in Dallas, Texas. The goal is to select comparison subjects with more similar baseline characteristics than was possible in previous studies to address potential selection bias and healthy worker effects. In addition, larger numbers and more detailed medical records are available for this group which should help to control for reporting bias.

A historical cohort study matching CCLS controls to NASA astronauts was undertaken. Controls met the cardiovascular qualifying standards (e.g. $VO_2\text{max} > 32.9 \text{ ml/kg/min}$) of the NASA astronaut corps and were matched retrospectively on sex, date of birth and age in a ratio of 5:1 to astronauts who completed at least one spaceflight mission prior to December 31, 2016 (Charvat et al. 2022). Importantly, this analysis including matching based upon maximal oxygen consumption as a measure of overall health because of its relationship to CVD mortality (DeFina et al. 2019). Astronauts and matched controls were followed for CVD mortality. A total of 303 NASA astronauts, alive as of 1970 (the beginning of the CCLS) were matched to 1514 CCLS controls (all astronauts had five well matched controls save one who had four). There were 11 deaths due to CVD among astronauts, and 46 deaths due to CVD among CCLS participants. The hazard ratio for the exposure variable of study group was 1.12 (95% CI: 0.57-2.22), indicating a slightly elevated, non-statistically significant risk of CVD death among astronauts compared to CCLS. When adjusting for baseline cardiovascular covariates of smoking status, cardiorespiratory fitness, systolic blood pressure, and total cholesterol, these results did not change (HR: 1.10 [95% CI: 0.50-2.45]).

The low number of CVD deaths in both the astronauts and CCLS participants is indicative of two healthy cohorts. These findings have important implications for the assessment of astronaut health and suggest that individuals in the astronaut corps are not at an increased risk of death due to CVD for the given spaceflight exposures to date. The United States has had over 20 years of continuous presence in space aboard ISS. The current mission profile for a NASA astronaut is six months in LEO; however, missions to Mars will last over three years with no ability for return to Earth, and further study of total cardiovascular risk is warranted.

In one of the few studies of CV morbidity, Ade et al. (2017) used data from the LSAH Study to compare cardiovascular diagnoses (including death) among astronauts to a matched

civil servant cohort from Johnson Space Center. In a 3:1 ratio of comparison subjects to astronauts, employees were recruited at the time of each astronaut class. Employees were matched to astronauts by sex, age, and body mass index. Data were derived from physical examinations of active astronauts (required annually), inactive or retired astronauts (voluntary participation), and the matched cohort (examinations offered every 1-3 years). Cause of death was confirmed by death certificate, with these reports supplemented by autopsy reports in astronauts.

A total of 16 astronauts had a clinical CVD endpoint over a follow-up period of 22 years while 46 non-astronaut comparisons developed a CVD endpoint. The incidence rate (per 1000 person-years) was similar between the two groups with astronauts having 2.34 events and comparisons having 2.15 events. There was no difference in the time to CVD event when controlling for cardiovascular covariates including age, cholesterol, body mass index (BMI), smoking status, and blood pressure. While the LSAH Study provided geographical control for the study participants, it had several other limitations. Despite best efforts at matching the cohort to the astronauts at baseline, astronauts attained higher educational levels (% graduate degrees) and had higher rates of smoking, but lower mean values of blood pressure, glucose, triglycerides, and hemoglobin, although these latter differences were likely not clinically relevant. Factors that could have influenced these results may have resulted from lifestyle factors; lifestyle data were not collected at baseline for most of the subjects and only were included in data collection procedures starting in 1994. Also, 11 of the comparison cohort had medical conditions (nine with hypertension, two with diabetes) at baseline that would have been disqualifying factors for astronauts. However, only one of these 11 comparison subjects died during the study period, and analyses excluding these 11 subjects did not substantially affect the outcomes. This analysis also did not differentiate mortal and morbid events, only looking at a single cumulative outcome. Further, all astronauts regardless of spaceflight experience were included in the analysis limiting the interpretation of results to the astronaut occupation.

In a second aim of the historical cohort study comparing astronauts to a cohort from CCLS, cardiovascular morbidity and mortality were studied through 2011 (Charvat et al. 2022). A total of 18 CVD events were seen among 300 astronauts and 64 among the 1472 controls. The crude incidence of CVD events through 2011 was 1.90 per 1,000 person-years of follow-up for astronauts versus 1.37 for CCLS. There was only one non-fatal CVD event among women. NASA astronauts had significantly increased risk of CVD events compared to their CCLS matched cohort (HR = 1.84 (95% CI 1.04-3.27)). This relationship persisted after adjustment for CVD covariates (adjusted HR = 2.41 (95% CI 1.26-4.63)). The etiology for this minor difference in CVD events is uncertain. Based on selection criteria, at their time of entrance into the Astronaut Corps, astronauts are considered very low risk for CVD. However, during spaceflight astronauts are exposed to an environment that induces behaviors that are known to increase risk of CVD, such as decreased total physical activity, increased stress, and decreased sleep.

All analyses of CVD mortality reported thus far have a significant limitation. Most studies of astronaut mortality have had higher than expected SMRs for all-cause mortality and accidental deaths. In general, accidental deaths have occurred at a younger average age than rates of death

due to natural causes (Reynolds and Day 2018). Consequently, it is impossible to determine whether the individuals who died of accidental causes would have developed CVD had they not died at an early age (Austin et al. 2016). One recent study (Reynolds and Day 2018) has looked at the possibility of competing risks in astronaut mortality, analyzing differences between natural causes of death (e.g., cancer, CVD) and external causes (e.g. accidents). When these competing risks are ignored, the assumption of independent censoring is violated, meaning that those who die at a younger age are not represented in the population who survive. When these competing risks are ignored, the mortality due to a particular cause may be underestimated. Reynolds and Day found that this was not necessarily the case among astronauts and cosmonauts (Reynolds and Day 2018); however, results should be interpreted with caution as the number of astronauts dying from either natural or external causes is low.

In summary, astronauts have lower rates of CVD mortality compared to the general United States populations and similar rates when compared with a well-matched cohort. Factors that likely contribute to the lower rates of CVD death among astronauts include extensive medical screening that is inherent in the astronaut selection process, the frequent re-evaluation of cardiovascular health throughout their career through annual and flight qualification physicals, prompt access to treatment should an event arise, and the generally high level of physical fitness that the astronauts are expected to maintain throughout their career (Reynolds and Day 2010). However, interpretation of the results from these particular studies are impaired by issues including lack of control for lifestyle factors (e.g., diet, smoking, alcohol consumption, stress, exercise habits) and competing risks of younger deaths due to external causes (Reynolds and Day 2017).

7. Long-term Health Consequences: Risk Prediction Algorithms

NASA flight surgeons currently use risk calculators to estimate risk in astronauts under their clinical care. A 2015 analysis by LSAH found differing scores between four different risk calculators: an EMR Generated Framingham score, Framingham General CVD risk score, Framingham 10-Year Coronary Heart Disease Risk Score and the PROCAM (Prospective Cardiovascular Münster study) risk score (Jain et al. 2015). Similar cardiovascular biomarkers across the four risk calculators included smoking status, diabetes diagnosis, age and systolic blood pressure. While other variables that comprised the risk score differed, a similar ability to predict the risk of cardiovascular events among astronauts was observed.

To improve CVD risk profiling among astronauts, an apparently healthy population, the Astronaut Cardiovascular Health and Risk Modification calculator (Astro-CHARM) was developed to improve risk prediction using more novel biomarkers (Khera et al. 2018). The American College of Cardiology and American Heart Association (ACC/AHA) Pooled Cohort Equation (PCE) was used as a baseline model and includes age, sex, race/ethnicity, total cholesterol, HDL cholesterol, systolic blood pressure, use of hypertension medication, current smoking status, and diabetes diagnosis. Investigators added coronary artery calcium scoring (CAC), family history of myocardial infarction, high-sensitivity C-reactive protein (hs-CRP), statin use, and glucose levels to see if risk prediction could be improved. The addition of CACs, family history and hs-CRP improved the

ability to predict incident CVD events over the ACC/AHA PCE (Khera et al. 2018). There have been no published reports on the use of Astro-CHARM in the general US population.

In order to improve the long 10 year time frame in which AstroCHARM adequately predicts the risk of CVD, investigators sought to create a shorter term (3-year) risk assessment for CVD events among adults without known CVD (Gore et al. 2020). This team once again used the PCE as a baseline model and tested additional cardiovascular-related biomarkers and imaging studies for their relationship to either global CVD or atherosclerotic CVD. Cardiovascular biomarkers added included N-terminal pro B-type natriuretic peptide (NT-proBNP), high-sensitivity cardiac troponin T (hs-cTnT) and hs-CRP, and ECG determination of cardiac hypertrophy (ECG-LVF). Imaging studies included either a cIMT or CACS, depending on which study the data were derived. The only variable found not to be related to either CVD outcome was ECG-LVF. The other markers were combined to form a score of 0-4 and a higher score was found to be related to increasing CVD risk among the cohort tested. This risk assessment could be used for evaluation of healthy individuals, such as astronauts, where short-term assessment of risk is needed. However, these findings have not been validated in additional cohorts.

Minimal data from these risk appraisals for astronauts have been published. After six months of spaceflight the estimated 10-year CVD risk was not different than preflight, whether calculated with the AHA 2018 Risk Calculator (Grundey et al. 2018) or Astro-CHARM (Khera et al. 2018), but risk calculated with the AHA 2018 calculator from inflight measures increased a small, but statistically significant amount compared to preflight (Lee et al. 2020d).

B. Ground-based Evidence (Ground Analogs and General Terrestrial Research)

1. Dysrhythmias

In general bed rest studies do not report increases in ventricular ectopy, although one study has suggested a potential for increased susceptibility based upon microvolt T-wave alternans (MTWA) assessed during a graded exercise test. After 9 to 16 days of bed rest, the number of subjects who were MTWA positive increased from 17% (4 of 24) before bed rest to 42% (10 of 24) after bed rest (Grenon et al. 2005); 2 of the 4 subjects who were categorized as MTWA positive before bed rest became MTWA negative, while 8 who were negative before bed rest were categorized as MTWA positive after bed rest. However, MTWA was not present in any subject who had a heart rate below 110 bpm, the threshold for which MTWA would be considered clinically significant. Clinically significant MTWA is associated with cardiac arrest, sudden death, and ventricular dysrhythmias in patient populations (Rosenbaum et al. 1994; Ikeda et al. 2002).

Multiple bed rest studies have reported decreased cardiac mass, particularly in the left ventricle, following chronic exposure to simulated spaceflight (Levine et al. 1997; Perhonen et al. 2001a; Spaak et al. 2005; Dorfman et al. 2007; Shibata et al. 2010; Hastings et al. 2012; Carrick-Ranson et al. 2013), with the magnitude of cardiac atrophy associated with the duration of exposure (Westby et al. 2016). Additionally, results from a small number of women suggest that sex does not affect the magnitude or time course of cardiac atrophy (Tuday and Berkowitz 2007;

Dorfman et al. 2007; Westby et al. 2016). Cardiac atrophy can be prevented by exercise alone (Shibata et al. 2010; Hastings et al. 2012; Carrick-Ranson et al. 2013) or in combination with orthostatic-like stress (Dorfman et al. 2007). Although decreases in left ventricular mass would be expected to contribute to orthostatic intolerance (Levine et al. 1997) and exercise capacity (Dorfman et al. 2007), to our knowledge no bed rest study has yet to report that testing was terminated as a result of dysrhythmias (Lee et al. 2010).

2. Venous Thromboembolism

Acute changes in blood flow in the arteries and veins in response to spaceflight can be modeled using altered gravity simulations such as parabolic flight. Increased IJV diameter and pressure are observed during reduced gravity parabolic flight (Martin et al. 2016; Lawley et al. 2017; Lee et al. 2020b), as has been reported during spaceflight (Arbeille et al. 2001, 2015, 2021; Marshall-Goebel et al. 2019). However, stagnant flow does not develop in the IJV upon entry into an altered gravity condition from 1-G when subjects are seated upright relative to the change in acceleration (Lee et al. 2020b). A research study to investigate IJV flow while subjects are supine upon entering weightlessness during parabolic flight to confirm these findings has been initiated. Importantly, when seated the flow pattern during reduced gravity parabolic flight is similar to that observed when subjects are supine, and stagnant flow does not develop until exposure to altered gravity is prolonged (Marshall-Goebel et al. 2019).

It is unclear whether partial gravity conditions, like those expected on the Moon and Mars, will reduce the incidence of stagnant flow compared to weightlessness. There are no data available from the Apollo missions that explored this area of venous structure and function. Results from a recent parabolic flight campaign suggest that > 0.50-G must be attained before restoration of flow similar to that experienced in upright posture in 1-G in a majority of subjects (Lee et al. 2020b). Thus, stays in reduced gravity during exploration missions still may include a concern for the development of venous thromboembolism.

3. Orthostatic Intolerance

a. Bed Rest

In 2004, slightly over 164,000 patients were hospitalized in the United States with a diagnosis of orthostatic hypotension (Shibao et al. 2007). Causes of these hospitalizations ranged from simple volume depletion to autonomic failure, with most hospitalizations occurring in those over age 65. Previous work has shown that the pattern of post-spaceflight orthostatic intolerance is similar to that seen in patients with autonomic failure (Fritsch-Yelle et al. 1996b). However, studies that involve ill subjects tend to make extrapolation to the astronaut corps difficult, thus a model that includes otherwise healthy individuals is preferable. Bed rest studies, particularly those at 6° head-down tilt, in which apparently healthy subjects participate are traditionally used as the best ground-based analog to spaceflight (Pavy-Le Traon et al. 2007a; Fortney et al. 2010; Lee et al. 2010). Like spaceflight, bed rest results in decreased body mass, reduced blood and plasma volume, decreased exercise capacity, and orthostatic intolerance. A recent bed rest study replicated the mild hypercapnic environment (ambient CO₂ of 0.05%) experienced by astronauts

on the International Space Station; this exposure did not appear to affect the incidence of orthostatic intolerance (Clément et al. 2022b).

Many bed rest studies have focused on elucidating the mechanisms of orthostatic hypotension. These mechanisms include cardiac atrophy, sympathetic dysfunction, and arterial and venous dysfunction. Numerous publications have demonstrated cardiac atrophy following bed rest (Levine et al. 1997; Iwasaki et al. 2000; Perhonen et al. 2001a), resulting in a stiffer left ventricle and decreased stroke volume (Levine et al. 1997). Micro-RNAs associated with cardiovascular structure and function were differentially expressed during 30 days of bed rest, elucidating pathways that contribute to cardiovascular deconditioning and possible targets for intervention (Ade and Bemben 2019).

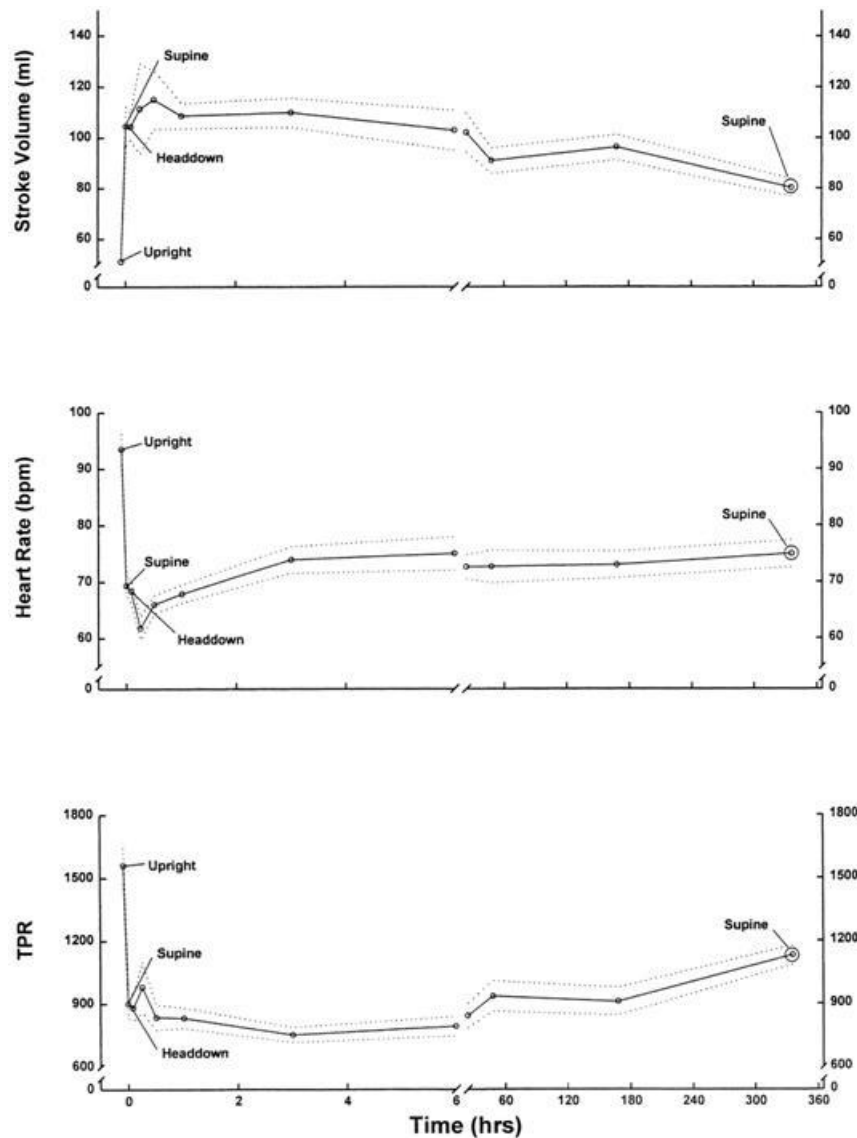


Figure 13. Stroke volume, heart rate, and total peripheral resistance measured during 14 d of 6° head-down bed rest (Levine et al. 1997).

Spaceflight-induced plasma volume losses (4-17%) have been replicated using 6° head-down tilt bed rest as an analog (Pavy-Le Traon et al. 2007b). Most of the loss occurs within the first week, and plasma volume then remains stable for the duration of bed rest. Recent bed rest studies have shown a markedly increased urine excretion upon initiation of bed rest (Trappe et al. 2006; Meck et al. 2009).

Many studies have shown that there is a disruption in the autonomic nervous regulation of the cardiovascular system following bed rest. Carotid baroreflex gain is reduced after short-duration bed rest (Convertino et al. 1990; Eckberg and Fritsch 1992) and is associated with an increase in orthostatic intolerance. Studies of muscle sympathetic nerve activity (MSNA), a

measurement of the signal sent from the nervous system to the blood vessels (sympathetic tone), have conflicting results. After 120 days of bed rest, resting heart rate and MSNA measured in the tibial nerve were increased, and the baroreflex slope during tilt were steeper. Surprisingly, there were no presyncopal subjects following this prolonged bed rest, and the authors concluded that the augmented MSNA response increased vasomotor tone and prevented presyncope. In a follow-up study, these same authors studied 22 male volunteers before and after 14 days of bed rest (Kamiya et al. 2003). In this study, 10 subjects became presyncopal during post-bed rest tilt testing. In the hypotensive subjects, MSNA was lower throughout the tilt and was suppressed during the last minute of tilt. This pattern was not seen in the subjects who were able to complete the tilt test. These data directly support the flight results that show a decreased norepinephrine response during postflight tilt testing. Pawelczyk et al. also measured MSNA during LBNP before and after bed rest (Pawelczyk et al. 2001). MSNA increased during LBNP following bed rest; however, this response was appropriate given the changes in stroke volume and cardiac filling pressure and thus reflex control of MSNA was not altered.

Nitric oxide has been hypothesized to contribute to orthostatic intolerance through its effects on the vascular smooth muscle. Bonnin et al. (Bonnin et al. 2001) showed that flow-dependent dilation of the brachial artery was increased following seven days of bed rest and that this increase was negatively correlated to post-bed rest orthostatic tolerance. There was no change in the response to nitroglycerin, implying an endothelium-dependent effect. Bleeker et al. (Bleeker et al. 2004) conducted a similar study in the femoral artery following horizontal bed rest. They found augmented arterial dilation in response to flow and nitroglycerin, implying an endothelium independent mechanism, likely an increased sensitivity to nitric oxide in the vascular smooth muscle. It is known that different vascular beds respond differently to the same stimuli, which may explain these differences.

b. Hypovolemia

Hypovolemia also has been employed as a laboratory model of post-spaceflight orthostatic intolerance, specifically using pharmaceuticals to induce a plasma volume loss similar to that experienced in astronauts. Kimmerly and Shoemaker used three days of spironolactone administration to induce a $15.5 \pm 1.7\%$ decrease in plasma volume (Kimmerly and Shoemaker 2002, 2003). While this model was useful for their purposes, spironolactone is known to have vasomotor effects, complicating interpretation of studies involving integrated cardiovascular responses. Fu et al. used a single dose of Lasix®, which decreased plasma volume by ~13% to study the effects of acute hypovolemia on orthostatic tolerance (Fu et al. 2005). Their results showed that orthostatic tolerance, as induced by LBNP, was markedly decreased in women, but not men, during hypovolemia; but they did not find any differences in norepinephrine responses between genders or between normovolemia and hypovolemia. Iwasaki et al. also used a single dose of Lasix® and found that the effects on cardiac filling pressures, stroke volume and high-frequency baroreflex sensitivity were similar between hypovolemia and two weeks of head-down tilt bed rest; however, they also found that vasomotor function differed between the two protocols (Iwasaki et al. 2000). Finally, Meck et al. used a single Lasix® (furosemide) infusion (0.5 mg/kg) followed by 36 hours of a very low sodium diet (10 mEq/day). This protocol induced a

plasma volume loss similar to that after spaceflight (Buckey et al. 1996b; Fritsch-Yelle et al. 1996b; Waters et al. 2002; Meck et al. 2004) (Figure 14). Those who became presyncopal after spaceflight and during hypovolemia exhibited the same etiology, a failure to release the extra amount of norepinephrine necessary to maintain standing arterial pressure when hypovolemic. The differences in these studies may be due to the hypovolemia protocol (acute vs. chronic) or in the orthostatic stimulus (tilt vs. LBNP). Regardless of these differences, pharmacologically-induced hypovolemia has been shown to reproduce the plasma volume losses seen following spaceflight. While obviously useful for some mechanistic studies and countermeasure development, this model is limited in that the deconditioning component of spaceflight (and bed rest) is not replicated.

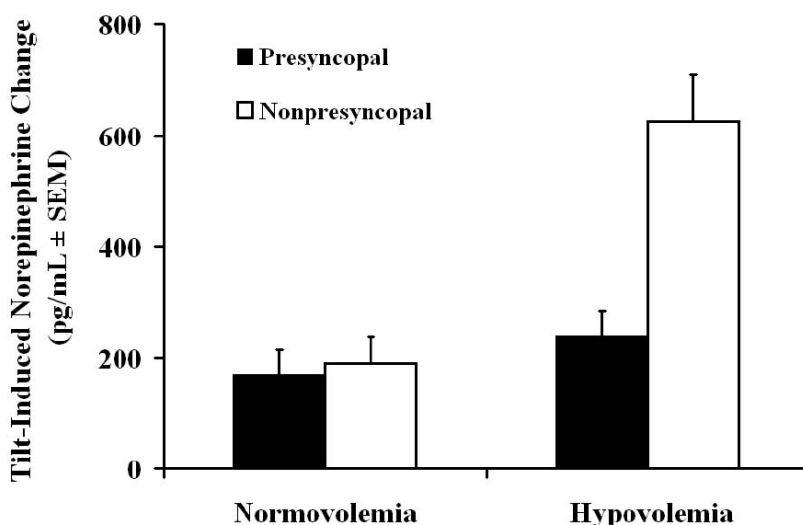


Figure 14. Norepinephrine responses of presyncopal (n=8) and non-presyncopal (n=9) test subjects during normovolemia and hypovolemia tilt tests

c. Cardiac Mass and Function

Cardiac muscle is responsive to changes in volume loading conditions. Decreased cardiac work during bed rest due to plasma volume loss, reduced end-diastolic volume, and reduced physical activity results initially in smaller cardiac volumes and later remodeling to a smaller cardiac muscle mass (Westby et al. 2016). Assumption of the head-down position initially increases stroke volume with a concurrent decrease in heart rate, with the highest stroke volumes achieved ~30 minutes after the start of bed rest. Thereafter, resting stroke volume decreases with the lowest value obtained by ~48 hours and remains depressed throughout the end of bed rest (Levine et al. 1997). Though head-down bed rest initially results in a cephalic shift of ~1 liter of blood from the legs to the upper torso, this is counteracted by humoral responses leading to diuresis and attainment of hemostasis at a level approximately half way between supine and upright about 24 to 48 hours after the start of bed rest (Levine et al. 1997).

After two weeks of 6° head-down tilt bed rest, cardiac mass and left ventricular end-diastolic volume measured by 2-d echocardiography was decreased by 5% ($p < 0.10$) and 7% ($p < 0.05$), respectively (Levine et al. 1997). This was concomitant with a 24% decrease in LBNP

tolerance measured using a ramp protocol. Longer duration bed rest results in further cardiac atrophy (Westby et al. 2016) (**Figure 15**). Left ventricular mass is reduced in men by 4.7, 8.0, and 15.6% after 2, 6, and 12 weeks, respectively, of horizontal bed rest (Perhonen et al. 2001a). Similarly, 60 days of 6° head-down bed rest resulted in decreased left ventricular mass of 9.5% in women (Dorfman et al. 2007). Reduced ventricular volume precedes cardiac muscle atrophy; left ventricular mass-volume ratio is initially reduced, followed by cardiac remodeling in response to decreased wall stress (Perhonen et al. 2001a). For example, left ventricular end-diastolic volume decreased by 2 weeks of bed rest, but left ventricular mass was not reduced until 6 weeks. Similar results have been obtained from measurements of the right ventricle, although the changes ventricular volume and mass are not as great (Perhonen et al. 2001a; Dorfman et al. 2007).

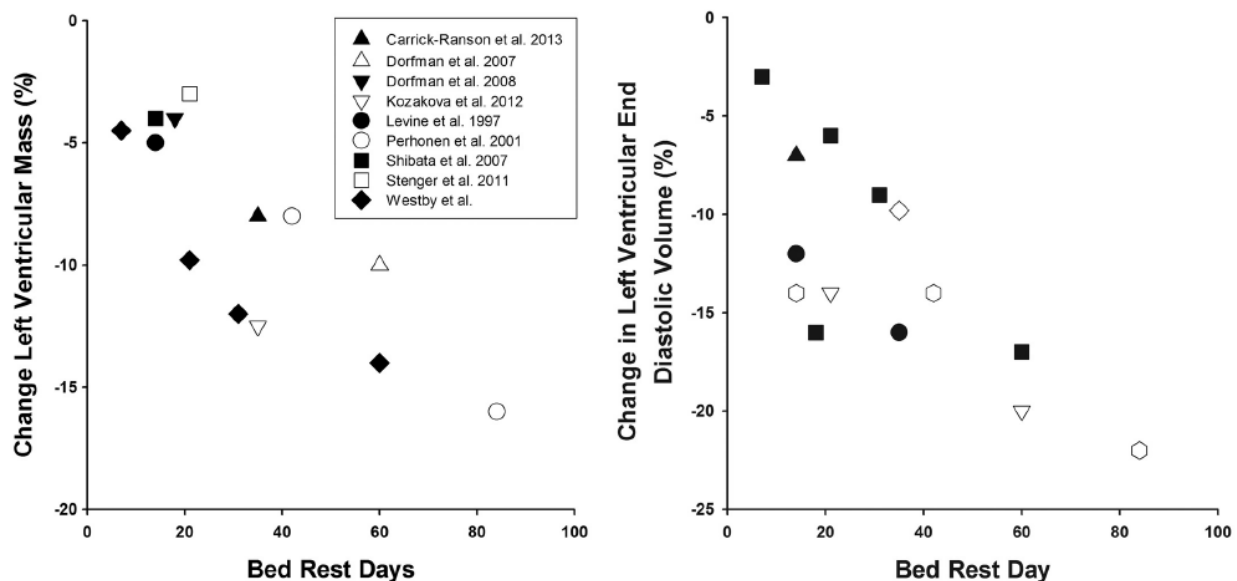


Figure 15. Mean percent change in left ventricular mass and end-diastolic volume in bed rest studies of varying duration. Reproduced from (Westby et al. 2016).

The decrease in cardiac mass is concomitant with a reduced stroke volume at any given filling pressure. In the supine resting condition, after two weeks of bed rest pulmonary capillary wedge pressure (PCWP) and left ventricular end-diastolic volume (LVEDV) are decreased by 18% and 7%, respectively (Levine et al. 1997). Starling Curves, constructed from PCWP and stroke volume measures during LBNP and with rapid warm saline infusion, indicated a change in the cardiac distensibility, particularly at lower filling pressures. Pre- to post-bed rest, PCWP was not changed during -15 mmHg and -30 mmHg of LBNP, yet stroke volume was reduced. Surprisingly, the relation between LVEDV and stroke volume was unchanged, suggesting that this duration of bed rest had no effect on cardiac contractility (same stroke volume per LVEDV); lower stroke volume after bed rest resulted from a decrease in diastolic volume. Thus, there is a leftward but parallel shift in the filling pressure-volume relation after bed rest. In total, bed rest-induced changes in cardiac distensibility leads to lower LVEDV and reduced stroke volume at filling pressures representative of low levels of orthostatic stress. The decreased distensibility is reflected in a decrease in left ventricular untwisting during diastole (Dorfman et al. 2008). Assuming that reduced cardiac distensibility continues to parallel the progression of cardiac

atrophy with longer bed rest durations, a progressive decrease in diastolic function would be one contributing factor to the decreased stroke volume during orthostatic stress and increasing frequency of orthostatic intolerance (Perhonen et al. 2001a).

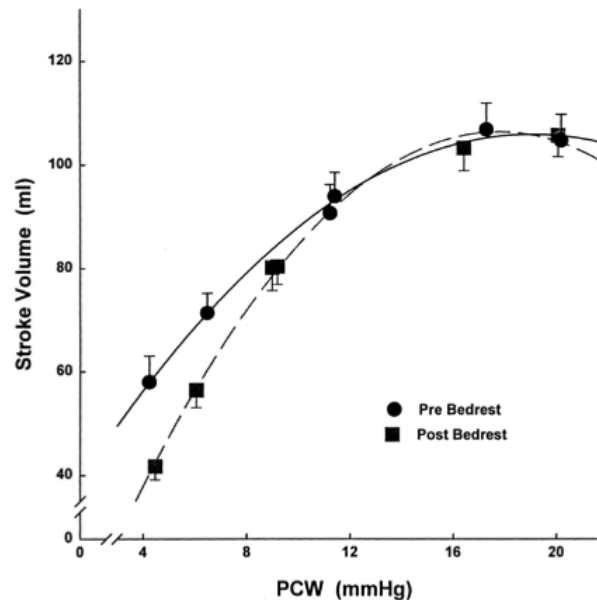


Figure 16. Starling Curves constructed from pulmonary capillary wedge pressure (PCW) and stroke volume measured at rest, during LBNP (-15 and -30 mmHg) and with rapid warm saline infusion (15 and 30 ml/kg) before and after 2 weeks of bed rest (Levine et al. 1997).

That these changes in diastolic cardiac performance result from cardiac remodeling, rather than just bed rest-induced hypovolemia is supported by the work of Perhonen et al. (Perhonen et al. 2001b). Subjects were studied before and after 18 day of bed rest as well as before and during furosemide-induced hypovolemia to compare changes in diastolic cardiac performance within the same subjects. Though bed rest and furosemide resulted in similar losses of plasma volume (-15% vs. -14%), LVEDV during supine rest was decreased by almost three times as much after bed rest as by hypovolemia (-20 vs. -7%) while pulmonary capillary wedge pressure (PCWP) at rest decreased by only ~50% more (-21 vs. -31%) after bed rest than during hypovolemia. Further, both bed rest and hypovolemia resulted in changes in the linear portion of the Starling Curves (PCWP-stroke volume, from baseline rest through LBNP at -30 mmHg), but the slope of the relation was increased approximately twice as much after bed rest as during hypovolemia. Additionally, while bed rest resulted in a leftward shift of the pressure-volume relation (PCWP-LVEDV), there was no change in this relation during hypovolemia. While the reduction LBNP tolerance after bed rest only tended to be more than during hypovolemia (-27 vs -18%), the combination of bed rest-induced cardiac remodeling and plasma volume loss resulted in more dramatic changes in diastolic cardiac function than hypovolemia alone.

d. Autonomic Control

Most studies examining HRV in bed rest have found a decrease in HF-RR during bed rest or immediately afterwards (Hughson et al. 1994b; Sigaudo et al. 1998; Hirayanagi et al. 2004b, a). This bed rest-induced decrease in HF-RR has additionally been evident in bed rest as short as 20 hours with a furosemide infusion (Patwardhan et al. 1995), during a controlled breathing protocol (Iwasaki et al. 2000, 2004), during LBNP (Pavy-Le Traon et al. 1998), and during vasoactive drug infusions (Arzeno et al. 2013). Despite some finding no difference in HF-RR with bed rest (Ferretti et al. 2009; Shoemaker et al. 2012), it is well accepted that the HRV results indicate diminished vagal modulation after bed rest. The majority of studies that examined LF/HF-RR as either an index of sympathovagal balance or an indicator of sympathetic activity, found LF/HF-RR increased with bed rest (Hughson et al. 1994b; Patwardhan et al. 1995; Pavy-Le Traon et al. 1998; Iwasaki et al. 2000, 2004; Hirayanagi et al. 2004b; Coupé et al. 2011). However, this increase in sympathovagal balance is not as explicit as the decrease in vagal modulation, since other studies did not find LF/HF-RR to change with bed rest (Goldberger et al. 1994; Sigaudo et al. 1998; Shoemaker et al. 2012). It should be noted that (Shoemaker et al. 2012) only studied a 24-hour bed rest, such that autonomic changes that become significant after the first 24 hours would not be detected. Approximation of sympathetic activity by blood pressure variability has yielded incongruent results, with increase (Patwardhan et al. 1995; Tanaka et al. 2013; Arzeno et al. 2013), no change (Hirayanagi et al. 2004b; Iwasaki et al. 2004; Ferretti et al. 2009), and decrease (Iwasaki et al. 2000; Hirayanagi et al. 2004a) in LF-SBP occurring after bed rests of various durations.

Other measures of HRV do not have established links to autonomic activity but can quantify overall variability or complexity. When studied throughout bed rest, these measures indicate that the total variability and complexity of heart rate decrease with bed rest. The total power in power spectral analysis of R-R interval quantifies the overall variability, and has been shown to decrease in bed rests ranging from 28 to 90 days (Hughson et al. 1994b; Pavy-Le Traon et al. 1998; Sigaudo et al. 1998; Ferretti et al. 2009). Heart rate complexity cannot be quantified by typical spectral analysis techniques such as methods based on the Fast Fourier Transform. Instead, coarse grain spectral analysis (Yamamoto and Hughson 1991) can be used to extract the harmonic and fractal components of HRV and characterize the complexity of HRV by the exponent of the $1/f$ noise in the signal. This exponent corresponds to the slope of the linear fit of the fractal component plotted in a log-power vs. log-frequency scale. The spectral exponent has been found to increase with bed rest (Hughson et al. 1994b; Fortrat et al. 2001), indicating a steeper power drop-off at higher frequencies and a decrease in heart rate complexity. Heart rate complexity has also been shown to decrease with bed rest when measured by approximate entropy, which quantifies the predictability of fluctuations in heart rate (Goldberger et al. 1994).

In summary, most of the studies indicate that bed rest causes autonomic dysfunction by diminishing vagal modulation, increasing sympathovagal balance, and decreasing the overall variability and complexity of heart rate. The effect of bed rest on the approximation of sympathetic activity by blood pressure spectral analysis is not clear due to conflicting results.

Analysis of the heart rate and blood pressure series can further yield information on baroreflex sensitivity. Baroreflex sensitivity has been broadly shown to decrease with bed rest,

as measured by examining the maximum slope of the baroreflex curve (Convertino et al. 1990), beat-to-beat sequences of R-R interval and SBP (Hughson et al. 1994a; Pavy-Le Traon et al. 1998; Sigaud et al. 1998; Hirayanagi et al. 2004b, a; Ferretti et al. 2009; Coupé et al. 2011; Arzeno et al. 2013), or the transfer function between the R-R interval and SBP (Iwasaki et al. 2000, 2004; Custaud et al. 2002; Hirayanagi et al. 2004a). Iwasaki et al. attributed the decrease in baroreflex sensitivity after a 14-day bed rest to hypovolemia, due to a reversal of the baroreflex change after plasma volume restoration. However, Arzeno et al. (Arzeno et al. 2013) found that the decrease in baroreflex sensitivity after a 60-day bed rest was not due to decreased plasma volume.

The bed rest-induced autonomic and baroreflex dysfunction can contribute to post-bed rest orthostatic intolerance, and pre-bed rest indices of autonomic function may help identify those at higher risk of post-bed rest orthostatic intolerance. In particular, Convertino et al. (Convertino et al. 1990) found presyncopal subjects suffered a larger decrease in baroreflex sensitivity during bed rest, and Pavy-Le Traon et al. (Pavy-Le Traon et al. 1998) found presyncopal subjects to have a smaller increase in LF/HF-RR during LBNP.

e. Sex Differences

As in spaceflight, a very low number of female subjects have been studied during bed rest. This is traditionally done to reduce the variability in the data and to eliminate the scheduling issues related to the menstrual cycle. The women studied have been found to be more susceptible to orthostatic intolerance than men after head-down tilt bed rest (Grenon et al. 2006) (**Figure 18**). While the incidence of orthostatic intolerance has been shown to be higher in women than in men (Montgomery et al. 1977; Gotshall et al. 1991; White et al. 1996; Shoemaker et al. 2001; Waters et al. 2002; Fu et al. 2004), the etiology has not been clearly defined. Women have greater heart rate responses than men during mental stress (Collins and Frankenhaeuser 1978), standing (Gotshall et al. 1991; Schondorf and Low 1992), infusions of pressor agents (Abdel-Rahman et al. 1994), and cold pressor tests (Girdler et al. 1993). It also has been shown that estrogen replacement therapy in postmenopausal women reduces muscle sympathetic nerve activity (Hunt et al. 2001; Vongpatanasin et al. 2001).

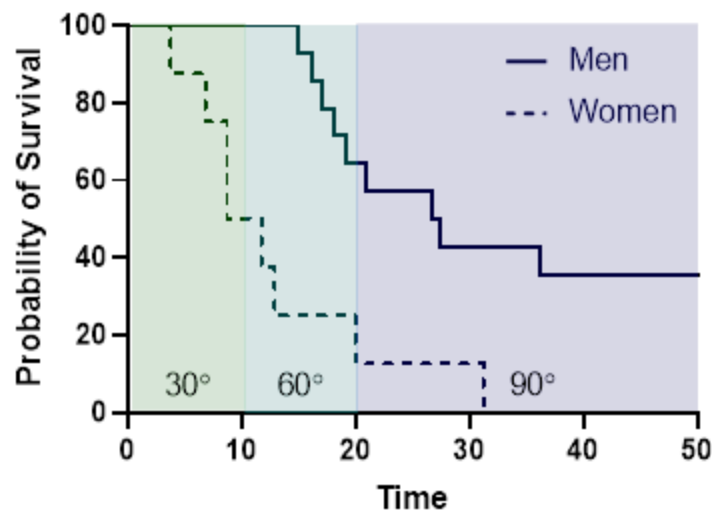


Figure 17. Women may not tolerate 0.50-G, a stressor similar to that experienced during Apollo missions, after weightlessness. Survival curves for women and men depicting tolerance to 10 minutes of 30° (0.50-G), 60° (0.85-G), and 90° (1-G) head-up tilt followed by 120 minutes of standing after 14-16 days of bed rest (Redrawn from (Grenon et al. 2006).

Differences in response to stress can be partially due to a dissemblance in autonomic regulation between the sexes. Women are often characterized by attenuated sympathetic response and enhanced parasympathetic reactivity, particularly during orthostatic stress (Convertino 1998; Shoemaker et al. 2001; Evans et al. 2001; Grenon et al. 2006; Barantke et al. 2008; Nugent et al. 2011; Cheng et al. 2011). These sex differences in autonomic dominance have been observed during tilt and stand tests (Barnett et al. 1999; Shoemaker et al. 2001; Grenon et al. 2006; Barantke et al. 2008; Cheng et al. 2011), lower body negative pressure (LBNP) (Convertino 1998), autonomic blockade (Evans et al. 2001), and vasoactive drug infusions (Arzeno et al. 2013). Autonomic control differences between the sexes are preserved throughout bed rest, with women also experiencing a decrease in baroreflex sensitivity within a shorter timeframe than men (Arzeno et al. 2013).

In addition, differences in vascular resistance and reactivity can contribute to the sex difference in orthostatic intolerance. Women have smaller increases in vascular resistance than men in response to LBNP (Frey and Hoffler 1988; White et al. 1996), standing (Frey et al. 1994), cold pressor and facial cooling tests (Kilgour and Carvalho 1994), and mental stress (McAdoo et al. 1990). Women also may have lower orthostatic tolerance because of increased splanchnic blood flow compared to men (Montgomery et al. 1977; Jarvis et al. 2006) and attenuated splanchnic constriction in women in the upright posture (Jarvis et al. 2010). The attenuated vascular constriction in women also has been observed as attenuated brachial and femoral vasoconstrictor responses to increases in transmural pressure (Lott et al. 2009). There could be several factors that contribute to the women's low vascular resistance, the most important of being estrogen. Several studies in humans demonstrate an augmentation of endothelium-dependent vasodilation with estrogen (Lieberman et al. 1994; Gilligan et al. 1995; Tagawa et al.

1997; Guetta et al. 1997; Arora et al. 1998). Low peripheral vascular resistance is considered one of the main drivers for post-spaceflight orthostatic intolerance (Waters et al. 2002). Increased arterial stiffness after spaceflight was linked to orthostatic tolerance in men (Tuday et al. 2007) but there was no significant increase in arterial stiffness was observed in women (Tuday et al. 2011).

Decreased cardiac filling also contributes to the higher incidence of orthostatic intolerance in women (Fu et al. 2004). Women have lower tolerance to LBNP, most likely due to a steeper Frank-Starling relationship. Women have larger decreases in stroke volume in response to decrements in cardiac filling pressure compared to men, perhaps due to a smaller and stiffer left ventricle.

The sex difference in propensity to orthostatic intolerance, observed during both spaceflight and bed rest, leads to an important consideration for countermeasure development. A single countermeasure may not be equally effective for both sexes. For example, midodrine was less effective in preventing orthostatic intolerance in women than men following simulated microgravity (Grenon et al. 2006).

f. Current Countermeasures: Exercise

Weightlessness induces cardiovascular deconditioning including cardiac structural and functional changes that result in a decreased stroke volume, which plays a key role in the risk of post spaceflight orthostatic intolerance. As such, exercise to mitigate this deconditioning is a current countermeasure used by crewmembers during spaceflight missions. Similar cardiovascular deconditioning occurs during bed rest as during spaceflight; thus, bed rest interventions provide an opportunity to assess the efficacy of countermeasures. Recently, Scott et al. (Scott et al. 2018) assessed the cardiovascular impact of sedentary bed rest compared daily exercise during 70 days of head-down tilt bed rest. The exercise routine combined submaximal and high intensity interval aerobic training and resistance exercise. Both the exercise and sedentary groups showed an initial decrease in cardiorespiratory fitness, likely due to the loss of plasma volume. However, after day 4 of bed rest the exercise group had no further decrements in fitness whereas the sedentary group experienced a progressive decline. Additionally, exercise was protected against left ventricular atrophy, while the sedentary group showed significant reductions in ventricular mass. Others have also demonstrated the utility of exercise to mitigate atrophy and maintain ventricular compliance during bed rest using various exercise modalities, durations, and intensities (Shibata et al. 2010; Hastings et al. 2012) However, exercise alone, regardless of the protocol specifics, does not prevent orthostatic intolerance after bed rest (Shibata et al. 2010).

g. Current Countermeasures: Fluid Loading

Fluid loading intended to expand plasma volume before re-entry and landing currently is implemented to mitigate the risk of post-spaceflight orthostatic hypotension upon return to a gravitational environment. Edgell et al. (Edgell et al. 2018) assessed plasma volume and the cardiovascular response to lower body negative pressure (LBNP) after three conditions in a cross-

over design: seated four hours with fluid loading and 28 hours of head-down tilt bed rest with and without fluid loading. The fluid loading protocol consisted of 1 g of salt per 125 mL water with total of 15 mL/kg of body weight consumed, which is consistent with the standard NASA protocol. In the seated posture fluid loading expanded plasma volume by an average of 152 mL. After 28 hours of head-down tilt bed rest without fluid loading plasma volume was reduced by 280 mL, and fluid loading did not significantly recover plasma volume to pre-bed rest levels. However, stroke volume index and central venous pressure were lower, and renin and angiotensin II levels were higher at rest and during LBNP after bed rest without fluid loading, suggesting moderate benefits by fluid loading. Interestingly, there was no difference in heart rate and mean arterial pressure after bed rest between subjects who performed fluid loading and those who did not at rest or during LBNP up to -40 mmHg. The lack of recovery seen in plasma volume may have been due to the ongoing diuresis in response to bed rest (Mauran et al. 1999). However, this cannot be ascertained clearly as urine output was not recorded after fluid loading (Edgell et al. 2018).

Conversely, Waters et al. (Waters et al. 2005) showed that fluid loading was effective at restoring plasma volume after 12 days of head-down tilt bed rest using the NASA fluid loading protocol. In this study, plasma volume decreased by ~200 mL from pre- to post-bed rest, and fluid loading restored plasma volume to pre-bed rest levels. The heart rate response to 30° head-up tilt after bed rest and fluid loading was augmented compared to pre-bed rest but arterial pressure was not different, and there were no incidences of orthostatic hypotension pre- or post-bed rest head-up tilt.

Both studies utilized the NASA fluid loading protocol, which is comprised of salt tablets and water to produce an isotonic solution after ingestion. Because some crew members reported low palatability and gastrointestinal symptoms, other solutions have been developed and tested. Fortney et al. (Fortney et al. 1994) compared the ability of salt tablets and water to expand plasma volume in supine subjects to an alternative solution, Astro-Ade developed by the NASA Ames Research Center. Plasma volume increased ~2% in both the salt tablets and water and the Astro-Ade conditions, peaking at 2.5 hours post fluid loading, and decreased by 6% in subjects who did not receive fluids. Symptoms of gastrointestinal distress were reported during and after consuming both fluids, though only the salt tablets and water resulted in vomiting in one subject. Recently, another isotonic electrolyte solution was developed by the NASA JSC Food Laboratory for use on commercial spaceflight vehicles and is being tested for efficacy and palatability in a ground-based study.

Attempts to restore blood and plasma volume status by oral fluid loading at the end of a spaceflight (Bungo et al. 1985) do not fully prevent orthostatic intolerance. Even restoring filling pressures in the face of decreased cardiac distensibility will not prevent the reduction in stroke volume (Levine et al. 1997; Perhonen et al. 2001b). Conversely, maintaining diastolic function alone through supine cycle exercise fails to prevent post-bed rest orthostatic intolerance (Shibata et al. 2010). Similarly, orthostatic tolerance was maintained in subjects who combined rowing and resistive exercise throughout bed rest with two days of fluid loading accomplished through

fludrocortisone and increased salt intake (Hastings et al. 2012). Thus, a combination of fluid loading and exercise are necessary.

h. Current Countermeasures: Lower Body Compression Garments

The Russian Kentavr and American anti-gravity suit (AGS; pressurized to ~78 mmHg) compression garments were highly effective at preventing orthostatic intolerance during a 15-minute tilt test in subjects who were in a hypovolemic state to mimic landing day plasma volume (Platts et al. 2009) (**Figure 19**). Both compression garments reduced the cardiovascular stress of tilt testing, suggested by lower heart rates, higher blood pressure, and higher stroke volumes compared to the control group. While small differences were seen in the cardiovascular responses between the Kentavr and the AGS these were not statistically significant.

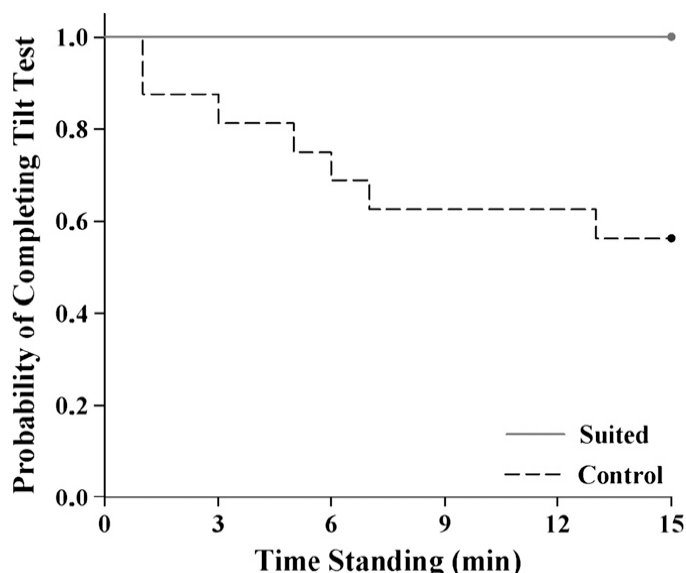


Figure 18. Probability of completing a 15-min tilt test under hypovolemic conditions with (solid line) and without a compression garment (dashed line, n=16) (Platts et al. 2009).

Neither garment is ideal; thus, recent laboratory research at the Johnson Space Center has been focused on developing a compression garment for exploration class missions that improves upon the beneficial features of the AGS and Kentavr. Because the Kentavr (~30 mmHg) was effective at only half the compression applied by the AGS (~75 mmHg), the AGS was evaluated at “one click”, or only ~26 mmHg (Unpublished data). In this study, orthostatic responses were evaluated in hypovolemic test subjects during control conditions (no garment), while utilizing the AGS at the lowest operational setting (~26 mmHg), and while utilizing commercially-available thigh-high compression garments that applied an average pressure of 25 mmHg from the ankle to the top of the leg. Neither the AGS at 26 mmHg nor the thigh-high compression garments were as effective as the AGS at ~75 mmHg or Kentavr in preventing signs and symptoms of orthostatic intolerance. However, in a separate evaluation of these thigh-high stockings compared to the AGS, it was found that the stockings did not increase the metabolic cost of walking compared to normal athletic attire, unlike the AGS (Lee et al. 2011). The metabolic

cost of an activity like walking is an important factor in egress capability (Bishop et al. 1999, p. 1999).

A gradient compression garment (GCG), extending from the feet to the lower torso, was tested after two weeks of 6° head-down tilt bed rest and completely prevented signs of orthostatic intolerance in 15 test subjects exposed to 15 minutes of 80° head-up tilt (Stenger et al. 2014). While utilizing these compression garments, blood pressure, heart rate, and stroke volume responses to head-up tilt were not different after two weeks of bed rest deconditioning. Furthermore, wearing some form of compression garment for 3 days of post-bed rest recovery did not diminish the rate of recovery compared to control subjects.

i. Countermeasures: Cooling

Skin-surface cooling is a current countermeasure used during re-entry to protect against orthostatic intolerance after spaceflight. Environmental heat stress, which results from the space vehicle cabin environment and the insulating properties of the re-entry suit, exacerbates the cardiovascular stress of orthostasis and attenuates the compensatory venoconstriction. Terrestrial data suggests that skin cooling improve orthostatic tolerance during head-up tilt (HUT) in normothermic (Durand et al. 2004) and hyperthermic states (Wilson et al. 2002) by increasing cerebral blood flow velocity, peripheral resistance, and mean arterial pressure. Skin cooling mitigated bed-rest induced orthostatic intolerance, assessed by LBNP, although tolerance was not restored to pre-bed rest levels (Keller et al. 2011). After 2 week of bed rest, forearm vascular resistance was higher and heart rate was lower when the skin was cooled during LBNP.

j. Countermeasures: Pharmacotherapy

Current pharmacological treatments for adrenergic failure may have application to spaceflight-induced orthostatic intolerance. Indeed, midodrine is FDA approved for orthostatic hypotension due to adrenergic failure and is discussed in depth under the countermeasure section. Such research done at NASA may also benefit the larger clinical community. Again using midodrine as an example, we uncovered a drug interaction between midodrine and promethazine that was previously unpublished (Platts et al. 2006). Healthy test subjects who received midodrine and promethazine together experienced a higher incidence of akathisia than controls or subjects with either drug alone. Anecdotal reports from emergency room physicians report similar symptoms in patients with diabetic neuropathy who present with nausea and are given promethazine while being treated for hypotension with midodrine. In the future, the knowledge of this interaction can help avoid unnecessary patient distress and hospital admissions in clinical practice.

k. Current Countermeasures: Lower Body Negative Pressure

Development of effective countermeasures to post-flight orthostatic intolerance was a high priority for NASA's in-house directed medical research during the Space Shuttle program. LBNP was one candidate countermeasure under investigation since the 1960s (Fortney 1991). Lower body negative pressure (LBNP) is a means of producing an orthostatic challenge during spaceflight and bed rest (Guell et al. 1990, 1992; Charles and Lathers 1994). By encasing the lower

body in a rigid container, venous return can be modulated by varying the level of vacuum. LBNP chamber decompression to 50 mmHg provides a cardiovascular stress similar to standing upright by pooling blood in the legs and abdomen (Wolthuis et al. 1974). This can be used to simulate standing on Earth or other gravity environments, depending on the magnitude of the LBNP.

Brief decompression challenges have been used as a gravity-independent test of orthostatic tolerance, while longer decompressions, either alone or in combination with salt and water ingestion, were used as a countermeasure to restore orthostatic responses in bed-rested subjects (Hyatt and West 1977; Fortney 1991). A number of LBNP protocols were effective in preventing or reversing orthostatic intolerance after bed rest but were prohibitively long (e.g., some requiring several days of application or many repetitions) for use during spaceflight (Fortney 1991). Hyatt and West (Hyatt and West 1977) demonstrated that a single application of only 4 h of LBNP, in combination with the ingestion of one liter of an isotonic bouillon solution, restored plasma volume and orthostatic responses to pre-bed rest levels after one week of bed rest. Subsequent studies confirmed these findings and also revealed that a shorter duration (2 h) LBNP countermeasure was not effective (Fortney 1991). These findings formed the basis for a countermeasure to spaceflight-induced orthostatic intolerance that combined LBNP and oral consumption of an isotonic solution tested during the Space Shuttle program. The duration and frequency of LBNP required to make it an efficient countermeasure, however, is not operationally feasible.

In a recent ground test, LBNP was studied acutely as a countermeasure to the venous congestion that might contribute to the risk of SANS and venous thromboembolism (Marshall-Goebel et al. 2021). Low level LBNP (-25 mmHg) reduced internal jugular vein cross-sectional area while supine by 48% compared to 93% when seated. Further, internal jugular vein pressure while supine decreased 5.5 mmHg with thigh cuffs and was 4.3 mmHg lower when seated than when supine.

I. Current Countermeasures: Veno-constrictive Thigh Cuffs

Acute application of high pressure (60-80 mmHg) thigh cuffs results shifts blood and fluid to the legs and away from the head, decreases in right atrial pressure and cardiac output, and can cause presyncope in some subjects (Ebert and Stead 1940; Warren et al. 1945). However, these pressures are not well-tolerated for long periods of time. Conversely, low pressures (≥ 30 mmHg) have no demonstrable benefit (Stevens et al. 1966). Most analog studies have employed 30-60 mmHg of compression.

Application of thigh cuffs during brief periods of head-down tilt and water immersion may mitigate plasma volume loss but most studies demonstrate that thigh cuffs are not an effective countermeasure to orthostatic intolerance after longer exposures (Vogt 1965; Stevens et al. 1966). For example, in a cross-over design, eight men participated in two 7-day head-down tilt bed rest periods separated by one month, once as a control and once as a countermeasure subject. Application of thigh cuffs (30 mmHg) for eight hours a day acutely reduced the internal jugular vein cross-sectional area and increased femoral and popliteal cross-sectional area when

head-down, but there was no difference in post-bed rest orthostatic intolerance between the two conditions. Further, application thigh cuffs for 10 hours per day during 5 day of dry immersion mitigated the reduction in plasma renin and increase in pro-hormone of brain natriuretic peptide (NT pro-BNP) during the first day but did not prevent the loss of plasma volume, LBNP tolerance, and exercise capacity after dry immersion compared to control subjects (Robin et al. 2020). In contrast, post-bed rest tolerance to 75° head-up tilt was not reduced six subjects who wore thigh cuffs at 40 mmHg for 10 hours per day during 10 days of bed rest (Yao et al. 2008). Before bed rest, all 12 subjects completed 20 minutes of head-up tilt, but after bed rest five of the six controls and only one of the thigh cuff countermeasure subjects became presyncopal before the end of the test.

In a recent ground test, thigh cuffs were studied acutely as a countermeasure to the venous congestion that might contribute to the risk of SANS and venous thromboembolism (Marshall-Goebel et al. 2021). Thigh cuffs tightened to produce 60 mmHg of compression reduced internal jugular vein cross-sectional area while supine by 31% compared to 93% when seated. Additionally, internal jugular vein pressure while supine decreased 4.8 mmHg with thigh cuffs and was 4.3 mmHg lower when seated than when supine. Importantly, subjects rated the thigh cuffs as the most comfortable countermeasure (compared to LBNP and impedance threshold breathing), and subjects that they could tolerate the thigh cuffs the longest amount of time.

m. Other Countermeasures: Artificial gravity

Artificial gravity (AG) via short radius centrifugation (SRC) has been suggested as a multi-system countermeasure to spaceflight deconditioning. While the idea of artificial gravity to prevent orthostatic intolerance is not new (White et al. 1966), the ideal combination of AG magnitude, frequency and duration needed to prevent cardiovascular deconditioning is yet to be determined. In a head-down bed rest study, two hours a day of passive standing was sufficient to prevent post-bed rest hypotension, although four hours a day of passive standing was required to prevent plasma volume losses (Vernikos 1997). Hastreiter and Young determined that 1.5 G_z (at the feet) on a short radius centrifuge was required to evoke calf blood flows that were similar to those when the subject was standing (Hastreiter and Young 1997). In a non-human primate model, Korolkov et al. showed that SRC AG was successful in preventing extracellular fluid loss and orthostatic hypotension resulting from bed rest. Furthermore, they determined that 1.2 G_z , three times a week was more effective than higher G_z levels, administered four to five times a week (Korolkov et al. 2001). Iwasaki et al. determined that daily 1 hour exposures to 2 G_z (at the heart) was sufficient to prevent the adverse effects of 6° head-down tilt bed rest on baroreflex function and plasma volume (Iwasaki et al. 2001; Katayama et al. 2004). Finally, combining exercise with centrifugation at 0.8, 1.2 and 1.6 G_z (at the feet) was previously shown to be effective in maintaining orthostatic tolerance after return from 3 to 28 days of simulated microgravity (Vil-Viliams 1994).

The utility of artificial gravity as a whole-body countermeasure has gained momentum in the last several years. Three weeks of daily short radius centrifugation (1 G_z at the heart, 2.5 G_z

at the feet) via the NASA Ames Research Center's Human Performance Centrifuge improved orthostatic tolerance in both ambulatory men and women (Evans et al. 2004; Stenger et al. 2007). These Ames studies utilized G_z stress in an oscillatory fashion – 2 minutes with 1 G_z at the feet followed by 2 minutes with 2.5 G_z at the feet, repeated for up to 45 minutes a day. At NASA Johnson Space Center, a similar study was conducted using a constant 2.5 G_z at the feet for up to an hour on 6° head-down tilt bed rested subjects. This two-week study evaluated the effectiveness of artificial gravity in eight treatment subjects compared to seven control subjects, and showed the utility of artificial gravity in preventing bed rest-induced orthostatic intolerance and decrements in aerobic capacity (Stenger et al. 2012). However, this study only enrolled men as women were unable to tolerate an hour of centrifugation at the intended study magnitudes (Fong et al. 2007). Similar results were shown in 11 men who participated in three different 5-day bed rest campaigns: as a control (no countermeasure), with 30 minutes of continuous centrifugation, and six sessions of 5-minute bouts of centrifugation. Orthostatic intolerance, exercise capacity, and plasma volume were mitigated but not preserved at pre-bed rest levels (Linnarsson et al. 2015). A more recent study, which included both male and female subjects, demonstrated that daily 30-minute sessions of either continuous or intermittent centrifugation producing 1-G at the center of mass and 2-G at the feet was tolerated during 60 days of strict head-down tilt bed rest (Kramer et al. 2020). Heart rate increased during daily centrifugation, but oxygen uptake was unchanged. However, there was no clear benefit to preserving orthostatic intolerance in the countermeasure groups (Clément et al. 2022a).

A recent partial gravity parabolic flight campaign investigated the effects of different G-levels on the headward fluid shift (Lee et al. 2020b) which has been associated with the loss of plasma volume and the decrease in orthostatic tolerance. Using IJV distension as an index of the headward fluid shift, all G-loads studied (0.25-, 0.50-, and 0.75-G) resulted in headward fluid shifts, suggesting that substantially high G-loads would need to be produced to prevent cardiovascular deconditioning. Further, the G-levels on the Moon and Mars by themselves are not likely to be protective.

n. Other: Countermeasures: Lower Body Negative Pressure and Exercise

In a series of bed rest investigations of increasing durations, a team of investigators sought to determine whether the combination of lower body negative pressure and exercise (LBNPex) would be an effective countermeasure to bed rest-induced deconditioning, including orthostatic intolerance, and by extension an appropriate countermeasure system for space flight (Lee et al. 2010). The investigator team hypothesized that LBNPex allow astronauts to perform treadmill exercise with axial and ground reaction forces (for protection of bone and muscle) similar to or greater than that experienced during walking and running in normal gravity while simulating the hydrostatic pressure gradients normally experienced by the cardiovascular system. Although most exercise performed during bed rest in the supine position fails to prevent orthostatic intolerance (Greenleaf et al. 1989b), LBNPex might preserve plasma volume (Greenleaf et al. 1992) and stimulate the cardiovascular reflexes used to promote venous return and maintain cerebral circulation.

In the initial study, 24 male subjects completed 5 days of 6° head-down tilt bed rest, with subjects divided into three groups. Control subjects performed no exercise during bed rest while the remaining subjects performed daily intense bouts of either exercise upright on a standard treadmill or supine treadmill exercise within the LBNPex device (Lee et al. 1997). The equivalent of one-body weight of loading was achieved by manipulating the chamber opening and the level of LBNP (Lee et al. 1997; Ashari et al. 2021). The exercise protocol used by both countermeasure groups consisted of 30 minutes of interval-style exercise with prescribed intensities up to 90% of pre-bed rest peak oxygen consumption. The exercise protocol previously had been shown to maintain peak oxygen consumption (Greenleaf et al. 1989a) and plasma volume (Greenleaf et al. 1992) during 30 day of bed rest when subjects exercised twice per day on a supine ergometer without LBNP. At the completion of the exercise protocol during this 5-day bed rest, subjects remained upright or with their legs positioned against the treadmill belt while LBNP was applied (Watenpaugh et al. 2007). Before and immediately after 5 days of bed rest all subjects participated in head-up tilt test for 30 minutes or until presyncope. After bed rest, tilt tolerance time decreased by ~18% in the control group while tolerance time tended to increase in the LBNPex group (~9%) (Watenpaugh et al. 1994, 2007).

Orthostatic intolerance, as assessed by cumulative stress index calculated from total time and negative pressures achieved during a LBNP ramp protocol, was not prevented in a subsequent 15-day bed rest study (Schneider et al. 2002). Seven men participated once as a control and once as a LBNPex subject. Although the sub-tolerance cardiovascular responses to LBNP were maintained in the LBNPex condition and not in the control condition, orthostatic tolerance was decreased in both conditions (LBNPex: -41%, control: -36%). There were two potentially important differences in the countermeasure protocol that may have influenced these results. First, the intensity of the exercise was reduced, but the duration lengthened, to accommodate the limitations on prescribed exercise intensities that was applied by NASA at the time for astronauts (Moore et al. 2010). The peak prescribed exercise intensity during this bed rest was 80% of pre-bed rest peak oxygen consumptions and the protocol duration was extended from 30 to 40 minutes. Second, the subjects in this bed rest study were not exposed to the 5-minute post-exercise LBNP as in the 5-d bed rest study. This additional orthostatic challenge, when the subjects were vasodilated from the previous exercise and thermal stress, may have been a key component to the success of the countermeasure in the prior study (Watenpaugh et al. 2007).

Subsequent 30-day bed rest studies in identical twins (16 men, 14 women) confirmed the potential efficacy of the LBNPex countermeasure against space flight-induced orthostatic intolerance (Watenpaugh et al. 2007). One twin was randomly assigned to the control group, and one was assigned to the LBNPex group. Control subjects performed no countermeasures throughout the duration of bed rest, while the LBNPex subjects performed the same exercise prescription as in the previous 15-day bed rest (Schneider et al. 2002) but with 5 minutes of post-exercise LBNP as was applied in the 5-day bed rest (Watenpaugh et al. 1994). Pre- to post-bed rest tolerance to a combined tilt and LBNP ramp test was not fully protected in the LBNPex group, but the 13% decrease was almost 1/3 of the decrease in orthostatic tolerance seen in control

subjects (-34%). Similar to previous observations in this series of studies, the cardiovascular responses to the sub-tolerance tilt and LBNP were maintained from pre- to post-bed rest in the LBNPex group but were exaggerated in the control group. These positive results following a longer duration bed rest study suggest that the repeated bouts of the post-exercise orthostatic challenge is an important aspect of the countermeasure effectiveness (Lightfoot et al. 1989),

Most recently, the combination of the LBNP and exercise countermeasure was tested in women during 60 days of bed rest (Guinet et al. 2009). This bed rest study was different from previous ones in this series of investigations in several important ways. First, the number of LBNPex sessions was reduced from 6 days per week to only 3 days per week. A resistive exercise protocol previously shown to maintain muscle strength and mass (Shackelford et al. 2004) also was performed 3 days per week, alternating days with the aerobic exercise sessions so that the two countermeasures were never performed on the same day. Second, although the exercise portion of the LBNPex protocol was the same as previous studies (Watenpaugh et al. 2000; Schneider et al. 2009), the duration of the post-exercise LBNP was extended to 10 minutes. It was hypothesized that a greater orthostatic challenge might be necessary to protect against bed rest-induced orthostatic challenge after this longer duration bed rest (Guinet et al. 2009), particularly in women who are more susceptible to space flight-induced orthostatic intolerance (Waters et al. 2002). Despite this modification, the countermeasures performed during bed rest only partially protected against orthostatic intolerance. Tolerance, measured by time to presyncope during a combined tilt and LBNP protocol, was reduced by 35% in the LBNPex subjects but by 50% in this control subjects. Similar to previous studies in this series, sub-tolerance cardiovascular responses to the orthostatic challenge were less in the LBNPex group than in the control group (Guinet et al. 2009). It was likely that the partial protection against bed-rest induced orthostatic intolerance resulted primarily from the LBNP and exercise protocol since the resistive exercise protocol alone is not effective after 90 d of bed rest (Belin de Chantemele et al. 2004), but the contributions of the respective countermeasures was not specifically elucidated.

In conclusion, the results from this series of studies suggest that LBNP during and after exercise is a potential countermeasure for protection against post-bed rest, and by extension, post-space flight orthostatic tolerance. Mechanisms for the protection against post-bed rest orthostatic tolerance after bed rest might include from protection of plasma volume (Lee et al. 1997; Watenpaugh et al. 2000; Pavy-Le Traon et al. 2007b) and left ventricular mass and function (Levine et al. 1997; Dorfman et al. 2007). The exclusion of the post-exercise LBNP in the one study in which post-bed rest orthostatic tolerance and sub-tolerance cardiovascular responses were not maintained suggests the importance of the post-exercise orthostatic challenge. Further refinement of the exercise and post-exercise protocols may elucidate the respective roles of the different portions of the LBNPex countermeasure and result in a more efficacious countermeasure for longer duration bed rest and space flight missions. It should be noted that in these studies it was occasionally necessary to interrupt post-exercise LBNP exposure due to presyncopal symptoms. This occurred more frequently at the start of bed rest and more frequently in women (Watenpaugh et al. 2007; Guinet et al. 2009).

4. Cardiovascular Disease from Radiation Exposure

The association between high doses of radiation exposure and cardiovascular damage is well established. Patients who have undergone radiotherapy for primary cancers of the head and neck and mediastinal regions have shown an increased risk of heart and vascular damage and long-term development of radiation-induced heart disease. These data are well-detailed in several reports (NCRP 2006, p. 2006, 2014, p. 2014; McMillan et al. 2010; Stewart et al. 2012)

The major driving evidence that proves a link between ionizing radiation exposure and the development of degenerative heart and vasculature changes comes from prospective studies that follow the long-term, radiotherapy-related effects in cancer survivors. These patients received relatively high therapeutic doses (~5–50 Gy) of low-LET thoracic radiation during therapy for cancers of the head, neck, and mediastinal regions, such as Hodgkin’s lymphoma and breast cancer (Prosnitz et al. 2005; Darby et al. 2005, 2013; Swerdlow et al. 2007; Carver et al. 2007; Mulrooney et al. 2009). There is a dose-dependent increase in the development of a wide variety of CVDs, including acute and chronic pericarditis, coronary artery disease, cardiomyopathy, valvular disease, and conduction abnormalities in these individuals. The study of Darby et al. reported the relationship of the risk of ischemic heart disease after breast cancer radiotherapy to radiation dose to the heart and to other cardiac risk factors (Darby et al. 2013).

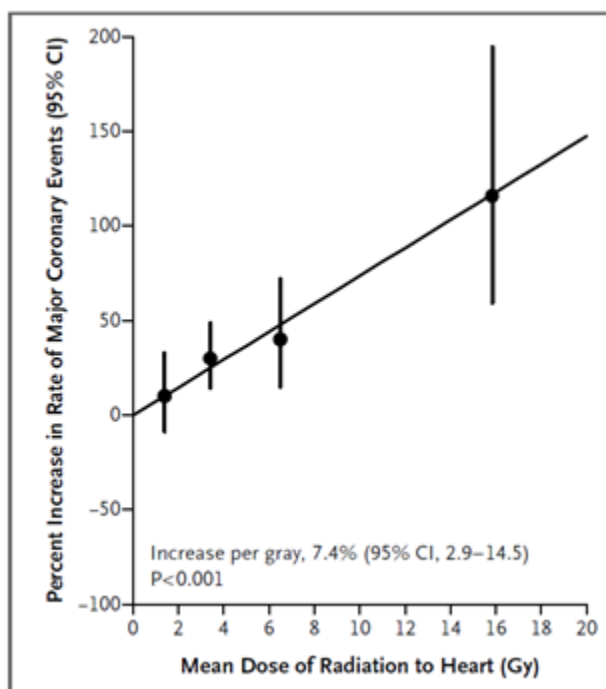


Figure 19. Rate of Major Coronary Events According to the Mean Radiation Dose to the Heart Compared with the Estimated Rate with No Radiation Exposure to the Heart (Darby et al. 2013). *Reproduced with permission from the New England Journal of Medicine - Copyright Massachusetts Medical Society.

Figure 20 shows the linear dose-dependent relationship between exposure and excess relative risk (ERR) based on their findings, with the risk increasing 7.4% for every gray of exposure at these high doses. Similar linear dose response relationship was shown by van Nimwegen et al., 2016 in

Hodgkin's lymphoma patients who received radiotherapy in the past and patients with 20 Gy mean heart doses of radiation showed a 2.5-fold increase in coronary heart disease.

Within 5 years after radiotherapy, the frequency of major coronary events starts to increase in a linear manner with no apparent threshold and continues for at least 20 years after radiation exposure (Zagar et al. 2016; Marks et al. 2017). Women with cardiac risk factors at the time of radiotherapy have greater absolute increases in risk from radiotherapy than those without (Darby et al. 2013). These high doses (>5 Gy exposures) are associated with damage to the structures of the heart and to the coronary, carotid, and other large arteries, including marked diffuse fibrotic damage, especially of the pericardium and myocardium, pericardial adhesions, microvascular damage, and stenosis of the valves (Little 2013). Mechanisms of damage involve cell killing or inactivation of large numbers of cells which cause functional impairment.

At moderate doses of radiation exposure, between 0.5 and 5 Gy, other evidence that supports a link between the occurrence of CVD and radiation exposure is derived from prospective studies of Japanese atomic bomb (A-bomb) survivors who received moderate doses of radiation (0–2 Gy), as well as from occupationally exposed workers who received continuous low-dose exposure (Hayashi et al. 2003; Preston et al. 2003; Yamada et al. 2004; Darby et al. 2005; McGale and Darby 2008; Shimizu et al. 2010; Gillies et al. 2017; Azizova et al. 2018, 2022; Anderson et al. 2021; Oslina et al. 2021). In atomic bomb survivors enrolled in the Life Span Study (LSS), the development of health effects has been extensively studied through continuous longitudinal health assessments starting in 1958. The average doses that were received by the survivors (Preston et al. 2003) are similar to the absorbed doses for an International Space Station (ISS) mission (30-120 mGy for six-month stays) and somewhat lower than the absorbed dose that is expected for a Mars mission (175-450 mGy). A significant dose-response relationship exists for hypertension, stroke, and heart attack in survivors who were exposed at less than 40 years of age, indicating an increase in risk with increasing dose. However, the existence of a threshold dose could not be excluded for risks associated with the lowest dose category in this analysis (Preston 2003) that are associated with doses that are less than 0.25 Sv (Table 1). However, cautious interpretation of the patterns of circulatory disease morbidity and mortality data is recommended because of the influence of other non-radiation related factors on CVD risk and discrepancies in disease classification records (Ozasa et al. 2012) (Wakeford 2022). Furthermore, the LSS cohort is comprised of the survivors of the Japanese population who were acutely exposed to radiation from a nuclear explosion, which is a drastically different experience than that of the astronaut population. Therefore, studies of workers exposed to radiation occupationally are of high value for space-related analyses because of the protracted nature of exposure to low dose-rate radiation, albeit of a different type. At these moderate doses, mechanisms of action are thought to involve atherosclerosis, potentially through inflammation, oxidative stress, endothelial dysfunction, DNA methylation, and cellular apoptosis and senescence (McMillan et al. 2010; Bernstein et al. 2020; Meerman et al. 2021).

For occupationally exposed workers, such as employees of nuclear power facilities, data are mixed at this range of moderate doses (0.5-5 Gy). A study of U.S. workers who were exposed

to chronic low doses of radiation at nuclear power plants showed a significant correlation between radiation dose and death from CVD (Howe et al. 2004). However, other occupational studies have shown either risks that are more consistent with estimated risks from the atomic bomb survivors or no increased risk is observed. The Mayak worker cohort has been subject to various recent analyses (Simonetto et al. 2014, 2015; Moseeva et al. 2014; Azizova et al. 2015a, b, 2016) and the data from the Mayak workers of a pooled analysis (with the Sellafield workers (Azizova et al. 2018) show a highly statistically significant trend in the dose-response relationship for ischemic heart disease (IHD) and cerebrovascular disease (CeVD) incidence, albeit with much lower and generally non-significant mortality trends for these endpoints. These studies have been adjusted for major lifestyle risk factors such as smoking and alcohol use; however, interpretation of results is complicated due to internal radiation exposures from internally deposited radionuclides such as plutonium, and heavily skewed doses to certain organs like liver and lung, predominantly from these internal exposures. External γ -ray exposures average 0.5-0.6 Gy, which is moderate in comparison to usually low doses for other occupational-exposure studies. The International Nuclear Worker study (INWORKS), comprised of three large national groups of workers (for France, UK, US), Gillies (Gillies et al. 2017) demonstrated significant risks of circulatory disease, IHD, acute myocardial infarction (AMI) and CeVD. They observed significant heterogeneity in all circulatory disease ERR by facility ($p=0.006$) although not by country ($p=0.089$), however, no such heterogeneity was observed for IHD, either by country or facility. A study of US uranium enrichment workers yielded very uncertain estimates of mortality ERR/mGy for IHD of 0.019 (95% CI $-0.077, 0.26$) and for CeVD of -0.13 (95% CI $-0.42, 0.44$) (Anderson et al. 2021). A cohort study of uranium processing workers at the Mallinckrodt plant yielded a significant ERR/Gy of 1.3 (95% CI $-0.1, 2.8$) for IHD mortality (Golden et al. 2022). A study of US military personnel exposed to eight atmospheric nuclear tests yielded an insignificant ERR/Gy of -0.01 (95% CI $-1.2, 1.1$) for IHD mortality (Boice et al. 2022a). A cohort of diagnostic medical radiation workers in South Korea yielded an all-circulatory morbidity ERR/Gy of 1.4 (95% CI $-5.7, 9.9$), an IHD morbidity ERR/Gy of 12.2 (95% CI $-7.1, 47.3$). However, the study suffered from low statistical power (Little et al. 2021). Finally, follow-up studies of the health risks in Chernobyl recovery workers also show excess CeVD and IHD morbidity, but, the cohort had very high rates of baseline circulatory disease, which may reflect the elevated rates of incidence and mortality from circulatory diseases in the Russian population compared to other developed countries (McGale and Darby 2005; Ivanov et al. 2006; Kashcheev et al. 2017). A study of the relationship between radiation exposure and atrial fibrillation among atomic bomb survivors did not yield a substantive evidence of radiation effects, even after adjusting for sex, city and atrial fibrillation risk factors (Haruta et al. 2021).

In a comprehensive and systematic review by Little et al. (Little et al. 2012), information was summarized on circulatory disease risks associated with moderate- and low-level whole-body ionizing radiation exposures with criteria of whole-body radiation exposures and a cumulative mean dose of < 0.5 Sv or exposure at a low dose rate (< 10 mSv/day). Although mean cumulative radiation doses were ≤ 0.2 Gy in most of the examined studies, the small numbers of participants exposed to high cumulative doses (≥ 0.5 Gy) drive the observed trends in most cohorts in these higher dose groups. Analyses supported an association between circulatory

disease mortality (excess risk of ischemic heart disease) and low and moderate doses of ionizing radiation above >0.5 Gy. Associations were not statistically significant at lower doses, which may be due to statistically significant heterogeneity across exposed populations (Japanese atomic-bomb survivors compared with a Western cohort of largely European/American populations). This lack of significance also may be partly due to uncontrolled confounding factors, such as lifestyle and genetic factors, within the many studies included in the meta-analyses. Little et al. (Little et al. 2021) updated their previous analyses with recent data from moderate and low dose studies, and the ERRs they observed remain consistent with the prior analyses (Little et al. 2012; Little 2016).

Molecular epidemiology strategies also are being utilized to supplement the body of epidemiological data and provide insight into relevant biological mechanisms and risk estimation at low doses (Abend et al. 2015; Kreuzer et al. 2015).

Currently, a large-scale effort known as Million Person Study (MPS) is underway among healthy American workers and veterans to evaluate radiation risks and health effects from low dose, low dose-rate exposures. In addition to being a more generalizable population to the astronaut corps, this cohort is more suitable to assess dose rate effects than the Japanese atomic bomb survivors who were exposed briefly to high-dose radiation in 1945. This epidemiology study of US radiation workers and veterans focuses primarily on cancer mortality but will also address causes of death due to cardiovascular and cerebrovascular disease (Boice et al. 2022c). Recent analysis of a major MPS cohort, the worker population at the Los Alamos National Laboratory (1943-2017) revealed no evidence of any association between ischemic heart disease risks and radiation dose (ERR at 100 weighted-mGy was -0.06 (95% CI $-0.16, 0.04$; $n=3043$) (Boice et al. 2022b). Increased ischemic heart disease risk from protracted radiation exposure from other MPS cohorts is yet to be observed as well. Pooled analysis of MPS cohorts is currently underway.

At low doses <0.5 Gy, the mechanisms of radiation action may involve non-targeted effects such as monocyte killing, altered oxygen metabolism and kidney dysfunction. It has been hypothesized that radiation may lead to changes in oxygen metabolism, resulting in long lasting elevations in reactive oxygen species and other aspects of an inflammatory cascade (Spitz et al. 2004); there is evidence for long-lasting elevations in vascular reactive oxygen species in the LSS (Nakachi et al. 2008). In a 2003 analysis of 992 patients treated for testicular cancer, those who received mediastinal radiotherapy showed a significant increase in their risk for a cardiac event (Huddart et al. 2003). More notably, patients who received intra-diaphragmatic radiotherapy (irradiation below the diaphragm, excluding the heart) still had higher risk compared to controls. With a low estimated heart dose of 0.75 Gy, the authors suggested radiation nephropathy, leading to hypertension, may be a partial cause for their elevated risk of CVD. Further analyses of LSS data by Adams et al. (Adams et al. 2012) and Sera et al. (Sera et al. 2013) show weak evidence of kidney as a target tissue and suggest that an increased risk of circulatory disease from acute exposures to γ -rays at low to moderate doses (up to 3 Gy) is related to renal dysfunction, with concomitant changes in blood pressure and inflammation. This is supported by experimental animal data (Lenarczyk et al. 2013, 2020). Adams et al. (Adams et al. 2012) also reported that a

quadratic dose response model best fits their data, which is consistent with the finding from Shimizu et al. (2010) (Shimizu et al. 2010) and Little et al. (Little et al. 2008), who reported that inflammation was not increased after low-dose radiation exposure (<0.5 Gy). Tapio et al. (Tapio et al. 2021) concluded that pro-atherogenic outcomes like inflammation and premature senescence can be observed in heart muscle if endothelial cells or vasculature are exposed to low radiation doses even below 0.5 Gy. However, if cells or vasculature are in an inflammatory state, low dose radiation exposure functions in an anti-inflammatory manner.

In summary, these observations demonstrate increased risk for circulatory disease from acutely delivered γ -rays and support the existence of a threshold dose, below which significant increases in risk cannot be detected. However, it is not known how the threshold might be modified for other radiation types, dose-rate effects, or individual sensitivity. For healthy workers who refrain from tobacco use and have a normal body mass index (BMI) and moderate alcohol use, it remains a possibility that the threshold dose may be higher. In contrast, epidemiological meta-studies (Schöllnberger et al. 2012; Little et al. 2012, 2021; Wakeford 2022) are consistent with an increased risk below a 0.5 Gy threshold. Additionally, the Health Protection Agency 2010 report identified several other possible mechanisms, such as mitochondrial dysfunction and DNA damage, through which radiation exposure may cause circulatory damage, leaving the existence of a threshold an open question. Additionally, the complex nature of high energy particle-induced DNA lesions makes uncovering the presence of a threshold dose for space-like radiation a major challenge. A recent meta-analysis shows a small but statistically significant increase in CVD mortality risk with increasing radiation exposure at doses lower than previously assumed, and given the very limited data available, it is not yet possible to quantitatively estimate a specific threshold value nor is there data available to resolve whether CVD risks are stochastic or deterministic in nature (EPRI Technical Report 2020).

C. Evidence from Other Organisms (Animal, Cells)

1. Dysrhythmias

Ground-based animal studies have determined the effects of simulated weightlessness on the cardiovascular system. Tachycardia has been observed in standing rats, after hindlimb-unloading for 28 days (Ray et al. 2001). Decreased cardiac mass also has been documented in studies of hindlimb-suspended rats (Bao et al. 1999). However, hemodynamics in humans differ from hemodynamics in quadrupeds; thus, the rat is not the most appropriate model in which to examine weightlessness-induced cardiovascular adaptations (Rowell 1993).

Recently, mice that were hindlimb suspended for 28-56 days were found to have enlarged hearts with decreased contractility as well as an increased susceptibility to pacing-induced dysrhythmias as detected by echocardiography (Respress et al. 2014). Further, cardiomyocytes harvested from the tail-suspended animals demonstrated an increased leakiness of calcium from the sarcoplasmic reticulum, as well as increased frequency of spontaneous calcium release that may have been caused by elevated phosphorylation of the ryanodine receptor (RyR2).

2. Radiation-Induced Cardiovascular Disease

Systematic studies on the progression of radiation-induced heart diseases were first conducted in rabbits (Fajardo and Stewart 1970) and rats (Lauk et al. 1985; Yeung and Hopewell 1985) with high doses of x-rays in the range of 10 to 20 Gy. Similar studies were conducted using heavy ions during the JANUS program at Argonne National Laboratory, in which ultrastructural studies of the mouse heart and vasculature were performed after the animals had been irradiated with neutrons. The results of the studies were compared with results from irradiating mice with low-LET radiation and showed vessel morphological changes, including marked fragmentation of vascular smooth muscle layers and an increase in deposition of extracellular matrix in vessel walls, similar to the changes observed in animals of advanced age (Yang et al. 1976, 1978; Stearner et al. 1979). Changes in the coronary arteries and aorta at 18 months post-exposure were greater in animals receiving fractionated neutron exposures compared to the single-exposure group. Opposite results were observed for the γ -ray-exposed animals, with single dose exposures being more effective than fractionated exposures. Relative biological effectiveness (RBEs) values for neutron effects increased with decreasing dose or dose fractionation, with RBE values exceeding 100 with protracted exposures (Yang et al. 1978). Similar results were found in early studies with low doses of neon (Ne) and argon (Ar) ions (Yang and Ainsworth 1982). Studies on the atherogenic changes that are associated with irradiation were conducted in dogs to compare the effects of fractionated doses of fast neutrons (15 MeV avg.) with those of low-LET photons. The RBE of neutrons was estimated at 4 to 5 from these studies (Bradley et al. 1981).

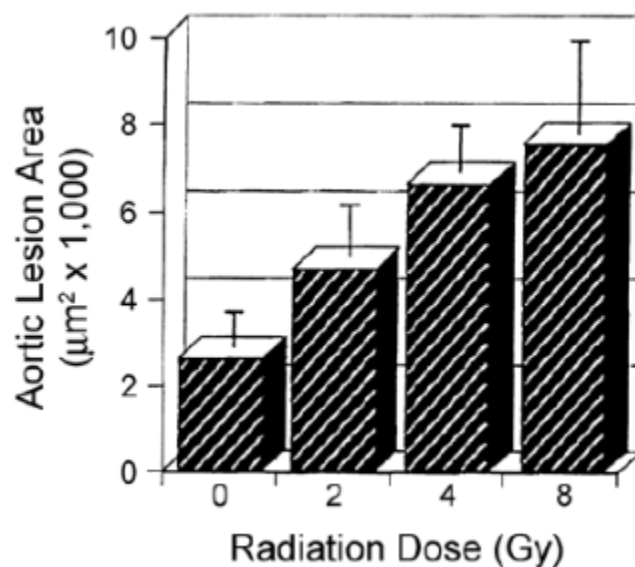


Figure 20. Dose-Dependent Effects of Ionizing Radiation on Aortic Lesion Formation in Fat-fed Mice (repeated-measures analysis of variance: $P = 0.02$) (Tribble et al. 1999).

More recently, studies aimed at defining the mechanisms by which radiation induces heart diseases were conducted using atherosclerosis-prone animal models. Increased oxidative

stress (from the formation of reactive oxygen species [ROS]) and promotion of inflammation have been implicated as possible mechanisms by which radiation promotes atherogenesis. For example, accelerated formation of aortic lesions occurred in a dose-dependent manner in x-ray-irradiated C57BL/6 mice fed a high-fat diet (**Figure 21**), while smaller lesions were observed in their irradiated transgenic littermates that over-expressed CuZn-superoxide dismutase, which is expected to decrease chronic oxidative stress and lead to a decreased atherosclerotic susceptibility (Tribble et al. 1999). The lowest dose in these studies was 2 Gy. Impairment in nitric oxide signal transduction may also contribute to degenerative vascular changes (Soloviev et al. 2003). In another study (Stewart et al. 2006), a single high dose exposure to γ -rays (14 Gy) was shown to accelerate the formation of macrophage-rich inflammatory atherosclerotic lesions in atherosclerosis-prone mice lacking the gene for apolipoprotein E (ApoE^{-/-}). These mice were given a single high dose of γ -rays (14 Gy) to the neck, supporting the notion hypothesis that radiation promotes degenerative heart diseases through (Khaled et al. 2012) [OB] showed that even in the absence of increased surface expression of VCAM and ICAM-1 on human aortic endothelial cells, radiation at doses of 15 Gy with x-rays induced chemokine-dependent signaling that resulted in increased adhesiveness of the cells.

At doses < 0.5 Gy, the response of the ApoE^{-/-} model is more complex. Mitchell et al. (2011) analyzed aortic lesion frequency, size, and severity as a function of γ -ray dose, dose rate, and disease state. This study revealed nonlinear responses with a complex combination of protective and detrimental effects depending on dose rate and disease stage, with maximum effects occurring between 25 and 50 mGy (Mitchell et al. 2011). Mitchell et al. (2013) followed that work with similar results (protective and detrimental effects) in an ApoE^{-/-} mouse model with reduced p53 function. Additional work in the same model using low and high dose rates again shows a non-linear dose response with moderate doses of 300 mGy exhibiting persistent vascular damage (Mancuso et al. 2015; Ebrahimian et al. 2018).

Exposure with high doses of iron (Fe) ions showed a similar pattern of accelerated lesion development in this ApoE^{-/-} mouse model (Yu et al. 2011), which is associated with early impairment of normal vascular reactivity (Yu et al. 2011; White et al. 2014). Yan et al. (Yan et al. 2014) examined the effects of protons and Fe irradiation on animals with acute myocardial ischemia (AMI), where 0.5 Gy of prior proton irradiation improved cardiac function and enhanced cardiac remodeling following Fe irradiation. Conversely, 0.15–150 mGy of prior Fe irradiation caused poorer cardiac function, adverse remodeling along with decreased angiogenesis and pro-survival factors in cardiac tissues following the second larger dose of 0.5 Gy. In a Yucatan minipig model used for investigating solar particle event (SPE) effects, a calculated heart dose of 350 mGy with 6 MeV electrons over three hours also resulted in impaired vascular relaxation (Sanzari et al. 2015).

Using wildtype animals, Soucy et al. (Soucy et al. 2011) showed chronic vascular dysfunction due to radiation-induced xanthine oxidase-dependent ROS production and nitroso-redox imbalance in rats after 1 Gy ⁵⁵Fe particle irradiation, a particularly damaging component of the galactic cosmic ray (GCR) spectrum. At doses as low as 100 mGy of silicon (Si) ions, changes in apoptosis and markers of chronic inflammation were observed in heart tissue of adult male

CBA/CaJ mice up to six months following exposure (Tungjai et al. 2013). Similarly, Sasi et al. (Sasi et al. 2015) irradiated C57BL/6NTac mice or bone-marrow- (BM-) derived endothelial progenitor cells (EPCs) with 0.9 Gy of protons or 0.150 mGy of ^{56}Fe ions and showed early (5-24h) and delayed (28 days) apoptosis in the mice and non-targeted effects in the cells. In addition to their experiments on animals with AMI, Yan et al. (Yan et al. 2014) reported an initial improvement in cardiac function (1 month after exposure) followed by a decline in cardiac function 10 months after exposure to 0.5 Gy of protons total body radiation (TBI) in WT C57BL/6 mice. This finding was in contrast, which was opposite to what was observed with 150 mGy of Fe irradiation where cardiac function decreased significantly after 1 month, but recovery was observed by 10 months post ^{56}Fe radiation. Using a similar mouse model (C57BL/6J), Seawright et al. (Seawright et al. 2019) observed no changes in cardiac function after either protons alone or protons followed one-hour later by oxygen (O) irradiation. Only minor cardiac function changes were observed with O alone, but only at later time points, unlike earlier observations with Fe ions. Sasi et al. (2017) used a sequential dosing methodology to examine cardiac function in C57BL/6NT mice, similar to the AMI surgery group mentioned earlier. No negative effects on cardiac function were observed after either three doses of fractionated protons every other day or three fractionated doses of protons followed by Fe two days later. Only animals that received Fe prior to three doses of protons exhibited depressed left ventricle (LV) function at one month and signs of cardiac hypertrophy at three months post radiation.

Sridharan et al. (2020) exposed male Long Evans rats to whole-body protons and O irradiation and did not observe any changes to cardiac function or indication of gross pathological changes in the heart or aorta but observed mild changes in vascular function and remodeling of existing collagens in the heart. Ramadan et al. (2016) showed differential effects of protons and ^{56}Fe irradiation on the hearts obtained from male C57BL/6 mice seven days post exposure. There was an increase in expression of α -smooth muscle actin, collagen type III and cleaved caspase 3, unlike protons that provoked only a mild increase in cleaved caspase 3 levels. Interestingly, the effect of Fe was lost when a priming dose of protons was given 24 hours prior to Fe irradiation. Nemec-Bakk et al. (Nemec-Bakk et al. 2022) observed no cardiac fibrosis or cardiac dysfunction in female C57BL/6J mice after 0.100 mGy or 250 mGy of O ions. Some increase in LV mass, diastolic volume and diameter was observed along with indications of minor cardiac remodeling, however, no changes to abdominal aorta was observed. Most recently, Bishawi et al. (Bishawi et al. 2022) observed decreased systolic function and increased arterial elastance one year after 1.5 Gy of GCR_{5-ion} exposure, but not after 0.50 cGy ^{56}Fe , ^{16}O , or 200 cGy of γ -radiation in male C57BL/6 mice. Changes in aortic elastic fiber organization and alterations in extracellular matrix of heart tissue was also observed.

Miousse et al. observed global DNA hypo-methylation in the 50-UTR LINE-1 14 days after exposure to 250 mGy and 1 Gy of O ions, whereas hypermethylation was observed in specific LINE-1 elements after 90 days (Miousse et al. 2019). Similarly, Koturbash et al. observed hypomethylation of LINE-1 in male C57BL/6J mice one week after exposure to 100 mGy of protons or 0.5 Gy of Fe ions (Koturbash et al. 2016). Comparable to Miousse et al. (2019), hypermethylation of LINE-1, and other retrotransposons as well as major satellite DNA was

observed 90 days after exposure to Fe ions. Metabolomics analyses revealed involvement of one-carbon metabolism pathway in both the studies. A bioinformatics comparative analysis of existing transcriptomic (microarray) data from cardiomyocytes isolated from male C57BL/6 mice 28 days after 0.9 Gy of protons or 150 mGy of Fe ions and human endothelial cells (HUVECs) cultured for seven days on ISS revealed common molecular pathways of oxidative stress with FYN as the central driver/hub for the cardiovascular response to space radiation (Beheshti et al. 2019). Interestingly, the transcriptomic signature resulting from exposure to protons also was much closer to the signature resulting from signature from spaceflight compared to that resulting from exposure to Fe ions. Recently, Garikipati et al. performed transcriptomic analysis on heart tissue harvested from female CB6F1/Hsd mice exposed to 0.4-1.6 Gy γ -rays, 40-320 mGy of Si and titanium ^{22}Ti (Ti) ions 16 months after exposure (Garikipati et al. 2021). Analysis revealed more profound effects with HZE particles at lower doses than γ -rays. Twelve common differentially expressed genes related to cardiovascular, pulmonary, and metabolic diseases, cancer, and aging across all three radiation types were identified, which showed an exclusive non-linear DEG pattern for Si and Ti ions, contrary to γ -rays. These analyses also revealed no clear radiation effect thresholds for lower doses of ions HZE or γ -rays.

In cellular models, the predominantly studied cell types of cardiovascular system are cardiomyocytes and vascular endothelial cells. While Heselich et al. (2018) demonstrated that avian cardiomyocytes exposed to Ti, Fe, or carbon (C) ions were radioresistant, Beck et al. (Beck et al. 2014a) and Baselet et al. (Baselet et al. 2017) showed that cultured coronary artery-derived human endothelial cells were radiosensitive after exposure to ^{56}Fe ions and x-rays and promoted distinct pro-atherosclerotic processes dependent on radiation type. Beck et al. showed that nickel (Ni) ions irradiation in human umbilical vein cells (HUVECs) at 0.5, 2, or 5 Gy resulted in persistent DNA damage, dysregulation of cell cycle, and increased secretion of inflammatory cytokines 24 hours after exposure, dysregulation of the cell cycle, and increased secretion of inflammatory cytokines (Beck et al. 2014b). Using 3-D micro-vessel models of human endothelial cells, Grabham et al. 2011 (2011) and Wu YR et al. (Wuu et al. 2020) showed that high LET charged particles ions inhibit angiogenesis by two distinct mechanisms depending on the LET of the particles.

In summary, substantial evidence from human epidemiology, animal, and cellular studies suggests that low-LET radiation strongly impacts the development of degenerative cardiac effects and CVDs at doses above 0.5 Gy, which may be related to the overall acceleration of age-related processes. Although recent epidemiological and experimental data points towards a sustained risk of CVD below the 0.5 Gy threshold, it open questions include how dose rate and radiation quality modify the overall response to radiation exposure. At these low doses, it can be difficult to separate effects of radiation from other CVD risk factors, such as smoking status, obesity and nutrition, alcohol consumption, and stress. Statistical analyses in human epidemiological studies often lack adjustments for these confounders, but newer sophisticated analysis methods are improving the confidence in the results from these investigations. These approaches are critical to interpreting long-term health effects in astronauts and should include of consideration a significant healthy worker effect (Cucinotta et al. 2013a). Unfortunately, ground-based animal

studies have not provided sufficient data to adequately characterize the risk to the cardiovascular system. Some studies have used doses higher than those relevant to spaceflight, lacked γ -ray controls to make estimates of biological effectiveness, and often used animal models that are pre-disposed to heart disease that were fed a high-fat diet; thus, the model was not reflective of the health status and diet of astronauts. However, it is important to note that observations from more recent use of wild-type animals have shown mostly mixed results after radiation exposures more relevant to the spaceflight experience. Future work will support better characterization of the mechanisms of CVD induction, uncover individual susceptibility, and accelerate the development of pharmacological or nutraceutical countermeasures to close knowledge gaps in space radiation-induced CVD research. This work will leverage data from multi-ion exposures (which is now available with the development of the GCR simulator at NASA Space Radiation Laboratory [NSRL]), elucidate sex differences and dose rate effects, identify and utilize better animal models, including larger species (e.g., mini-pigs) to improve translation to humans, and include newer in-vitro technologies (e.g., tissue-chips, induced pluripotent stem cells (iPSCs)).

D. Computer-Based Models and Simulation

1. Venous Thromboembolism

Lan et al. (2021) developed a novel model of the cranial circulation, accounting for the removal of tissue weight in weightlessness, to explore changes in internal jugular vein flow during spaceflight. The initial model was based upon experimental data from their laboratory as well as published work from others and considered jugular flow as a single flexible tube (Lan et al. 2021). The model was later modified to consider two jugular veins with right-side dominance (Lan et al. 2022). The model describes a shift in blood flow from the internal jugular veins to the vertebral plexus in weightlessness and the lowest flow in the left (non-dominant) internal jugular vein, corresponding to the location of a venous thrombosis in an ISS astronaut (Marshall-Goebel et al. 2019; Auñón-Chancellor et al. 2020). Left internal jugular vein flow decreased from 3.3 to 0.5 mL/min and right-side flow decreased from 7.4 to 1.2 mL/min when transitioning from 1-G supine to weightlessness. The model, however, did not consider the reduction in blood volume during spaceflight and is yet to be validated with spaceflight data.

2. Cardiovascular Disease

Cucinotta et al. developed a cardiovascular risk model using the same basic model structure previously used for cancer (Cucinotta et al. 2013b). A linear no-threshold dose response based on the meta-study of Little et al. (2012) was used. Due to a lack of experimental and human data, a quality factor based on endpoints that were not CVD specific was used, and no dose rate effect modification was assumed since the meta study used occupational and other low dose rate cohorts. ICRP and other recommendation for exposure limits for radiation protection assume a dose response threshold at 0.5 Gy or Gy-eq whereas model of Cucinotta et al. does not assume this threshold. However, there is little direct evidence of a threshold in the human data with very few studies showing any evidence of curvature in the dose response that would be indicative of threshold-like behavior (Wakeford 2022). While some studies showed a statistically significant response above that threshold but not below, this type of behavior is to

be expected even in the case of a linear dose response as the signal becomes smaller than the statistical fluctuations in the data. A critical issue in developing risk models is the shape of the cardiovascular dose response. To address this question, Schöllnberger et al. developed a low LET risk model using the LSS data based on multi-model inference (Schöllnberger et al. 2018). This model includes the possibility of different shaped dose responses, including the possibility of a threshold dose, to represent the uncertainty in the dose response. This allows a probabilistic estimate of the risk even in the case where there is some probability of threshold. Current analysis of experimental data to model quality effects is ongoing but has not yet been published. As stated above, the risk of CVD is also highly individualized, dependent on factors such as age, cholesterol levels, and calcium score. Recent clinical guidelines (Arnett et al. 2019) directly uses CVD risk estimates and part of that decision making process. Huff et al. lays out a modeling framework to begin to account for these individual factors in radiation risk assessment (Huff et al. 2022). Such an approach can better determine risk both for informed consent before spaceflight as well as to help inform medical management decisions related to primary prevention of CVD. Recent clinical guidelines (Arnett et al. 2019) directly uses CVD risk estimates along with the guidelines suggested by Huff et al. (Huff et al. 2022) to include a framework for space radiation risks.

As pointed out in NCRP commentary 24 (NCRP 2015), biologically-based dose response (BBDR) models, which rely on a more mechanistic understanding of the underlying biological processes, are a way forward to provide more reliable risk projections when extrapolating beyond the available epidemiological data. That report was focused on low dose radiation exposures, but this issue can be even more important for space radiation since the radiation environment is quite different and more complex than for exposure received terrestrially and analyses of the astronaut cohort are quite limited by statistical power. Tapio et al. recently reviewed progress in this area (Tapio et al. 2021). Significant progress has been made but a comprehensive BBDR model for CVD remains to be developed.

X. RISK IN CONTEXT OF EXPLORATION OPERATIONAL SCENARIOS

A. Mission-Related Factors Contributing to Risk

1. Altered Gravity

Exposure to weightlessness is associated with the development of orthostatic intolerance during re-exposure to gravity. While orthostatic intolerance may present in some astronauts after landing on Earth, the risk of orthostatic intolerance also may be present during descent to and ascent from Lunar or Martian surface. The orientation of the astronauts in the vehicle and the acceleration profile (magnitude, duration, and frequency) will have profound effects on the risk, as will the provision of countermeasures.

There are no data suggesting that astronauts and cosmonauts will experience orthostatic intolerance on the surface of the Moon or Mars. There were no reports of orthostatic intolerance during the Apollo missions (Pattarini et al. 2020). Additionally, there currently are no data to suggest that exposure to partial gravity environments of the Moon and Mars will be protective.

Measurements of the IJV from partial gravity parabolic flight (Martin et al. 2016; Lee et al. 2020b) suggest that the gravity levels on the Moon and Mars will not be sufficient to prevent the headward fluid shift and venous congestion that may contribute to the development of a deep vein thrombosis or thromboembolism.

2. Mission Duration

Even short stays in weightlessness are associated with post-spaceflight orthostatic intolerance (Hoffler et al. 1974; Bungo and Johnson 1983; Bungo et al. 1985), and the prevalence is increased in longer missions (Meck et al. 2001; Lee et al. 2015). Exploration missions will be longer in duration, and it is plausible that even longer duration missions (>6 to 8 months) will increase the risk profile.

Currently, there is no evidence that mission duration contributes to the risk of dysrhythmias or deep vein thrombosis during spaceflight. The cardiac chambers and IJVs are distended during even brief exposures to weightlessness, and this persists throughout long-duration missions (Khine et al. 2018; Marshall-Goebel et al. 2019).

Analyses conducted thus far have been insufficiently powered to determine an effect of mission duration on the risk of CVD or CVD death after exposure to weightlessness (Charvat et al. 2022) but increased mission duration will contribute to total radiation dosage, particularly beyond LEO.

3. Distance from Earth

Distance from Earth is a critical factor affecting the consequences of a cardiovascular health-related event. Astronauts and cosmonauts in LEO are in regular communication with ground personnel, with ~2-second delay (Martin et al. 2012), and can be returned to Earth in the event of a medical emergency. All ISS crew receive training on independently initiating a medical emergency response and at least two crewmembers on each ISS increment receive additional medical training to enable support of less emergent issues with assistance from ground medical personnel (i.e., flight surgeon) through private voice and video communication. Crewmembers and flight surgeons regularly participate in scheduled private medical conferences. Exploration missions beyond LEO will experience longer delays in video and verbal communication as the distance from Earth increases, necessitating the use of “just-in-time” or virtual training. During missions to Mars, the communication delay will be long enough that astronauts must have sufficient knowledge, skills, and resources to diagnose and treat medical contingencies autonomously. In addition, rapid Earth return will not be possible necessitating that crew be able to identify, treat, and stabilize a broader range of medical issues.

Increasing distance of travel from Earth also will likely constrain overall mass, volume, power, and bandwidth resources resulting in decreased medical diagnostic and intervention capabilities unless additional resources can be prepositioned. Also, the increased mission duration will extend the time across which hardware must function and during which medications would have to remain safe and effective (limited shelf life).

4. Radiation Exposures

The environment outside of the shield-like atmosphere and magnetosphere of the Earth contains several types of radiation. Galactic cosmic rays (GCR) represent an additional particle constituent of space radiation and originate from outside our solar system. GCR is comprised of mostly highly energetic protons with a small component of high atomic number (Z), high-energy (HZE) nuclei. Because HZE nuclei are the components of space radiation that have the highest biological effectiveness, they are a large concern for astronaut safety. HZE nuclei produce highly ionizing tracks as they pass through matter. In addition, they leave columns of damage at the molecular level when they traverse a biological system, damage that is different in severity and complexity from the damage from low-linear energy transfer (LET) radiation sources such as γ - and X-rays (Durante and Cucinotta 2008). HZE nuclei impart damage through the primary energetic particle and secondary delta-ray electrons as well as from fragmentation events that produce a spectrum of other energetic nuclei, protons, neutrons, and heavy fragments (Wilson et al. 1995). In low Earth orbit, astronauts will be exposed to GCR only in the polar regions of the trajectory due to geomagnetic protection from the Earth's magnetosphere. Beyond the protection of the planet's magnetosphere and atmosphere, GCR represents a chronic low-dose exposure that accumulates over the entire span of a mission. Protons and electron trapped within the magnetosphere in toroidal belts around the Earth are another major source of space radiation exposure dependent of the mission trajectory. Intermittent solar particle events (SPEs) can accelerate large quantities of highly energetic protons into the solar system and are the largest contributor to local changes in the space radiation environment. Each of these sources are modulated by the sun's approximately 11-year solar cycle (Tascione 2010).

(Durante and Cucinotta 2008); Wilson et al. 1995) The effects of protracted exposure to low dose rates (<20 mGy/hour) of protons, HZE particles, and neutrons of the relevant energies for doses up to ≈ 0.5 to 1 Gy (corresponding to exposures estimated for design reference missions [DRM] in deep space) on degenerative conditions of the circulatory and other organ systems are of concern. Current Mars DRM exposure estimates vary between 0.3 and 0.5 Gy from GCR, with shielded SPE exposures on the order of 0.15 to 0.5 Gy to internal body organs within a typically shielded spacecraft. Approximate relative dose (Gy) contributions to total organ exposure from GCR include protons delivering approximately 66% of the dose, approximately 14% to 16% from helium, approximately 4% to 5% from particles of $3 \leq Z \leq 9$, approximately 2% to 3% from particles of $Z \geq 10$, and approximately 10% to 15% from neutrons, pion, muon, and electromagnetic (π /EM) components (Simonsen et al. 2020).

Degenerative tissue effects including CVD are documented following exposures to terrestrial sources of ionizing radiation (e.g., γ -rays and x-rays). Degenerative conditions of concern that potentially could result from exposures to ionizing radiation in the form of GCR or SPEs expected during long-duration spaceflight include CVD (ischemic heart disease, atherosclerosis), cardiomyopathy, and cerebrovascular and peripheral arterial diseases leading to stroke. However, the existence of low-dose thresholds and dose-rate and radiation quality effects, as well as mechanisms and major risk pathways, are not well-characterized. Further, the epidemiological evidence regarding lower dose rate has changed in recent years showing that

lower doses can promote CVDs through microvascular damage and inflammation. As additional evidence is pointing to lower, space-relevant thresholds for CVD, additional research with cell and animal studies is required to quantify the magnitude of this risk, understand mechanisms, and determine if additional protection strategies are required. Accumulated exposures from multiple missions resulting in doses above 0.5 Gy-Eq to heart tissue should be considered elevated above the threshold for increased risk due to lack of evidence to support differences between acute, chronic, and fractionated doses.

Permissible exposure limits (PELs) for short-term and career exposures to space radiation, based on existing human epidemiology data, have been approved by the NASA Chief Health and Medical Officer, who also sets the requirements and standards for mission design and crew selection. Table 1 and Table 2, recreated directly from NASA-STD-3001, Volume 1, rev. B (NASA 2022), list the current short- and long-term PELs for non-cancer effects (**Table 1**; in mGy-Equivalents or mGy) and relevant relative biological effectiveness (RBE) values (**Table 2**). The lifetime limits for cataracts and heart disease are imposed to limit or prevent risks of clinically significant degenerative tissue outcomes. The current PELs are based on recommendations from NCRP 2000, which defines a threshold as an exposure below which clinically significant effects do not occur. However, the ICRP redefined their notion of a threshold dose as the dose required to cause a 1% incidence of an observable effect (ICRP 2007; Stewart et al. 2012). In particular, ICRP 2012 included newer evidence and noted a lower threshold for cardiovascular effects from ground-based radiation exposure, with the previous threshold of 2 Gy lowered down to 0.5 Gy. Further, a review and analysis of data for cardiovascular risk from low dose radiation exposure by Electric Power Research Institute (EPRI) in a recent technical report posited that limited available data prevents a quantitative estimation of a specific threshold value for CVD from either acute or chronic exposures and suggested no immediate need for changes to the current system of protection (Bernstein et al. 2020).

Table 1. Dose Limits for Short-Term or Career Non-Cancer Effects (in mGy-Eq. or mGy) (NASA STD 3001 Vol 1 Rev B) Note: RBEs for specific risks are distinct as described below.

Organ	30-Day Limit (mGy-eq)	1-Year Limit (mGy-eq)	Career Non-Cancer Limit (mGy-eq)
Lens of the eye	1000	2000	4000
Skin	1500	3000	6000
Blood-forming organs	250	500	n/a
Circulatory system	250	500	1000
Central Nervous System*	500 mGy	1000 mGy	1500 mGy
Central nervous system (heavy particles, $Z \geq 10$)*	n/a	100 mGy	250 mGy

**Lens limits are intended to prevent early (<5 years) severe cataracts, e.g., from a solar particle event. An additional cataract risk exists at lower doses from cosmic rays for sub-clinical cataracts, which may progress to severe types after long latency (>5 years) and are not preventable by existing mitigation measures; however, they are deemed an acceptable risk to the program.*

***Circulatory system doses calculated as average over heart muscle and adjacent arteries.*

****Central Nervous System limits should be calculated at the hippocampus.*

#Reference: National Council on Radiation Protection and Measurements. 2000. Recommendations of Dose Limits for Low Earth Orbit. NCRP Report 132, Bethesda MD.

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Table 2. Relative Biological Effectiveness (RBE) for non-cancer effects of the lens, skin, blood forming organs, and circulatory systems

<i>Radiation Type</i>	<i>Recommended RBE^b</i>	<i>Range</i>
1 to 5 MeV neutrons	6.0	(4-8)
5 to 50 MeV neutrons	3.5	(2-5)
Heavy Ions	2.5 ^c	(1-4)
Protons > 2 MeV	1.5	-

a RBE values for late deterministic effects are higher than for early effects in some tissues and are influenced by the doses used to determine the RBE.

b There are not sufficient data on which to base RBE values for early or late effects by neutrons of energies <1 MeV or greater than about 25 MeV.

c There are few data for the tissue effects of ions with a Z>18, but the RBE values for iron ions (Z=26) are comparable to those of argon (Z=18). One possible exception is cataract of the lens of the eye because high RBE values for cataracts in mice have been reported.

Reference: National Council on Radiation Protection and Measurements. 2000. Recommendations of Dose Limits for Low Earth Orbit. NCRP Report 132, Bethesda MD.

Currently, no existing biophysical model projects all degenerative risks for the entire range of particle types and energies in space. The large RBEs that are found in the few studies that have been performed suggest that an organ dose-equivalent based on radiation quality factors can be used to make a first approximation for risk estimates; however, the shape of the dose-response curve for specific diseases and dose-rate modifiers are unknown. Dose-rate modifiers could be higher than those observed for cancer risks because of the possibility of threshold effects. Cucinotta et al. presented risk assessments for extended LEO and exploratory missions, including the prediction of a combined cancer and circulatory disease risk for a Mars mission (Cucinotta et al. 2013b). The authors used the excess relative risk (ERR) results of Little et al. (Little et al. 2012) and noted that the inclusion of the circulatory disease risk increased the overall risk by 40% from cancer alone and reduced the overall age at exposure dependence of the risk.

NCRP 2014 (NCRP 2014, p. 2014) notes that this approach for combining cancer and circulatory disease risks is comprehensive but should be considered “preliminary”, largely due to the assumptions “of risk in the low-dose domain for circulatory disease that are far from conclusive”. The estimates for ERR per Sv, which were provided by studies of the atomic bomb survivors, are not sufficient to estimate risk for astronauts because a risk transfer approach is needed together with estimates of RBE and dose-rate modifiers. Additionally, there is significant heterogeneity between different populations; the baseline risk of coronary heart disease is several-fold larger in the U.S. than in Japan, while the risk of stroke is comparable. NCRP 2014 (NCRP 2014, p. 2014) suggests that “it would be preferred to have less heterogeneous and more comparable populations to generate risk coefficients.” To determine the cancer risks, the NCRP suggests using multiplicative and additive transfer models to transfer risks between populations.” Thus, “it is essential that additional experimental data with HZE particles be acquired using relevant model systems and relevant doses to provide robust input to refine the current model of cardiovascular risk from space radiation” (NCRP 2014, p. 2014). These data will help to quantify the magnitude of this risk to decrease the uncertainty in current permissible exposure limits and to determine if additional protection strategies are required.

5. Medical Care Capabilities

Current capabilities to diagnose and treat cardiovascular conditions on the ISS include an automated external defibrillator, crew medical restraint system to enable providing chest compressions in microgravity, and a limited scope of medications and consumables such as epinephrine, beta blockers, intravenous fluids, anti-coagulants, and anti-hypertensives (NASA 2016). Recently, wearables such as Apple watches have been flown and may provide limited single-lead cardiac rhythm monitoring capabilities. Onboard ultrasound devices combined with real-time ground subject matter expert support enable crew to perform and acquire high-quality, diagnostic sonographic images (Sargsyan et al. 2006). For exploration missions, increasing distance from Earth and communication delays combined with reduced or absent resupply and crew return capabilities will necessitate more robust onboard medical capabilities, including medical hardware and consumables as well as resources to enhance crew autonomy to mitigate risk. Specifically, management of *de novo* dysrhythmias may require the ability to perform chemical and/or mechanical cardioversion and cardiac monitoring, to provide long-term thromboembolic prophylaxis, and to manage complications such as hypotension. Due to communications delays, decision support tools and just-in-time training will be needed to aid crews in medical decision making and to assist with performance of imaging or other procedures. *In situ* intravenous fluid generation may be required for Mars missions given that mission duration may exceed established shelf-life of intravenous fluid launched from Earth and that medical system mass/volume constraints are likely to prohibit carrying a significant quantity of intravenous fluid. Development of an exploration formulary that includes relevant medications for treatment of CVD and that addresses concerns regarding medication safety and effectiveness over the duration and exposures relevant to exploration missions will be needed.

6. Vehicle Design

Orientation of the astronauts and cosmonauts relative to the acceleration that they will experience during the dynamic phases of flight contributes to the risk of orthostatic intolerance. Except for crewmembers on the Space Shuttle and the Lunar Excursion Module, astronauts and cosmonauts have been reclined or recumbent to improve orthostatic tolerance during dynamic phases of flight. Space Shuttle astronauts experienced as much as 1.7-G during re-entry, and the Apollo astronauts experience 0.4-to 0.6-G (Pattarini et al. 2020). Apollo astronauts on the Lunar Excursion Module were standing during descent to and ascent from the Lunar surface to increase their field of vision of the landing site and to reduce weight of the vehicle (i.e., no seats).

Protection for solar particle events is an important function of the vehicle design. Radiation shielding is described in detail below (**Section C.5**).

7. Suit Design

The protective garments designed for wear during launch and landing on the Space Shuttle, the Launch and Entry Suit and the Advanced Crew Escape Suit, were intended to be protective in the event of a bail-out, particularly over cold water (Bishop et al. 1999, p. 1999). Consequently, without liquid or air cooling, the astronaut might be prone to overheating and heat injury. Elevated core temperature could increase the incidence of orthostatic intolerance during re-entry and landing on a Lunar or Martian surface. Even with liquid cooling, core temperature was as much as 0.5 °C higher during re-entry while wearing the re-entry suit than the same time of day inflight without the garment (Rimmer et al. 1999). Current evidence suggests that thermoregulation at rest (Stahn et al. 2017) and during exercise (Fortney et al. 1998) is impaired after 6-months of spaceflight. Given that these changes occurred even though crewmembers were performing inflight exercise, if countermeasures are not available during an exploration mission, it is possible that impaired thermoregulation may be exacerbated and tolerance to standing and exercise will be further reduced.

The extravehicular suit is a closed system in which it is difficult to provide food and fluids to the astronauts. Given that the preparation for extravehicular activities includes significant logistical support and other resources, historically these excursions are prolonged (4 to 8 hours). Thus, astronauts are likely to become dehydrated, further impairing thermoregulation and potentially leading to heat injuries.

8. Ground Support Availability

Nominally astronauts and cosmonauts returning from ISS are met by ground support and medical personnel soon after landing. For crew returning on the Soyuz capsule, ground support and medical personnel are transported by helicopter and set-up shelter at the landing site in Kazakhstan for crew medical examinations (Lee et al. 2020c). Currently, for SpaceX-NASA missions, the crew land and are recovered in the Gulf of Mexico or in the Atlantic Ocean off the Eastern coast of Florida, receive initial care including medical examinations onboard the SpaceX recovery ship, and, once they are stable, are transported by helicopter to the rally airport in Florida. Should cardiovascular complications arise, prompt access to standard medical interventions is available. Landing and the safe recovery of crewmembers on SpaceX vehicles

continues to provide valuable data, experience, and lessons learned that should be applied to similarly planned landings in the Orion vehicle.

In off-nominal conditions, such as a ballistic re-entry or landing at an unsupported landing site, crew may not be reached by the ground support and medical personnel for several hours to days, in which case the crew must extricate themselves from the capsule after landing. Orthostatic intolerance in this context can delay or impair the ability to egress from the capsule and participate in survival activities (Pettit 2010). This situation can be challenging in the first 24 hours after landing, particularly after long-duration missions (Meck et al. 2001; Lee et al. 2015). It is currently unclear how the motion of the waves in the case of a water landing will impact egress, orthostatic intolerance, and motion sickness.

Teams in mission control are in constant contact with crews aboard the ISS. There is a ~2-second delay in video and audio communications that may complicate the performance of some procedures, such as ultrasound. There is breadth of experience in remotely guiding the crew on ISS (Hamilton et al. 2011; Martin et al. 2012; Pavela et al. 2022), but plans are being made to improve procedures and provide just-in-time training (e.g. computer-based instruction, video review of procedures) as communication delays increase with increasing distance from Earth and reliance on remote guidance becomes impractical.

B. Risks Related to Design Reference Mission (DRM)

1. Short- and long-duration mission in LEO

The primary in-mission risks are dysrhythmias and venous thromboembolism. There is no clear evidence of increased incidence of dysrhythmias, and the incidence of deep vein thrombosis or thromboembolism is yet to be defined. Neither dysrhythmia or venous thromboembolism have been monitored systematically so asymptomatic cases may have been missed; however, with the recent findings of a venous thromboembolism during an International Space Station mission, Space Medicine is actively monitoring crewmembers for venous thromboembolism.

Orthostatic intolerance remains the primary concern for crew health during re-exposure to gravity and in the first few days after landing. Countermeasures are employed in these astronauts during re-entry and landing, and care is given to limit their activities based upon their clinical presentation. The prevalence of orthostatic intolerance appears to increase with mission duration, particularly for missions greater than a few months (Meck et al. 2001; Lee et al. 2015). The risk of orthostatic intolerance during re-entry is exacerbated when crew are in an upright or seated posture, and when adequate cooling is not supplied (Perez et al. 2003). It has become common practice to provide isotonic intravenous saline in the hours after landing following a long-duration mission, and food and fluid consumption is encouraged as tolerated (Mulavara et al. 2018; Lee et al. 2020c). Some astronauts wear lower body compression garments for several days after landing to control symptoms of orthostatic intolerance and manage swelling in the legs. Although recovery after spaceflight may be increased after longer spaceflight missions (Lee et al. 2015), there appear to be no long-term health consequences related to orthostatic intolerance after these missions.

There is no evidence that spaceflights in LEO result in an increased risk of death due to CVD but a recent retrospective analysis suggests that the lifetime risk of a cardiovascular event is higher in astronauts than in matched control subjects with no spaceflight experience (Charvat et al. 2022). Current estimations of radiation exposure for these missions suggest that the risk of radiation-related CVD is not increased as a result of missions in LEO but epidemiological data are limited.

2. Short- and Long-Duration Missions to the Moon

There currently is no evidence that the risk of dysrhythmias and venous thromboembolism are increased in missions to the Moon though it is likely that oxidative stress and inflammation will be increased during missions beyond LEO concomitant with an increase in radiation exposure. Recent data from partial gravity simulated by parabolic flight suggests that the headward fluid shift experienced during weightlessness will not be greatly reduced by lunar gravity (0.16-G) (Lee et al. 2020b).

The risk of orthostatic intolerance is not expected to be greater during re-exposure to gravity after lunar missions than after missions in LEO but there likely is an in-mission risk of orthostatic intolerance during descent to and ascent from the lunar surface if astronauts are oriented in the vehicle such that they experience head-to-foot accelerations (Pattarini et al. 2020). It is unknown what magnitude and duration of acceleration will be tolerated by the astronauts during the dynamic phases of flight, but it appears unlikely that stays in the partial gravity environment of the Moon will be protective against orthostatic intolerance.

Based upon experiences of the Apollo astronauts, there is no evidence that short-duration missions to the Moon increases the lifetime risk of CVD (Cucinotta et al. 2016; Elgart et al. 2018) but the longest stay on the Moon was only three days. Current estimations of radiation exposure during longer missions (up to one year) suggest that the risk of exploration-related CVD risk is not increased as a result of missions to the Moon.

3. Missions to Mars and the Martian Surface

There currently is no evidence that the risks of dysrhythmias and venous thromboembolism are increased in these missions to Mars and the Martian surface though it is likely that oxidative stress and inflammation will be increased during missions beyond LEO concomitant with an increase in radiation exposure. Additionally, the length of the Mars missions will be greater than any previous human spaceflight experience. Current data from partial gravity simulated by parabolic flight suggests that the headward fluid shift experienced during weightlessness will not be greatly reduced by Martian gravity (0.38-G) (Lee et al. 2020b).

The incidence of post-spaceflight orthostatic intolerance is likely to be higher than after missions in LEO and to the Moon because of the duration of the transit to and from Mars. The risk of orthostatic hypotension during descent to and ascent from the Martian surface will be present and perhaps higher due to the long transit time. While astronauts and cosmonauts have completed one-year missions (Grigoriev et al. 1991; Charles and Pietrzyk 2019; Garrett-Bakelman

et al. 2019), there is no experience up to three years. Further, the acceleration profiles and the orientation of the crew relative to the acceleration vectors are not defined.

Risk to crew health and mission success may be impacted by the availability of medical monitoring and intervention capabilities due to size, mass, and power limitations as well as complications with early crew return due to orbital mechanics. Medical interventions are further complicated by the availability (stowage) and efficacy (shelf-life) of pharmaceuticals and communication delays due to the distance from Earth. Crew selection will strongly influence the in-mission risk of a cardiovascular event (Gore et al. 2020) and the skillset of the crew to respond to a medical emergency.

The long-term health consequences of extended travel to, stays on, and return from an extraterrestrial destination is unknown given the limited human experience, including stays in partial gravity. Estimations of radiation exposure during these Martian missions will approach and likely exceed the level at which the risk of CVD will increase.

C. Countermeasures

1. Inflight Exercise

During short-duration spaceflight, exercise reduces the heart rate response to standing but does not appear to prevent presyncope (Lee et al. 1999). Exercise prevents cardiovascular deconditioning (Dorfman et al. 2007; Hastings et al. 2012), but orthostatic intolerance is not prevented without the inclusion of fluid loading (Shibata et al. 2010).

2. Fluid Loading

Fluid loading was not performed by NASA astronauts until the Space Shuttle program. The efficacy of the end-of-mission fluid loading was demonstrated in astronauts returning from Space Shuttle missions (Bungo et al. 1985), although the countermeasure may be less effective in preventing signs of orthostatic intolerance after longer missions. Spaceflight-induced plasma volume loss is one factor contributing to the development of orthostatic intolerance, but other cardiovascular factors contributing to blood pressure regulation during orthostatic stressor also may be impaired (Tuday et al. 2007; Behnke et al. 2008).

The efficacy of the fluid loading countermeasure is affected by timing of ingestion relative to the time when orthostatic intolerance protection is targeted. In a ground-based analog, Fortney et al. (Fortney et al. 1994) demonstrated that fluid retention peaks about 2 to 3 hours after ingestion and plasma volume decreases thereafter. To maximize the benefit of this countermeasure, crew time must be allotted for this activity. Future work will need to address if distributing the fluid ingestion over several hours, as in the Russian protocol, maintains the fluid retention peak profile.

Fluid loading continues to be used routinely by astronauts and cosmonauts returning from International Space Station missions and is planned for use on vehicles from commercial providers. Stowage has to be allocated for potable water and salt tablets or isotonic drink mixes.

The need to fluid loading prior to descent to and ascent from the lunar or Mars surface has not been investigated. Based upon results from a 2-week bed rest study (Stenger et al. 2014), fluid loading may not be required for short-duration missions when lower body compression garments are utilized.

3. Lower Body Compression Garments

The efficacy of lower body compression garments has been demonstrated after exposure to weightlessness (Vil-Viliams et al. 1998; Perez et al. 2003) and spaceflight analogs (Platts et al. 2009; Stenger et al. 2014). After short-duration exposures (two weeks), compression garments are effective even without the implementation of other countermeasures (e.g., exercise, fluid loading) (Stenger et al. 2014). Lower body compression garments also appear to be effective after short- and long-duration spaceflight (Stenger et al. 2013; Lee et al. 2020c). However, astronauts and cosmonauts may not tolerate the upright posture due to motion sickness (Jennings 1998) even when wearing compression garments (Lee et al. 2020c). If elastic garments are used for longer missions, a further complication arises from the need to custom-build compression garments before spaceflight without the insight into anthropometric changes that may occur during the mission before the compression garment is needed. The effectiveness of the garments will depend upon the appropriate sizing (e.g., adjustments) required to achieve the desired level of compression at the end of the mission.

Stowage must be allocated for the compression garments on any vehicle in which astronauts will return to Earth. The need to wear lower body compression garments during descent to and ascent from the Lunar or Martian surface has not been demonstrated. If required, stowage requirements would increase if different garments are used for different phases of the mission (e.g., one garment for Lunar landing and ascent, and one garment for Earth return).

Lower body compression garments are regularly worn by astronauts and cosmonauts during re-entry and in the hours after landing. Some crewmembers will wear lower body compression garments for several days after landing to manage symptoms of orthostatic intolerance as well as lower leg swelling (Stenger et al. 2014). It is unlikely that compression garments will be needed while on the surface of the Moon. Should they be required for use in any protective garment (intra- or extravehicular), their integration into the suit ensemble will have to be tested.

4. Lower Body Negative Pressure

Use of lower body negative pressure for 4 hours, when combined with fluid loading, has been shown to temporarily reverse the development of orthostatic intolerance following seven days of bed rest (Hyatt and West 1977) and signs of deconditioning during short-duration spaceflight (Charles et al. 1999), possibly through the restoration of plasma volume. The protocol tested during the Space Shuttle program required four hours of LBNP and was considered too complicated and required too much crew time to be used operationally for crews returning from space. Additionally, the LBNP protocol would have to be repeated if landing was delayed.

However, Russian crewmembers regularly participate in LBNP countermeasures at the end of their spaceflight missions in preparation for landing.

LBNP has been shown to acutely decrease IJV cross-sectional and volume (Marshall-Goebel et al. 2021; Arbeille et al. 2021) and restore normal flow patterns in some individuals (Marshall-Goebel et al. 2019). However, there is not sufficient evidence to demonstrate its efficacy as a countermeasure to the development of a venous thromboembolism.

Cosmonauts routinely participate in LBNP countermeasure sessions using the Russian Chibis Suit in the three to four weeks before landing. The Chibis suit consists of a consist of waist seal that is positioned at the level of the iliac crest, a hard shell covering the pelvis, two corrugated tubes for the legs, with hard shoes. The plantar surfaces of the crewmember's feet are in contact with the insole of the shoes, and the cosmonauts are encouraged to perform knee bends and simulate walking to prevent hypotension. A vacuum pump with a control panel is attached to the suit, and the maximum decompression that can be applied 60–70 mm Hg, though the training protocol does not exceed -45 mmHg (Kozlovskaya et al. 2010). The amount of decompression increased is a staged fashion within a LBNP training session, and across days the duration of decompression is increased.

Currently LBNP countermeasure training is monitored by fellow crewmembers and remotely by medical specialists using continuous ECG and intermittent brachial blood pressure. In addition to the size and power requirements for exploration-class vehicles, there will need to be a way to maintain sufficient medical monitoring capabilities by the crew themselves because time delays with increased distance from Earth will prevent practical medical monitoring by mission control. Presyncopal episodes were reported during Space Shuttle and Skylab missions at or near pressures that provide a similar cardiovascular stress to standing (Henry et al. 1977; Charles et al. 1999). Depending on the nature of LBNP use during exploration-class missions, the duration and frequency will need to be optimized for targeted countermeasures.

5. Radiation Shielding

The space radiation environment comprises mainly of GCR and high energy solar protons from trapped radiation and SPEs. (NCRP 1989, 2006). Health risks associated with protracted exposures to the ubiquitous space radiation environment poses as one of the most critical challenges facing planners of long duration exploration class missions and possibly a future sustained habitat in space. These health risks range from risk of cancer, degenerative diseases such as diseases of circulatory system, central nervous system, cataract, and immune system risks, to acute radiation syndromes. There are huge uncertainties in estimating these risks, mainly due to large uncertainties in the radiobiology of heavy particles.

In space, traditional terrestrial-based radiation strategies of managing duration, distance, and shielding are not effective in keeping doses “As Low as Reasonably Achievable” or ALARA. In space, no single point source of radiation exists to move away from and effectively shielding heavy particles moving at near relativistic speeds is not currently feasible. Therefore, time is the

only means to limit astronaut radiation exposure, which is contrary to the goal of extended deep space missions beyond low Earth orbit.

While it currently is not feasible to protect against ionizing radiation from GCR due to their high penetration depth that require impractical shielding mass, protecting crew from infrequent large SPEs is feasible due to their shorter penetration depth and limited duration. However, to reduce the high costs of launching mass into space, optimization of the spacecraft design, shielding and placement of equipment inside the spacecraft is necessary. Shielding optimization is an important component in the application of radiation risk models. It is performed by ray tracing, a technique in which 3-dimensional (3D) vehicle geometry is represented as the input for the space radiation transport code and subsequent risk calculations. The proper quantification of spacecraft shielding through ray tracing is a prerequisite to the calculation of health risks from space radiation (Nounu et al. 2009).

To limit radiation exposure and protect crewmembers from large SPEs, NASA has implemented a short-term effective dose PEL of 250 mSv (NASA 2022), specifying the requirements for SPE shielding as “[V1 4031] The program shall protect crewmembers from exposure to the design reference SPE environment proton energy spectrum (sum of the October 1989 events) to less than a NASA effective dose of 250 mSv.”

Excerpt from NASA-STD-3001 “SPE shielding should be an inherent part of the vehicle design and/or reconfigured components within the vehicle to minimize the addition of mass. To be most effective, it is critical that the shielding surrounds the crew. The design solution which includes considerations for the vehicle/habitat should minimize exposure as much as possible utilizing the ALARA principle. For solar particle event shielding designs, an iterative approach should be taken for determining shielding designs that continue to iterate the design until less than a 10 mSv is achieved from the previous iteration. If a reconfigurable shelter is deployed, environmental control life support system (ECLSS) impacts need to be considered. Refer to Table 1, Recommended Shielding Guidelines for SPEs, for shielding recommendations based on mission duration and location.”

Table 3. Recommended shielding guidelines for solar particle events (SPEs)

<i>Mission Location and Duration</i>	<i>Shielding*</i>	<i>Type(s) of Shielding</i>	<i>Comments</i>
<i>Celestial surface any duration</i>	<i>10 cm (or g/cm²) water equivalent surrounding the astronaut; Considers celestial surface shielding contribution</i>	<i>Reconfigurable shielding already within the vehicle</i>	<i>Timeline of SPEs allows for reconfiguration</i>

<i>Beyond low Earth orbit < 6 Months</i>	<i>15 cm (or g/cm²) water equivalent surrounding the astronaut</i>	<i>Reconfigurable shielding already within the vehicle; Shielding may include personal protective equipment (PPE)</i>	<i>Timeline of SPEs allows for reconfiguration</i>
<i>Beyond low Earth orbit > 6 Months</i>	<i>20 cm (or g/cm²) water equivalent surrounding the astronaut</i>	<i>Integrated vehicle and/or reconfigurable Shielding which may include PPE</i>	<i>Long-duration missions increase the probability of the crew being exposed to SPEs</i>

**The shielding required to meet the standard (utilizing existing mass when feasible).*

Because of the lack of adequate GCR shielding possibilities, emphasis is on developing pharmacological countermeasures to limit the biological effects of space radiation, which is discussed in the radiation countermeasures section.

6. Radiation Countermeasures

Traditional compound-based approaches to reduce radiation health impacts include either administering radioprotectors, medical countermeasures administered prior to a known radiation exposure that counter the immediate radiation damage or radiation mitigators following radiation exposure, targeting molecular processes that cause damage, or enhancing pathways that promote repair to minimize the damage associated with radiation exposure.

Much of the indirect cell damage produced by ionizing radiation is mediated through reactive oxygen species (ROS) that are immediately generated from the interaction of the charged particle with water and other cell components (Li et al. 2014). Excessive production of free radicals produces oxidative damage to cellular structures, which includes proteins, DNA, and lipids, and contributes to the radiation-induced degenerative changes that are associated with aging, CVD, and cataract formation. This extensive ROS profile also can induce persistent metabolic changes that result in a chronic inflammatory response. The subsequent recruitment of inflammatory cells, such as macrophages, which also generate ROS and pro-inflammatory signals, feeds into a loop of oxidative stress, inflammation, and cell damage (Zhao and Robbins 2009).

Two main types of countermeasures have been used to protect vasculature from ionizing radiation: sulfhydryl or thiol compounds and antioxidants. Both classes of compounds function by scavenging the free radicals produced by the interaction of ionizing radiation with water. WR2721, known as Amifostine/Gammaphos/Ethyol, is the best-described member of the sulfhydryl class and is the only drug that is approved by the Food and Drug Administration (FDA) to help prevent excess damage to normal tissues during radiotherapy. The mechanism of action of this drug is thought to be the scavenging of free radicals produced by radiation and H-atom

donation to protect against the damage caused by free radicals. This compound has been tested as a countermeasure for both cataract formation and vascular damage (Kador 1983; Plotnikova et al. 1988; Warfield et al. 1990; Mooteri et al. 1996). However, the utility of amifostine in the space radiation environment is limited for several reasons, including an unacceptable safety profile in healthy populations and a short half-life (Singh and Seed 2018, 2019, 2021; Singh et al. 2018). Radical scavenging vitamins such as C and E have also been shown to protect the lens and vascular system (Bantseev et al. 1997; Jacques et al. 1997; Taylor and Hobbs 2002). In addition, growth factor treatments have been shown to decrease blood vessel stenosis (Fuks et al. 1994). Gamma tocotrienol (GT3), an isoform of vitamin E, is being tested as a mitigator for space radiation-induced effects on the cardiovascular system. γ -tocotrienol is a potent inhibitor of the cholesterol biosynthesis pathway and has previously been shown to act as a protective agent against vascular radiation injury due to high doses of whole-body γ -ray exposures (Berbée et al. 2009), with a suggested mechanism of action of protection of endothelial cell functions (Berbée et al. 2012). Small animal studies with GT3 showed radio-protective effects on the intestine potentially by accelerating intestinal immune cell recovery (Garg et al. 2019). GT3 was shown to attenuate radiation-induced cytogenetic damage in human umbilical vein endothelial cells (HUVECs) (Pathak et al. 2016). GT3 was also recently shown to mitigate the effects of O ions irradiation on cardiac function, expression of a collagen type III peptide, and markers of mast cells, T-cells and monocytes/macrophages in the left ventricle in male C57BL/6 J mice (Nemec-Bakk et al. 2021). Given the chronic inflammatory response that may occur from the low dose and dose rates experienced in space, the use of anti-inflammatory drugs for the prevention of radiation-induced effects also is a possibility. Wilson et al. (Wilson et al. 2011) used corticosteroid therapy to improve the conditions of radiation-induced pneumonitis and pneumonopathy, but it should be approached with caution due to evidence pointing to immune dysregulation in weightlessness (Crucian et al. 2018). The use of a long-term anti-inflammatory drug may synergize with the depressed immune function to significantly increase the risk of infection and even CVD (Hsu and Katelaris 2009).

Another potential avenue for biological countermeasures is the use of statins, a class of pharmaceuticals currently used to slow or prevent CVD. A recent study has utilized simvastatin to mitigate the effects of radiation-induced CVD (Lenarczyk et al. 2015), albeit after very high doses of whole-body irradiation (10 Gy γ - rays). Aside from biological countermeasures, aerobic training has also been suggested as a safe approach for potential mitigation of CVD after radiotherapy or chemotherapy (Berkman and Lakoski 2015; Yu and Jones 2015). In general, suggested cardioprotective strategies for radiation-induced CVD include early and frequent monitoring of cardiovascular health through modern approaches like clonal hematopoiesis of indeterminate potential (CHIP) mutations and other CVD biomarker monitoring, non-invasive imaging techniques (e.g., electrocardiography, echocardiography), management of traditional risk factors (e.g., hypertension and dyslipidemia) through diet, exercise, and pharmaceuticals, and surgical intervention as required (Cutter et al. 2013).

Identifying or developing an effective medical countermeasure that addresses multiple indications and provides cross-risk mitigation by targeting common pathways for each health

impact is desirable but challenging. This becomes even more important due to the gaps in knowledge regarding the cardiovascular system response to space-relevant radiation types, doses, and dose rates. Undoubtedly, the recently upgraded GCR simulator at NSRL, that provides a more accurate ground-based representation of the space radiation environment, will play a critical role in evaluation of effective medical countermeasures to mitigate space radiation effects (Sishc et al. 2022).

XI. DIRECTED ACYCLIC GRAPH REVIEW AND INTEGRATION WITH OTHER RISKS

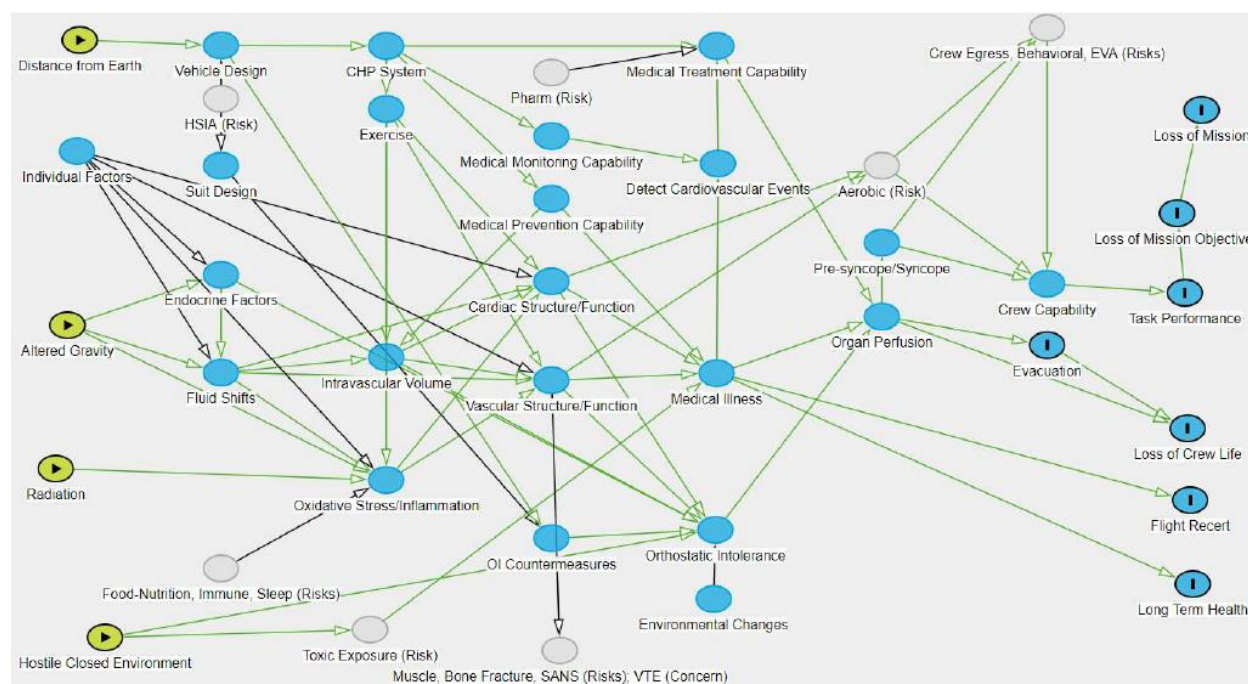


Figure 21. Directed Acyclic Graph (DAG) leading from spaceflight-related stressors to physiologic and anatomical adaptations, contributing to performance and health outcomes. Approved by HSRB, May 2021

A. Directed Acyclic Graph Narrative

Altered gravity and radiation are considered the two main hazards contributing to this risk (**Figure 22**). Altered gravity results in a headward fluid shift, which sets off a cascade of events that include plasma volume loss, decreased red cell mass, cardiac atrophy, and vascular adaptations. These are modified by individual factors (e.g., preflight fitness, existing medical conditions), performance of exercise countermeasures, dietary practices (i.e., available food choices and personal preferences), and circadian rhythm disturbances. Changes in blood flow during spaceflight may contribute to skeletal muscle atrophy, bone demineralization, and decreased exercise capacity and tolerance. Orthostatic intolerance during re-exposure to gravity results from physiological adaptations combined with human integration factors, including vehicle design (e.g., body orientation relative to acceleration, cabin environment heat load) and

suit design (e.g., insulating properties of the garment, provision of cooling) as well as the use of countermeasures (e.g., lower body compression garment, end-of-mission fluid loading). Decreased or loss of consciousness or symptoms associated with reduced cerebral brain perfusion can impact crew activities (e.g., piloting, egress) and contribute to the decreased capacity or failure to complete mission objectives or critical tasks.

Should a medical condition arise during a spaceflight mission, successful treatment is reliant upon the skill set of the crew (i.e., prior training, access to just-in-time training), communication capabilities with medical support personnel (i.e., audio and visual communication, data transfer between ground teams and vehicle), communication delays (that increase with increasing distance from Earth), medical monitoring capabilities (e.g., electrocardiogram, ultrasound or other imaging, biological sample analysis, radiation exposure monitoring), medical treatment capabilities (e.g., pharmaceutical or nutraceutical agents, defibrillation or cardioversion, cardiopulmonary resuscitation), and the ability to return the crewmember to Earth or abort the mission for more extreme circumstances.

Radiation exposure results in a long-term health risk, influenced by the amount and type of radiation and individual factors (i.e., non-modifiable [family history/genetics, age, sex] and modifiable [diet, exercise, smoking habit] risk factors). Radiation exposures result in the generation of reactive oxidate species and inflammation that contribute to the development of many chronic diseases, including CVD. Primary and secondary tissue damage lead to cardiac and vascular dysfunction and disease. Long-term health outcomes for the astronauts are dependent upon recognition of symptoms, appropriate occupational surveillance activities, routine medical monitoring, and access to medical personnel and facilities for diagnosis and treatment.

B. Integration with Other Risks

Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity

- Cardiovascular structure and function will have a direct impact on aerobic exercise capacity. Cardiovascular deconditioning is associated with lower maximal and submaximal exercise tolerance (Lee et al. 2010).
- Countermeasures to prevent aerobic capacity deconditioning also will be protective of cardiovascular structure and function (Dorfman et al. 2007, 2008; Hastings et al. 2012) and may control modifiable risk factors associated with the long-term health of the astronauts after spaceflight.
- Orthostatic intolerance may impair an astronaut's ability to perform an emergency egress without the assistance of other crewmembers or ground support personnel. However, lower body compression garments worn to protect against orthostatic intolerance can impair mobility and increase the metabolic cost of egress (Lee et al. 2001, 2011). Success in simulated egress from the Space Shuttle was correlated with aerobic capacity (Bishop et al. 1999, p. 1999).

Risk of Impaired Performance Due to Reduced Muscle Size, Strength, and Endurance

- Changes in skeletal muscle performance during unloading are preceded by vascular adaptations (Trinity et al. 2022). Alterations in blood flow and distribution to skeletal muscle may contribute to reduced exercise tolerance.
- Change in muscle size and volume, particularly during long-duration spaceflight (Thornton et al. 1977), will alter the level of compression provided by elastic lower body compression garments, unless the sizing of the garment is adjustable and can be changed by the astronaut during the mission to accommodate changes in anthropometry.

Risk of Ineffective or Toxic Medications During Long-Duration Exploration Spaceflight

- Medications for cardiovascular interventions must have sufficient shelf-life and be resistant to radiation exposure for the appropriate treatment of expected cardiovascular complications during an extended spaceflight mission (e.g., three-year round-trip to Mars). If unanticipated cardiovascular events occur, the appropriate medications may not be available, and thus crew health and performance may be compromised.

Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition

- Insufficient fluid intake during spaceflight could exacerbate the impaired thermoregulation associated with spaceflight (Fortney et al. 1998) and spaceflight analogs (Lee et al. 2002), leading to reduced exercise tolerance and the potential for heat injury.
- Insufficient fluid intake (i.e., fluid loading) at the end of the mission also may contribute to orthostatic intolerance.
- Inappropriate food choices during a mission also might be a contributing factor to unfavorable changes in modifiable CVD risk factors (e.g., blood lipids).

Risk of Spaceflight Associated Neuro-ocular Syndrome

- The headward fluid shift that occurs in weightlessness is hypothesized to be one of the contributors to the development of spaceflight-associated neuro-ocular syndrome (SANS). Although the exact mechanisms and long-term outcomes are not well defined, one sign of SANS is the engorgement of the choroid behind the eye. This might be associated with venous congestion, as suggested by changes in IJV dimension and pressure (Marshall-Goebel et al. 2019). It has been suggested that endothelial dysfunction (Zwart et al. 2017) and changes to the glycocalyx (Zwart et al. 2022) contribute to alterations in the blood brain barrier, possibly another factor in the development of SANS.
- Recently, Mao et al. examined potential disruption of the blood retinal barrier in mice through monitoring key proteins in the structure and function of the blood-retinal barrier after whole body exposure to a mixed energy source (Mao et al. 2022). Indices of increased blood-retinal barrier permeability were observed after an exposure that replicates the anticipated dosage experienced during a Mars mission (0.5 Gy), suggesting that radiation could contribute to ocular and brain changes in spaceflight. Though the animals were not hindlimb unloaded, the findings were similar to previous results from spaceflight.

Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload

- Increased psychological stress and circadian misalignment (Guo et al. 2013, 2014) are associated oxidative stress, inflammation, and the development of CVD.

Risk of Adverse Health Event Due to Altered Immune Response

- Biomarkers during spaceflight are suggestive of a low-level chronic inflammatory state that may increase the risk of long-term health consequences, including CVD.

Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks

- Sensorimotor disturbances after spaceflight may contribute to cardiovascular responses to orthostatic stress and result in symptoms similar to orthostatic intolerance. The combined effects could impair safe egress performance.

Risk of Bone Fracture due to Spaceflight-induced Changes to Bone

- Changes in bone blood supply may contribute to bone demineralization and impair fracture healing.

XII. KNOWLEDGE BASE

A. Gaps in Knowledge

CV-101: Determine whether long-duration weightlessness induces cardiovascular structural and functional changes and/or oxidative stress & damage (OSaD)/inflammation, that can contribute to development of disease.

CV-102: Determine whether space radiation induces cardiovascular structural and functional changes and/or oxidative stress & damage (OSaD)/inflammation, that can contribute to development of disease.

CV-103: Determine whether the combined effects of relevant deep-space radiation and weightlessness induce additive or synergistic effects on the cardiovascular system, and whether it is of concern for development of disease.

CV-201: Select countermeasures and procedures to be tested based on previous objectives.

CV-202: Develop and ground-test countermeasures against the spaceflight-induced changes in the cardiovascular system of importance for development of disease.

CV-203: Test countermeasures on the ISS against the spaceflight-induced changes in the cardiovascular system of importance for development of disease.

CV-301: Test the integrated countermeasure suite in deep-space missions and on the moon, based on previous tests of countermeasures on the ground and ISS.

CV4: Is 1/6-g exposure protective of 1-g orthostatic tolerance?

B. State of Knowledge/Future Work

Exposure to weightlessness alters blood volume, flow, and pressure distribution, manifesting in changes in vascular and cardiac structure and function. Cardiovascular deconditioning is a well-recognized consequence of exposure to weightlessness, contributing to orthostatic intolerance and decreased exercise capacity. In-mission countermeasures have been instituted, and available evidence from four to six-month missions in LEO suggest that functional impairments recover in the weeks after landing with the assistance of a reconditioning program. Less is known following missions >1 year, and there is minimal experience with missions beyond LEO longer than 14 days or in partial gravity longer than 3 days. Epidemiological evidence does not support an increased lifetime risk of CVD death in astronauts, but mission durations have been limited largely to less than 6 months and there has been limited long term follow-up for those crewmembers. However, a recent report suggest that the lifetime risk of cardiovascular event is greater in astronauts with at least one spaceflight mission when compared to a well-matched cohort, though the number of events is low in both populations compared to general population (Charvat et al. 2022).

Terrestrially in humans, a statistical increase in atherosclerosis-related cardiovascular pathologies is observed at late time points when exposed to ionizing radiation (e.g., γ -rays and x-rays) at and above 500 mGy. 2018 NCRP guidance states “there is not sufficient evidence that absorbed doses in the heart ≤ 0.5 Gy cause CVDs” (in the context of terrestrial radiation protection) and that the underlying mechanisms are still unclear which makes modeling these effects difficult. Age, sleep, exercise, diet, and stress are known contributors to CVD. Resource limitations will affect medical capability and clinical outcomes in missions, particularly in missions at a greater distance from Earth.

C. Mitigation Strategy

The strategy for understanding and mitigating spaceflight-induced cardiovascular effects is stepwise and includes: 1) Understanding and characterizing of effects of long duration weightlessness and radiation – separately and in combination, 2) developing of countermeasures by first ground and thereafter low-earth orbit (LEO) tests, and finally 3) testing in the deep space (Gateway) environment.

For characterizing and understanding the effects of each of the major spaceflight environmental factors (weightlessness, radiation, isolation, confinement, and stress) in isolation and in combination, ground-based analogs using *in vitro* methods, animal models, and human subjects (parabolic flight) will be used, as well as long-duration flights of cells/animals/humans in LEO. Because experimental radiation studies are generally limited to *in vitro* and animal models, the effects of space radiation (thresholds, dose-responses and the effects of chronic or

protracted dose-rates) in isolation and in combination with other spaceflight stressors will be characterized using a multifaceted approach. This approach includes long-term occupational surveillance; leveraging data from human radiotherapy patients and previously flown astronauts of various durations; tracking of potential new biomarkers of CVD susceptibility such as CHIP (clonal hematopoiesis of indeterminate potential) mutations, using/developing animal, cell culture, and organ-on-a-chip models for ground-based laboratory studies using space radiation analogs (e.g., NSRL). Monitoring of astronauts and rodents in Gateway and on the Lunar surface will also be conducted and data will be analyzed by translational and computational modeling.

Countermeasures to orthostatic intolerance have been used for many years by both NASA and the Russian Space Agency. Future work may address verification of the countermeasures in an operational setting and refinement of the countermeasures to address commercial crew and exploration missions. Current work includes characterization tolerance to G-loads less than 1-G in women and men who are hypovolemic, mimicking the plasma volume loss experienced during spaceflight. Specifically, this addresses concerns for orthostatic hypotension for the Artemis missions and will be used for definition of the acceleration limits (G-load and duration of exposure) during descent to and ascent from the Lunar surface. Further, this study intends to define whether a lower body compression garment is required during these dynamic phases of flight for the Human Landing System.

Countermeasure development for non-orthostatic intolerance concerns will be stepwise in that firstly, data mining, emerging research and developments highlighted at conferences, literature review, the NASA process of Request for Information (RFI), studies using human induced pluripotent stem cells (iPSCs), and workshops with key experts will constitute the basis for selection of the candidates to be tested. Then, tests will be conducted on the ground in primarily animal models followed by countermeasure efficacy tests in ISS astronauts. Finally, data collection on astronauts during various Gateway and lunar missions using the suggested countermeasures will complete the mitigation strategy.

XIII. SECTION V: CONCLUSIONS

Although some aspects of cardiovascular system-related risk are well-characterized (e.g., orthostatic intolerance) and operational countermeasures are in place (e.g., lower body compression garments), evidence regarding other potential concerns is emerging (e.g., venous thromboembolism). Continued surveillance of cardiovascular biomarkers and long-term health outcomes are required to define the risk to crew health and mission success, particularly during and after long exposures to weightlessness and partial gravity as well as space radiation exposures. Evidence from ground-based research, terrestrial medicine, and spaceflight analogs will continue to be critical. Development of countermeasures during and providing health care after the missions is a duty of the national and international space agencies that engage these missions, in cooperation with the commercial partners.

XIV. REFERENCES

- Abdel-Rahman AR, Merrill RH, Wooles WR (1994) Gender-related differences in the baroreceptor reflex control of heart rate in normotensive humans. *J Appl Physiol* 77:606–613
- Abdulla J, Nielsen JR (2009) Is the risk of atrial fibrillation higher in athletes than in the general population? A systematic review and meta-analysis. *Europace* 11:1156–1159. <https://doi.org/10.1093/europace/eup197>
- Abend M, Azizova T, Müller K, et al (2015) Association of radiation-induced genes with noncancer chronic diseases in Mayak workers occupationally exposed to prolonged radiation. *Radiat Res* 183:249–261. <https://doi.org/10.1667/RR13758.1>
- Abraham P, Leftheriotis G, Desvaux B, et al (1994) Diameter and blood velocity changes in the saphenous vein during thermal stress. *Eur J Appl Physiol Occup Physiol* 69:305–308. <https://doi.org/10.1007/BF00392035>
- Adams MJ, Grant EJ, Kodama K, et al (2012) Radiation dose associated with renal failure mortality: a potential pathway to partially explain increased cardiovascular disease mortality observed after whole-body irradiation. *Radiat Res* 177:220–228. <https://doi.org/10.1667/rr2746.1>
- Ade CJ, Bemben DA (2019) Differential MicroRNA expression following head-down tilt bed rest: implications for cardiovascular responses to microgravity. *Physiol Rep* 7:e14061. <https://doi.org/10.14814/phy2.14061>
- Ade CJ, Broxterman RM, Charvat JM, Barstow TJ (2017) Incidence Rate of Cardiovascular Disease End Points in the National Aeronautics and Space Administration Astronaut Corps. *J Am Heart Assoc* 6:. <https://doi.org/10.1161/JAHA.117.005564>
- Alfrey CP, Udden MM, Leach-Huntoon C, et al (1996) Control of red blood cell mass in spaceflight. *J Appl Physiol* 81:98–104
- Amino M, Yoshioka K, Fujibayashi D, et al (2010) Year-long upregulation of connexin43 in rabbit hearts by heavy ion irradiation. *Am J Physiol Heart Circ Physiol* 298:H1014-1021. <https://doi.org/10.1152/ajpheart.00160.2009>
- Amino M, Yoshioka K, Tanabe T, et al (2006) Heavy ion radiation up-regulates Cx43 and ameliorates arrhythmogenic substrates in hearts after myocardial infarction. *Cardiovasc Res* 72:412–421. <https://doi.org/10.1016/j.cardiores.2006.09.010>
- Anderson JL, Bertke SJ, Yiin J, et al (2021) Ischaemic heart and cerebrovascular disease mortality in uranium enrichment workers. *Occup Environ Med* 78:105–111. <https://doi.org/10.1136/oemed-2020-106423>
- Anzai T, Frey MA, Nogami A (2014) Cardiac arrhythmias during long-duration spaceflights. *Journal of Arrhythmia* 30:139–149. <https://doi.org/10.1016/j.joa.2013.07.009>

- Arbeille P, Fomina G, Roumy J, et al (2001) Adaptation of the left heart, cerebral and femoral arteries, and jugular and femoral veins during short- and long-term head-down tilt and spaceflights. *Eur J Appl Physiol* 86:157–168
- Arbeille P, Provost R, Zuj K (2016) Carotid and Femoral Artery Intima-Media Thickness During 6 Months of Spaceflight. *Aerosp Med Hum Perform* 87:449–453. <https://doi.org/10.3357/AMHP.4493.2016>
- Arbeille P, Provost R, Zuj K (2017) Carotid and Femoral Arterial Wall Distensibility During Long-Duration Spaceflight. *Aerosp Med Hum Perform* 88:924–930. <https://doi.org/10.3357/AMHP.4884.2017>
- Arbeille P, Provost R, Zuj K, Vincent N (2015) Measurements of jugular, portal, femoral, and calf vein cross-sectional area for the assessment of venous blood redistribution with long duration spaceflight (Vessel Imaging Experiment). *Eur J Appl Physiol* 115:2099–2106. <https://doi.org/10.1007/s00421-015-3189-6>
- Arbeille P, Zuj KA, Macias BR, et al (2021) Lower body negative pressure reduces jugular and portal vein volumes and counteracts the elevation of middle cerebral vein velocity during long-duration spaceflight. *J Appl Physiol* (1985) 131:1080–1087. <https://doi.org/10.1152/jappphysiol.00231.2021>
- Arnett DK, Blumenthal RS, Albert MA, et al (2019) 2019 ACC/AHA Guideline on the Primary Prevention of Cardiovascular Disease: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol* 74:e177–e232. <https://doi.org/10.1016/j.jacc.2019.03.010>
- Arora S, Veves A, Caballero AE, et al (1998) Estrogen improves endothelial function. *Journal of Vascular Surgery* 27:1141–1147. [https://doi.org/10.1016/S0741-5214\(98\)70016-3](https://doi.org/10.1016/S0741-5214(98)70016-3)
- Arzeno NM, Stenger MB, Lee SMC, et al (2013) Sex differences in blood pressure control during 6° head-down tilt bed rest. *Am J Physiol Heart Circ Physiol* 304:H1114–1123. <https://doi.org/10.1152/ajpheart.00391.2012>
- Ashari N, Kong M, Poudel A, et al (2021) Generating waist area-dependent ground reaction forces for long-duration spaceflight. *J Biomech* 118:110272. <https://doi.org/10.1016/j.jbiomech.2021.110272>
- Auñón-Chancellor SM, Pattarini JM, Moll S, Sargsyan A (2020) Venous Thrombosis during Spaceflight. *N Engl J Med* 382:89–90. <https://doi.org/10.1056/NEJMc1905875>
- Austin PC, Lee DS, Fine JP (2016) Introduction to the Analysis of Survival Data in the Presence of Competing Risks. *Circulation* 133:601–609. <https://doi.org/10.1161/CIRCULATIONAHA.115.017719>

- Azizova TV, Bannikova MV, Grigorieva ES, et al (2016) Risk of lower extremity arterial disease in a cohort of workers occupationally exposed to ionizing radiation over a prolonged period. *Radiat Environ Biophys* 55:147–159. <https://doi.org/10.1007/s00411-016-0645-6>
- Azizova TV, Bannikova MV, Grigoryeva ES, et al (2022) Mortality from various diseases of the circulatory system in the Russian Mayak nuclear worker cohort: 1948-2018. *J Radiol Prot* 42:. <https://doi.org/10.1088/1361-6498/ac4ae3>
- Azizova TV, Batistatou E, Grigorieva ES, et al (2018) An Assessment of Radiation-Associated Risks of Mortality from Circulatory Disease in the Cohorts of Mayak and Sellafield Nuclear Workers. *Radiat Res* 189:371–388. <https://doi.org/10.1667/RR14468.1>
- Azizova TV, Grigorieva ES, Hunter N, et al (2015a) Risk of mortality from circulatory diseases in Mayak workers cohort following occupational radiation exposure. *J Radiol Prot* 35:517–538. <https://doi.org/10.1088/0952-4746/35/3/517>
- Azizova TV, Grigoryeva ES, Haylock RGE, et al (2015b) Ischaemic heart disease incidence and mortality in an extended cohort of Mayak workers first employed in 1948-1982. *Br J Radiol* 88:20150169. <https://doi.org/10.1259/bjr.20150169>
- Baevsky RM, Baranov VM, Funtova II, et al (2007a) Autonomic cardiovascular and respiratory control during prolonged spaceflights aboard the International Space Station. *J Appl Physiol* 103:156–161. <https://doi.org/10.1152/japplphysiol.00137.2007>
- Baevsky RM, Funtova II, Diedrich A, et al (2007b) Autonomic function testing on board the ISS—update on “Pneumocard.” *Acta Astronautica* 61:672–675. <https://doi.org/10.1016/j.actaastro.2006.11.017>
- Bantseev V, Bhardwaj R, Rathbun W, et al (1997) Antioxidants and cataract: (cataract induction in space environment and application to terrestrial aging cataract). *Biochem Mol Biol Int* 42:1189–1197. <https://doi.org/10.1080/15216549700203661>
- Bao JX, Zhang LF, Shang HH, et al (1999) [Echocardiographic assessment of left ventricular structure and function after simulated weightlessness in rats]. *Space Med Med Eng (Beijing)* 12:88–91
- Barantke M, Krauss T, Ortak J, et al (2008) Effects of gender and aging on differential autonomic responses to orthostatic maneuvers. *J Cardiovasc Electrophysiol* 19:1296–1303. <https://doi.org/10.1111/j.1540-8167.2008.01257.x>
- Barger LK, Flynn-Evans EE, Kubey A, et al (2014) Prevalence of sleep deficiency and use of hypnotic drugs in astronauts before, during, and after spaceflight: an observational study. *Lancet Neurol* 13:904–912. [https://doi.org/10.1016/S1474-4422\(14\)70122-X](https://doi.org/10.1016/S1474-4422(14)70122-X)

- Barnett SR, Morin RJ, Kiely DK, et al (1999) Effects of Age and Gender on Autonomic Control of Blood Pressure Dynamics. *Hypertension* 33:1195–1200. <https://doi.org/10.1161/01.HYP.33.5.1195>
- Barr YR (2010) Atrial Arrhythmia Summit: Summary Report, NASA/TP–2010– 216124. NASA Center for AeroSpace Information, Hanover, MD
- Barrios V, Calderón A, Escobar C, et al (2012) Patients with atrial fibrillation in a primary care setting: Val-FAAP study. *Rev Esp Cardiol (Engl Ed)* 65:47–53. <https://doi.org/10.1016/j.recesp.2011.08.008>
- Baselet B, Azimzadeh O, Erbelinger N, et al (2017) Differential Impact of Single-Dose Fe Ion and X-Ray Irradiation on Endothelial Cell Transcriptomic and Proteomic Responses. *Front Pharmacol* 8:570. <https://doi.org/10.3389/fphar.2017.00570>
- Basso C, Thiene G, Corrado D, et al (1996) Arrhythmogenic right ventricular cardiomyopathy. Dysplasia, dystrophy, or myocarditis? *Circulation* 94:983–991
- Beck M, Moreels M, Quintens R, et al (2014a) Chronic exposure to simulated space conditions predominantly affects cytoskeleton remodeling and oxidative stress response in mouse fetal fibroblasts. *Int J Mol Med*. <https://doi.org/10.3892/ijmm.2014.1785>
- Beck M, Rombouts C, Moreels M, et al (2014b) Modulation of gene expression in endothelial cells in response to high LET nickel ion irradiation. *Int J Mol Med* 34:1124–1132. <https://doi.org/10.3892/ijmm.2014.1893>
- Beheshti A, McDonald JT, Miller J, et al (2019) GeneLab Database Analyses Suggest Long-Term Impact of Space Radiation on the Cardiovascular System by the Activation of FYN Through Reactive Oxygen Species. *Int J Mol Sci* 20:E661. <https://doi.org/10.3390/ijms20030661>
- Behnke BJ, Zawieja DC, Gashev AA, et al (2008) Diminished mesenteric vaso- and venoconstriction and elevated plasma ANP and BNP with simulated microgravity. *J Appl Physiol* 104:1273–1280. <https://doi.org/10.1152/jappphysiol.00954.2006>
- Belin de Chantemele E, Blanc S, Pellet N, et al (2004) Does resistance exercise prevent body fluid changes after a 90-day bed rest? *Eur J Appl Physiol* 92:555–564. <https://doi.org/10.1007/s00421-004-1121-6>
- Berbée M, Fu Q, Boerma M, et al (2009) gamma-Tocotrienol ameliorates intestinal radiation injury and reduces vascular oxidative stress after total-body irradiation by an HMG-CoA reductase-dependent mechanism. *Radiat Res* 171:596–605. <https://doi.org/10.1667/RR1632.1>

- Berbée M, Fu Q, Boerma M, et al (2012) Mechanisms underlying the radioprotective properties of γ -tocotrienol: comparative gene expression profiling in tocol-treated endothelial cells. *Genes Nutr* 7:75–81. <https://doi.org/10.1007/s12263-011-0228-8>
- Berkman AM, Lakoski SG (2015) The potential of aerobic exercise training in mitigating cardiovascular side effects of cancer therapy. *Future Cardiol* 11:407–411. <https://doi.org/10.2217/FCA.15.41>
- Bernstein J, Dauer L, Dauer Z, et al (2020) Cardiovascular Risks from Low Dose Radiation Exposure: Review and Scientific Appraisal of the Literature. EPRI, Palo Alto, CA
- Berry CA (1968) The medical legacy of Gemini. *Life Sci Space Res* 6:1–19
- Bishawi M, Lee FH, Abraham DM, et al (2022) Late onset cardiovascular dysfunction in adult mice resulting from galactic cosmic ray exposure. *iScience* 25:104086. <https://doi.org/10.1016/j.isci.2022.104086>
- Bishop PA, Lee SM, Conza NE, et al (1999) Carbon dioxide accumulation, walking performance, and metabolic cost in the NASA launch and entry suit. *Aviat Space Environ Med* 70:656–665
- Bleeker MWP, Groot PCED, Pawelczyk JA, et al (2004) Effects of 18 days of bed rest on leg and arm venous properties. *J Appl Physiol* 96:840–847. <https://doi.org/10.1152/japplphysiol.00835.2003>
- Blomqvist CG, Nixon JV, Johnson RL Jr, Mitchell JH (1980) Early cardiovascular adaptation to zero gravity simulated by head-down tilt. *Acta Astronaut* 7:543–553
- Bogomolov VV, Duncan JM, Sargsyan AE, Pool SL (2009) Medical Support for Crewmembers of the International Space Station. In: Sawin CF, Hanson SI, House NG, Pestov ID (eds) *Space Biology and Medicine V: US and Russian Cooperation in Space Biology and Medicine*. American Institutes of Aeronautics and Astronautics, Inc., pp 331–394
- Boice JD, Cohen SS, Mumma MT, et al (2022a) Mortality among U.S. military participants at eight aboveground nuclear weapons test series. *Int J Radiat Biol* 98:679–700. <https://doi.org/10.1080/09553002.2020.1787543>
- Boice JD, Cohen SS, Mumma MT, et al (2022b) Mortality among workers at the Los Alamos National Laboratory, 1943-2017. *Int J Radiat Biol* 98:722–749. <https://doi.org/10.1080/09553002.2021.1917784>
- Boice JD, Quinn B, Al-Nabulsi I, et al (2022c) A million persons, a million dreams: a vision for a national center of radiation epidemiology and biology. *Int J Radiat Biol* 98:795–821. <https://doi.org/10.1080/09553002.2021.1988183>

- Bonnin P, Ben Driss A, Benessiano J, et al (2001) Enhanced flow-dependent vasodilatation after bed rest, a possible mechanism for orthostatic intolerance in humans. *Eur J Appl Physiol* 85:420–426
- Bradley EW, Zook BC, Casarett GW, Rogers CC (1981) Coronary arteriosclerosis and atherosclerosis in fast neutron or photon irradiated dogs. *Int J Radiat Oncol Biol Phys* 7:1103–1108. [https://doi.org/10.1016/0360-3016\(81\)90169-3](https://doi.org/10.1016/0360-3016(81)90169-3)
- Brainard J, Gobel M, Scott B, et al (2015) Health implications of disrupted circadian rhythms and the potential for daylight as therapy. *Anesthesiology* 122:1170–1175. <https://doi.org/10.1097/ALN.0000000000000596>
- Buchheim J-I, Matzel S, Rykova M, et al (2019) Stress Related Shift Toward Inflammaging in Cosmonauts After Long-Duration Space Flight. *Frontiers in Physiology* 10:
- Buckey JC Jr, Gaffney FA, Lane LD, et al (1996a) Central venous pressure in space. *J Appl Physiol* 81:19–25
- Buckey JC, Lane LD, Levine BD, et al (1996b) Orthostatic intolerance after spaceflight. *J Appl Physiol* 81:7–18
- Budoff MJ, Young R, Lopez VA, et al (2013) Progression of coronary calcium and incident coronary heart disease events: MESA (Multi-Ethnic Study of Atherosclerosis). *J Am Coll Cardiol* 61:1231–1239. <https://doi.org/10.1016/j.jacc.2012.12.035>
- Bungo MW, Charles JB, Johnson PC Jr (1985) Cardiovascular deconditioning during space flight and the use of saline as a countermeasure to orthostatic intolerance. *Aviat Space Environ Med* 56:985–990
- Bungo MW, Johnson PC (1983) Cardiovascular examinations and observations of deconditioning during the space shuttle orbital flight test program. *Aviat Space Environ Med* 54:1001–1004
- Caiani EG, Sugeng L, Weinert L, et al (2006) Objective evaluation of changes in left ventricular and atrial volumes during parabolic flight using real-time three-dimensional echocardiography. *J Appl Physiol* 101:460–468. <https://doi.org/10.1152/japplphysiol.00014.2006>
- Carrick-Ranson G, Hastings JL, Bhella PS, et al (2013) The effect of exercise training on left ventricular relaxation and diastolic suction at rest and during orthostatic stress after bed rest. *Exp Physiol* 98:501–513. <https://doi.org/10.1113/expphysiol.2012.067488>
- Carver JR, Shapiro CL, Ng A, et al (2007) American Society of Clinical Oncology clinical evidence review on the ongoing care of adult cancer survivors: cardiac and pulmonary late effects. *J Clin Oncol* 25:3991–4008. <https://doi.org/10.1200/JCO.2007.10.9777>

- Charles JB, Frey MAB, Fritsch-Yelle JM, Fortner GW (1996) Cardiovascular and Cardiorespiratory Function. In: Dietlein LF, Pestov ID (eds) Space Biology Medicine: Humans in Spaceflight. American Institutes of Aeronautics and Astronautics, Inc., Reston, Virginia, pp 63–88
- Charles JB, Fritsch-Yelle JM, Whitson PA, et al (1999) Cardiovascular Deconditioning. In: Extended Duration Orbiter Medical Project: Final Report (1989-1995). National Aeronautics and Space Administration, pp 1-1-1–19
- Charles JB, Lathers CM (1994) Summary of lower body negative pressure experiments during space flight. *J Clin Pharmacol* 34:571–583
- Charles JB, Pietrzyk RA (2019) A Year on the International Space Station: Implementing a Long-Duration Biomedical Research Mission. *Aerosp Med Hum Perform* 90:4–11. <https://doi.org/10.3357/AMHP.5178.2019>
- Charvat JM, Lee SMC, Stenger MB, Van Baalen M (2021) Atrial fibrillation among NASA astronauts. Galveston, TX
- Charvat JM, Leonard D, Barlow CE, et al (2022) Long-term Cardiovascular Risk in Astronauts: Comparing NASA Mission Astronauts With a Healthy Cohort From the Cooper Center Longitudinal Study. *Mayo Clinic Proceedings* 97:1237–1246. <https://doi.org/10.1016/j.mayocp.2022.04.003>
- Cheng Y-C, Vyas A, Hymen E, Perlmutter LC (2011) Gender differences in orthostatic hypotension. *Am J Med Sci* 342:221–225. <https://doi.org/10.1097/MAJ.0b013e318208752b>
- Chi G, Gibson CM, Liu Y, et al (2019) Inverse relationship of serum albumin to the risk of venous thromboembolism among acutely ill hospitalized patients: Analysis from the APEX trial. *Am J Hematol* 94:21–28. <https://doi.org/10.1002/ajh.25296>
- Clément G, Rittweger J, Nitsche A, et al (2022a) Assessing the effects of artificial gravity in an analog of long-duration spaceflight: The protocol and implementation of the AGBRESA bed rest study. *Frontiers in Physiology* 13:
- Clément GR, Crucian BE, Downs M, et al (2022b) International standard measures during the VaPER bed rest study. *Acta Astronautica* 190:208–217. <https://doi.org/10.1016/j.actaastro.2021.10.017>
- Coleman MA, Sasi SPP, Onufrak J, et al (2015) Low Dose Radiation Affects Cardiac Physiology: Gene Networks and Molecular Signaling in Cardiomyocytes. *American Journal of Physiology - Heart and Circulatory Physiology* ajpheart.00050.2015. <https://doi.org/10.1152/ajpheart.00050.2015>
- Collins A, Frankenhaeuser M (1978) Stress responses in male and female engineering students. *J Human Stress* 4:43–48. <https://doi.org/10.1080/0097840X.1978.9934986>

- Conard J (1999) Clinical implications. Biological coagulation findings in third-generation oral contraceptives. *Human Reproduction Update* 5:672–680. <https://doi.org/10.1093/humupd/5.6.672>
- Convertino VA (1998) Gender differences in autonomic functions associated with blood pressure regulation. *Am J Physiol* 275:R1909-20
- Convertino VA (2009) Status of cardiovascular issues related to space flight: Implications for future research directions. *Respir Physiol Neurobiol* 169 Suppl 1:S34-37. <https://doi.org/10.1016/j.resp.2009.04.010>
- Convertino VA, Cooke WH (2005) Evaluation of cardiovascular risks of spaceflight does not support the NASA bioastronautics critical path roadmap. *Aviat Space Environ Med* 76:869–876
- Convertino VA, Doerr DF, Eckberg DL, et al (1990) Head-down bed rest impairs vagal baroreflex responses and provokes orthostatic hypotension. *J Appl Physiol* 68:1458–1464
- Convertino VA, Doerr DF, Ludwig DA, Vernikos J (1994) Effect of simulated microgravity on cardiopulmonary baroreflex control of forearm vascular resistance. *Am J Physiol Regul Integr Comp Physiol* 266:R1962–R1969
- Convertino VA, Engelke KA, Ludwig DA, Doerr DF (1996) Restoration of plasma volume after 16 days of head-down tilt induced by a single bout of maximal exercise. *Am J Physiol* 270:R3-10
- Corsi C, Saracino G, Lamberti C, et al (2002) Changes in left ventricular size during parabolic flights by two-dimensional echocardiography and level set method. *Comput Cardiol* 29:73–76
- Coupé M, Yuan M, Demiot C, et al (2011) Low-magnitude whole body vibration with resistive exercise as a countermeasure against cardiovascular deconditioning after 60 days of head-down bed rest. *Am J Physiol Regul Integr Comp Physiol* 301:R1748-1754. <https://doi.org/10.1152/ajpregu.00234.2011>
- Crandall CG, Engelke KA, Convertino VA, Raven PB (1994) Aortic baroreflex control of heart rate after 15 days of simulated microgravity exposure. *J Appl Physiol* 77:2134–2139
- Crucian BE, Choukèr A, Simpson RJ, et al (2018) Immune System Dysregulation During Spaceflight: Potential Countermeasures for Deep Space Exploration Missions. *Front Immunol* 9:1437. <https://doi.org/10.3389/fimmu.2018.01437>
- Crucian BE, Zwart SR, Mehta S, et al (2014) Plasma cytokine concentrations indicate that in vivo hormonal regulation of immunity is altered during long-duration spaceflight. *J Interferon Cytokine Res* 34:778–786. <https://doi.org/10.1089/jir.2013.0129>

- Cucinotta FA, Hamada N, Little MP (2016) No evidence for an increase in circulatory disease mortality in astronauts following space radiation exposures. *Life Sci Space Res (Amst)* 10:53–56. <https://doi.org/10.1016/j.lssr.2016.08.002>
- Cucinotta FA, Kim M-HY, Chappell LJ (2013a) Space radiation cancer risk projections and uncertainties-2012, NASA/TP-2013-217375. NASA Center for AeroSpace Information, Hanover, MD
- Cucinotta FA, Kim M-HY, Chappell LJ, Huff JL (2013b) How Safe Is Safe Enough? Radiation Risk for a Human Mission to Mars. *PLoS One* 8:e74988. <https://doi.org/10.1371/journal.pone.0074988>
- Custaud M-A, de Souza Neto EP, Abry P, et al (2002) Orthostatic tolerance and spontaneous baroreflex sensitivity in men versus women after 7 days of head-down bed rest. *Auton Neurosci* 100:66–76
- Cutter D, Taylor C, Rahimi K, et al (2013) Effects of radiation therapy on the cardiovascular system. In: Ewer M, Yeh E (eds) *Cancer and the Heart*, Second. People's Medical Publishing House, Shelton, pp 83–131
- Darby SC, Ewertz M, McGale P, et al (2013) Risk of ischemic heart disease in women after radiotherapy for breast cancer. *N Engl J Med* 368:987–998. <https://doi.org/10.1056/NEJMoa1209825>
- Darby SC, McGale P, Taylor CW, Peto R (2005) Long-term mortality from heart disease and lung cancer after radiotherapy for early breast cancer: prospective cohort study of about 300,000 women in US SEER cancer registries. *Lancet Oncol* 6:557–565. [https://doi.org/10.1016/S1470-2045\(05\)70251-5](https://doi.org/10.1016/S1470-2045(05)70251-5)
- D'Aunno DS, Dougherty AH, DeBlock HF, Meck JV (2003) Effect of short- and long-duration spaceflight on QTc intervals in healthy astronauts. *Am J Cardiol* 91:494–497
- DeFina LF, Radford NB, Barlow CE, et al (2019) Association of All-Cause and Cardiovascular Mortality With High Levels of Physical Activity and Concurrent Coronary Artery Calcification. *JAMA Cardiol* 4:174–181. <https://doi.org/10.1001/jamacardio.2018.4628>
- Delp MD, Collieran PN, Wilkerson MK, et al (2000) Structural and functional remodeling of skeletal muscle microvasculature is induced by simulated microgravity. *Am J Physiol Heart Circ Physiol* 278:H1866–H1873
- Delp MD, Holder-Binkley T, Laughlin MH, Hasser EM (1993) Vasoconstrictor properties of rat aorta are diminished by hindlimb unweighting. *J Appl Physiol* 75:2620–8

- Denq JC, Opfer-Gehrking TL, Giuliani M, et al (1997) Efficacy of compression of different capacitance beds in the amelioration of orthostatic hypotension. *Clin Auton Res* 7:321–326
- Deshpande N, Laurie SS, Lee SMC, et al (2020) Vestibular and Cardiovascular Responses After Long-Duration Spaceflight. *Aerosp Med Hum Perform* 91:621–627. <https://doi.org/10.3357/AMHP.5502.2020>
- Dijk DJ, Neri DF, Wyatt JK, et al (2001) Sleep, performance, circadian rhythms, and light-dark cycles during two space shuttle flights. *Am J Physiol Regul Integr Comp Physiol* 281:R1647–1664
- Dorfman TA, Levine BD, Tillery T, et al (2007) Cardiac atrophy in women following bed rest. *Journal of Applied Physiology* 103:8–16. <https://doi.org/10.1152/japplphysiol.01162.2006>
- Dorfman TA, Rosen BD, Perhonen MA, et al (2008) Diastolic suction is impaired by bed rest: MRI tagging studies of diastolic untwisting. *J Appl Physiol* 104:1037–1044. <https://doi.org/10.1152/japplphysiol.00858.2006>
- Dos Santos RQ, Smidt L, Suzigan BH, et al (2013) Efficacy of lower limb compression in the management of vasovagal syncope--randomized, crossover study. *Pacing Clin Electrophysiol* 36:451–455. <https://doi.org/10.1111/pace.12069>
- Durand S, Cui J, Williams KD, Crandall CG (2004) Skin surface cooling improves orthostatic tolerance in normothermic individuals. *Am J Physiol Regul Integr Comp Physiol* 286:R199–205. <https://doi.org/10.1152/ajpregu.00394.2003>
- Durante M, Cucinotta FA (2008) Heavy ion carcinogenesis and human space exploration. *Nat Rev Cancer* 8:465–472. <https://doi.org/10.1038/nrc2391>
- Ebert RV, Stead EA (1940) THE EFFECT OF THE APPLICATION OF TOURNIQUETS ON THE HEMODYNAMICS OF THE CIRCULATION. *J Clin Invest* 19:561–567. <https://doi.org/10.1172/JCI101159>
- Ebrahimian TG, Beugnies L, Surette J, et al (2018) Chronic Exposure to External Low-Dose Gamma Radiation Induces an Increase in Anti-inflammatory and Anti-oxidative Parameters Resulting in Atherosclerotic Plaque Size Reduction in ApoE^{-/-} Mice. *Radiat Res* 189:187–196. <https://doi.org/10.1667/RR14823.1>
- Eckberg DL (1997) Sympathovagal Balance A Critical Appraisal. *Circulation* 96:3224–3232. <https://doi.org/10.1161/01.CIR.96.9.3224>
- Eckberg DL, Fritsch JM (1992) Influence of ten-day head-down bedrest on human carotid baroreceptor-cardiac reflex function. *Acta Physiol Scand Suppl* 604:69–76

- Edgell H, Grinberg A, Beavers KR, et al (2018) Efficacy of fluid loading as a countermeasure to the hemodynamic and hormonal changes of 28-h head-down bed rest. *Physiol Rep* 6:e13874. <https://doi.org/10.14814/phy2.13874>
- Elgart SR, Little MP, Chappell LJ, et al (2018) Radiation Exposure and Mortality from Cardiovascular Disease and Cancer in Early NASA Astronauts. *Sci Rep* 8:8480. <https://doi.org/10.1038/s41598-018-25467-9>
- El-Sherif N (2001) Mechanism of ventricular arrhythmias in the long QT syndrome: on hermeneutics. *J Cardiovasc Electrophysiol* 12:973–976
- Engelke KA, Doerr DF, Crandall CG, Convertino VA (1996) Application of acute maximal exercise to protect orthostatic tolerance after simulated microgravity. *Am J Physiol Regul Integr Comp Physiol* 271:R837–R847
- English KL, Downs M, Goetchius E, et al (2020) High intensity training during spaceflight: results from the NASA Sprint Study. *NPJ Microgravity* 6:21. <https://doi.org/10.1038/s41526-020-00111-x>
- Evans JM, Stenger MB, Moore FB, et al (2004) Centrifuge training increases presyncopal orthostatic tolerance in ambulatory men. *Aviat Space Environ Med* 75:850–858
- Evans JM, Ziegler MG, Patwardhan AR, et al (2001) Gender differences in autonomic cardiovascular regulation: spectral, hormonal, and hemodynamic indexes. *J Appl Physiol* 91:2611–2618
- Fajardo LF, Stewart JR (1970) Experimental radiation-induced heart disease. I. Light microscopic studies. *Am J Pathol* 59:299–316
- Ferretti G, Iellamo F, Pizzinelli P, et al (2009) Prolonged head down bed rest-induced inactivity impairs tonic autonomic regulation while sparing oscillatory cardiovascular rhythms in healthy humans. *J Hypertens* 27:551–561. <https://doi.org/10.1097/HJH.0b013e328322ca2f>
- Fomina G, Kotovskaya A, Arbeille F, et al (2004) Changes in hemodynamic and post-flights orthostatic tolerance of cosmonauts under application of the preventive device--thigh cuffs bracelets in short-term flights. *J Gravit Physiol* 11:P229-230
- Fomina GA, Kotovskaia AR, Temnova EV (2007) [Effect of prophylactic device “Braslet” on hemodynamic changes during long-term missions to orbital station Mir]. *Aviakosm Ekolog Med* 41:8–13
- Fong KJ, Arya M, Paloski WH (2007) Gender differences in cardiovascular tolerance to short radius centrifugation. *J Gravit Physiol* 14:P15-19

- Fortney SM (1991) Development of lower body negative pressure as a countermeasure for orthostatic intolerance. *J Clin Pharmacol* 31:888–892
- Fortney SM, Mikhaylov V, Lee SM, et al (1998) Body temperature and thermoregulation during submaximal exercise after 115-day spaceflight. *Aviat Space Environ Med* 69:137–141
- Fortney SM, Schneider VS, Greenleaf JE (2010) The Physiology of Bed Rest. In: *Comprehensive Physiology*. John Wiley & Sons, Inc.
- Fortney SM, Steinmann L, Young JA, et al (1994) Fluid-Loading Solutions and Plasma Volume: Astro-Ade and Salt Tablets with Water, NASA Technical Paper 3456. NASA Center for Aerospace Information, Washington, D.C.
- Fortrat JO, Sigaud D, Hughson RL, et al (2001) Effect of prolonged head-down bed rest on complex cardiovascular dynamics. *Auton Neurosci* 86:192–201. [https://doi.org/10.1016/S1566-0702\(00\)00212-5](https://doi.org/10.1016/S1566-0702(00)00212-5)
- Frey MA, Hoffler GW (1988) Association of sex and age with responses to lower-body negative pressure. *J Appl Physiol* 65:1752–1756
- Frey MA, Tomaselli CM, Hoffler WG (1994) Cardiovascular responses to postural changes: differences with age for women and men. *J Clin Pharmacol* 34:394–402
- Fritsch JM, Charles JB, Bennett BS, et al (1992) Short-duration spaceflight impairs human carotid baroreceptor-cardiac reflex responses. *J Appl Physiol* 73:664–671
- Fritsch-Yelle JM, Charles JB, Jones MM, et al (1994) Spaceflight alters autonomic regulation of arterial pressure in humans. *J Appl Physiol* 77:1776–1783
- Fritsch-Yelle JM, Charles JB, Jones MM, Wood ML (1996a) Microgravity decreases heart rate and arterial pressure in humans. *J Appl Physiol* 80:910–914
- Fritsch-Yelle JM, Leuenberger UA, D'Aunno DS, et al (1998) An episode of ventricular tachycardia during long-duration spaceflight. *Am J Cardiol* 81:1391–1392
- Fritsch-Yelle JM, Whitson PA, Bondar RL, Brown TE (1996b) Subnormal norepinephrine release relates to presyncope in astronauts after spaceflight. *J Appl Physiol* 81:2134–41. <https://doi.org/8941538>
- Fu Q, Arbab-Zadeh A, Perhonen MA, et al (2004) Hemodynamics of orthostatic intolerance: implications for gender differences. *Am J Physiol Heart Circ Physiol* 286:H449–457. <https://doi.org/10.1152/ajpheart.00735.2002>

- Fu Q, Shibata S, Hastings JL, et al (2019) Impact of Prolonged Spaceflight on Orthostatic Tolerance During Ambulation and Blood Pressure Profiles in Astronauts. *Circulation* 140:729–738. <https://doi.org/10.1161/CIRCULATIONAHA.119.041050>
- Fu Q, Witkowski S, Okazaki K, Levine BD (2005) Effects of gender and hypovolemia on sympathetic neural responses to orthostatic stress. *Am J Physiol Regul Integr Comp Physiol* 289:R109-116. <https://doi.org/10.1152/ajpregu.00013.2005>
- Fuks Z, Persaud RS, Alfieri A, et al (1994) Basic fibroblast growth factor protects endothelial cells against radiation-induced programmed cell death in vitro and in vivo. *Cancer Res* 54:2582–2590
- Garg S, Sadhukhan R, Banerjee S, et al (2019) Gamma-Tocotrienol Protects the Intestine from Radiation Potentially by Accelerating Mesenchymal Immune Cell Recovery. *Antioxidants (Basel)* 8:E57. <https://doi.org/10.3390/antiox8030057>
- Garikipati VNS, Arakelyan A, Blakely EA, et al (2021) Long-Term Effects of Very Low Dose Particle Radiation on Gene Expression in the Heart: Degenerative Disease Risks. *Cells* 10:387. <https://doi.org/10.3390/cells10020387>
- Garrett-Bakelman FE, Darshi M, Green SJ, et al (2019) The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight. *Science* 364:eaau8650. <https://doi.org/10.1126/science.aau8650>
- Gillies M, Richardson DB, Cardis E, et al (2017) Mortality from Circulatory Diseases and other Non-Cancer Outcomes among Nuclear Workers in France, the United Kingdom and the United States (INWORKS). *Radiat Res* 188:276–290. <https://doi.org/10.1667/RR14608.1>
- Gilligan DM, Badar DM, Panza JA, et al (1995) Effects of estrogen replacement therapy on peripheral vasomotor function in postmenopausal women. *Am J Cardiol* 75:264–268
- Girdler SS, Hinderliter AL, Light KC (1993) Peripheral adrenergic receptor contributions to cardiovascular reactivity: influence of race and gender. *J Psychosom Res* 37:177–193
- Goldberger AL, Mietus JE, Rigney DR, et al (1994) Effects of head-down bed rest on complex heart rate variability: response to LBNP testing. *J Appl Physiol* 77:2863–2869
- Golden AP, Ellis ED, Cohen SS, et al (2022) Updated mortality analysis of the Mallinckrodt uranium processing workers, 1942-2012. *Int J Radiat Biol* 98:701–721. <https://doi.org/10.1080/09553002.2019.1569773>
- Goldstein DS, Benth O, Park M-Y, Sharabi Y (2011) Low-frequency power of heart rate variability is not a measure of cardiac sympathetic tone but may be a measure of modulation of cardiac autonomic outflows by baroreflexes. *Exp Physiol* 96:1255–1261. <https://doi.org/10.1113/expphysiol.2010.056259>

- Goldstein DS, Pechnik S, Holmes C, et al (2003) Association between supine hypertension and orthostatic hypotension in autonomic failure. *Hypertension* 42:136–142. <https://doi.org/10.1161/01.HYP.0000081216.11623.C3>
- Gontcharov IB, Kovachevich IV, Pool SL, et al (2005) In-flight medical incidents in the NASA-Mir program. *Aviat Space Environ Med* 76:692–696
- Gore MO, Ayers CR, Amit K, et al (2020) Combining Biomarkers and Imaging for Short-Term Assessment of Cardiovascular Disease Risk in Apparently Healthy Adults. *Journal of the American Heart Association* 9:e015410. <https://doi.org/10.1161/JAHA.119.015410>
- Gotshall RW, Tsai PF, Frey MA (1991) Gender-based differences in the cardiovascular response to standing. *Aviat Space Environ Med* 62:855–859
- Grabham P, Hu B, Sharma P, Geard C (2011) Effects of ionizing radiation on three-dimensional human vessel models: differential effects according to radiation quality and cellular development. *Radiat Res* 175:21–28. <https://doi.org/10.1667/RR2289.1>
- Greaves DK, Arbeille P, Hughson RL (2007) WISE 2005: altered cerebrovascular autoregulation after 60 day head-down bed rest. *J Gravit Physiol* 14:P61-62
- Greenisen MC, Hayes JC, Siconolfi S, Moore AD (1999) Functional Performance Evaluation. In: *Extended Duration Orbiter Medical Project: Final Report (1989-1995)*. National Aeronautics and Space Administration, pp 3-1-3–24
- Greenleaf JE, Bernauer EM, Ertl AC, et al (1989a) Work capacity during 30 days of bed rest with isotonic and isokinetic exercise training. *J Appl Physiol* 67:1820–1826
- Greenleaf JE, Vernikos J, Wade CE, Barnes PR (1992) Effect of leg exercise training on vascular volumes during 30 days of 6 degrees head-down bed rest. *J Appl Physiol* 72:1887–1894
- Greenleaf JE, Wade CE, Leftheriotis G (1989b) Orthostatic responses following 30-day bed rest deconditioning with isotonic and isokinetic exercise training. *Aviat Space Environ Med* 60:537–542
- Grenon SM, Xiao X, Hurwitz S, et al (2005) Simulated microgravity induces microvolt T wave alternans. *Ann Noninvasive Electrocardiol* 10:363–370. <https://doi.org/10.1111/j.1542-474X.2005.00654.x>
- Grenon SM, Xiao X, Hurwitz S, et al (2006) Why is orthostatic tolerance lower in women than in men? Renal and cardiovascular responses to simulated microgravity and the role of midodrine. *J Investig Med* 54:180–190
- Grigoriev AI, Bugrov SA, Bogomolov VV, et al (1991) Medical results of the Mir year-long mission. *Physiologist* 34:S44-48

- Grundey SM, Stone NJ, Bailey AL, et al (2018) 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Journal of the American College of Cardiology* 25709. <https://doi.org/10.1016/j.jacc.2018.11.003>
- Guell A, Braak L, Pavy Le Traon A, Gharib C (1990) Cardiovascular deconditioning during weightlessness simulation and the use of lower body negative pressure as a countermeasure to orthostatic intolerance. *Acta Astronaut* 21:667–672
- Guell A, Cornac A, Faurat MM, et al (1992) Lower body negative pressure as a countermeasure against orthostatic intolerance for long term space flight. *Acta Astronaut* 27:103–107
- Guetta V, Quyyumi AA, Prasad A, et al (1997) The role of nitric oxide in coronary vascular effects of estrogen in postmenopausal women. *Circulation* 96:2795–2801
- Guinet P, Schneider SM, Macias BR, et al (2009) WISE-2005: effect of aerobic and resistive exercises on orthostatic tolerance during 60 days bed rest in women. *Eur J Appl Physiol* 106:217–227. <https://doi.org/10.1007/s00421-009-1009-6>
- Guo J-H, Qu W-M, Chen S-G, et al (2014) Keeping the right time in space: importance of circadian clock and sleep for physiology and performance of astronauts. *Mil Med Res* 1:23. <https://doi.org/10.1186/2054-9369-1-23>
- Guo X, Zheng L, Wang J, et al (2013) Epidemiological evidence for the link between sleep duration and high blood pressure: a systematic review and meta-analysis. *Sleep Med* 14:324–32. <https://doi.org/10.1016/j.sleep.2012.12.001>
- Hamilton DR, Murray JD, Ball CG (2006) Cardiac health for astronauts: coronary calcification scores and CRP as criteria for selection and retention. *Aviat Space Environ Med* 77:377–387
- Hamilton DR, Murray JD, Kapoor D, Kirkpatrick AW (2005) Cardiac health for astronauts: current selection standards and their limitations. *Aviat Space Environ Med* 76:615–626
- Hamilton DR, Sargsyan AE, Garcia K, et al (2012) Cardiac and vascular responses to thigh cuffs and respiratory maneuvers on crewmembers of the International Space Station. *J Appl Physiol* 112:454–462. <https://doi.org/10.1152/japplphysiol.00557.2011>
- Hamilton DR, Sargsyan AE, Martin DS, et al (2011) On-orbit prospective echocardiography on International Space Station crew. *Echocardiography* 28:491–501. <https://doi.org/10.1111/j.1540-8175.2011.01385.x>
- Hamm PB, Nicogossian AE, Pool SL, et al (2000) Design and current status of the longitudinal study of astronaut health. *Aviat Space Environ Med* 71:564–570

- Harm DL, Jennings RT, Meck JV, et al (2001) Invited review: gender issues related to spaceflight: a NASA perspective. *J Appl Physiol* 91:2374–2383
- Harris K, Laws JM, Elias A, et al (2022) Search for Venous Endothelial Biomarkers Heraldng Venous Thromboembolism in Space: A Qualitative Systematic Review of Terrestrial Studies. *Frontiers in Physiology* 13:
- Haruta D, Landes RD, Hida A, et al (2021) Relationship Between Radiation Exposure and Incident Atrial Fibrillation Among Atomic Bomb Survivors. *Circ Rep* 3:381–387. <https://doi.org/10.1253/circrep.CR-21-0059>
- Hastings JL, Krainski F, Snell PG, et al (2012) Effect of rowing ergometry and oral volume loading on cardiovascular structure and function during bed rest. *J Appl Physiol* 112:1735–1743. <https://doi.org/10.1152/jappphysiol.00019.2012>
- Hastreiter D, Young LR (1997) Effects of a gravity gradient on human cardiovascular responses. *J Gravit Physiol* 4:P23-26
- Haverkamp W, Breithardt G, Camm AJ, et al (2000) The potential for QT prolongation and proarrhythmia by non-antiarrhythmic drugs: clinical and regulatory implications. Report on a policy conference of the European Society of Cardiology. *Eur Heart J* 21:1216–1231. <https://doi.org/10.1053/euhj.2000.2249>
- Hayashi T, Kusunoki Y, Hakoda M, et al (2003) Radiation dose-dependent increases in inflammatory response markers in A-bomb survivors. *Int J Radiat Biol* 79:129–136
- Hayes JC, Guillems ME, Lee SMC, et al (2013) Exercise: Developing Countermeasure Systems for Optimizing Astronaut Performance in Space. In: Risin D, Stepaniak PC (eds) *Biomedical Results of the Space Shuttle Program, NASA/SP-2013-607*. US Government Printing Office, Washington, D.C., pp 289–314
- Henry W, Epstein SE, Griffith JM, et al (1977) Effect of Prolonged Space Flight on Cardiac Function and Dimensions. In: *Biomedical results from Skylab. Scientific and Technical Information Office, National Aeronautics and Space Administration : for sale by the Supt. of Docs., U.S. Govt. Print. Off., Washington, D.C., pp 366–371*
- Herault S, Fomina G, Alferova I, et al (2000) Cardiac, arterial and venous adaptation to weightlessness during 6-month MIR spaceflights with and without thigh cuffs (bracelets). *Eur J Appl Physiol* 81:384–390
- Heselich A, Frieß JL, Ritter S, et al (2018) High LET radiation shows no major cellular and functional effects on primary cardiomyocytes in vitro. *Life Sci Space Res (Amst)* 16:93–100. <https://doi.org/10.1016/j.lssr.2018.01.001>

- Hirayanagi K, Iwase S, Kamiya A, et al (2004a) Functional changes in autonomic nervous system and baroreceptor reflex induced by 14 days of 6 degrees head-down bed rest. *Eur J Appl Physiol* 92:160–167. <https://doi.org/10.1007/s00421-004-1067-8>
- Hirayanagi K, Kamiya A, Iwase S, et al (2004b) Autonomic cardiovascular changes during and after 14 days of head-down bed rest. *Autonomic Neuroscience* 110:121–128. <https://doi.org/10.1016/j.autneu.2004.01.001>
- Hoeldtke RD, Davis KM, Joseph J, et al (1991) Hemodynamic effects of octreotide in patients with autonomic neuropathy. *Circulation* 84:168–176
- Hoeldtke RD, Horvath GG, Bryner KD, Hobbs GR (1998) Treatment of orthostatic hypotension with midodrine and octreotide. *J Clin Endocrinol Metab* 83:339–343
- Hoffler GW, Johnson RL (1975) Apollo flight crew cardiovascular evaluations. In: *Biomedical Results of Apollo*. Scientific and Technical Information Office, National Aeronautics and Space Administration, Washington, D.C., pp 227–264
- Hoffler GW, Wolthuis RA, Johnson RL (1974) Apollo space crew cardiovascular evaluations. *Aerosp Med* 45:807–823
- Hoffmann F, Möstl S, Luchitskaya E, et al (2019) An oscillometric approach in assessing early vascular ageing biomarkers following long-term space flights. *Int J Cardiol Hypertens* 2:100013. <https://doi.org/10.1016/j.ijchy.2019.100013>
- Howe GR, Zablotska LB, Fix JJ, et al (2004) Analysis of the mortality experience amongst U.S. nuclear power industry workers after chronic low-dose exposure to ionizing radiation. *Radiat Res* 162:517–526. <https://doi.org/10.1667/rr3258>
- Hsu DC, Katelaris CH (2009) Long-term management of patients taking immunosuppressive drugs. *Aust Prescr* 32:68–71. <https://doi.org/10.18773/austprescr.2009.035>
- Huddart RA, Norman A, Shahidi M, et al (2003) Cardiovascular disease as a long-term complication of treatment for testicular cancer. *J Clin Oncol* 21:1513–1523. <https://doi.org/10.1200/JCO.2003.04.173>
- Huff JL, Plante I, Blattnig SR, et al (2022) Cardiovascular Disease Risk Modeling for Astronauts: Making the Leap From Earth to Space. *Front Cardiovasc Med* 9:873597. <https://doi.org/10.3389/fcvm.2022.873597>
- Hughson RL, Maillet A, Gharib C, et al (1994a) Reduced spontaneous baroreflex response slope during lower body negative pressure after 28 days of head-down bed rest. *J Appl Physiol* 77:69–77

- Hughson RL, Robertson AD, Arbeille P, et al (2016) Increased postflight carotid artery stiffness and inflight insulin resistance resulting from 6-mo spaceflight in male and female astronauts. *Am J Physiol Heart Circ Physiol* 310:H628-638. <https://doi.org/10.1152/ajpheart.00802.2015>
- Hughson RL, Yamamoto Y, Blaber AP, et al (1994b) Effect of 28-day head-down bed rest with countermeasures on heart rate variability during LBNP. *Aviat Space Environ Med* 65:293–300
- Hunt BE, Taylor JA, Hamner JW, et al (2001) Estrogen replacement therapy improves baroreflex regulation of vascular sympathetic outflow in postmenopausal women. *Circulation* 103:2909–2914
- Hyatt KH, Johnson PC, Hoffler GW, et al (1975) Effect of potassium depletion in normal males: an Apollo 15 simulation. *Aviat Space Environ Med* 46:11–15
- Hyatt KH, West DA (1977) Reversal of bedrest-induced orthostatic intolerance by lower body negative pressure and saline. *Aviat Space Environ Med* 48:120–124
- ICRP (2007) The 2007 Recommendations of the International Commission on Radiological Protection. ICRP publication 103. *Ann ICRP* 37:1–332. <https://doi.org/10.1016/j.icrp.2007.10.003>
- Ikeda T, Saito H, Tanno K, et al (2002) T-wave alternans as a predictor for sudden cardiac death after myocardial infarction. *Am J Cardiol* 89:79–82
- Isaacs AW, Macaluso F, Smith C, Myburgh KH (2019) C-Reactive Protein Is Elevated Only in High Creatine Kinase Responders to Muscle Damaging Exercise. *Front Physiol* 10:. <https://doi.org/10.3389/fphys.2019.00086>
- Ishida S, Takahashi N, Nakagawa M, et al (1995) Relation between QT and RR intervals in patients with bradyarrhythmias. *Br Heart J* 74:159–162
- Ishii K, Kano T, Ando J (1983) Pharmacological effects of flurazepam and diazepam on isolated canine arteries. *Jpn J Pharmacol* 33:65–71. <https://doi.org/10.1254/jjp.33.65>
- Ivanov VK, Maksioutov MA, Chekin SY, et al (2006) The risk of radiation-induced cerebrovascular disease in Chernobyl emergency workers. *Health Phys* 90:199–207. <https://doi.org/10.1097/01.HP.0000175835.31663.ea>
- Iwasaki K, Levine BD, Zhang R, et al (2007) Human cerebral autoregulation before, during and after spaceflight. *J Physiol (Lond)* 579:799–810. <https://doi.org/10.1113/jphysiol.2006.119636>

- Iwasaki KI, Sasaki T, Hirayanagi K, Yajima K (2001) Usefulness of daily +2Gz load as a countermeasure against physiological problems during weightlessness. *Acta Astronaut* 49:227–235
- Iwasaki K-I, Zhang R, Perhonen MA, et al (2004) Reduced baroreflex control of heart period after bed rest is normalized by acute plasma volume restoration. *Am J Physiol Regul Integr Comp Physiol* 287:R1256-1262. <https://doi.org/10.1152/ajpregu.00613.2002>
- Iwasaki KI, Zhang R, Zuckerman JH, et al (2000) Effect of head-down-tilt bed rest and hypovolemia on dynamic regulation of heart rate and blood pressure. *Am J Physiol Regul Integr Comp Physiol* 279:R2189-2199
- Jacob G, Robertson D, Mosqueda-Garcia R, et al (1997) Hypovolemia in syncope and orthostatic intolerance role of the renin-angiotensin system. *Am J Med* 103:128–133
- Jacques PF, Taylor A, Hankinson SE, et al (1997) Long-term vitamin C supplement use and prevalence of early age-related lens opacities. *Am J Clin Nutr* 66:911–916. <https://doi.org/10.1093/ajcn/66.4.911>
- Jain I, Charvat JM, Van Baalen M, et al (2015) Evaluation of Cardiovascular Risk Scores Applied to NASA's Astronaut Corps. In: Human Research Program Investigators' Workshop. Galveston, TX
- Jain V, Ploutz-Snyder R, Young M, et al (2020) Potential Venous Thromboembolism Risk in Female Astronauts. *Aerosp Med Hum Perform* 91:432–439. <https://doi.org/10.3357/AMHP.5458.2020>
- Jarvis SS, Florian JP, Curren MJ, Pawelczyk JA (2012) A somatostatin analog improves tilt table tolerance by decreasing splanchnic vascular conductance. *J Appl Physiol* 112:1504–1511. <https://doi.org/10.1152/jappphysiol.01475.2010>
- Jarvis SS, Florian JP, Curren MJ, Pawelczyk JA (2006) Sex Differences in Splanchnic Hemodynamics during 70 degrees Head-Up Tilt. *Medicine and Science in Sports and Exercise* 38:S4
- Jarvis SS, Florian JP, Curren MJ, Pawelczyk JA (2010) Sex differences in vasoconstrictor reserve during 70 deg head-up tilt. *Exp Physiol* 95:184–193. <https://doi.org/10.1113/expphysiol.2009.048819>
- Jennings RT (1998) Managing space motion sickness. *J Vestib Res* 8:67–70
- Johnson P, Driscoll T, LeBlanc A (1977a) Blood Volume Changes. In: Biomedical Results from Skylab. Scientific and Technical Information Office, National Aeronautics and Space Administration, Washington, D.C., pp 235–241

- Johnson RL, Bergman SA Jr (1977) Evaluation of the Electromechanical Properties of the Cardiovascular System After Prolonged Weightlessness. In: Biomedical results from Skylab. Washington, D.C.: Scientific and Technical Office National Aeronautics and Space Administration, pp 351–365
- Johnson RL, Hoffler GW, Nicogossian AE, et al (1977b) Lower Body Negative Pressure: Third Manned Skylab Mission. In: Biomedical results from Skylab. Scientific and Technical Information Office, National Aeronautics and Space Administration : [for sale by the Supt. of Docs., U.S. Govt. Print. Off.], Washington, D.C., pp 284–312
- Johri AM, Nambi V, Naqvi TZ, et al (2020) Recommendations for the Assessment of Carotid Arterial Plaque by Ultrasound for the Characterization of Atherosclerosis and Evaluation of Cardiovascular Risk: From the American Society of Echocardiography. *J Am Soc Echocardiogr* 33:917–933. <https://doi.org/10.1016/j.echo.2020.04.021>
- Kador PF (1983) Overview of the current attempts toward the medical treatment of cataract. *Ophthalmology* 90:352–364. [https://doi.org/10.1016/s0161-6420\(83\)34562-0](https://doi.org/10.1016/s0161-6420(83)34562-0)
- Kamiya A, Michikami D, Fu Q, et al (2003) Pathophysiology of orthostatic hypotension after bed rest: paradoxical sympathetic withdrawal. *Am J Physiol Heart Circ Physiol* 285:H1158-1167. <https://doi.org/10.1152/ajpheart.00965.2002>
- Kashcheev VV, Chekin SY, Karpenko SV, et al (2017) Radiation Risk of Cardiovascular Diseases in the Cohort of Russian Emergency Workers of the Chernobyl Accident. *Health Phys* 113:23–29. <https://doi.org/10.1097/HP.0000000000000670>
- Katayama K, Sato K, Akima H, et al (2004) Acceleration with exercise during head-down bed rest preserves upright exercise responses. *Aviat Space Environ Med* 75:1029–1035
- Keller DM, Low DA, Davis SL, et al (2011) Skin surface cooling improves orthostatic tolerance following prolonged head-down bed rest. *J Appl Physiol* 110:1592–1597. <https://doi.org/10.1152/jappphysiol.00233.2010>
- Kemmeren JM, Algra A, Grobbee DE (2001) Third generation oral contraceptives and risk of venous thrombosis: meta-analysis. *BMJ* 323:131. <https://doi.org/10.1136/bmj.323.7305.131>
- Khaled S, Gupta KB, Kucik DF (2012) Ionizing radiation increases adhesiveness of human aortic endothelial cells via a chemokine-dependent mechanism. *Radiat Res* 177:594–601. <https://doi.org/10.1667/rr2557.1>
- Khan IA (2002) Clinical and therapeutic aspects of congenital and acquired long QT syndrome. *Am J Med* 112:58–66

- Khera A, Budoff MJ, O'Donnell CJ, et al (2018) Astronaut Cardiovascular Health and Risk Modification (Astro-CHARM) Coronary Calcium Atherosclerotic Cardiovascular Disease Risk Calculator. *Circulation* 138:1819–1827. <https://doi.org/10.1161/CIRCULATIONAHA.118.033505>
- Khine HW, Steding-Ehrenborg K, Hastings JL, et al (2018) Effects of Prolonged Spaceflight on Atrial Size, Atrial Electrophysiology, and Risk of Atrial Fibrillation. *Circ Arrhythm Electrophysiol* 11:e005959. <https://doi.org/10.1161/CIRCEP.117.005959>
- Kilgour RD, Carvalho J (1994) Gender differences in cardiovascular responses to the cold hand pressor test and facial cooling. *Can J Physiol Pharmacol* 72:1193–1199
- Kim DS, Vaquer S, Mazzolai L, et al (2021) The effect of microgravity on the human venous system and blood coagulation: a systematic review. *Exp Physiol* 106:1149–1158. <https://doi.org/10.1113/EP089409>
- Kim Y-H, Kim H-B, Kim D-H, et al (2018) Use of hypnotics and the risk of or mortality from heart disease: a meta-analysis of observational studies. *Korean J Intern Med* 33:727–736. <https://doi.org/10.3904/kjim.2016.282>
- Kimmerly DS, Shoemaker JK (2003) Hypovolemia and MSNA discharge patterns: assessing and interpreting sympathetic responses. *Am J Physiol Heart Circ Physiol* 284:H1198-1204. <https://doi.org/10.1152/ajpheart.00229.2002>
- Kimmerly DS, Shoemaker JK (2002) Hypovolemia and neurovascular control during orthostatic stress. *Am J Physiol Heart Circ Physiol* 282:H645-655. <https://doi.org/10.1152/ajpheart.00535.2001>
- Korolkov VI, Kozlovskaya IB, Kotovskaya AR, et al (2001) Efficacy of periodic centrifugation of primates during 4-week head-down tilt. *Acta Astronaut* 49:237–242
- Korth DW (2015) Exercise Countermeasure Hardware Evolution on ISS: The First Decade. *Aerospace Medicine and Human Performance* 86:A7–A13. <https://doi.org/10.3357/AMHP.EC02.2015>
- Kotovskaia AR, Koloteva MI, Luk'ianiuk VI, et al (2008) [Individual prediction by the analysis of preflight ECG data of cardiac function disorders in cosmonauts during standard deorbit after long-term space flights and in the period of postflight observation]. *Aviakosm Ekolog Med* 42:14–20
- Kotovskaia AR, Vil'-Villiams IF (2007) [Peculiar physiological reactions to +Gx loads of nonprofessional members of short-term missions to the ISS]. *Aviakosm Ekolog Med* 41:7–13

- Koturbash I, Miousse IR, Sridharan V, et al (2016) Radiation-induced changes in DNA methylation of repetitive elements in the mouse heart. *Mutat Res* 787:43–53. <https://doi.org/10.1016/j.mrfmmm.2016.02.009>
- Kozlovskaya I, Pestov I, Egorov A (2010) The system of preventive measures in long space flights. *Human Physiology* 36:773–779. <https://doi.org/10.1134/S0362119710070066>
- Kozlovskaya IB, Grigoriev AI, Stepantsov VI (1995) Countermeasure of the negative effects of weightlessness on physical systems in long-term space flights. *Acta Astronaut* 36:661–668
- Kramer A, Venegas-Carro M, Mulder E, et al (2020) Cardiorespiratory and Neuromuscular Demand of Daily Centrifugation: Results From the 60-Day AGBRESA Bed Rest Study. *Front Physiol* 11:. <https://doi.org/10.3389/fphys.2020.562377>
- Kreuzer M, Auvinen A, Cardis E, et al (2015) Low-dose ionising radiation and cardiovascular diseases--Strategies for molecular epidemiological studies in Europe. *Mutat Res Rev Mutat Res* 764:90–100. <https://doi.org/10.1016/j.mrrev.2015.03.002>
- Kunz H, Quiariarte H, Simpson RJ, et al (2017) Alterations in hematologic indices during long-duration spaceflight. *BMC Hematology* 17:12. <https://doi.org/10.1186/s12878-017-0083-y>
- Lan M, Phillips SD, Archambault-Leger V, et al (2021) Proposed mechanism for reduced jugular vein flow in microgravity. *Physiol Rep* 9:e14782. <https://doi.org/10.14814/phy2.14782>
- Lan M, Van Akin M, Anderson A, Buckey JC (2022) Microgravity-induced reduced jugular vein flow is more pronounced on the non-dominant side. *Acta Astronautica* 198:746–755. <https://doi.org/10.1016/j.actaastro.2022.05.048>
- Lauk S, Kizel Z, Buschmann J, Trott KR (1985) Radiation-induced heart disease in rats. *Int J Radiat Oncol Biol Phys* 11:801–808. [https://doi.org/10.1016/0360-3016\(85\)90314-1](https://doi.org/10.1016/0360-3016(85)90314-1)
- Lawley JS, Petersen LG, Howden EJ, et al (2017) Effect of gravity and microgravity on intracranial pressure. *J Physiol (Lond)* 595:2115–2127. <https://doi.org/10.1113/JP273557>
- Leach C, Alexander W, Johnson P (1975) Endocrine, Electrolyte, and Fluid Volume Changes Associated with Apollo Missions. In: *Biomedical results of Apollo. Scientific and Technical Information Office, National Aeronautics and Space Administration: for sale by the Supt. of Docs., U.S. Govt. Print. Off., pp 163–184*
- Leach CS, Alfrey CP, Suki WN, et al (1996) Regulation of body fluid compartments during short-term spaceflight. *J Appl Physiol* 81:105–116
- Leach CS, Inners LD, Charles JB (1991) Changes in total body water during spaceflight. *J Clin Pharmacol* 31:1001–1006

- Lee SM, Bennett BS, Hargens AR, et al (1997) Upright exercise or supine lower body negative pressure exercise maintains exercise responses after bed rest. *Med Sci Sports Exerc* 29:892–900
- Lee SM, Bishop PA, Schneider SM, Greenisen MC (2001) Simulated shuttle egress: comparison of two Space Shuttle protective garments. *Aviat Space Environ Med* 72:110–114
- Lee SM, Moore AD Jr, Fritsch-Yelle JM, et al (1999) Inflight exercise affects stand test responses after space flight. *Med Sci Sports Exerc* 31:1755–1762
- Lee SMC, Feiveson AH, Stein SP, et al (2015) Orthostatic intolerance after International Space Station and Space Shuttle missions. *Aerosp Med Human Perf* 86:A54–A67
- Lee SMC, Guined JR, Brown AK, et al (2011) Metabolic consequences of garments worn to protect against post-spaceflight orthostatic intolerance. *Aviat Space Environ Med* 82:648–653
- Lee SMC, Laurie SS, Macias BR, et al (2020a) Reply to Greaves et al. *J Appl Physiol* (1985) 129:1113. <https://doi.org/10.1152/jappphysiol.00757.2020>
- Lee SMC, Martin DS, Miller C, et al (2020b) Venous and arterial responses to partial gravity. *Front Physiol* 11:. <https://doi.org/10.3389/fphys.2020.00863>
- Lee SMC, Moore AD, Everett ME, et al (2010) Aerobic exercise deconditioning and countermeasures during bed rest. *Aviat Space Environ Med* 81:52–63
- Lee SMC, Ribeiro LC, Laurie SS, et al (2020c) Efficacy of gradient compression garments in the hours after long-duration spaceflight. *Front Physiol* 11:784. <https://doi.org/10.3389/fphys.2020.00784>
- Lee SMC, Ribeiro LC, Martin DS, et al (2020d) Arterial structure and function during and after long-duration spaceflight. *J Appl Physiol* 129:108–123. <https://doi.org/10.1152/jappphysiol.00550.2019>
- Lee SMC, Scheuring RA, Guilliams ME, Kerstman EL (2019) Physical Performance, Countermeasures, and Postflight Reconditioning. In: Barratt MR, Baker ES, Pool SL (eds) *Principles of Clinical Medicine for Space Flight*, Second Edition. Springer Science+Business Media LLC, New York, NY, pp 609–658
- Lee SMC, Stenger MB, Laurie SS, Macias BR (2017) Risk of Cardiac Rhythm Problems during Spaceflight. National Aeronautics and Space Administration, Lyndon B. Johnson Space Center, Houston, TX
- Lee SMC, Williams WJ, Schneider SM (2002) Role of skin blood flow and sweating rate in exercise thermoregulation after bed rest. *J Appl Physiol* 92:2026–2034. <https://doi.org/10.1152/jappphysiol.00105.2001>

- Leguay G, Seigneuric A (1981) Cardiac arrhythmias in space. Role of vagotonia. *Acta Astronaut* 8:795–801
- Lehmann N, Erbel R, Mahabadi AA, et al (2018) Value of Progression of Coronary Artery Calcification for Risk Prediction of Coronary and Cardiovascular Events: Result of the HNR Study (Heinz Nixdorf Recall). *Circulation* 137:665–679. <https://doi.org/10.1161/CIRCULATIONAHA.116.027034>
- Lenarczyk M, Laiakis EC, Mattson DL, et al (2020) Irradiation of the kidneys causes pathologic remodeling in the nontargeted heart: A role for the immune system. *FASEB Bioadv* 2:705–719. <https://doi.org/10.1096/fba.2020-00071>
- Lenarczyk M, Lam V, Jensen E, et al (2013) Cardiac injury after 10 Gy total body irradiation: indirect role of effects on abdominal organs. *Radiat Res* 180:247–258. <https://doi.org/10.1667/RR3292.1>
- Lenarczyk M, Su J, Haworth ST, et al (2015) Simvastatin mitigates increases in risk factors for and the occurrence of cardiac disease following 10 Gy total body irradiation. *Pharmacol Res Perspect* 3:e00145. <https://doi.org/10.1002/prp2.145>
- Levine BD, Buckey JC, Fritsch JM, et al (1991) Physical fitness and cardiovascular regulation: mechanisms of orthostatic intolerance. *J Appl Physiol* 70:112–122
- Levine BD, Bungo MW (2015) Integrated Cardiovascular Experiment (E377): Final Report to NASA
- Levine BD, Pawelczyk JA, Ertl AC, et al (2002) Human muscle sympathetic neural and haemodynamic responses to tilt following spaceflight. *J Physiol (Lond)* 538:331–340
- Levine BD, Zuckerman JH, Pawelczyk JA (1997) Cardiac atrophy after bed-rest deconditioning: a nonneural mechanism for orthostatic intolerance. *Circulation* 96:517–525
- Li M, Gonon G, Buonanno M, et al (2014) Health risks of space exploration: targeted and nontargeted oxidative injury by high-charge and high-energy particles. *Antioxid Redox Signal* 20:1501–1523. <https://doi.org/10.1089/ars.2013.5649>
- Lieberman EH, Gerhard MD, Uehata A, et al (1994) Estrogen improves endothelium-dependent, flow-mediated vasodilation in postmenopausal women. *Ann Intern Med* 121:936–941
- Lightfoot JT, Febles S, Fortney SM (1989) Adaptation to repeated presyncopal lower body negative pressure exposures. *Aviat Space Environ Med* 60:17–22
- Linnarsson D, Hughson RL, Fraser KS, et al (2015) Effects of an artificial gravity countermeasure on orthostatic tolerance, blood volumes and aerobic power after short-term bed rest (BR-AG1). *J Appl Physiol* (1985) 118:29–35. <https://doi.org/10.1152/jappphysiol.00061.2014>

- Little MP (2013) A review of non-cancer effects, especially circulatory and ocular diseases. *Radiat Environ Biophys* 52:435–449. <https://doi.org/10.1007/s00411-013-0484-7>
- Little MP (2016) Radiation and circulatory disease. *Mutat Res Rev Mutat Res* 770:299–318. <https://doi.org/10.1016/j.mrrev.2016.07.008>
- Little MP, Azizova TV, Bazyka D, et al (2012) Systematic review and meta-analysis of circulatory disease from exposure to low-level ionizing radiation and estimates of potential population mortality risks. *Environ Health Perspect* 120:1503–1511. <https://doi.org/10.1289/ehp.1204982>
- Little MP, Azizova TV, Hamada N (2021) Low- and moderate-dose non-cancer effects of ionizing radiation in directly exposed individuals, especially circulatory and ocular diseases: a review of the epidemiology. *Int J Radiat Biol* 97:782–803. <https://doi.org/10.1080/09553002.2021.1876955>
- Little MP, Tawn EJ, Tzoulaki I, et al (2008) A systematic review of epidemiological associations between low and moderate doses of ionizing radiation and late cardiovascular effects, and their possible mechanisms. *Radiat Res* 169:99–109. <https://doi.org/10.1667/RR1070.1>
- Lo SS, Mathias CJ, Sutton MS (1996) QT interval and dispersion in primary autonomic failure. *Heart* 75:498–501
- Lott MEJ, Hogeman C, Herr M, et al (2009) Sex differences in limb vasoconstriction responses to increases in transmural pressures. *Am J Physiol Heart Circ Physiol* 296:H186–H194. <https://doi.org/10.1152/ajpheart.00248.2008>
- Lux RL, Hilbel T, Brockmeier K (2001) Electrocardiographic measures of repolarization revisited: why? what? how? *J Electrocardiol* 34 Suppl:259–264
- Mallat Z, Tedgui A, Fontaliran F, et al (1996) Evidence of apoptosis in arrhythmogenic right ventricular dysplasia. *N Engl J Med* 335:1190–1196. <https://doi.org/10.1056/NEJM199610173351604>
- Malliani A, Pagani M, Lombardi F, Cerutti S (1991) Cardiovascular neural regulation explored in the frequency domain. *Circulation* 84:482–492. <https://doi.org/10.1161/01.CIR.84.2.482>
- Mancia G, Fagard R, Narkiewicz K, et al (2013) 2013 ESH/ESC Guidelines for the management of arterial hypertension: the Task Force for the management of arterial hypertension of the European Society of Hypertension (ESH) and of the European Society of Cardiology (ESC). *J Hypertens* 31:1281–1357. <https://doi.org/10.1097/01.hjh.0000431740.32696.cc>

- Mancuso M, Pasquali E, Braga-Tanaka I, et al (2015) Acceleration of atherogenesis in ApoE-/- mice exposed to acute or low-dose-rate ionizing radiation. *Oncotarget* 6:31263–31271. <https://doi.org/10.18632/oncotarget.5075>
- Mao XW, Stanbouly S, Chieu B, et al (2022) Low dose space radiation-induced effects on the mouse retina and blood-retinal barrier integrity. *Acta Astronautica* 199:412–419. <https://doi.org/10.1016/j.actaastro.2022.07.029>
- Marks LB, Zagar TM, Kaidar-Person O (2017) Reassessing the Time Course for Radiation-Induced Cardiac Mortality in Patients With Breast Cancer. *Int J Radiat Oncol Biol Phys* 97:303–305. <https://doi.org/10.1016/j.ijrobp.2016.10.036>
- Marshall-Goebel K, Laurie SS, Alferova IV, et al (2019) Assessment of Jugular Venous Blood Flow Stasis and Thrombosis During Spaceflight. *JAMA Netw Open* 2:e1915011. <https://doi.org/10.1001/jamanetworkopen.2019.15011>
- Marshall-Goebel K, Macias BR, Laurie SS, et al (2021) Mechanical countermeasures to headward fluid shifts. *J Appl Physiol* (1985) 130:1766–1777. <https://doi.org/10.1152/jappphysiol.00863.2020>
- Martin DS, Caine TL, Matz T, et al (2012) Virtual guidance as a tool to obtain diagnostic ultrasound for spaceflight and remote environments. *Aviat Space Environ Med* 83:995–1000
- Martin DS, Lee SMC, Matz TP, et al (2016) Internal jugular pressure increases during parabolic flight. *Physiol Rep* 4:. <https://doi.org/10.14814/phy2.13068>
- Mauran P, Sediame S, Traon AP, et al (1999) Effects of a three-day head-down tilt on renal and hormonal responses to acute volume expansion. *Am J Physiol* 277:R1444-1452. <https://doi.org/10.1152/ajpregu.1999.277.5.R1444>
- McAdoo WG, Weinberger MH, Miller JZ, et al (1990) Race and gender influence hemodynamic responses to psychological and physical stimuli. *J Hypertens* 8:961–967
- McGale P, Darby SC (2008) Commentary: A dose-response relationship for radiation-induced heart disease--current issues and future prospects. *Int J Epidemiol* 37:518–523. <https://doi.org/10.1093/ije/dyn067>
- McGale P, Darby SC (2005) Low doses of ionizing radiation and circulatory diseases: a systematic review of the published epidemiological evidence. *Radiat Res* 163:247–257. <https://doi.org/10.1667/rr3314>
- McMillan T, Bennett M, Bridges B, et al (2010) Circulatory Disease Risk, Report of the independent Advisory Group on Ionising Radiation. In: Documents of the Health Protection Agency (RCE-16). Health Protection Agency, Loondon, UK

- Meck JV, Dreyer SA, Warren LE (2009) Long-duration head-down bed rest: project overview, vital signs, and fluid balance. *Aviat Space Environ Med* 80:A1-8. <https://doi.org/10.3357/asem.br01.2009>
- Meck JV, Reyes CJ, Perez SA, et al (2001) Marked exacerbation of orthostatic intolerance after long- vs. short-duration spaceflight in veteran astronauts. *Psychosom Med* 63:865–873
- Meck JV, Waters WW, Ziegler MG, et al (2004) Mechanisms of postspaceflight orthostatic hypotension: low alpha1-adrenergic receptor responses before flight and central autonomic dysregulation postflight. *Am J Physiol Heart Circ Physiol* 286:H1486-1495. <https://doi.org/10.1152/ajpheart.00740.2003>
- Meerman M, Bracco Gartner TCL, Buikema JW, et al (2021) Myocardial Disease and Long-Distance Space Travel: Solving the Radiation Problem. *Front Cardiovasc Med* 8:631985. <https://doi.org/10.3389/fcvm.2021.631985>
- Miousse IR, Skinner CM, Sridharan V, et al (2019) Changes in one-carbon metabolism and DNA methylation in the hearts of mice exposed to space environment-relevant doses of oxygen ions (16O). *Life Sci Space Res (Amst)* 22:8–15. <https://doi.org/10.1016/j.lssr.2019.05.003>
- Mitchel REJ, Hasu M, Bugden M, et al (2011) Low-dose radiation exposure and atherosclerosis in ApoE^{-/-} mice. *Radiat Res* 175:665–676. <https://doi.org/10.1667/RR2176.1>
- Mitchel REJ, Hasu M, Bugden M, et al (2013) Low-dose radiation exposure and protection against atherosclerosis in ApoE(-/-) mice: the influence of P53 heterozygosity. *Radiat Res* 179:190–199. <https://doi.org/10.1667/RR3140.1>
- Mitchell BM, Meck JV (2004) Short-duration spaceflight does not prolong QTc intervals in male astronauts. *Am J Cardiol* 93:1051–1052. <https://doi.org/10.1016/j.amjcard.2003.12.060>
- Montano N, Ruscone TG, Porta A, et al (1994) Power spectrum analysis of heart rate variability to assess the changes in sympathovagal balance during graded orthostatic tilt. *Circulation* 90:1826–1831
- Montgomery LD, Kirk PJ, Payne PA, et al (1977) Cardiovascular responses of men and women to lower body negative pressure. *Aviat Space Environ Med* 48:138–145
- Moore AD, Downs ME, Lee SMC, et al (2014) Peak exercise oxygen uptake during and following long-duration spaceflight. *J Appl Physiol* 117:231–238. <https://doi.org/10.1152/jappphysiol.01251.2013>
- Moore AD Jr, Lee SM, Charles JB, et al (2001) Maximal exercise as a countermeasure to orthostatic intolerance after spaceflight. *Med Sci Sports Exerc* 33:75–80

- Moore AD, Lee SMC, Stenger MB, Platts SH (2010) Cardiovascular exercise in the U.S. space program: Past, present and future. *Acta Astronautica* 66:974–988. <https://doi.org/10.1016/j.actaastro.2009.10.009>
- Mooteri SN, Podolski JL, Drab EA, et al (1996) WR-1065 and radioprotection of vascular endothelial cells. II. Morphology. *Radiat Res* 145:217–224
- Morris CJ, Purvis TE, Hu K, Scheer FAJL (2016) Circadian misalignment increases cardiovascular disease risk factors in humans. *Proc Natl Acad Sci U S A* 113:E1402–1411. <https://doi.org/10.1073/pnas.1516953113>
- Moseeva MB, Azizova TV, Grigoryeva ES, Haylock R (2014) Risks of circulatory diseases among Mayak PA workers with radiation doses estimated using the improved Mayak Worker Dosimetry System 2008. *Radiat Environ Biophys* 53:469–477. <https://doi.org/10.1007/s00411-014-0517-x>
- Mulavara AP, Peters BT, Miller CA, et al (2018) Physiological and Functional Alterations after Spaceflight and Bed Rest. *Med Sci Sports Exerc* 50:1961–1980. <https://doi.org/10.1249/MSS.0000000000001615>
- Mullington JM, Haack M, Toth M, et al (2009) Cardiovascular, inflammatory, and metabolic consequences of sleep deprivation. *Prog Cardiovasc Dis* 51:294–302. <https://doi.org/10.1016/j.pcad.2008.10.003>
- Mulrooney DA, Yeazel MW, Kawashima T, et al (2009) Cardiac outcomes in a cohort of adult survivors of childhood and adolescent cancer: retrospective analysis of the Childhood Cancer Survivor Study cohort. *BMJ* 339:b4606. <https://doi.org/10.1136/bmj.b4606>
- Nakachi K, Hayashi T, Hamatani K, et al (2008) Sixty years of follow-up of Hiroshima and Nagasaki survivors: current progress in molecular epidemiology studies. *Mutat Res* 659:109–117. <https://doi.org/10.1016/j.mrrev.2008.02.001>
- NASA (2016) National Aeronautics and Space Administration (NASA) Emergency Medical Procedures Manual for the International Space Station (ISS)
- NASA (2022) NASA SPACE FLIGHT HUMAN SYSTEM STANDARD 3001, Revision B. National Aeronautics and Space Administration, Washington, D.C.
- NCRP (2006) Information needed to make radiation protection recommendations for space missions beyond Low Earth Orbit. NCRP Report No. 153. NCRP, Bethesda, MD
- NCRP (2014) Radiation protection for space activities: supplement to previous recommendations. NCRP Commentary No. 23. NCRP, Bethesda, MD

- NCRP (2015) NCRP Commentary No. 24 – Health Effects of Low Doses of Radiation: Perspectives on Integrating Radiation Biology and Epidemiology. NCRP, Bethesda, MD
- NCRP (1989) Guidance on radiation received in space activities. NCRP Report No. 98. NCRP, Bethesda, MD
- Negishi K, Borowski AG, Popović ZB, et al (2017) Effect of Gravitational Gradients on Cardiac Filling and Performance. *J Am Soc Echocardiogr* 30:1180–1188. <https://doi.org/10.1016/j.echo.2017.08.005>
- Nemec-Bakk AS, Sridharan V, Landes RD, et al (2022) Effects of low-dose oxygen ions on cardiac function and structure in female C57BL/6J mice. *Life Sci Space Res (Amst)* 32:105–112. <https://doi.org/10.1016/j.lssr.2021.12.004>
- Nemec-Bakk AS, Sridharan V, Landes RD, et al (2021) Mitigation of late cardiovascular effects of oxygen ion radiation by γ -tocotrienol in a mouse model. *Life Sci Space Res (Amst)* 31:43–50. <https://doi.org/10.1016/j.lssr.2021.07.006>
- Nicogossian A, Pool S, Sawin C (1995) Status and efficacy of countermeasures to physiological deconditioning from space flight. *Acta Astronaut* 36:393–398. [https://doi.org/10.1016/0094-5765\(95\)00123-9](https://doi.org/10.1016/0094-5765(95)00123-9)
- Nixon JV, Murray RG, Bryant C, et al (1979) Early cardiovascular adaptation to simulated zero gravity. *J Appl Physiol* 46:541–548
- Norsk P (2005) Cardiovascular and fluid volume control in humans in space. *Curr Pharm Biotechnol* 6:325–330
- Norsk P, Asmar A, Damgaard M, Christensen NJ (2015) Fluid shifts, vasodilatation and ambulatory blood pressure reduction during long duration spaceflight. *J Physiol (Lond)* 593:573–584
- Nounu HN, Kim M-HY, Ponomarev AL, Cucinotta FA (2009) The Use of Pro/Engineer CAD Software and Fishbowl Tool Kit in Ray-tracing Analysis
- Nugent AC, Bain EE, Thayer JF, et al (2011) Sex differences in the neural correlates of autonomic arousal: a pilot PET study. *Int J Psychophysiol* 80:182–191. <https://doi.org/10.1016/j.ijpsycho.2011.03.001>
- Odutayo A, Wong CX, Hsiao AJ, et al (2016) Atrial fibrillation and risks of cardiovascular disease, renal disease, and death: systematic review and meta-analysis. *BMJ* 354:i4482. <https://doi.org/10.1136/bmj.i4482>
- O’Neal WT, Efird JT, Qureshi WT, et al (2015) Coronary Artery Calcium Progression and Atrial Fibrillation: The Multi-Ethnic Study of Atherosclerosis. *Circ Cardiovasc Imaging* 8:e003786. <https://doi.org/10.1161/CIRCIMAGING.115.003786>

- Oslina D, Rybkina V, Adamova G, et al (2021) Biomarkers of Atherosclerotic Vascular Disease in Workers Chronically Exposed to Ionizing Radiation. *Health Phys* 121:92–101. <https://doi.org/10.1097/HP.0000000000001416>
- Ozasa K, Shimizu Y, Suyama A, et al (2012) Studies of the mortality of atomic bomb survivors, Report 14, 1950-2003: an overview of cancer and noncancer diseases. *Radiat Res* 177:229–243. <https://doi.org/10.1667/rr2629.1>
- Pagani M, Lombardi F, Guzzetti S, et al (1986) Power spectral analysis of heart rate and arterial pressure variabilities as a marker of sympatho-vagal interaction in man and conscious dog. *Circ Res* 59:178–193
- Pathak R, Bachri A, Ghosh SP, et al (2016) The Vitamin E Analog Gamma-Tocotrienol (GT3) Suppresses Radiation-Induced Cytogenetic Damage. *Pharm Res* 33:2117–2125. <https://doi.org/10.1007/s11095-016-1950-0>
- Pattarini JM, Somers JT, Charvat JM, et al (2020) Artemis Sustained Translational Acceleration Limits: Human Tolerance Evidence from Apollo to International Space Station. National Aeronautics and Space Administration, Houston, TX
- Patwardhan AR, Evans JM, Berk M, et al (1995) Spectral indices of cardiovascular adaptations to short-term simulated microgravity exposure. *Integr Physiol Behav Sci* 30:201–214
- Pavela J, Sargsyan A, Bedi D, et al (2022) Surveillance for jugular venous thrombosis in astronauts. *Vasc Med* 1358863X221086619. <https://doi.org/10.1177/1358863X221086619>
- Pavy-Le Traon A, Heer M, Narici MV, et al (2007a) From space to Earth: advances in human physiology from 20 years of bed rest studies (1986-2006). *Eur J Appl Physiol* 101:143–194. <https://doi.org/10.1007/s00421-007-0474-z>
- Pavy-Le Traon A, Heer M, Narici MV, et al (2007b) From space to Earth: advances in human physiology from 20 years of bed rest studies (1986-2006). *Eur J Appl Physiol* 101:143–194. <https://doi.org/10.1007/s00421-007-0474-z>
- Pavy-Le Traon A, Sigaud D, Vasseur P, et al (1998) Cardiovascular responses to orthostatic tests after a 42-day head-down bed-rest. *Eur J Appl Physiol Occup Physiol* 77:50–59
- Pawelczyk JA, Zuckerman JH, Blomqvist CG, Levine BD (2001) Regulation of muscle sympathetic nerve activity after bed rest deconditioning. *Am J Physiol Heart Circ Physiol* 280:H2230–2239
- Penev PD, Kolker DE, Zee PC, Turek FW (1998) Chronic circadian desynchronization decreases the survival of animals with cardiomyopathic heart disease. *Am J Physiol* 275:H2334–2337. <https://doi.org/10.1152/ajpheart.1998.275.6.H2334>

- Perez SA, Charles JB, Fortner GW, et al (2003) Cardiovascular effects of anti-G suit and cooling garment during space shuttle re-entry and landing. *Aviat Space Environ Med* 74:753–757
- Perhonen MA, Franco F, Lane LD, et al (2001a) Cardiac atrophy after bed rest and spaceflight. *J Appl Physiol* 91:645–653
- Perhonen MA, Zuckerman JH, Levine BD (2001b) Deterioration of left ventricular chamber performance after bed rest : “cardiovascular deconditioning” or hypovolemia? *Circulation* 103:1851–1857
- Peterson LE, Pepper LJ, Hamm PB, Gilbert SL (1993) Longitudinal study of astronaut health: mortality in the years 1959-1991. *Radiat Res* 133:257–264
- Pettit D (2010) Mars Landing on Earth: An Astronaut’s Perspective. *J Cosmology* 12:3529–3536
- Platts SH, Bairey Merz CN, Barr Y, et al (2014) Effects of Sex and Gender on Adaptation to Space: Cardiovascular Alterations. *J Womens Health (Larchmt)* 23:950–955. <https://doi.org/10.1089/jwh.2014.4912>
- Platts SH, Shi S-J, Meck JV (2006) Akathisia with combined use of midodrine and promethazine. *JAMA* 295:2000–2001. <https://doi.org/10.1001/jama.295.17.2000-b>
- Platts SH, Tuxhorn JA, Ribeiro LC, et al (2009) Compression garments as countermeasures to orthostatic intolerance. *Aviat Space Environ Med* 80:437–442
- Platts SH, Ziegler MG, Waters WW, et al (2004) Midodrine prescribed to improve recurrent post-spaceflight orthostatic hypotension. *Aviat Space Environ Med* 75:554–556
- Plotnikova ED, Levitman MK, Shaposhnikova VV, et al (1988) Protection of microvasculature in rat brain against late radiation injury by gammaphos. *Int J Radiat Oncol Biol Phys* 15:1197–1201. [https://doi.org/10.1016/0360-3016\(88\)90204-0](https://doi.org/10.1016/0360-3016(88)90204-0)
- Pomeranz B, Macaulay RJ, Caudill MA, et al (1985) Assessment of autonomic function in humans by heart rate spectral analysis. *Am J Physiol* 248:H151-153
- Preiss G, Polosa C (1974) Patterns of sympathetic neuron activity associated with Mayer waves. *Am J Physiol* 226:724–730
- Preston DL, Shimizu Y, Pierce DA, et al (2003) Studies of mortality of atomic bomb survivors. Report 13: Solid cancer and noncancer disease mortality: 1950-1997. *Radiat Res* 160:381–407
- Previtali E, Bucciarelli P, Passamonti SM, Martinelli I (2011) Risk factors for venous and arterial thrombosis. *Blood Transfus* 9:120–138. <https://doi.org/10.2450/2010.0066-10>

- Prisk GK, Guy HJ, Elliott AR, et al (1993) Pulmonary diffusing capacity, capillary blood volume, and cardiac output during sustained microgravity. *J Appl Physiol* 75:15–26
- Prosnitz RG, Chen YH, Marks LB (2005) Cardiac toxicity following thoracic radiation. *Semin Oncol* 32:S71-80. <https://doi.org/10.1053/j.seminoncol.2005.03.013>
- Protheroe CL, Dikareva A, Menon C, Claydon VE (2011) Are compression stockings an effective treatment for orthostatic presyncope? *PLoS ONE* 6:e28193. <https://doi.org/10.1371/journal.pone.0028193>
- Purdy RE, Duckles SP, Krause DN, et al (1998) Effect of simulated microgravity on vascular contractility. *J Appl Physiol* 85:1307–1315
- Putcha L, Berens KL, Marshburn TH, et al (1999) Pharmaceutical use by U.S. astronauts on space shuttle missions. *Aviat Space Environ Med* 70:705–8
- Radford NB, DeFina LF, Leonard D, et al (2018) Cardiorespiratory Fitness, Coronary Artery Calcium, and Cardiovascular Disease Events in a Cohort of Generally Healthy Middle-Age Men: Results From the Cooper Center Longitudinal Study. *Circulation* 137:1888–1895. <https://doi.org/10.1161/CIRCULATIONAHA.117.032708>
- Raggi P, Callister TQ, Shaw LJ (2004) Progression of coronary artery calcium and risk of first myocardial infarction in patients receiving cholesterol-lowering therapy. *Arterioscler Thromb Vasc Biol* 24:1272–1277. <https://doi.org/10.1161/01.ATV.0000127024.40516.ef>
- Ramadan SS, Sridharan V, Koturbash I, et al (2016) A priming dose of protons alters the early cardiac cellular and molecular response to (56)Fe irradiation. *Life Sci Space Res (Amst)* 8:8–13. <https://doi.org/10.1016/j.lssr.2015.12.001>
- Ramsdell CD, Mullen TJ, Sundby GH, et al (2001) Midodrine prevents orthostatic intolerance associated with simulated spaceflight. *J Appl Physiol* 90:2245–2248
- Ray CA, Vasques M, Miller TA, et al (2001) Effect of short-term microgravity and long-term hindlimb unloading on rat cardiac mass and function. *J Appl Physiol* 91:1207–1213
- Respress JL, Gershovich PM, Wang T, et al (2014) Long-term simulated microgravity causes cardiac RyR2 phosphorylation and arrhythmias in mice. *Int J Cardiol* 176:994–1000. <https://doi.org/10.1016/j.ijcard.2014.08.138>
- Reyes Del Paso GA, Langewitz W, Mulder LJM, et al (2013) The utility of low frequency heart rate variability as an index of sympathetic cardiac tone: A review with emphasis on a reanalysis of previous studies. *Psychophysiology* 50:477–487. <https://doi.org/10.1111/psyp.12027>
- Reynolds RJ, Day SM (2010) Mortality among U.S. astronauts: 1980-2009. *Aviat Space Environ Med* 81:1024–1027

- Reynolds RJ, Day SM (2017) Mortality Due to Cardiovascular Disease Among Apollo Lunar Astronauts. *Aerosp Med Hum Perform* 88:492–496. <https://doi.org/10.3357/AMHP.4757.2017>
- Reynolds RJ, Day SM (2019) Mortality of US astronauts: comparisons with professional athletes. *Occup Environ Med* 76:114–117. <https://doi.org/10.1136/oemed-2018-105304>
- Reynolds RJ, Day SM (2018) The effect of competing risks on astronaut and cosmonaut mortality. *Life Sci Space Res (Amst)* 18:35–41. <https://doi.org/10.1016/j.lssr.2018.05.006>
- Reynolds RJ, Day SM, Nurgalieva ZZ (2014) Mortality among Soviet and Russian cosmonauts: 1960-2013. *Aviat Space Environ Med* 85:750–754
- Rimmer D, Dijk D, Rhonda J, et al (1999) Efficacy of liquid cooling garments to minimize heat strain during Space Shuttle deorbit and landing. *Medicine and Science in Sports and Exercise* 31:S305
- Robertson D, Convertino VA, Vernikos J (1994) The sympathetic nervous system and the physiologic consequences of spaceflight: a hypothesis. *Am J Med Sci* 308:126–132
- Robin A, Auvinet A, Degryse B, et al (2020) DI-5-CUFFS: Venoconstrictive Thigh Cuffs Limit Body Fluid Changes but Not Orthostatic Intolerance Induced by a 5-Day Dry Immersion. *Front Physiol* 11:. <https://doi.org/10.3389/fphys.2020.00383>
- Romanov EM, Artamonova NP, Golubchikova ZA, et al (1987) [Results of long-term electrocardiographic examinations of cosmonauts]. *Kosm Biol Aviakosm Med* 21:10–14
- Romero LM, Platts SH, Schoech SJ, et al (2015) Understanding stress in the healthy animal - potential paths for progress. *Stress* 18:491–497. <https://doi.org/10.3109/10253890.2015.1073255>
- Rosenbaum DS, Jackson LE, Smith JM, et al (1994) Electrical alternans and vulnerability to ventricular arrhythmias. *N Engl J Med* 330:235–241. <https://doi.org/10.1056/NEJM199401273300402>
- Rossum AC, Wood ML, Bishop SL, et al (1997) Evaluation of cardiac rhythm disturbances during extravehicular activity. *Am J Cardiol* 79:1153–1155
- Rossum AC, Ziegler MG, Meck JV (2001) Effect of spaceflight on cardiovascular responses to upright posture in a 77-year-old astronaut. *Am J Cardiol* 88:1335–1337
- Rowe WJ (1998) The Apollo 15 Space Syndrome. *Circulation* 97:119–120. <https://doi.org/10.1161/01.CIR.97.1.119>
- Rowell LB (1993) *Human Cardiovascular Control*. Oxford University Press, New York, NY

- Sanzari JK, Billings PC, Wilson JM, et al (2015) Effect of electron radiation on vasomotor function of the left anterior descending coronary artery. *Life Sci Space Res (Amst)* 4:6–10. <https://doi.org/10.1016/j.lssr.2014.11.002>
- Sargsyan A, Hamilton D, Melton S, Young J (2006) The International Space Station ultrasound imaging capability overview for prospective users. NASA/TP-2006-213731. NASA Center for AeroSpace Information, Hanover, MD
- Sasi SP, Park D, Muralidharan S, et al (2015) Particle Radiation-Induced Nontargeted Effects in Bone-Marrow-Derived Endothelial Progenitor Cells. *Stem Cells Int* 2015:496512. <https://doi.org/10.1155/2015/496512>
- Sasi SP, Yan X, Zuriaga-Herrero M, et al (2017) Different Sequences of Fractionated Low-Dose Proton and Single Iron-Radiation-Induced Divergent Biological Responses in the Heart. *Radiat Res* 188:191–203. <https://doi.org/10.1667/RR14667.1>
- Savelieva I, Yap YG, Yi G, et al (1999) Relation of ventricular repolarization to cardiac cycle length in normal subjects, hypertrophic cardiomyopathy, and patients with myocardial infarction. *Clin Cardiol* 22:649–654
- Sawin CF, Baker E, Black FO (1998) Medical investigations and resulting countermeasures in support of 16-day Space Shuttle missions. *J Gravit Physiol* 5:1–12
- Schneider SM, Lee SMC, Macias BR, et al (2009) WISE-2005: exercise and nutrition countermeasures for upright VO₂pk during bed rest. *Med Sci Sports Exerc* 41:2165–2176. <https://doi.org/10.1249/MSS.0b013e3181aa04e5>
- Schneider SM, Watenpaugh DE, Lee SMC, et al (2002) Lower-body negative-pressure exercise and bed-rest-mediated orthostatic intolerance. *Med Sci Sports Exerc* 34:1446–1453. <https://doi.org/10.1249/01.MSS.0000027761.31366.06>
- Schöllnberger H, Eidemüller M, Cullings HM, et al (2018) Dose-responses for mortality from cerebrovascular and heart diseases in atomic bomb survivors: 1950–2003. *Radiat Environ Biophys* 57:17–29. <https://doi.org/10.1007/s00411-017-0722-5>
- Schöllnberger H, Kaiser JC, Jacob P, Walsh L (2012) Dose-responses from multi-model inference for the non-cancer disease mortality of atomic bomb survivors. *Radiat Environ Biophys* 51:165–178. <https://doi.org/10.1007/s00411-012-0410-4>
- Schondorf R, Low PA (1992) Gender related differences in the cardiovascular responses to upright tilt in normal subjects. *Clin Auton Res* 2:183–187
- Scott JM, Martin D, Ploutz-Snyder R, et al (2018) Efficacy of Exercise and Testosterone to Mitigate Atrophic Cardiovascular Remodeling. *Medicine & Science in Sports & Exercise* 50:1940–1949. <https://doi.org/10.1249/MSS.0000000000001619>

- Seawright JW, Sridharan V, Landes RD, et al (2019) Effects of low-dose oxygen ions and protons on cardiac function and structure in male C57BL/6J mice. *Life Sci Space Res (Amst)* 20:72–84. <https://doi.org/10.1016/j.lssr.2019.01.003>
- Sera N, Hida A, Imaizumi M, et al (2013) The association between chronic kidney disease and cardiovascular disease risk factors in atomic bomb survivors. *Radiat Res* 179:46–52. <https://doi.org/10.1667/RR2863.1>
- Shackelford LC, LeBlanc AD, Driscoll TB, et al (2004) Resistance exercise as a countermeasure to disuse-induced bone loss. *J Appl Physiol* 97:119–129. <https://doi.org/10.1152/japplphysiol.00741.2003>
- Shaw E, Tofler GH (2009) Circadian rhythm and cardiovascular disease. *Curr Atheroscler Rep* 11:289. <https://doi.org/10.1007/s11883-009-0044-4>
- Shi S-J, South DA, Meck JV (2004) Fludrocortisone does not prevent orthostatic hypotension in astronauts after spaceflight. *Aviat Space Environ Med* 75:235–239
- Shibao C, Grijalva CG, Raj SR, et al (2007) Orthostatic hypotension-related hospitalizations in the United States. *Am J Med* 120:975–980. <https://doi.org/10.1016/j.amjmed.2007.05.009>
- Shibata S, Perhonen M, Levine BD (2010) Supine cycling plus volume loading prevent cardiovascular deconditioning during bed rest. *J Appl Physiol* 108:1177–1186. <https://doi.org/10.1152/japplphysiol.01408.2009>
- Shimizu Y, Kodama K, Nishi N, et al (2010) Radiation exposure and circulatory disease risk: Hiroshima and Nagasaki atomic bomb survivor data, 1950-2003. *BMJ* 340:b5349. <https://doi.org/10.1136/bmj.b5349>
- Shkolnikov VM, Cornia GA, Leon DA, Meslé F (1998) Causes of the Russian mortality crisis: Evidence and interpretations. *World Dev* 26:1995–2011. [https://doi.org/10.1016/S0305-750X\(98\)00102-8](https://doi.org/10.1016/S0305-750X(98)00102-8)
- Shoemaker JK, Hogeman CS, Khan M, et al (2001) Gender affects sympathetic and hemodynamic response to postural stress. *Am J Physiol Heart Circ Physiol* 281:H2028–H2035
- Shoemaker JK, Usselman CW, Rothwell A, Wong SW (2012) Altered cortical activation patterns associated with baroreflex unloading following 24 hours of physical deconditioning. *Exp Physiol*. <https://doi.org/10.1113/expphysiol.2012.065557>
- Shusterman V, Goldberg A, London B (2006) Upsurge in T-wave alternans and nonalternating repolarization instability precedes spontaneous initiation of ventricular tachyarrhythmias in humans. *Circulation* 113:2880–2887. <https://doi.org/10.1161/CIRCULATIONAHA.105.607895>

- Shykoff BE, Farhi LE, Olszowka AJ, et al (1996) Cardiovascular response to submaximal exercise in sustained microgravity. *J Appl Physiol* 81:26–32
- Sigaud D, Fortrat JO, Allevard AM, et al (1998) Changes in the sympathetic nervous system induced by 42 days of head-down bed rest. *Am J Physiol* 274:H1875-1884
- Simonetto C, Azizova TV, Grigoryeva ES, et al (2014) Ischemic heart disease in workers at Mayak PA: latency of incidence risk after radiation exposure. *PLoS One* 9:e96309. <https://doi.org/10.1371/journal.pone.0096309>
- Simonetto C, Schöllnberger H, Azizova TV, et al (2015) Cerebrovascular Diseases in Workers at Mayak PA: The Difference in Radiation Risk between Incidence and Mortality. *PLoS One* 10:e0125904. <https://doi.org/10.1371/journal.pone.0125904>
- Simonsen LC, Slaba TC, Guida P, Rusek A (2020) NASA's first ground-based Galactic Cosmic Ray Simulator: Enabling a new era in space radiobiology research. *PLoS Biol* 18:e3000669. <https://doi.org/10.1371/journal.pbio.3000669>
- Singh VK, Santiago PT, MacVittie TJ (2018) Opportunities and challenges with animal models for acute radiation syndrome drug discovery. *Expert Opin Drug Discov* 13:987–992. <https://doi.org/10.1080/17460441.2018.1526172>
- Singh VK, Seed TM (2018) An update on sargramostim for treatment of acute radiation syndrome. *Drugs Today (Barc)* 54:679–693. <https://doi.org/10.1358/dot.2018.54.11.2899370>
- Singh VK, Seed TM (2019) The efficacy and safety of amifostine for the acute radiation syndrome. *Expert Opin Drug Saf* 18:1077–1090. <https://doi.org/10.1080/14740338.2019.1666104>
- Singh VK, Seed TM (2021) Radiation countermeasures for hematopoietic acute radiation syndrome: growth factors, cytokines and beyond. *Int J Radiat Biol* 97:1526–1547. <https://doi.org/10.1080/09553002.2021.1969054>
- Sishc BJ, Zawaski J, Saha J, et al (2022) The Need for Biological Countermeasures to Mitigate the Risk of Space Radiation-Induced Carcinogenesis, Cardiovascular Disease, and Central Nervous System Deficiencies. *Life Sciences in Space Research*. <https://doi.org/10.1016/j.lssr.2022.06.003>
- Smit AAJ, Wieling W, Fujimura J, et al (2004) Use of lower abdominal compression to combat orthostatic hypotension in patients with autonomic dysfunction. *Clin Auton Res* 14:167–175. <https://doi.org/10.1007/s10286-004-0187-x>
- Smith MG, Kelley M, Basner M (2020) A brief history of spaceflight from 1961 to 2020: An analysis of missions and astronaut demographics. *Acta Astronaut* 175:290–299. <https://doi.org/10.1016/j.actaastro.2020.06.004>

- Smith SM, Heer MA, Shackelford LC, et al (2012) Benefits for bone from resistance exercise and nutrition in long-duration spaceflight: Evidence from biochemistry and densitometry. *J Bone Miner Res* 27:1896–1906. <https://doi.org/10.1002/jbmr.1647>
- Smith SM, Zwart SR (2015) Magnesium and Space Flight. *Nutrients* 7:10209–10222. <https://doi.org/10.3390/nu7125528>
- Smith SM, Zwart SR, Block G, et al (2005) The nutritional status of astronauts is altered after long-term space flight aboard the International Space Station. *J Nutr* 135:437–443
- Smith SM, Zwart SR, Kloeris V, Heer M (2009) *Nutritional Biochemistry of Space Flight*. Nova Science Publishers, Inc., New York, NY
- Soloviev AI, Tishkin SM, Parshikov AV, et al (2003) Mechanisms of endothelial dysfunction after ionized radiation: selective impairment of the nitric oxide component of endothelium-dependent vasodilation. *Br J Pharmacol* 138:837–844. <https://doi.org/10.1038/sj.bjp.0705079>
- Soucy KG, Lim HK, Kim JH, et al (2011) HZE ⁵⁶Fe-ion irradiation induces endothelial dysfunction in rat aorta: role of xanthine oxidase. *Radiat Res* 176:474–485. <https://doi.org/10.1667/rr2598.1>
- Spaak J, Montmerle S, Sundblad P, Linnarsson D (2005) Long-term bed rest-induced reductions in stroke volume during rest and exercise: cardiac dysfunction vs. volume depletion. *J Appl Physiol* 98:648–654. <https://doi.org/10.1152/jappphysiol.01332.2003>
- Spitz DR, Azzam EI, Li JJ, Gius D (2004) Metabolic oxidation/reduction reactions and cellular responses to ionizing radiation: a unifying concept in stress response biology. *Cancer Metastasis Rev* 23:311–322. <https://doi.org/10.1023/B:CANC.0000031769.14728.bc>
- Sridharan V, Seawright JW, Landes RD, et al (2020) Effects of single-dose protons or oxygen ions on function and structure of the cardiovascular system in male Long Evans rats. *Life Sci Space Res (Amst)* 26:62–68. <https://doi.org/10.1016/j.lssr.2020.04.002>
- Stahn AC, Werner A, Opatz O, et al (2017) Increased core body temperature in astronauts during long-duration space missions. *Sci Rep* 7:16180. <https://doi.org/10.1038/s41598-017-15560-w>
- Stearner SP, Yang VV, Devine RL (1979) Cardiac injury in the aged mouse: comparative ultrastructural effects of fission spectrum neutrons and gamma rays. *Radiat Res* 78:429–447
- Stein TP, Schuler MD, Boden G (1994) Development of insulin resistance by astronauts during spaceflight. *Aviat Space Environ Med* 65:1091–1096

- Stenger MB, Brown AK, Lee SMC, et al (2010) Gradient compression garments as a countermeasure to post-spaceflight orthostatic intolerance. *Aviat Space Environ Med* 81:883–887
- Stenger MB, Evans JM, Knapp CF, et al (2012) Artificial gravity training reduces bed rest-induced cardiovascular deconditioning. *Eur J Appl Physiol* 112:605–616. <https://doi.org/10.1007/s00421-011-2005-1>
- Stenger MB, Evans JM, Patwardhan AR, et al (2007) Artificial gravity training improves orthostatic tolerance in ambulatory men and women. *Acta Astronautica* 60:267–272. <https://doi.org/10.1016/j.actaastro.2006.08.008>
- Stenger MB, Lee SMC, Ribeiro LC, et al (2014) Gradient compression garments protect against orthostatic intolerance during recovery from bed rest. *Eur J Appl Physiol* 114:597–608. <https://doi.org/10.1007/s00421-013-2787-4>
- Stenger MB, Platts SH, Lee SMC, et al (2015) Evidence Report:Risk of Orthostatic Intolerance During Re-exposure to Gravity. National Aeronautics and Space Administration Lyndon B. Johnson Space Center, Houston, TX
- Stevens PM, Lynch TN, Johnson RL, Lamb LE (1966) Effects of 9-alphaflurohydrocortisone and venous occlusive cuffs on orthostatic deconditioning of prolonged bed rest. *Aerosp Med* 37:1049–1056
- Stewart FA, Akleyev AV, Hauer-Jensen M, et al (2012) ICRP publication 118: ICRP statement on tissue reactions and early and late effects of radiation in normal tissues and organs--threshold doses for tissue reactions in a radiation protection context. *Ann ICRP* 41:1–322. <https://doi.org/10.1016/j.icrp.2012.02.001>
- Stewart FA, Heeneman S, Te Poele J, et al (2006) Ionizing radiation accelerates the development of atherosclerotic lesions in ApoE-/- mice and predisposes to an inflammatory plaque phenotype prone to hemorrhage. *Am J Pathol* 168:649–658. <https://doi.org/10.2353/ajpath.2006.050409>
- Summers RL, Martin DS, Meck JV, Coleman TG (2007) Computer systems analysis of spaceflight induced changes in left ventricular mass. *Comput Biol Med* 37:358–363. <https://doi.org/10.1016/j.combiomed.2006.04.003>
- Summers RL, Martin DS, Platts SH, et al (2010) Ventricular chamber sphericity during spaceflight and parabolic flight intervals of less than 1 G. *Aviat Space Environ Med* 81:506–510

- Sun H, Gusdon AM, Qu S (2016) Effects of melatonin on cardiovascular diseases: progress in the past year. *Curr Opin Lipidol* 27:408–413. <https://doi.org/10.1097/MOL.0000000000000314>
- Swerdlow AJ, Higgins CD, Smith P, et al (2007) Myocardial infarction mortality risk after treatment for Hodgkin disease: a collaborative British cohort study. *J Natl Cancer Inst* 99:206–214. <https://doi.org/10.1093/jnci/djk029>
- Tagawa H, Shimokawa H, Tagawa T, et al (1997) Short-term estrogen augments both nitric oxide-mediated and non-nitric oxide-mediated endothelium-dependent forearm vasodilation in postmenopausal women. *J Cardiovasc Pharmacol* 30:481–488
- Tanaka K, Nishimura N, Sato M, et al (2013) Arterial pressure oscillation and muscle sympathetic nerve activity after 20days of head-down bed rest. *Auton Neurosci*. <https://doi.org/10.1016/j.autneu.2013.02.025>
- Tapio S, Little MP, Kaiser JC, et al (2021) Ionizing radiation-induced circulatory and metabolic diseases. *Environ Int* 146:106235. <https://doi.org/10.1016/j.envint.2020.106235>
- Tascione TF (2010) Introduction to the Space Environment. Krieger Publishing Company
- Task Force of the European Society of Cardiology the North American Society of Pacing (1996) Heart Rate Variability Standards of Measurement, Physiological Interpretation, and Clinical Use. *Circulation* 93:1043–1065. <https://doi.org/10.1161/01.CIR.93.5.1043>
- Taylor A, Hobbs M (2002) The 2001 assessment of nutritional influences on risk of cataract. In: Rosenberg I, Sastre A (eds) Nestlé Nutrition Workshop Series Clinical & Performance Program. Nestec Ltd. Vevey/S. Karger AG, Basel, Switzerland, pp 163–191
- Thornton WE, Hoffler GW, Rummel J (1977) Anthropometric Changes and Fluid Shifts. In: Biomedical Results from Skylab. Scientific and Technical Information Office, National Aeronautics and Space Administration, Washington, D.C., pp 330–339
- Thornton WE, Moore TP, Pool SL (1987) Fluid shifts in weightlessness. *Aviat Space Environ Med* 58:A86-90
- Thosar SS, Butler MP, Shea SA (2018) Role of the circadian system in cardiovascular disease. *J Clin Invest* 128:2157–2167. <https://doi.org/10.1172/JCI80590>
- Trappe T, Trappe S, Lee G, et al (2006) Cardiorespiratory responses to physical work during and following 17 days of bed rest and spaceflight. *J Appl Physiol* 100:951–957. <https://doi.org/10.1152/japplphysiol.01083.2005>

- Tribble DL, Barcellos-Hoff MH, Chu BM, Gong EL (1999) Ionizing radiation accelerates aortic lesion formation in fat-fed mice via SOD-inhibitable processes. *Arterioscler Thromb Vasc Biol* 19:1387–1392. <https://doi.org/10.1161/01.atv.19.6.1387>
- Trinity JD, Drummond MJ, Fermoy CC, et al (2022) Cardiovasomobility: an integrative understanding of how disuse impacts cardiovascular and skeletal muscle health. *Journal of Applied Physiology* 132:835–861. <https://doi.org/10.1152/jappphysiol.00607.2021>
- Trudel G, Shahin N, Ramsay T, et al (2022) Hemolysis contributes to anemia during long-duration space flight. *Nat Med* 1–4. <https://doi.org/10.1038/s41591-021-01637-7>
- Tsao CW, Aday AW, Almarzooq ZI, et al (2022) Heart Disease and Stroke Statistics-2022 Update: A Report From the American Heart Association. *Circulation* 145:e153–e639. <https://doi.org/10.1161/CIR.0000000000001052>
- Tuday EC, Berkowitz DE (2007) Microgravity and cardiac atrophy: no sex discrimination. *J Appl Physiol* 103:1–2. <https://doi.org/10.1152/jappphysiol.00010.2007>
- Tuday EC, Meck JV, Nyhan D, et al (2007) Microgravity-induced changes in aortic stiffness and their role in orthostatic intolerance. *J Appl Physiol* 102:853–858. <https://doi.org/10.1152/jappphysiol.00950.2006>
- Tuday EC, Platts SH, Nyhan D, et al (2011) A Retrospective Analysis on Gender Differences in the Arterial Stiffness Response to Microgravity Exposure. *Gravitational and Space Biology* 25:
- Tungjai M, Whorton EB, Rithidech KN (2013) Persistence of apoptosis and inflammatory responses in the heart and bone marrow of mice following whole-body exposure to ²⁸Silicon (²⁸Si) ions. *Radiat Environ Biophys* 52:339–350. <https://doi.org/10.1007/s00411-013-0479-4>
- Udden MM, Driscoll TB, Pickett MH, et al (1995) Decreased production of red blood cells in human subjects exposed to microgravity. *J Lab Clin Med* 125:442–449
- Vernikos J (1997) Artificial gravity intermittent centrifugation as a space flight countermeasure. *J Gravit Physiol* 4:P13-16
- Videbaek R, Norsk P (1997) Atrial distension in humans during microgravity induced by parabolic flights. *J Appl Physiol* 83:1862–1866
- Vil-Viliams IF (1994) Principle approaches to selection of the short-arm centrifuge regimens for extended space flight. *Acta Astronaut* 33:221–229
- Vil-Viliams IF, Kotovskaya AR, Gavrilova LN, et al (1998) Human +Gx tolerance with the use of anti-G suits during descent from orbit of the Soyuz space vehicles. *J Gravit Physiol* 5:P129-130

- Vogt FB (1965) Effect of extremity cuff tourniquets on tilt table tolerance after water immersion. *Aerospace Medicine* 36:442–446
- Vongpatanasin W, Tuncel M, Mansour Y, et al (2001) Transdermal estrogen replacement therapy decreases sympathetic activity in postmenopausal women. *Circulation* 103:2903–2908
- Wakeford R (2022) Risk of diseases of the circulatory system after low-level radiation exposure—an assessment of evidence from occupational exposures. *J Radiol Prot* 42:. <https://doi.org/10.1088/1361-6498/ac6275>
- Warfield ME, Schneidkraut MJ, Ramwell PW, Kot PA (1990) WR2721 ameliorates the radiation-induced depression in reactivity of rat abdominal aorta to U46619. *Radiat Res* 121:63–66
- Warren JV, Brannon ES, Stead EA, Merrill AJ (1945) THE EFFECT OF VENESECTOMY AND THE POOLING OF BLOOD IN THE EXTREMITIES ON THE ATRIAL PRESSURE AND CARDIAC OUTPUT IN NORMAL SUBJECTS WITH OBSERVATIONS ON ACUTE CIRCULATORY COLLAPSE IN THREE INSTANCES. *J Clin Invest* 24:337–344. <https://doi.org/10.1172/JCI101611>
- Watenpaugh DE, Ballard RE, Schneider SM, et al (2000) Supine lower body negative pressure exercise during bed rest maintains upright exercise capacity. *J Appl Physiol* 89:218–227
- Watenpaugh DE, Fortney SM, Ballard RE, et al (1994) Lower body negative pressure exercise during bed rest maintains orthostatic tolerance. *FASEB J* 8:A261
- Watenpaugh DE, O’Leary DD, Schneider SM, et al (2007) Lower body negative pressure exercise plus brief postexercise lower body negative pressure improve post-bed rest orthostatic tolerance. *J Appl Physiol* 103:1964–1972. <https://doi.org/10.1152/japplphysiol.00132.2007>
- Waters WW, Platts SH, Mitchell BM, et al (2005) Plasma volume restoration with salt tablets and water after bed rest prevents orthostatic hypotension and changes in supine hemodynamic and endocrine variables. *Am J Physiol Heart Circ Physiol* 288:H839–847. <https://doi.org/10.1152/ajpheart.00220.2004>
- Waters WW, Ziegler MG, Meck JV (2002) Postspaceflight orthostatic hypotension occurs mostly in women and is predicted by low vascular resistance. *J Appl Physiol* 92:586–94. <https://doi.org/10.1152/japplphysiol.00676.2015>
- Westby CM, Martin DS, Lee SMC, et al (2016) Left ventricular remodeling during and after 60 days of sedentary head-down bed rest. *Journal of Applied Physiology* 120:956–964. <https://doi.org/10.1152/japplphysiol.00676.2015>
- White CR, Yu T, Gupta K, et al (2014) Heavy-ion (⁵⁶Fe) irradiation leads to impaired aortic relaxation prior to atherosclerotic plaque formation in ApoE^{-/-} mice. *J Radiat Res* 55 Suppl 1:i42–i43. <https://doi.org/10.1093/jrr/rrt190>

- White DD, Gotshall RW, Tucker A (1996) Women have lower tolerance to lower body negative pressure than men. *J Appl Physiol* 80:1138–1143
- White P, Nyberg J, Finney L, White W (1966) Influence of periodic centrifugation on cardiovascular functions of man during bed rest. Douglas Aircraft, co, Inc., San Monica, CA
- White RH (2003) The epidemiology of venous thromboembolism. *Circulation* 107:14-8. <https://doi.org/10.1161/01.CIR.0000078468.11849.66>
- Wilhelm M (2014) Atrial fibrillation in endurance athletes. *Eur J Prev Cardiol* 21:1040–1048. <https://doi.org/10.1177/2047487313476414>
- Wilhelm M, Roten L, Tanner H, et al (2011a) Atrial remodeling, autonomic tone, and lifetime training hours in nonelite athletes. *Am J Cardiol* 108:580–585. <https://doi.org/10.1016/j.amjcard.2011.03.086>
- Wilhelm M, Roten L, Tanner H, et al (2011b) Gender differences of atrial and ventricular remodeling and autonomic tone in nonelite athletes. *Am J Cardiol* 108:1489–1495. <https://doi.org/10.1016/j.amjcard.2011.06.073>
- Wilson JM, Sanzari JK, Diffenderfer ES, et al (2011) Acute biological effects of simulating the whole-body radiation dose distribution from a solar particle event using a porcine model. *Radiat Res* 176:649–659. <https://doi.org/10.1667/rr2541.1>
- Wilson JW, Kim M, Schimmerling W, et al (1995) Issues in space radiation protection: galactic cosmic rays. *Health Phys* 68:50–58. <https://doi.org/10.1097/00004032-199501000-00006>
- Wilson TE, Cui J, Zhang R, et al (2002) Skin cooling maintains cerebral blood flow velocity and orthostatic tolerance during tilting in heated humans. *J Appl Physiol* 93:85–91. <https://doi.org/10.1152/japplphysiol.01043.2001>
- Wolthuis RA, Bergman SA, Nicogossian AE (1974) Physiological effects of locally applied reduced pressure in man. *Physiol Rev* 54:566–595
- Wu B, Wang Y, Wu X, et al (2018) On-orbit sleep problems of astronauts and countermeasures. *Mil Med Res* 5:17. <https://doi.org/10.1186/s40779-018-0165-6>
- Wuu Y-R, Hu B, Okunola H, et al (2020) LET-Dependent Low Dose and Synergistic Inhibition of Human Angiogenesis by Charged Particles: Validation of miRNAs that Drive Inhibition. *iScience* 23:101771. <https://doi.org/10.1016/j.isci.2020.101771>
- Yamada M, Wong FL, Fujiwara S, et al (2004) Noncancer disease incidence in atomic bomb survivors, 1958-1998. *Radiat Res* 161:622–632. <https://doi.org/10.1667/rr3183>

- Yamamoto Y, Hughson RL (1991) Coarse-graining spectral analysis: new method for studying heart rate variability. *J Appl Physiol* 71:1143–1150
- Yan X, Sasi SP, Gee H, et al (2014) Cardiovascular risks associated with low dose ionizing particle radiation. *PLoS One* 9:e110269. <https://doi.org/10.1371/journal.pone.0110269>
- Yang VC (1993) Late structural changes in mouse coronary arteries after iron-particle irradiation of the orbital region. *Radiat Res* 134:390–393
- Yang VV, Ainsworth EJ (1982) Late effects of heavy charged particles on the fine structure of the mouse coronary artery. *Radiat Res* 91:135–144
- Yang VV, Stearner SP, Ainsworth EJ (1978) Late ultrastructural changes in the mouse coronary arteries and aorta after fission neutron or ⁶⁰Co gamma irradiation. *Radiat Res* 74:436–456
- Yang VV, Stearner SP, Tyler SA (1976) Radiation-induced changes in the fine structure of the heart: comparison of fission neutrons and ⁶⁰Co gamma rays in the mouse. *Radiat Res* 67:344–360
- Yao Y-J, Jiang S-Z, Jiang C-L, et al (2008) Effect of thigh cuffs on haemodynamic changes of the middle cerebral artery and on orthostatic intolerance induced by 10 days head-down bed rest. *Clin Exp Pharmacol Physiol* 35:1178–1182. <https://doi.org/10.1111/j.1440-1681.2008.04989.x>
- Yeung TK, Hopewell JW (1985) Effects of single doses of radiation on cardiac function in the rat. *Radiother Oncol* 3:339–345. [https://doi.org/10.1016/s0167-8140\(85\)80047-5](https://doi.org/10.1016/s0167-8140(85)80047-5)
- Yu AF, Jones LW (2015) Modulation of cardiovascular toxicity in Hodgkin lymphoma: potential role and mechanisms of aerobic training. *Future Cardiol* 11:441–452. <https://doi.org/10.2217/fca.15.29>
- Yu T, Parks BW, Yu S, et al (2011) Iron-ion radiation accelerates atherosclerosis in apolipoprotein E-deficient mice. *Radiat Res* 175:766–773. <https://doi.org/10.1667/RR2482.1>
- Zagar TM, Cardinale DM, Marks LB (2016) Breast cancer therapy-associated cardiovascular disease. *Nat Rev Clin Oncol* 13:172–184. <https://doi.org/10.1038/nrclinonc.2015.171>
- Zhao W, Robbins MEC (2009) Inflammation and chronic oxidative stress in radiation-induced late normal tissue injury: therapeutic implications. *Curr Med Chem* 16:130–143. <https://doi.org/10.2174/092986709787002790>
- Zhu W-G, Wan R, Din Y, et al (2016) Sex Differences in the Association Between Regular Physical Activity and Incident Atrial Fibrillation: A Meta-analysis of 13 Prospective Studies. *Clin Cardiol* 39:360–367. <https://doi.org/10.1002/clc.22531>

Zwart SR, Auñón-Chancellor SM, Heer M, et al (2022) Albumin, oral contraceptives, and venous thromboembolism risk in astronauts. J Appl Physiol (1985) 132:1232–1239. <https://doi.org/10.1152/japplphysiol.00024.2022>

Zwart SR, Gibson CR, Gregory JF, et al (2017) Astronaut ophthalmic syndrome. FASEB J 31:3746–3756. <https://doi.org/10.1096/fj.201700294>

Additional Resources

- Human Research Roadmap: <https://humanresearchroadmap.nasa.gov/>
- NASA Task Book: <https://taskbook.nasaprs.com/tbp/welcome.cfm>
- NASA Technical Report Server: <https://ntrs.nasa.gov/>
- NASA PubSpace: <https://www.ncbi.nlm.nih.gov/pmc/funder/nasa/>

XV. TEAM

Stuart M. C. Lee, PhD

Dr. Lee (KBR, Houston, TX) currently serves as the Human Research Program Health and Human Countermeasures Cardiovascular Discipline Lead, is one of the Human Systems Risk Board's Cardiovascular Risk Custodians, and is a Lead Scientist in the Johnson Space Center Cardiovascular and Vision Laboratory. Dr. Lee has supported NASA research and operations as a contractor since 1992 in the areas of exercise and cardiovascular physiology as well as countermeasure development and implementation. Dr. Lee is the author or co-author of more than 80 peer-reviewed publications, five book chapters, and eight Human Research Program Evidence Reports (four original versions and four revisions) and serves as a reviewer for multiple physiology and aerospace medicine journals. He has served as principal investigator or co-investigator for numerous NASA-sponsored studies, including NASA's Twins Study and international collaborations with the Russian Institute of Biomedical Problems ("Field Test" and "Fluid Shifts" Studies).

Janapriya Saha, PhD

Dr. Saha (Leidos, Houston, TX) serves as the Degenerative Tissues and Diseases Discipline Lead for the NASA Human Research Program's Space Radiation Element and as senior scientist and lead subject matter expert for Space Radiation Analysis Group. Dr. Saha has over 15 years of research experience in ionizing radiation-induced DNA damage signaling and repair with a strong emphasis on the characterization of space radiation-induced biological effects and charged particle radiotherapy. Prior to his current roles at NASA, Dr. Saha was a Faculty at Department of Radiation Oncology, University of Texas Southwestern Medical Center, Dallas. Currently, he manages the cardiovascular disease risk portfolio for the Space Radiation Element, providing the Element and the Human Research Program with management and scientific support. As a senior scientist and lead subject matter expert in the Space Radiation Analysis Group, he performs crew radiation risk analysis for the Radiation Health Office and critically reviews scientific literature on radiation health outcomes to support translation of research findings into evidence-based operational strategies.

Jacqueline Charvat, PhD

Dr. Charvat (KBR, Houston, TX) has served as the Senior Epidemiologist in the Lifetime Surveillance of Astronaut Health (LSAH), the occupational surveillance program for the NASA astronaut corps, for the last 10 years. She is responsible for understanding long term health implications of spaceflight, especially as related to cardiovascular disease and cancer, and serves as one of the Human Systems Risk Board's Cardiovascular Risk Custodians. She is author or co-author of multiple peer-review scientific publications, including several key epidemiological investigations cited in this evidence report. Dr. Charvat received her PhD in Epidemiology from Case Western Reserve University.

Rahul Suresh, MD, MS, MPH

Dr. Suresh (NASA Johnson Space Center) has been a Flight Surgeon at the NASA Johnson Space Center since 2018. During this time, he has served as the deputy crew surgeon for Expeditions 61-62 (Jessica Meir) and the prime crew surgeon for Expeditions 66-67 (Raja Chari, Tom Marshburn, Kayla Barron). He provides ongoing clinical care to astronauts and flight directors at the JSC Flight Medicine Clinic. Apart from his role as a Human Systems Risk Custodian, his duties include serving as a certified SpaceX Hawthorne Mission Control Specialist for SpaceX Crew Dragon launches and landings, the Flight Surgeon Lead for integration of Private Astronaut Missions and LEO Commercialization Programs, and the Deputy Lead of the Exploration Medical Integrated Product Team, for which he helps to identify and fund development of medical capabilities for future exploration missions. Dr. Suresh is dual board certified in Internal Medicine and Aerospace Medicine. Dr. Suresh provided his expertise in clinical and aerospace medicine as well as spaceflight operations in the development of this evidence report.

Steve Blattnig, PhD

Dr. Blattnig (NASA Langley Research Center) has been working on a wide variety of different aspects of space radiation research for over 20 years. He graduated from UW-Milwaukee with a Ph.D. in Physics, and his graduate work comprised the development of a pion and muon radiation transport code. More recently, his focus has been on the development of probabilistic risk methodology and radiation biology modeling for effects including acute radiation syndrome, cancer, cardiovascular disease, and degenerative central nervous system diseases. He was the project manager for the space radiation transport and measurement project and was the principal investigator of the space radiation risk assessment project.

Jason Lytle, PhD

Dr. Lytle (KBR, Houston, TX) currently is a senior scientist in the Cardiovascular and Vision Laboratory at NASA Johnson Space Center. Dr. Lytle provides his expertise in the development and testing of countermeasures in ground-based and analog studies and is a co-investigator in spaceflight investigations. Dr. Lytle received his PhD in applied exercise physiology from Texas A&M University where he studied the effects of diet on cardiovascular health, published multiple peer-reviewed manuscripts, and served as a reviewer for several scientific journals.

Steven S. Laurie, PhD

Dr. Laurie (KBR, Houston, TX) received a Ph.D. in Human Physiology from the University of Oregon and currently serves as the Human Research Program Health and Human Countermeasures SANS Discipline Lead, is one of the Human Safety Risk Board's SANS Risk Custodians, and provides subject matter expertise to the Cardiovascular Risk and the Venous Thromboembolism Concern. He is the Technical Lead in the Johnson Space Center Cardiovascular and Vision Laboratory, overseeing a team of scientists, biomedical engineers, sonographers, and technical support staff in the development, execution, and reporting of spaceflight, analog, and

ground-based studies. Since joining KBR in 2014, Dr. Laurie has led multiple spaceflight, analog, and ground research studies, including the “Fluid Shifts” ISS spaceflight study and the “VaPER” bed rest study, which was the first spaceflight analog study to elicit ocular changes similar to those observed in astronauts during spaceflight.

Brandon R. Macias, PhD

Dr. Macias (NASA Johnson Space Center) currently is the Director of the Cardiovascular and Vision Laboratory at NASA Johnson Space Center. He is responsible for leading a diverse team of scientists, bioengineers, and technical support staff to address human space flight risks and gaps. He serves as a subject matter expert for the Spaceflight Associated Neuro-ocular Syndrome (SANS) Risk, the Cardiovascular Risk, and the Venous Thromboembolism Concern. Dr. Macias received his Ph.D. in Kinesiology-Exercise Physiology with a certificate in the Space Life Sciences in 2012 from Texas A&M University. Over the course of his career, he has developed a broad space physiology research experience base from molecular and cell biology to human systems physiology. He currently is the principal investigator of three ISS flight projects, and co-investigator on multiple ground and analog studies, including a SANS bed rest analog countermeasures project, a parabolic flight study, and ground studies investigating fluid loading and compression garment countermeasures. Other past international collaborative projects include that with Russian Institute of Biomedical Problems to conduct experiments on the BION-M1 mission and on the “Fluid Shifts” ISS study where he served as co-investigator. Dr. Macias has developed a strong experience base in the space physiology field as evidenced by his 90 peer-reviewed articles and several career awards.

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XVI. ACRONYMS

1-G	one-gravity (Earth equivalent)
ACES	Advanced Crew Escape Suit (NASA)

AG	artificial gravity
AGS	anti-gravity suit (NASA garment)
AGBRESA	Artificial Gravity Bed Rest with the European Space Agency
AMI	acute myocardial infarction
BBDR	biologically-based dose response
BFO	blood forming organs
BMI	body mass index
bpm	beats per minute
CCLS	Cooper Center Longitudinal Study
CeVD	cerebrovascular disease
cGy	centi Gray (absorbed dose)
cIMT	carotid intima-media thickness
cm	centimeter
CVD	cardiovascular disease
CVP	central venous pressure
DAG	directed acyclic graph
DI	dry immersion
DRM	design reference mission
DVT	deep vein thrombosis
ECG	Electrocardiogram
EVA	extravehicular activity
FD	flight day
G	gravity
G-level	gravity level

GCG	Gradient Compression Garment
GCR	galactic cosmic radiation
Gy	Gray (absorbed dose of radiation)
HR	heart rate (bpm)
HRR	Human Research Roadmap
HRV	heart rate variability
hs-CRP	high-sensitivity C-reactive protein
hs-TnT	high-sensitivity troponin
HSRB	Human Systems Risk Board
HZE	high-energy heavy ions
ICRP	International Commission on Radiological Protection
IJV	internal jugular vein
ISS	International Space Station
LBNP	lower body negative pressure
LCG	liquid cooling garment
LEO	low Earth orbit
LES	Launch and Entry Suit (NASA)
LET	linear energy transfer
LSAH	Lifetime Surveillance of Astronaut Health (formerly Longitudinal Study of Astronaut Health)
LVEDV	left ventricular end-diastolic volume
LV	Left ventricle
LVM	left ventricular mass
mGy	milli Gray (absorbed dose of radiation)

mmHg	millimeters of mercury (pressure)
MI	myocardial infarction
MRI	magnetic resonance imaging
MTWA	microvolt T-wave alternans
NASA	National Aeronautics and Space Administration
NCRP	National Council on Radiation Protection & Measurements
NSRL	NASA Space Radiation Laboratory
OI	orthostatic intolerance
OIG	Orthostatic Intolerance Garment
PAC	premature atrial contraction
PCWP	pulmonary capillary wedge pressure
PE	pulmonary embolism
PVC	premature ventricular contraction
PY	person-year
RBE	relative biological effectiveness
REID	risk of exposure-induced death
SANS	spaceflight associated neuro-ocular syndrome
SBP	systolic blood pressure
SMR	standardized mortality rate
SMS	space motion sickness
SPE	solar particle event
STS	Space Transportation System (Space Shuttle)
THIR	Targeted heavy ion irradiation
TTE	Transthoracic Echocardiogram

VTE venous thromboembolism

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XVII. Appendix A: Epidemiological Evidence of Cardiovascular Disease in Astronauts

Authors	Title	Sample	Cardiovascular Disease Outcomes	Results	Conclusion
(Peterson et al. 1993)	Longitudinal Study of Astronaut Health: mortality in years 1959-1991	Astronauts selected through 1991 compared to US general population	Deaths through 9/30/1991 SMR for Circulatory Disease SMR for Ischemic Heart Disease	2 deaths (of 195 astronauts) (of 20 deaths) due to circulatory disease 47 64	Small number of CVD deaths among astronauts and astronauts had a lower rate of CVD deaths than the general US population.
(Hamm et al. 2000)	Design and Current Status of the Longitudinal Study of Astronaut Health	Astronauts selected through 1991 compared to JSC Civil Service sample	Crude Mortality Rate	3 CVD deaths for 3901 PY among astronauts compared to 7 deaths for 12471 PY for comparisons (Adjusted RR = 1.20, 0.27-5.28)	Small number of CVD deaths among both study groups. Astronauts had a higher relative risk than the comparison group, but this was not statistically significant.
(Reynolds and Day 2010)	Mortality Among US Astronauts 1980-2009	Astronauts selected 1980 – 2009 compared to Harris County TX general population.	Circulatory Deaths SMR for Circulatory Disease Deaths	5 CVD deaths for 6145 PY 27 (9-63)	Small number of CVD deaths in the astronauts. Astronauts had a lower rate of CVD death compared to the general population of Harris County, TX (the

			(compared to Harris County TX)		location of Johnson Space Center)
(Reynolds et al. 2014)	Mortality among Soviet and Russian cosmonauts: 1960-2013	Russian Cosmonauts compared to NASA Astronauts	SMR for Circulatory Disease Deaths in cosmonauts compared to astronauts	364 (225-557)	Cosmonauts at greater risk of all-cause mortality in comparison to U.S. astronauts (SMR = 190, 95% C.I. 154-239), as well as circulatory disease (SMR = 364, 95% C.I. 225-557) compared to U.S. astronauts. However, cosmonauts had a lower risk of fatal accidents (SMR = 88, 95% C.I. = 54-136) than their U.S. counterparts.
(Reynolds and Day 2017)	Mortality Due to Cardiovascular Disease among Apollo Lunar Astronauts	All NASA male astronauts selected through 2013 + US Air Force astronauts; all followed until 2015 for vital status	CVD Deaths across study period (1968-2015) <u>SMR</u> Apollo : Nonflight Apollo : LEO Apollo : Non-lunar Apollo : US	117 (24-343) 72 (15-210) 81 (17-237) 15 (3-43)	Apollo astronauts compared with LEO, non-Lunar and nonflight astronauts had similar rates of CVD death. Apollo astronauts had a lower rate of CVD death than the general US population.
(Reynolds and Day 2019)	Mortality of US astronauts: comparisons with professional athletes	All male NASA astronauts selected through 2013 compared to	Explored difference in mortality between astronauts and		Astronauts had lower rates of CVD death than the general US population along with professional NBA basketball players. (MLB

		professional baseball and basketball players along with US General population. Followed through 5/31/2018	professional athletes <u>SMR CVD</u> US General population NBA Basketball players	25 (11-50) 36 (16-72)	players could only be compared to astronauts on all-cause mortality).
(Elgart et al. 2018)	Radiation Exposure and Mortality from Cardiovascular Disease and Cancer in Early NASA Astronauts	Astronauts selected between 1959-1969 (homogenous cohort) compared to US white male general population	<u>SMR CVD</u>	33 (14-65)	34/73 had died as of 2/16/17 with XXXX due to CVD. Mortality rate was lower among astronauts compared with general population. There was no excess risk of CVD death due to radiation exposure.
(Charvat et al. 2022)	Long-term cardiovascular risk in astronauts: Comparing NASA mission astronauts to a healthy cohort from the Cooper Center Longitudinal Study	All astronaut selected between 1959 and 2009 who flew at least one spaceflight mission compared to matched CCLS participants	Crude Mortality	1.25 deaths [astronauts]/ 1000 PY vs. 1.00 deaths/1000 PY [CCLS]	Similar unadjusted mortality rates for astronauts compared with a well-matched cohort

XVIII. Appendix B: Summary of Animal and Cellular Studies for Cardiovascular Effects from Space-Relevant Radiation (Adapted and recreated from: Davis CM et al. 2021)

Radiation dose (Gy); Exposure type	Animal/Cellular Models and Methods	Key Findings/Conclusions	Reference
ANIMAL/FUNCTIONAL PHYSIOLOGY			
¹ H 0.5 Gy, 1 GeV ⁵⁶ Fe 0.15 Gy, 1 GeV	Male C57Bl/6NT mice, 8-10 months old, with/without MI, TTE evaluation	Healthy mice • ¹H- Initial improvement in cardiac function (determined by echocardiography and hemodynamic measurements) at 1 month that declines by 10-month post-irradiation. • ⁵⁶Fe- significant decline in cardiac function at 1 month and 3 months, but recovery by 10 months. MI-mice • ¹H prior to MI improved cardiac function restoration and enhanced cardiac remodeling. • ⁵⁶Fe prior to MI led to poorer cardiac function and more adverse remodeling.	(Yan et al. 2014)
Sham-irradiation, 0.1 Gy of ¹ H (150 MeV), 0.5 Gy of ⁵⁶ Fe (600 MeV/n), or 0.1 Gy of ¹ H 24 hours prior to 0.5 Gy of ⁵⁶ Fe	Male C57BL/6 mice, 10 weeks old, examined selected proteins.	• ⁵⁶Fe caused an increase in expression of α-smooth muscle cell actin, collagen type III, the inflammatory cell markers mast cell • Tryptase, CD2 and CD68, the endothelial glycoprotein thrombomodulin, and cleaved caspase 3. • ¹H induced a small increase only in cleaved caspase 3 levels. • Exposure to ¹H 24 hours before ⁵⁶Fe prevented all of the responses to ⁵⁶Fe. Low dose of ¹H may prime the heart to • respond differently to a subsequent challenge dose of heavy ions.	(Ramadan et al. 2016)
¹⁶ O, 600 MeV/n 0, 0.05, 0.1, 0.25, or 1 Gy)	Male C57BL/6J mice, 6 months old, echocardiography	• ¹⁶O -Mild changes in cardiac function (ejection fraction and fractional shortening) and mild tissue remodeling determined by echocardiography.	(Seawright et al. 2019)

^1H (150 MeV, 0, 0.5, or 1 Gy), ^1H (150 MeV, 0.5 Gy) followed by ^{16}O (0.1 Gy).	evaluation of cardiac function and blood velocity at 3, 5, 7, and 9 months after irradiation	^1H or ^1H followed by ^{16}O- No change in cardiac function.	
0.1, 0.25Gy, 600MeV/n. Dose Rate = 0.25 – 0.26Gy/min.	Female C57BL/6J mice, 6 months old	Overall, no cardiac fibrosis or cardiac dysfunction was observed in female mice. Some increase in LV mass, diastolic volume, and diameter was observed; however, no changes to abdominal aorta. Indications of minor cardiac remodeling in the form of increased 75 kDa peptide of left ventricular collagen type III and α-smooth cell actin observed.	(Nemec-Bakk et al. 2022)
2 and 5 Gy ^{56}Fe , 600 MeV/n, targeted to specific arterial sites (upper aortic tree)	Male apoE $^{-/-}$ mice, 10 weeks old	^{56}Fe accelerated development of atherosclerosis in irradiated portions of the aorta. - Larger necrotic cores associated with greater numbers of apoptotic macrophages and reduced lesional collagen. - Intima media thickening of the carotid arteries was exacerbated.	(Yu et al. 2011)
3-fractionated doses of 17 cGy ^1H every other day ($^1\text{H} \times 3$). 3-fractionated doses of 17 cGy ^1H every other day followed by 15 cGy ^{56}Fe two days after the final ^1H dose ($^1\text{H} \times 3 + ^{56}\text{Fe}$). Single low dose of 15 cGy ^{56}Fe followed (after	Male C57Bl/6NT mice, 8 – 10 months old. Subgroup underwent MI surgery 28 days post IR Cardiac structure and function were assessed in all animals at days 7, 14 and 28 after MI surgery was performed	- Healthy: No negative effect on cardiac function or structure. - MI: No negative effect post MI/post radiation exposure. No negative effect on cardiac function or structure. - After MI: 24% increase in mortality, significant decrease in LV function, and 35% increase in infarct size. - Healthy: depressed LV functions at one month with concomitant enhancement in cardiac fibrosis and induction of cardiac hypertrophy signaling after 3 months. - MI: No negative effect post MI/post radiation	(Sasi et al. 2017)

2 days) by three fractionated doses of 17 cGy ^1H every other day.			
0.1Gy, 150 MeV ^1H + 0.5Gy, 600 MeV/n ^{56}Fe ; WBE	Male C57Bl/6J mice, 10 weeks old.	No changes suggestive of myocardial remodeling	(Ramadan et al. 2016)
50, 100, or 200 cGy ^{137}Cs γ -rays (0.662 MeV), 15, 25, or 50 cGy ^{16}O (600 MeV/n), 15, 25, or 50 cGy ^{56}Fe (1,000 MeV/n), or 150 cGy GCR 5-ion (^1H 1000 MeV/n, ^{28}Si 600 MeV/n, ^4He 250 MeV/n, ^{16}O 350 MeV/n, ^{56}Fe 600 MeV/n, and ^1H 250 MeV/n.	Male C57BL/6 mice, 6 months old	- Decreased systolic function and increased arterial elastance one year after GCR_{5-ion} only, not after ^{56}Fe, ^{16}O, or γ-radiation. Changes in aortic elastic fiber organization. - Alterations in extracellular matrix of heart tissue.	(Bishawi et al. 2022)
Whole-body ^1H (250 MeV, 0.5 Gy) or ^{16}O , 600 MeV/n, 0.5 Gy.	Male Long Evans rats, ultrasonography and blood flow pulsed wave Doppler at 3, 5, 9, and 12 months post rad	No significant change in cardiac function with ^1H or ^{16}O (decrease in LV posterior wall thickness at 3-5 months, but no change in echocardiographic measures). Mild changes in aortic vascular function following ^{16}O irradiation.	(Sridharan et al. 2020)
^{56}Fe 1.0 Gy, 1GeV/n; targeted exposure of the aorta.	Male Wistar rats, 3- 4 months old, pulse	Rats exposed to 1 Gy ^{56}Fe exhibited significantly increased aortic stiffness and impaired endothelial- dependent relaxation consistent with endothelial dysfunction.	(Soucy et al. 2011)

	wave velocity, aortic ring assays		
Carbon (5–15 Gy) targeted to heart 2 weeks after MI.	Rabbit, MI model, In-vivo epicardial potential mapping on the free wall.	• In the context of MI model, carbon ion improved conduction, decreased the spatial heterogeneity of repolarization, and reversed vulnerability to ventricular arrhythmias	(Amino et al. 2006)
0.1–0.2Gy; targeted exposure of the orbital region	Female B6CF1 mice, 4 months old	• Degenerative changes in coronary arteries: smooth muscle degeneration with fibrosis and ECM accumulation in the tunica media.	(Yang 1993)
Targeted heavy ion radiation (THIR) (5–15 Gy), carbon ions 290 MeV/u.	Rabbit, New Zealand White, Longitudinal echocardiography (UCG) and electrocardiography (ECG) before, 2 wk., and 3, 6, and 12 mo. after 15 Gy Carbon. In vivo electrophysiological study (EPS) was performed 1 yr after carbon irradiation.	• No significant changes in cardiac function 1 year post 15 Gy carbon ion irradiation.	(Amino et al. 2010)
MOLECULAR BIOLOGY/BIOMARKERS/PATHWAYS FROM TISSUES OR CELLS			
0.1, 0.25, or 1 Gy of ^{16}O (600 MeV/n).	Male C57BL/6J mice, heart, LV, cardiac methylome and one-carbon metabolism	• Dynamic changes in cardiac epigenome and metabolome were until 90 days post ^{16}O exposure. • At 14 days, 0.25 and 1 Gy ^{16}O elicited global DNA hypomethylation in the 50-UTR of LINE-1. At 90 days, specific LINE-1 elements were hypermethylated. Gene expression showed first a decrease followed by an increase in transcript abundance.	(Miousse et al. 2019)

		• Metabolomics analysis homocysteine re-methylation were unaffected by radiation, but the trans-sulfuration pathway was impacted after 90 days.	
^{16}O , 600 MeV/n 0, 0.05, 0.1, 0.25, or 1 Gy) ^1H (150 MeV, 0, 0.5, or 1 Gy), ^1H (150 MeV, 0.5 Gy) followed by ^{16}O (0.1 Gy).	Male C57BL/6 J mice, 6 months old, cardiac tissue was collected to assess apoptosis, tissue remodeling, and markers of immune cells immune cell infiltration, histology.	• Evidence of mild apoptosis observed with each radiation type as well as histological evidence of cardiac tissue remodeling. • Increased evidence of immune cell infiltration was observed in heart. • ^{16}O was more damaging than ^1H and ^1H followed by ^{16}O was least damaging.	(Seawright et al. 2019)
THIR (5–15 Gy), Carbon ion.	Rabbit	• Cx43 protein and mRNA significantly upregulated in the ventricular myocardium from 2 week up to 1 year. • All regions of the LV, RV, and septum were similarly affected in terms of Cx43 upregulation. Cx43 was present in intercalated disk region, lateral surface of ventricular myocytes, and sarcolemma.	(Amino et al. 2010)
THIR (5–15 Gy) targeted to heart 2 weeks after MI.	New Zealand White Rabbits, MI model.	• THIR was associated with an increase of Cx43mRNA and protein levels in the LV in control as well as in MI rabbits. • THIR also increased lateralization of Cx43, which was no longer colocalized with cadherins. In MI hearts, immunoreactive Cx43 signals were reduced in the peri-infarct zone, and the reduction was reversed by THIR.	(Amino et al. 2006)
^1H (0.1 Gy) ^{56}Fe (0.5 Gy)	C57BL/6J male mice	• Cardiac DNA methylation associated with repetitive elements detected. • Modest hypomethylation of retrotransposon LINE-1 was observed at day 7 after irradiation with either ^1H or ^{56}Fe. LINE-1, and other retrotransposons, ERV2 and SINE B1,	(Koturbash et al. 2016)

		other major satellite DNA was hypermethylated at day 90 post ^{56}Fe. • Alterations in the expression of DNA methylation machinery that involve the one-carbon metabolism pathway. Substantial accumulation of mRNA transcripts, associated with major satellites seen at day 90.	
Sham-irradiation, 0.1 Gy of ^1H (150 MeV), 0.5 Gy of ^{56}Fe (600 MeV/n), or 0.1 Gy of ^1H 24 hours prior to 0.5 Gy of ^{56}Fe	Male C57BL/6 mice at 10 weeks of age, examined selected proteins.	• ^{56}Fe caused an increase in expression of α-smooth muscle cell actin, collagen type III, the inflammatory cell markers mast cell tryptase, CD2 and CD68, the endothelial glycoprotein thrombomodulin, and cleaved caspase 3. • ^1H induced a small increase only in cleaved caspase 3 levels. • Exposure to ^1H 24 hours before ^{56}Fe prevented all the responses to ^{56}Fe. Low dose of ^1H may prime the heart to respond differently to a subsequent challenge dose of heavy ions.	(Ramadan et al. 2016)
900 mGy of 1 GeV ^1H 150 mGy of 1 GeV/n ^{56}Fe	Cardiomyocytes from male C57BL/6 mice followed-up for 28 days human endothelial cells (HUVECs) cultured for 7 days on the International Space Station (ISS). Bioinformatics comparison of mouse cells on ground versus cells	• Common molecular pathways (FYN being central driver/hub) between simulated space radiation and HUVECs flown on the ISS were found. • Known oxidative stress induced immediately following radiation would only be transient and would upregulate FYN, which in turn would reduce reactive oxygen species (ROS) levels, protecting the cardiovascular system. • The transcriptomic signature of exposure to ^1H was closer to the spaceflight signature than ^{56}Fe signature.	(Beheshti et al. 2019)

	in space. <i>Re-analysis of Coleman et al. 2015.</i>		
γ -rays: ^{137}Cs (40-160 cGy, 0.662 MeV), ^{14}Si -IR (4-32 cGy, 260 MeV/n), ^{22}Ti -IR (3-26 cGy, 1 GeV/n).	Female CB6F1/Hsd mice, 3- to 4 months old.	• Transcriptomic analysis 16-month post IR revealed more profound effects with HZE particles at lower doses than γ-rays; also dependent upon quality and quantity of radiation. • 12 common DEGs across all 3 IR types. These genes revealed an exclusive non-linear DEG pattern in Si and Ti exposed hearts, contrary to a linear pattern of DEGs observed in a majority of γ-rays exposed hearts. • Evidence of lack of a threshold dose for biological effects from ionizing radiation exposures.	(Garikipati et al. 2021)
Whole body exposure. 0, 0.1, 0.25, or 0.5 Gy of 300 MeV/n ^{28}Si (two exposures, 15 days apart)	Male CBA/CaJ mice measured selected proteins in heart and bone marrow: cleaved poly (ADP-ribose) polymerase (cleaved PARP, activated nuclear factor-kappa B (NF-KB) and selected NF-KB regulated cytokines	• Up to 6 months post-radiation, cell death and inflammatory responses in tissues from the heart and BM from exposed mice were statistically higher than those in sham controls. • Overall suggestive of chronic apoptotic cell death and inflammation.	(Tungjai et al. 2013)
Whole-body ^1H (250 MeV, 0.5 Gy) or ^{16}O , 600 MeV/n, 0.5 Gy or	Tissues from male Long Evans rats.	• No significant changes in histopathology or histological quantification of total collagens in heart or aorta. An increase in a 75 kDa peptide of collagen type III in LV of	(Sridharan et al. 2020)

^{16}O (600 MeV/n and 1 GeV/n).		exposed to ^1H and ^{16}O . ^{16}O irradiation increased left ventricular protein levels of immune cell markers CD2, CD4, CD8, and CD68.	
90 cGy, 1 GeV ^1H 15 cGy, 1 GeV/ ^{56}Fe 28 days after exposure.	Cardiomyocytes from mice exposed to radiation. Re-analyzed in Beheshti et al. 2019	• Molecular responses and gene expression to ^{56}Fe were unique and long lasting. ^{56}Fe showed the greatest level of gene modulation. ^1H little differential transcript modulation. Major networks affected cell cycle, oxidative responses, and transcriptional regulation functional groups. • ^{56}Fe regulates ERK1/2, p38 MAPK, NFATc4, GATA4, STAT3, and NF-KB.	(Coleman et al. 2015)Coleman et al. 2015.
^1H 0.5 Gy, 1 GeV ^{56}Fe 0.15 Gy, 1 GeV	C57Bl/6NT male, with/without MI. Western blot for proteins in key pathways.	• ^1H- increased pro-survival factors in cardiac tissues long term ^{56}Fe decreased angiogenesis and pro-survival factors in cardiac tissues long term.	(Yan et al. 2014)
CULTURED CELLS			
Ti, ^{56}Fe , C up to 7 Gy	Avian cardiomyocyte cultures, electrophysiology	• Capacity to repair induced DNA damage within 24 h up to 7 Gy complete recovery in proliferative behavior. No significant effects on apoptosis, beat rate remained unaffected.	(Heselich et al. 2018)
(Ni) (LET, 183 keV/um) 0.5, 2 and 5 Gy.	Human endothelial cell line (EA.hy926). DNA damage 2 and 24 h following irradiation by γ -H2AX foci. Gene expression changes (microarrays at 8 and 24 h post-radiation).	• Persistent DNA damage response up to 24 h after treatment. • Downregulation of genes involved in cell cycle regulation. • Upregulation of genes involved in cell cycle checkpoint, DNA damage response, oxidative stress, apoptosis and cell-cell signaling (cytokines). • E2F and NFkB may be involved.	(Beck et al. 2014b)

2 Gy of X-ray. ^{56}Fe ions (1 GeV/u, 155 keV/um)	ECs (human coronary artery) transduced with retroviruses bearing the est2 gene. Proteomics, transcriptomics, and cytokine analysis, evaluated 1 and 7 days post radiation	Cell cycle, cell adhesion, and caveolar mediated endocytosis signaling affected in time and radiation type dependent manner. Transcriptome and proteome effects were more pronounced and longer lasting for ^{56}Fe ions than for X-rays. ^{56}Fe and x-Ray decreased the expression of genes involved in cell-cell adhesion and enhanced the expression of proteins involved in caveolar mediated endocytosis signaling.	(Baselet et al. 2017)
Low-LET ^1H and high-LET ^{56}Fe .	3D human vessel model created with human endothelial cells in a gel matrix.	^{56}Fe more damaging than ^1H. ^{56}Fe- significant reduction in the length of intact vessels in both developing and mature vessels. ^1H-no effect on mature vessels but inhibited vessel formation. Inhibition of vessel development or deterioration of mature vessels was not due to apoptosis.	(Grabham et al. 2011)Grabham et al. 2011.